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Preface

The study of the psychological aspects of sports is the focus of the emerging field of sports psychology. It is a distinct field with its own theories, approaches to doing research, issues, and procedures. Due to its emphasis on comprehending the teaching and learning processes that take place in formal situations, educational psychology differs from other branches of psychology.

The goals of educational psychology include the children's righteous behaviour and healthy development. It strives to foster "continuous growth" and "wholesome personality development." It seeks to support the instructor in his duty of assisting the student in developing a harmonious personality by offering facts and generalisations. The psychological underpinnings of different teaching pedagogies, educational innovations and experiments, educational goals, mental health and hygiene, and special accommodations for slow learners, gifted, disabled, and underprivileged kids are all productively used to educational development.

As a result, they are integral to educational psychology. The educational psychologists conduct research investigations and experiments in a variety of fields with the use of psychological tools, techniques, methodologies, and approaches. They also create new strategies and procedures for gathering, interpreting, and analysing data. These things together make up educational psychology's domain.

The study of the relationship between psychological factors and athletic and physical performance is known as sport psychology. This book's overarching goal is to familiarise readers with psychological theory and practical abilities that affect athletic performance.

The fields of psychology and sports science are combined in this book. Aspiring grads have a variety of options for their educational and professional paths. Sports psychology job prospects include coaching, teaching, research, counseling/therapy, and other related fields.

Author

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Sports Psychology

INTRODUCTION

Sport psychology is the study of a person's behaviour in sport. It is also a specialization within the brain psychology and kinesiology that seeks to understand psychological/mental factors that affect performance in sports, physical activity, and exercise and apply these to enhance individual and team performance. It deals with increasing performance by managing emotions and minimizing the psychological effects of injury and poor performance.

HISTORY

The study of psychology in philosophical context dates back to the ancient civilizations of Egypt, Greece, China, India, and Persia. Historians point to the writings of ancient Greek philosophers, such as Thales, Plato, and Aristotle, as the first significant body of work in the West to be rich in psychological thought.

STRUCTURALISM

German physician Wilhelm Wundt is credited with introducing psychological discovery into a laboratory setting. Known as the "father of experimental psychology", he founded the first psychological laboratory, at Leipzig University, in 1879. Wundt focused on breaking down mental processes into the most basic components, starting a school of psychology that is called structuralism. Edward Titchener was another major structuralist thinker.

Functionalism

Functionalism formed as a reaction to the theories of the structuralist school of thought and was heavily influenced by the work of the American philosopher and psychologist William James. James felt that psychology should have practical value, and that psychologists should find out how the mind can function to a person's benefit. Major functionalist thinkers included John Dewey and Harvey Carr. Other 19th-century contributors to the field include the German psychologist Hermann Ebbinghaus, a pioneer in the experimental study of memory, who developed quantitative models of learning and forgetting at the University of Berlin; and the Russian-Soviet physiologist Ivan Pavlov, who discovered in dogs a learning process that was later termed "classical conditioning" and applied to human beings. Starting in the 1950s, the experimental techniques set forth by Wundt, James, Ebbinghaus, and others would be reiterated as experimental psychology became increasingly cognitive—concerned with information and its processing—and, eventually, constituted a part of the wider cognitive science. In its early years, this development had been seen as a "revolution", as it both responded to and reacted against strains of thought—including psychodynamics and behaviourism—that had developed in the meantime.

Psychoanalysis

From the 1890s until his death in 1939, the Austrian physician Sigmund Freud developed psychoanalysis, a method of investigation of the mind and the way one thinks; a systematized set of theories about human behaviour; and a form of psychotherapy to treat psychological or emotional distress, especially unconscious conflict. Freud's psychoanalytic theory was largely based on interpretive methods, introspection and clinical observations. It became very well-known, largely because it tackled subjects such as sexuality, repression, and the unconscious mind as general aspects of psychological development. These were largely considered taboo subjects at the time, and Freud provided a catalyst for them to be openly discussed in polite society. Clinically, Freud helped to pioneer the method of free association and a therapeutic interest in dream interpretation.

Freud had a significant influence on Swiss psychiatrist Carl Jung, whose analytical psychology became an alternative form of depth psychology. Other well-known psychoanalytic scholars of the mid-20th century included psychoanalysts, psychologists, psychiatrists, and philosophers. Among these thinkers were Erik Erickson, Melanie Klein, D. W. Winnicott, Karen Horney, Erich Fromm, John Bowlby and Sigmund Freud's daughter, Anna Freud. Throughout the 20th century, psychoanalysis evolved into diverse schools of thought, most of which may be classed. Psychoanalytic theory and therapy were criticized by psychologists and philosophers such as B. F. Skinner, Hans Eysenck, and Karl Popper. Popper, a philosopher of science, argued that Freud's, as well as Alfred Adler's, psychoanalytic theories included enough ad hoc safeguards against empirical contradiction to keep the theories outside the realm of scientific enquiry. By contrast, Eysenck maintained that although Freudian ideas could be subjected to experimental science, they had not withstood experimental tests. By the 20th century, psychology departments in American universities had become experimentally oriented, marginalizing Freudian theory and regarding it as a "desiccated and dead" historical artifact. Meanwhile, however, researchers in the emerging field of neuro-psychoanalysis defended some of Freud's ideas on scientific grounds while scholars of the humanities maintained that Freud was not a "scientist at all, but... an interpreter."

Behaviourism

In the United States, behaviourism became the dominant school of thought during the 1950s. Behaviourism was founded in the early 20th century by John B. Watson, and embraced and extended by Edward Thorndike, Clark L. Hull, Edward C. Tolman, and later B. F. Skinner. Theories of learning emphasized the ways in which people might be predisposed, or conditioned, by their environments to behave in certain ways. Classical conditioning was an early behaviourist model. It posited that behavioural tendencies are determined by immediate associations between various environmental stimuli and the degree of pleasure or pain that follows.

Behavioural patterns, then, were understood to consist of organisms' conditioned responses to the stimuli in their environment. The stimuli were held to exert influence in proportion to their prior repetition or to the previous intensity of their associated pain or pleasure. Much research consisted of laboratory-based animal experimentation, which was increasing in popularity as physiology grew more sophisticated. Skinner's behaviourism shared with its predecessors a philosophical inclination towards positivism and determinism.

He believed that the contents of the mind were not open to scientific scrutiny and that scientific psychology should emphasize the study of observable behaviour. He focused on behaviour–environment relations and analysed overt and covert behaviour as a function of the organism interacting with its environment. Behaviourists usually rejected or deemphasized dualistic explanations such as "mind" or "consciousness"; and, in lieu of probing an "unconscious mind" that underlies unawareness, they spoke of the "contingency-shaped behaviours" in which unawareness becomes outwardly manifest.

Among the behaviourists' most famous creations are John B. Watson's Little Albert experiment, which applied classical conditioning to the developing human child, and Skinner's notion of operant conditioning, which acknowledged that human agency could affect patterns and cycles of environmental stimuli and behavioural responses. Linguist Noam Chomsky's critique of the behaviourist model of language acquisition is widely regarded as a key factor in the decline of behaviourism's prominence. Martin Seligman and colleagues discovered that the conditioning of dogs led to outcomes that opposed the predictions of behaviourism. But Skinner's behaviourism did not die, perhaps in part because it generated successful practical applications. The fall of behaviourism as an overarching model in psychology, however, gave way to a new dominant paradigm: cognitive approaches.

Humanism

Humanistic psychology was developed in the 1950s in reaction to both behaviourism and psychoanalysis. By using phenomenology, intersubjectivity and first-person categories, the humanistic approach sought to glimpse the whole person—not just the fragmented parts of the personality or cognitive functioning. Humanism focused on fundamentally and uniquely human issues, such as individual free will, personal growth, self-actualization, self-identity, death, aloneness, freedom, and meaning. The humanistic approach was distinguished by its emphasis on subjective meaning, rejection of determinism, and concern for positive growth rather than pathology. Some of the founders of the humanistic school of thought were American psychologists Abraham Maslow, who formulated a hierarchy of human needs, and Carl Rogers, who created and developed client-centred therapy. Later, positive psychology opened up humanistic themes to scientific modes of exploration.

Gestalt

Wolfgang Kohler, Max Wertheimer and Kurt Koffka co-founded the school of Gestalt psychology. This approach is based upon the idea that individuals experience things as unified wholes. This approach to psychology began in Germany and Austria during the late 19th century in response to the molecular approach of structuralism. Rather than breaking down thoughts and behaviour to their smallest element, the Gestalt position maintains that the whole of experience is important, and the whole is different than the sum of its parts. Gestalt psychology should not be confused with the Gestalt therapy of Fritz Perls, which is only peripherally linked to Gestalt psychology.

Existentialism

Influenced largely by the work of German philosopher Martin Heidegger and Danish philosopher Søren Kierkegaard, psychoanalytically trained American psychologist Rollo May pioneered an existential breed of psychology, which included existential therapy, in the 1950s and 1960s. Existential psychologists differed from others often classified as humanistic in their comparatively neutral view of human nature and in their relatively positive assessment of anxiety. Existential psychologists emphasized the humanistic themes of death, free will, and meaning, suggesting that meaning can be shaped by myths, or narrative patterns, and that it can be encouraged by an acceptance of the free will requisite to an authentic, albeit often anxious, regard for death and other future prospects.

Austrian existential psychiatrist and Holocaust survivor Viktor Frankl drew evidence of meaning's therapeutic power from reflections garnered from his own internment, and he created a variety of existential psychotherapy called logotherapy. In addition to May and Frankl, Swiss psychoanalyst Ludwig Binswanger and American psychologist George Kelly may be said to belong to the existential school.

Cognitivism

Cognitive psychology is the branch of psychology that studies mental processes including how people think, perceive, remember, and learn. As part of the larger field of cognitive science, this branch of psychology is related to other disciplines including neuroscience, philosophy, and linguistics. Noam Chomsky helped to ignite a "cognitive revolution" in psychology when he criticized the behaviourists' notions of "stimulus", "response", and "reinforcement", arguing that such ideas—which Skinner had borrowed from animal experiments in the laboratory—could be applied to complex human behaviour, most notably language acquisition, in only a vague and superficial manner. The postulation that humans are born with the instinct or "innate facility" for acquiring language posed a challenge to the behaviourist position that all behaviour is contingent upon learning and reinforcement.

Social learning theorists such as Albert Bandura argued that the child's environment could make contributions of its own to the behaviours of an observant subject. Meanwhile, accumulating technology helped to renew interest and belief in the mental states and representations—*i.e.,* the cognition—that had fallen out of favour with behaviourists. English neuroscientist Charles Sherrington and Canadian psychologist Donald O. Hebb used experimental methods to link psychological phenomena with the structure and function of the brain. With the rise of computer science and artificial intelligence, analogies were drawn between the processing of information by humans and information processing by machines.

Research in cognition had proven practical since World War II, when it aided in the understanding of weapons operation. By the late 20th century, though, cognitivism had become the dominant paradigm of mainstream psychology, and cognitive psychology emerged as a popular branch. Assuming both that the covert mind should be studied and that the scientific method should be used to study it, cognitive psychologists set such concepts as "subliminal processing" and "implicit memory" in place of the psychoanalytic "unconscious mind" or the behaviouristic "contingency-shaped behaviours". Elements of behaviourism and cognitive psychology were synthesized to form the basis of cognitive behavioural therapy, a form of psychotherapy modified from techniques developed by American psychologist Albert Ellis and American psychiatrist Aaron T. Beck. Cognitive psychology was subsumed along with other disciplines, such as philosophy of mind, computer science, and neuroscience, under the umbrella discipline of cognitive science.

ROLE OF SPORTS PSYCHOLOGY

The specialised field of sports psychology has developed rapidly in recent years. The importance of a sports psychologist as an integral member of the coaching and health care teams is widely recognised. Sports psychologists can teach skills to help athletes enhance their learning process and motor skills, cope with competitive pressures, fine-tune the level of awareness needed for optimal performance, and stay focused amid the many distractions of team travel and in the competitive environment. Psychological training should be an integral part of an athlete's holistic training process, carried out in conjunction with other training elements. This is best accomplished by a collaborative effort among the coach, the sport psychologist, and the athlete; however, a knowledgeable and interested coach can learn basic psychological skills and impart them to the athlete, especially during actual practice.

THE MEDICAL STAFF AND PSYCHOSOMATIC DISORDERS

The health professional often plays a major role in supporting the emotional health of athletes. An athlete's psychological stresses may be manifested as somatic complaints, such as sleep disturbances, irritability, fatigue, gastrointestinal disturbances, muscle tension, or even injury. Athletes often turn to a therapist or physician for relief, either because they do not recognise the psychological basis of the physical complaint, or because they fear the services of a mental health practitioner due to the perceived stigma, or because no psychologist is available. Therapists must be aware of the possibility of an underlying psychological basis for a complaint and enquire into the emotional status of the athlete as part of the medical history. Careful, non-judgemental questioning may reveal inter-personal problems with a coach, teammate, family member, or other individuals, or anxiety concerning an upcoming competition. In these situations, a sports psychologist is invaluable. If none is available, the physician or therapist may need to assume the role of sounding board, intermediary, or stress-management advisor. At times, being a patient listener and confidant may be all that is required. If mediation between parties is required, a neutral, non-judgemental stance must be maintained to help the parties air and resolve differences.

Preparing for Competition

Simple psychological skills to help the athlete manage the competitive performance environment include:

- Learning relaxation skills; slow, controlled, deep abdominal breathing; or autogenic training;
- Mastering all of the attentional styles;
- Imagery;
- Appropriate self-talk; and

- Developing a precompetition mental routine to be employed immediately prior to competition on game day.

The Injured Athlete

Athletes have a strong sense of body awareness, and take great pride in the capabilities of their bodies. Thus, injuries can be psychologically as well as physically devastating. The ability to train and compete well involves enormous ego. Athletes often identify themselves by who they are as an athlete. Thus, an injury places considerable stress on this self-identification. The more severe the injury, and the longer the recovery-rehabilitation period, the more prolonged and profound the mood disturbance may be.

Injured athletes commonly experience at least three emotional responses: isolation, frustration, and disturbances of mood:

1. The injury forces the athlete to become separated from teammates and coaches. Other team members may provide little support, and in fact they may shun their injured teammate to avoid reminders of their own potential frailty.
2. The athlete becomes frustrated because he or she perceives the loss of months of training and skills mastery, although there are many instances where athletes have used the recovery period to master mental and other physical skills to return successfully to competition.
3. Mood disturbances are common. The athlete may be temporarily depressed, or become upset by minor annoyances.

SPORTS: PSYCH-DOWN TECHNIQUES

It's natural to feel some increase in your intensity in a competition. You're putting yourself to the test and want to do your best. But when that increase intensity turns to anxiety that can hurt your performances, that can be a problem. But rather than just resigning yourself to feeling nervous and performing poorly, you can take active steps to reach and maintain your prime intensity so you can performing your best.

There are a number of simple "psych-down" techniques you can use to get your intensity back under control. Deep breathing. When you experience overintensity, one of the first things that's disrupted is your breathing. It becomes short and choppy and you don't get the oxygen your body needs to perform its best. The most basic way to lower your intensity then is to take control of your breathing again by taking slow, deep breaths.

Deep breathing has several important benefits. It ensures that you get enough oxygen so your body can function well. By getting more oxygen into your body, you will relax, feel better, and it will give you a greater sense of control. This increased comfort will give you more confidence and enable you to more easily combat negative thoughts (which are often the cause of the overintensity). It will also help you let go of negative emotions such as fear or frustration, and allow you to regain positive emotions such as excitement. Focusing on your breathing also acts to take your mind off of things that may be interfering causing your overintensity.

For athletes who participate in sports that involved a series of short performances, such as baseball, football, tennis, and golf, deep breathing should be a part of your between-performance routine. One place in particular where deep breathing can be especially valuable to reduce intensity is before you begin another performance. If you take two deep breaths at this point, you ensure that your body will be more relaxed, comfortable, and prepared for the upcoming performance.

Muscle relaxation. Muscle tension is the most common symptom of overintensity. This is the most crippling physical symptom because if your muscles are tight and stiff, you simply won't be able to perform at your highest level. There are two muscle-relaxation techniques you can use away from your sport and, in a shortened form, during competitions: passive relaxation and active relaxation. Similar to deep breathing, muscle relaxation is beneficial because it allows you to regain control of your body and to make you feel more comfortable physically. It also offers the same mental and emotional advantages as does deep breathing.

Passive relaxation involves imagining that tension is a liquid that fills your muscles creating discomfort that interferes with your body performing its best. To prepare for passive relaxation, lie down in a comfortable position in a quiet place where you won't be disturbed. As you go through the passive relaxation procedure, focus on your breathing, allow the tension to drain out of your muscles, and, at the end, focus on your overall state of mental calmness and physical relaxation.

Active relaxation is used when your body is very tense and you can't relax your muscles with passive relaxation. When your intensity is too high and your muscles are tight, it's difficult to just relax them. So instead of trying to relax your muscles, do just the opposite. Tighten them more, then relax them.

For example, before a competition, your muscle tension might be at an 8, where 1 is totally relaxed and 10 is very tense, but you perform best at a 4. By further tightening your muscles up to a 10, the natural reaction is for your muscles to rebound back past 8 towards a more relaxed 4. So, making your muscles more tense at first then results in them becoming more relaxed.

Active relaxation typically involves tightening and relaxing four major muscle groups: face and neck, arms and shoulders, chest and back, and buttocks and legs. It can also be individualized to focus on particular muscles that trouble you the most.

To prepare for passive relaxation, lie down in a comfortable position in a quiet place where you won't be disturbed. For each muscle group, tighten your muscles for five seconds, release, and repeat. As you go through the active relaxation procedure, focus on the differences between tension and relaxation, be aware of how you are able to induce a greater feeling of relaxation, and, at the end, focus on your overall state of mental calmness and physical relaxation. These two relaxation procedures can also be used during a competition (for those sports comprised of a series of short performances) in an abbreviated form. Between performances, you can stop for five seconds and allow the tension to drain out of tense parts of your body (passive relaxation) or tighten and relax the tense muscles (active relaxation).

Slow the pace of competition. A common side effect of overintensity is that athletes tend to speed up the tempo of competition. Athletes in sports such as tennis, golf, baseball, and football can rush between performances almost as if they want to get the competition over with as soon as possible. So, to lower your intensity, slow your pace between performances. Simply slowing your pace and giving yourself time to slow your breathing and relax your muscles will help you lower your intensity to its prime level.

Process focus. One of the primary causes of overintensity is focusing on the outcome of the competition. If you're worried about whether you will win or lose, you're bound to get nervous. The prospect of losing is threatening, so that will make you anxious. The thought of winning, especially if it's against an opponent you have never defeated before, can also be anxiety provoking because it may be unfamiliar or unexpected to you.

To reduce the anxiety caused by an outcome focus, redirect your focus onto the process. Ask yourself, what do I need to do to perform my best? This process focus can include paying attention to your technique or tactics. Or it might involve focusing on mental skills such as positive thinking or the psych-down strategies I am currently describing. You can also shift your focus onto your breathing which will take your mind off of the outcome and will directly relax your body by providing more oxygen to your system.

A process focus takes your mind off things that cause your over-intensity and shifts your focus onto things that will reduce your anxiety, build your confidence, and give you a greater sense of control over your sport.

Keywords. Another focusing technique for lowering your intensity is to use what I call intensity keywords, such as calm, easy, and relax. These words act as reminders of what you need to do with your intensity to perform your best. Keywords are especially important in the heat of a tight competition when you can get so wrapped up in the pressure that you forget to do the things you need to do in order to perform your best.

By saying the keyword between performances, you'll be reminded to use the psych-down techniques when your intensity starts to go up. I also recommend that you write one or two keywords on a piece of tape which you then put on a piece of your equipment. Looking at the equipment acts as a further reminder to follow the keyword and lower your intensity.

Music. Music is one of the most common tools athletes in many sports use to control their intensity. We all know that music has a profound physical and emotional impact on us. Music has the ability to make us happy, sad, inspired, and motivated. Music can also excite or relax us. Many world-class and professional athletes listen to music before they compete to help them reach their prime intensity.

Music is beneficial in several ways. It has a direct effect on you physically. Calming music slows your breathing and relaxes your muscles. Simply put, it makes you feel good. Mentally, it makes you feel positive and motivated. It also generates positive emotions such as joy and contentment. Finally, calming music takes your mind off of aspects of the competition that may cause doubt or anxiety. The overall sensation of listening to relaxing music is a generalized sense of peace and well-being.

Smile. The last technique for lowering intensity is one of the strangest and most effective I've ever come across. A few years ago, I was working with a young professional athlete who was having a terrible practice session. She was performing very poorly and her coach was getting frustrated with her. She approached me during a break feeling angry and depressed, and her body was in knots. She asked me what she could do. I didn't have a good answer until an idea just popped into my head. I told her to smile. She said, I don't want to smile. I told her to smile. She said she was not happy and didn't want to smile. I told her again to smile. This time, just to get me off her back, she smiled.

I told her to hold the smile. During the next two minutes there was an amazing transformation. As she stood there with the smile on her face, the tension began to drain out of her body. Her breathing became slow and deep. She said that she was feeling better. In a short time, she was looking more relaxed and happier. She returned to practice, her performance improved, and she made some progress during the remainder of the practice session.

Her response was so dramatic that I wanted to learn how such a change could occur. When I returned to my office, I looked at the research related to smiling and learned two things. First, as we grow up, we become conditioned to the positive effects of smiling. In other words, we learn that when we smile, it means we're happy and life is good. Second, there's been some fascinating research looking at the effects of smiling on ourbrain chemistry. What this research has found is that when we smile, it releases brain chemicals called endorphins which have an actual physiologically relaxing effect.

For all of these psych-down techniques to be effective, you should rehearse them in practice and less important competitions.

The goal is to ingrain them so well that when you get to a major competition where you are likely to feel nervous, you will automatically use them, your intensity will decrease to a more comfortable level, and you will be better prepared to performance your best.

THE HUMANISTIC MODEL IN SPORTS PSYCHOLOGY

"Man is the measure of all things", reasoned the philosophers of the Renaissance. "Be all that you can be", reasons the U.S. Army of today. In either case, the emphasis is on celebrating the individual, and the development of the *whole* person.

For that reason, this is sometimes known as "Person-Centered" theory. The goal of the Humanistic model is for every person to become their absolute best, whatever that may be. It accepts as fact that we cannot all excel at everything we attempt, but every one of us will excel at some things. We are viewed as strong, capable people with the innate ability to grow, develop, and resolve our own problems. The counselor is expected to be compassionate, a sensitive listener, and helpful as the subject works through their difficulties. *Reality* is seen not as an objective situation that can be measured or quantified, but as a condition that is somewhat different for each person. Thus, two teammates, involved in the same game, having statistically the same performance, may still have experienced a different *reality*. One may be confident that they have succeeded marvelously, while the other feels like an abject failure. Humanistic psychology endeavors to help people define their reality more clearly, and in a way that will help them feel good, and perform at a high level.

At the forefront of the Humanistic movement was Carl Rogers, though ideas from Abraham Maslow and Karen Horney are incorporated in the process. Like the other models, later theorists branched off from the main model to emphasize certain aspects of the model. The *Gestalt* branch is named for the German word which means "whole". It looks at the whole person, and believes that problems occur when we become separated from important parts of ourselves. Fritz and Laura Perls and Erving and Miriam Polster were the primary developers of this branch. The *Existential* branch is sometimes considered to have picked up where Carl Rogers left off, and has a broad range of proponents, including authors (Camus, Sartre, Kafka) and philosophers (Kierkegaard, Nietzsche), but the therapeutic approach owes its development to Victor Frankl, Rollo May, and Irvin Yalom.

In Humanistic psychology, everything that we *are* contributes to our thoughts and feelings about everything we encounter. Physically, we may be in seemingly perfect *homeostasis*, while emotionally or mentally we are a wreck. The object is to help people become *self-actualized* - capable of achieving their complete best in every possible way. With this model, we are seeking internal changes that will have external results - inner changes in thoughts and feelings that result in positive changes in behaviour.

This theory teaches that all people are unique and special, and therefore not easily studied or categorised (as Behaviourists might think). You are not a lab rat or guinea pig, but are expected to be an active participant in your entire therapy process. Because it takes into account the whole person, the Humanistic model is more concerned with the overall quality of a person's life than with statistics or other measureable data. Unfortunately, this also means that many other schools of thought downplay the value of the Humanistic model, because it does not lend itself well to the kinds of measurement scientific study usually demands.

When Carl Rogers began Humanistic psychology, he was, like most who delve into the human mind, heavily influenced by his own personal history and values. As he grew older, he changed from a person with strict "rules" for behaviour and a highly judgemental attitude about others' motivations and values to a caring, compassionate individual, seeking out the good in others and sincerely trying to help them live better lives. He was renowned as an excellent listener, and learned to carefully sort out the mixed and overwhelming emotions some folks were experiencing to help them find the keys they needed to repair their dysfunctions.

Rogers claimed that he could see the need for *self-actualisation* even in small children as early as when they were learning to walk. He saw it as one of the earliest examples of our inner desire to reach for the full extent of our potential skills. Why lie still if you can crawl? -crawl if you can stand? -stand if you can walk? -walk if you can run? Little children are constantly changing because, as Rogers would explain it, they are constantly moving towards *self-actualization*, and the changes are obvious because they have so far to go. As we become older, we begin to think that we have conquered all of our problems and accomplished all our skills, but the truth is that they have just become more complex and difficult to figure out. If we continue to strive for *self-actualization*, we will have a goal that will carry on throughout our lives.

According to Humanists, it is our concept of *Self* that most influences our behaviours. Our *Self* develops as we interact with our environment and struggle to maintain what we have achieved while moving on to conquer more. Happiness, then, is the result of how closely our ideal (best) *Self* corresponds to our behaving *Self.* In other words, if we always imagine ourself serving an ace to win a key match, but in actual competition our serve always goes "out", we are unhappy. If the real-life serve does become an ace, we are very happy, because our ideal matches the reality.

It is also shown that we are happier when our *locus of evaluation* is internal rather than external. When we have our own standards of performance, we know exactly what we need to do in order to feel fulfilled. As long as we set our goals realistically and give an appropriate effort to accomplish them, we can find satisfaction with our *Self*. If we attempt to find satisfaction through *social comparison*however, we are usually likely to end up unhappy and dissatisfied, because we have looked to external motives, which are much more inconsistent.

Humanism also contributes to the Cognitive concept of *achievement orientation*, by showing that *task orientation* is usually more satisfying than *ego orientation*. When we focus on "trying hard" and "getting better" in our sport, we are more likely to succeed according to our own measurements than we are if our only emphases are on *ego-oriented* standards like "winning" and "outperforming that other player". One of the basic problems with *ego orientation*, as pointed out by this, is that we can achieve our goals yet still end up unsatisfied because we have left out the "how" the method. (For example, if "win the game" is your only goal, and the team wins but you fumble 6 times, it's pretty unlikely that you'll feel fully satisfied.) It is in the approach to sports that the Humanistic model is most different from other models. To achieve true *self-actualization*, the quality of our life must almost continually improve. Sport is seen as a wonderful vehicle towards this end, if we use it as an expression of our joy at being alive. By playfully expressing our happiness, and through balancing the technical and expressive aspects of our sport, we get self-fulfillment and a liberation of our spirit. A significant study (Fitts) shows that when two athletes of nearly equal ability compete, the one with the better *self-concept* nearly always performs better, because they have more positive expectation, fewer negative emotions, and less anxiety. Humanists would say that those athletes are *experiencing* life much more fully.

A really radical concept is Humanism's view of contests as a completely "win-win" proposition, where everyone profits from the competition. First, is the idea that if we compete primarily against our own past achievements and strive for our *personal best*, we will be more *self-actualized* and therefore happier than other athletes, regardless of the outcome of the game. (As an added plus, imagine the outcome of a game in which your entire team reached its personal best!)

The game itself is viewed as a cooperative activity, in which the players are all working together so that everyone will perform their best. The reasoning is that none of us can create our best possible performance alone, otherwise we would simply play by ourselves all the time. Those of us who have competed know that games have much more intensity and a much higher level of technical performance than practices or pick-up games. It is for that reason that so many coaches try to make practices as game-like as possible. If we wish to reach our full potential, we actually require a capable opponent to draw that performance out of us. In an environment regulated by rules and officials, each participant reaches to develop their full potential. In this larger sense, every person present at the game is an agent in helping you become the most successful player you can possibly be.

Several Humanistic concepts have been applied closely to sports situations:

- The idea of *Peak Experience* comes from this theory, and is applicable to many aspects of life. When we are fully focused on the task at hand, firmly connected to our present, feeling a sense of control over ourselves and our environment, and feel that we are transcending our self-imposed limits, we are having a *peak experience.* Thus, when we are having the feeling of a creative outburst, spiritual epiphany, the euphoria of love, or sensory maximization, we are having a *peak experience* in other areas of our life.

- During a *peak experience*, we may be fortunate enough to reach a *Personal Best*. To reach this level of performance, several elements must be in place: we must be clearly focused on our task; we must have an intrinsic interest in success; we must *intend* to perform at a high level and act strongly on our intentions; we must be absorbed in our personal involvement; we must spontaneously adapt to activity situations and change our strategy in response to game conditions; and we must experience a superior performance.
- Mihaly Csikszentmihalyi developed the concept of *Flow*, to describe the feeling we experience when "everything comes together." It is the ultimate moment, when all our physical, intellectual, and emotional skills are fluid, coordinated, and operating at peak efficiency. Other theorists have used terms like "The White Moment" to describe such times, but it is an experience no one easily forgets. Unfortunately, *flow* is easier to describe than it is to duplicate. In fact, there is no evidence that any athlete, no matter how accomplished, can summon *flow* at will. We can, however, learn to recognise *flow*and perhaps help it flourish when the time is right.

  There are considered to be elements to flow:
  - A balance between the challenge and our skill level ["I know I can do this."];
  - Smooth coordination of body and mind with our environment ["I feel fluid and totall aware."];
  - Knowledge of exactly what needs to be done, minute-by-minute ["I can see everything that's going to happen."];
  - Clear, objective feedback ["I need to do that more smoothly next time."];
  - Unwavering concentration ["It's as though it's only me and the ball."];
  - An effortless sense of control over our environment ["Everything is going exactly how I intended it to."];
  - The loss of all self-consciousness ["I'm insulated from everything."];
  - There are distortions in time or other measurements ["It's like everything is happening in slow motion. The game is almost over so fast? The hoop seems so big tonight!"];
  - Autotelic Experience ["This is worth doing just for the fun of it!"].
- Humanists recognise some truisms about *arousal* too. Athletes who are most often successful are the ones who are most able to attain their personal preferred level of *arousal* before, during, and after competition. Additionally, we can use Humanistic techniques to discover our preferred emotional state, by asking, "How did I feel during my best ever performance? How did I feel during my worst?" Then we work on strategies that will put us in the proper state.

As with all models, there are criticisms of Humanistic theory too. One of the biggest ones is that it is *qualitative* rather than *quantitative*. Since the goal is to get you to have the highest *quality* experience, Humanists believe that your interpretation of your experience is the most valid way to measure your success. More scientific thinkers prefer specific data that can be recorded, measured, and tested. Since*flow* and *peak experience* cannot be measured without the subjective view of the individual to which it occurred, many scientists consider the data invalid. Further, some criticize Humanism because subjects are expected to help themselves, while the counselor acts mostly as a companion, rather than a guide or director as other models suggest.

The reality is that most of us would like to believe that we are rational, normal people who can solve our own problems if we are given the proper tools to do so. (And statistically, most of us are.) Humanism suggests ways that we can take charge of the situations we face, and our thoughts and emotions in facing them. By doing so, we can overcome obstacles that prevent us from being our very best, and succeed in ways we previously had not thought possible.

AGGRESSION IN SPORT

The relationship between sport and aggression has been studied extensively for decades, yet investigators still have only an incomplete understanding of the link between the two. That there is a link seems certain, and researchers in various disciplines continue trying to refine their understanding of it in ways that will illuminate both sport and society. In the first half of the 20th century, many psychologists assumed that participation in sports might allow individuals to vent their aggressive tendencies. Generally, these assumptions arose from the view that aggression is an internal drive based on frustration and/or instinct.

However, more recent research shows the opposite-participation in sports is likely to increase an individual's aggression. Sport psychologists distinguish between hostile and instrumental aggression. The primary purpose of hostile aggression is to inflict physical or psychological injury on another; the main aim of instrumental aggression is to attain an approved goal, such as winning a game. These two forms of aggression can be distinguished clearly in most sport situations, although not necessarily in extreme contact sports such as boxing and ice hockey. Recent research suggests that instrumental aggression in sport may spill over into hostile aggression outside of sport, for example, male athletes involved in sexual assault against women.

BACKDROP

Some argued that sport developed as a constraint on aggression, or at least as a means to channel aggression into culturally acceptable forms. Others have contended that sports do not necessarily increase aggression, but rather reflect and enhance the dominant values and attitudes of the broader culture. Yet another school of thought has proposed that sport creates a separate moral sphere, distinct from the real world, in which the goal of winning is more important than the rules of the game.

Others consider that when athletes are overly aggressive; they are overconforming to what they see as acceptable within the sport. Display of machismo, playing with pain, or intentionally injuring an opponent may be "grounded in athletes' uncritical acceptance of and commitment to what they have been told by important people in their lives ever since, they began participating in competitive programmes. Where winning is valued above all else, athletes may use aggression to show their total commitment to sport or to winning in sport.

AGGRESSION AND THE INDIVIDUAL

Individuals who participate in sports seem to exhibit higher levels of aggression than those who do not. However, this may be because sports attract people who are naturally more aggressive than non-athletes. Some sports are more likely to be associated with violence and inappropriate aggression. When provoked, for example, participants in contact sports reveal much higher levels of aggression than those in non-contact sports. Research also shows that aggression may give players an edge when used early in a contest, or they may show aggression if they fail in the sport. Other factors also influence aggression during sports events. For example, the presence of officials in organized sports increases the number of fouls since, the athletes assume it is the referees' job to control inappropriate aggression.

Studies of martial arts suggest that sport participation does not necessarily promote aggression. For example, one study showed that among 13-to-17-yearold delinquents, the group that was taught the philosophical elements of Tae Kwon Do-respect for others, maintaining a sense of responsibility, for example-along with the physical component lowered their aggression levels, compared to those who were not taught the philosophy or engaged in activities other than Tae Kwon Do. Aggression and the Group Some scholars have argued that games are models of culturally relevant activities and provide the greatest opportunity to practice and to learn these activities. American football is an unmistakable model of warfare, for example, with its "men in the trenches", "field generals", efforts by teams to move the ball into "enemy territory" and, ultimately, scoring by "invading" the opponent's end zone. Cross-cultural studies too show a positive association between the existence of combative sports and the prevalence of warfare in particular cultures.

Not all sports fit this model, though. Baseball, in contrast, cannot be so directly linked with any single culturally relevant activity, although running, clubbing, and missile throwing were all important activities in human evolutionary history. Aggression is appropriate, even essential for success in war, but what happens to individuals with heightened levels of aggression in peacetime or when there is no active war in which to channel their aggression? Recently, this question-still controversial-has been raised about violence towards women. Several studies indicate that athletes are disproportionately represented among rapists and others who abuse women physically.

Other investigators suggest that sports contribute to male dominance by linking maleness with acceptable aggression while belittling women and their activities. Some researchers believe that athletes are unfairly stereotyped because they are more visible and are typically held to higher standards. Aggression and Fan Violence by sport spectators or fans has become an issue of considerable concern. Soccer hooliganism in Great Britain has received much attention as have violent confrontations between European soccer fans.

What is it about sports that excites spectators to violent aggression? One theory is that it directly results from observing athletes' aggression; another links it to fans' desire to establish their own social identity; a third proposes that spectator violence is a kind of ritual.

Drugs, especially alcohol, are another common element in spectator aggression. During the 1995 U.S. National Football League season, a national audience was treated to a game-long spectacle of fans throwing snowballs and ice at players, coaches, and officials on the field, as well as at each other, during a game between the New York Giants and the San Diego Chargers.

Alcoholic beverages were subsequently banned from Giants Stadium for the next home game to be played there. Drugs have also been implicated in aggressiveness by players. In particular, steroids, usually taken surreptitiously by athletes, appear to heighten aggressiveness.

Aggression, Sport, and Mass Media Instant replays have brought an interesting but chilling phenomenon in modern sport spectatorship. Scoring plays or other exciting or exceptional plays are commonly replayed. But also commonly replayed are tactics that involve exceptional aggressiveness, such as a "good hit" or a particularly devastating down-field block in American football.

This replaying occurs even when the violent moves have little apparent effect on the outcome of the game or the particular play. Aggressive acts that lead to actual violence-fights among players-are frequently replayed or rebroadcast on sports shows. Spectators seem to enjoy exhibitions of aggression and even violence, while players in many sports believe that aggressive play is instrumental in winning. Precisely how sports and aggression are linked is unclear, but that they are linked seems certain. Sports may be one way to teach young people how and when to use violent forms of aggressive behaviour.

Young athletes observe the behaviour of role models and learn from interactions with coaches, parents, and others. This may well have long-lasting consequences for individuals and for society. Can there be sport without increased aggression? Studies suggest that sport could be reformed so that it would not necessarily lead to increases in aggression. Spectators and players both would experience sport in a different way. Nevertheless, it seems likely that sport could be enjoyed without the promotion of inappropriate aggression.

GOAL SETTING

Goal Setting involves establishing specific, measurable and time-targeted objectives. The ideal-taught-is a disposition that causes an intrinsic drive to be delivered in a professional manner. The parameters of professionalism indicate a time continuum-without continuity-but exhaustion and anxiety: an irony. It is a conundrum out mania, to cause professionals to work backward in order to get to the ultimate goal. It is without faith, but grander thought in moments of observation in hopes to serve need; whether it be self or others. As goal-setters, time-targets become real and more so sensational as the end draws near—a sense of urgency falls afoot. The time is precise and specific; in which is set by the goal-setter. Goal setting is a major component of personal development literature. Compulsions, fantasy and dreams that serve an observed inequity-or perhaps it is a personal bout against inequity-is the cause of personality; which can be determined by the value one puts on extrinsic and intrinsic drives usually defined by the individual or goal-setter. Many times the definition of goal-setting comes from free literary expression that serves a needed fulfillment of goals one must fulfill to become satisfied. This expression-written-may bring to hindsight a goal grander that must be made to self in order to fulfill that need, and even desire.

At times satisfaction may come from serving others, and at times it comes from serving the self as a result of personal development. It is true that one can not identify another as being like them; for the other has distinctive traits, skills and abilities as a result of individual sovereignty attributed to individual development. It may be difficult to warrant the cause, but it is not difficult to warrant the need when under marginal disillusionment one becomes focused again. What may come to hindsight as a result of sovereignty is relative according to each individual. In other words: goals are different from person to person.

Effective goals should be tangible, specific, realistic and have a time targeted for completion. There must be realistic plans to achieve the intended goal. For example, setting a goal to go to Mars on a shoe string budget is not a realistic goal, while setting a goal to go to Hawaii as a backpacker is a possible goal with possible, realistic plans. In setting goals that are unrealistic-one may never obtain-loses any sort satisfaction; therefore can not continue on the goal setter’s path. This path may come to an end abruptly; for it can not be achieved before one’s own lifetime. However setting one’s mark on the children of tomorrow may bring hope beyond the mania of unmaintainable goals for one person. Realistically, goals are persistent of an individuals intrinsic drives designed by their general environment-defining socialization.

Of course in a group environment-where two minds are better than one-goals can be obtained much quicker. Work on the theory of goal-setting suggests that it’s an effective tool for making progress by ensuring that participants in a group with a common goal are clearly aware of what is expected from them if an objective is to be achieved. On a personal level, setting goals is a process that allows people to specify then work towards their own objectives-most commonly with financial or career-based goals. However, some say that much of what is currently taught about goal setting is incomplete. Prominent speakers on goal setting such as Jim Rohn or Zig Ziglar have suggested that goal setting is more than writing something down, setting a date and working towards that end. In order to make the success or achievement a lasting value the person must become something different in the process.

There are significant differences in how a person accomplishes a "be" goal versus a "Have" goal. Some people feel that one possible drawback of goal setting is that implicit learning may be inhibited. This is because goal setting may encourage simple focus on an outcome without openness to exploration, understanding or growth. "Goals provide a sense of direction and purpose". Locke *et al.,* examined the behavioral effects of goal-setting, concluding that 90% of laboratory and field studies involving specific and challenging goals led to higher performance than easy or no goals. In business, goal setting has the advantages of encouraging participants to put in substantial effort; and, because every member has defined expectations set upon him or her (high role perception), little room is left for inadequate effort going unnoticed. While some managers would believe it is sufficient to urge employees to 'do their best', Locke and Latham have a clear contradicting view on this. The authors state that people who are told to 'do their best' will not do so. 'Doing your best' has no external referent which implies that it is useless in eliciting specific behaviour.

To elicit some specific form of behaviour from others, it is important that this person has a clear view of what is expected from him/her. A goal is thereby of vital importance because it facilitates an individual in focusing their efforts in a specified direction. In other words; goals canalize behaviour. However when goals are established at a management level and thereafter solely laid down, employee motivation with regard to achieving these goals is rather suppressed. In order to increase motivation the employees not only need to be allowed to participate in the goal setting process but the goals have to be challenging as well. Managers cannot be constantly able to drive motivation and keep track of an employee's work on a continuous basis. Goals are therefore an important tool for managers since, goals have the ability to function as a self-regulatory mechanism that acquires an employee a certain amount of guidance have distilled four mechanisms through which goal setting is able to affect individual performance:

1. Goals focus attention towards goal-relevant activities and away from goal-irrelevant activities.
2. Goals serve as an energizer; higher goals will induce greater effort while low goals induce lesser effort.
3. Goals affect persistence; constraints with regard to resources will affect work pace.
4. Goals activate cognitive knowledge and strategies which allows employees to cope with the situation at hand.

Through an understanding of the effect of goal setting on individual performance organizations are able to use goal setting to benefit organizational performance.

Have therefore indicated three moderators which indicate the success of goal setting:

- *Goal commitment*: People will perform better when they are committed to achieve certain goals. Goal commitment is dependent of:
  - The importance of the expected outcomes of goal attainment and;
  - Self-efficacy-one's belief that they are able to achieve the goals;
  - Commitment to others-promises or engagements to others can strongly improve commitment
- *Feedback*: Keep track of performance to allow employees to see how effective they have been in attaining the goals. Without proper feedback channels it is impossible to adapt or adjust to the required behaviour.
- *Task complexity*: More difficult goals require more cognitive strategies and well developed skills. The more difficult the tasks ahead, a smaller group of people will possess the necessary skills and strategies. From an organizational perspective it is thereby more difficult to successfully attain more difficult goals since, resources become more scarce.
- *Employee motivation*: The more employees are motivated, the more they are stimulated and interested in accepting goals.
- *Macro-economical characteristics*: The position of the economy in the conjucture puts pressure or simply relieves the organization. This means that some goals are easier set in specific macro-economical surroundings. Depression is for instance the least successful conjecturial phase for goal setting.

These success factors are not to be seen independently. For example the expected outcomes of goals are positively influenced when employees are involved in the goal setting process.

Not only does participation increase commitment in attaining the goals that are set, participation influences self-efficacy as well. In addition to this feedback is necessary to monitor one's progress. When this is left aside, an employee might think (s)he is not making enough progress. This can reduce self-efficacy and thereby harm the performance outcomes in the long run.

PERSONAL GOAL SETTING

Goal setting is a powerful process for thinking about your ideal future, and for motivating yourself to turn this vision of the future into reality. The process of setting goals helps you choose where you want to go in life. By knowing precisely what you want to achieve, you know where you have to concentrate your efforts. You'll also quickly spot the distractions that would otherwise lure you from your course. More than this, properly-set goals can be incredibly motivating, and as you get into the habit of setting and achieving goals, you'll find that your self-confidence builds fast.

Achieving More With Focus

Goal setting techniques are used by top-level athletes, successful business-people and achievers in all fields. They give you long-term vision and short-term motivation. They focus your acquisition of knowledge and help you to organize your time and your resources so that you can make the very most of your life. By setting sharp, clearly defined goals, you can measure and take pride in the achievement of those goals. You can see forward progress in what might previously have seemed a long pointless grind. By setting goals, you will also raise your self-confidence, as you recognize your ability and competence in achieving the goals that you have set.

Starting to Set Personal Goals

Goals are set on a number of different levels: First you create your "big picture" of what you want to do with your life, and decide what large-scale goals you want to achieve. Second, you break these down into the smaller and smaller targets that you must hit so that you reach your lifetime goals. Finally, once you have your plan, you start working to achieve it.

Key Points

Goal setting is an important method of:

- Deciding what is important for you to achieve in your life.
- Separating what is important from what is irrelevant, or a distraction.
- Motivating yourself.
- Building your self-confidence, based on successful achievement of goals.

If you don't already set goals, do so, starting now. As you make this technique part of your life, you'll find your career accelerating, and you'll wonder how you did without it!

STRESS MANAGEMENT

A lot of research has been conducted into stress over the last hundred years. Some of the theories behind it are now settled and accepted; others are still being researched and debated. During this time, there seems to have been something approaching open warfare between competing theories and definitions: Views have been passionately held and aggressively defended. What complicates this is that intuitively we all feel that we know what stress is, as it is something we have all experienced. A definition should therefore be obvious…except that it is not.

DEFINITIONS

Hans Selye was one of the founding fathers of stress research. His view in 1956 was that "stress is not necessarily something bad–it all depends on how you take it. The stress of exhilarating, creative successful work is beneficial, while that of failure, humiliation or infection is detrimental.” Selye believed that the biochemical effects of stress would be experienced irrespective of whether the situation was positive or negative. Since, then, a great deal of further research has been conducted, and ideas have moved on.

Stress is now viewed as a “bad thing”, with a range of harmful biochemical and long-term effects. These effects have rarely been observed in positive situations. The most commonly accepted definition of stress is that stress is a condition or feeling experienced when a person perceives that “demands exceed the personal and social resources the individual is able to mobilize.” In short, it’s what we feel when we think we’ve lost control of events.

FIGHT-OR-FLIGHT

Some of the early research on stress established the existence of the well-known “fight-or-flight” response. His work showed that when an organism experiences a shock or perceives a threat, it quickly releases hormones that help it to survive. In humans, as in other animals, these hormones help us to run faster and fight harder. They increase heart rate and blood pressure, delivering more oxygen and blood sugar to power important muscles. They increase sweating in an effort to cool these muscles, and help them stay efficient.

They divert blood away from the skin to the core of our bodies, reducing blood loss if we are damaged. As well as this, these hormones focus our attention on the threat, to the exclusion of everything else. All of this significantly improves our ability to survive life-threatening events. Not only life-threatening events trigger this reaction: We experience it almost any time we come across something unexpected or something that frustrates our goals.

When the threat is small, our response is small and we often do not notice it among the many other distractions of a stressful situation. Unfortunately, this mobilization of the body for survival also has negative consequences. In this state, we are excitable, anxious, jumpy and irritable.

This actually reduces our ability to work effectively with other people. With trembling and a pounding heart, we can find it difficult to execute precise, controlled skills. The intensity of our focus on survival interferes with our ability to make fine judgements by drawing information from many sources. We find ourselves more accident-prone and less able to make good decisions.

There are very few situations in modern working life where this response is useful. Most situations benefit from a calm, rational, controlled and socially sensitive approach. In the short term, we need to keep this fight-or-flight response under control to be effective in our jobs. In the long term we need to keep it under control to avoid problems of poor health and burnout.

MANAGING STRESS

There are very many proven skills that we can use to manage stress. These help us to remain calm and effective in high pressure situations, and help us avoid the problems of long term stress. Keeping a Stress Diary or carrying out the Burnout Self-Test will help you to identify your current levels of stress, so you can decide what action, if any, you need to take. Job Analysis and Performance Planning will help you to get on top of your workload. While the emotionally-oriented skills of Imagery, Physical Techniques and Rational Positive Thinking will help you change the way you see apparently stressful situations. Finally, the stage on Anger Management will help you to channel your feelings into performance. This is a much-abridged excerpt from the 'Understanding Stress and Stress Management' module of the Mind Tools Stress Management Masterclass.

It also discusses:

- *Long-term stress*: The General Adaptation Syndrome and Burnout
- The Integrated Stress Response
- Stress and Health
- Stress and its Affect on the Way We Think
- *Pressure and Performance*: Flow and the 'Inverted-U'

Warning: Stress can cause severe health problems and, in extreme cases, can cause death. While these stress management techniques have been shown to have a positive effect on reducing stress, they are for guidance only, and readers should take the advice of suitably qualified health professionals if they have any concerns over stress-related illnesses or if stress is causing significant or persistent unhappiness. Health professionals should also be consulted before any major change in diet or levels of exercise.

PERFORMANCE PLANNING

We all know the feeling of sickness in our stomach before an important presentation or performance. We have all experienced the sweaty palms, the raised heart rate, and the sense of agitation that we feel as these events approach. We have probably all also experienced how much worse this becomes when things go wrong in the run up to an event. This stage helps you deal with this by helping you to prepare well for future performances. The Thought Awareness, Rational Thinking and Positive Thinking technique that we look at later may be enough to help you manage the fears, anxieties and negative thoughts that may arise in a small performance. For larger events, it is worth preparing a Performance Plan. This is a pre-prepared plan that helps you to deal effectively with any problems or distractions that may occur, and perform in a positive and focused frame of mind.

How to Use the Tool: To prepare your Performance Plan, begin by making a list all of the steps that you need to do from getting prepared for a performance through to its conclusion. Start far enough in advance to sort out any equipment problems.

List all of the physical and mental steps that you need to take to:

- Prepare and check your equipment, and repair or replace it where it does not work;
- Make travel arrangements;
- Pack your equipment and luggage;
- Travel to the site of your performance;
- Set up equipment;
- Wait and prepare for your performance; and
- Deliver your performance.

Next, work through each of these steps.

Think though:

- Everything that could reasonably go wrong at each step with equipment and arrangements; and
- Any distractions and negative thinking that could undermine your confidence or stop you having a positive, focused frame of mind at the start of and during your performance.

Work through all of the things that could go wrong. Look at the likelihood of the problem occurring. Many of the things you have listed may be extremely unlikely. Where appropriate, strike these out and ignore them from your planning. Look at each of the remaining contingencies.

These will fall into three categories:

1. Things you can eliminate by appropriate preparation, including making back-up arrangements and acquiring appropriate additional or spare equipment;
2. Things you can manage by avoiding unnecessary risk; and
3. Things you can manage with a pre-prepared action or with an appropriate stress management technique

For example, if you are depending on using a data projector for a presentation, you can arrange for a back up projector to be available, purchase a replacement bulb, and/or print off paper copies of the presentation in case all else fails. You can leave earlier than strictly necessary so that you have time for serious travel delays. You can also think through appropriate alternatives if your travel plans are disrupted. If you are forced to wait before your event in an uncomfortable or unsuitably distracting place, prepare the relaxation techniques you can use to keep a calm, positive frame of mind. Research all of the information you will need to take the appropriate actions quickly, and ensure that you have the appropriate resources available. Also, prepare the positive thinking you will use to counter fears and negative thoughts both before the event and during it. Use stress anticipation skills to ensure that you are properly prepared to manage stress. Then use thought awareness, rational thinking and positive thinking skills to prepare the positive thoughts that you will use to protect and build your confidence. Write your plan down on paper in a form that is easy to read and easy to refer to. Keep it with you as you prepare for, and deliver, your performance. Refer to it whenever you need it in the time leading up to the event, and during it. Performance Plans help you to prepare for an important performance. They bring together practical contingency planning with mental preparation to help you prepare for situations and eventualities that may realistically occur. This gives you the confidence that comes from knowing you are as well prepared for an event as is practically possible to be. It also helps you to avoid the unpleasant stresses that come from poor preparation, meaning that you can deliver your performance in a relaxed, positive and focused frame of mind, whatever problems or upsets may have occurred. This stage is an abridged version of just one of the techniques used to manage performance stress explained in “Managing Stress for Career Success”, Mind Tools’ Stress Management Masterclass. The ‘Managing Performance Stress’ module explains how to prepare for the event, how to manage negative thinking leading up to it and how to learn lessons from your experience of stress. As well as this, it shows you how to use a range of useful adrenaline management techniques for controlling the anxiety you will inevitably feel just before your performance.

IMAGERY

Sometimes we are not able to change our environment to manage stress–this may be the case where we do not have the power to change a situation, or where we are about to give an important performance. Imagery is a useful skill for relaxing in these situations. Imagery is a potent method of stress reduction, especially when combined with physical relaxation methods such as deep breathing. You will be aware of how particular environments can be very relaxing, while others can be intensely stressful. The principle behind the use of imagery in stress reduction is that you can use your imagination to recreate and enjoy a situation that is very relaxing. The more intensely you imagine the situation, the more relaxing the experience will be. This sounds unlikely. In fact, the effectiveness of imagery can be shown very effectively if you have access to biofeedback equipment. By imagining a pleasant and relaxing scene you can objectively see the measured stress in your body reduce. By imagining an unpleasant and stressful situation, you can see the stress in your body increase. This very real effect can be quite alarming when you see it happen the first time! How to Use the Tool: Two situations where imagery can be very effective are when you’re trying to relax and when you’re preparing or rehearsing for a performance.

Imagery in Relaxation

One common use of imagery in relaxation is to imagine a scene, place or event that you remember as safe, peaceful, restful, beautiful and happy. You can bring all your senses into the image with, for example, sounds of running water and birds, the smell of cut grass, the taste of cool white wine, the warmth of the sun, etc., Use the imagined place as a retreat from stress and pressure. Scenes can involve complex images such as lying on a beach in a deserted cove. You may “see” cliffs, sea and sand around you, “hear” the waves crashing against rocks, “smell” the salt in the air, and “feel” the warmth of the sun and a gentle breeze on your body. Other images might include looking at a mountain view, swimming in a tropical pool, or whatever you want. You will be able to come up with the most effective images for yourself. Other uses of imagery in relaxation involve creating mental pictures of stress flowing out of your body, or of stress, distractions and everyday concerns being folded away and locked into a padlocked chest.

Imagery in Preparation and Rehearsal

You can also use imagery in rehearsal before a big event, allowing you to run through the event in your mind. Aside from allowing you to rehearse mentally, imagery also allows you to practice in advance for anything unusual that might occur, so that you are prepared and already practiced in handling it. This is a technique used very commonly by top sports people, who learn good performance habits by repeatedly rehearsing performances in their imagination. When the unusual eventualities they have rehearsed using imagery occur, they have good, pre-prepared, habitual responses to them. Imagery also allows you to pre-experience achievement of your goals, helping to give you the self-confidence you need to do something well. This is another technique used by successful athletes.

With imagery, you substitute actual experience with scenes from your imagination. Your body reacts to these imagined scenes almost as if they were real, calming you down and letting adrenaline disperse. To relax with imagery, imagine a warm, comfortable, safe and pleasant place, and enjoy it in your imagination. Imagery can be shown to work by using biofeedback devices that measure body stress. By imagining pleasant and unpleasant scenes, you can actually see or hear the changing levels of stress in your body diminish. This is an excerpt from “Managing Stress for Career Success”, the Mind Tools Stress Management Masterclass. Imagery is just one of the important mental relaxation techniques that you learn with this course. Not only does the course show you how to use these techniques, it also explains the sound practical psychology that lies behind them.

PHYSICAL RELAXATION TECHNIQUES

Physical relaxation techniques are as effective as mental techniques in reducing stress. In fact, the best relaxation is achieved by using physical and mental techniques together. These three useful physical relaxation techniques can help you reduce muscle tension and manage the effects of the fight-or-flight response on your body. This is particularly important if you need to think clearly and perform precisely when you are under pressure. The techniques we will look at are Deep Breathing, Progressive Muscular Relaxation and “The Relaxation Response”.

Deep Breathing

Deep breathing is a simple, but very effective, method of relaxation. It is a core component of everything from the "take ten deep breaths" approach to calming someone down, right through to yoga relaxation and Zen meditation. It works well in conjunction with other relaxation techniques such as Progressive Muscular Relaxation, relaxation imagery and meditation to reduce stress. To use the technique, take a number of deep breaths and relax your body further with each breath. That's all there is to it!

Progressive Muscular Relaxation

Progressive Muscular Relaxation is useful for relaxing your body when your muscles are tense. The idea behind PMR is that you tense up a group of muscles so that they are as tightly contracted as possible. Hold them in a state of extreme tension for a few seconds. Then, relax the muscles normally. Then, consciously relax the muscles even further so that you are as relaxed as possible. By tensing your muscles first, you will find that you are able to relax your muscles more than would be the case if you tried to relax your muscles directly. Experiment with PMR by forming a fist, and clenching your hand as tight as you can for a few seconds. Relax your hand to its previous tension, and then consciously relax it again so that it is as loose as possible. You should feel deep relaxation in your hand muscles.

The Relaxation Response

'The Relaxation Response' is the name of a book published by Dr Herbert Benson of Harvard University in 1968. In a series of experiments into various popular meditation techniques, Dr Benson established that these techniques had a very real effect on reducing stress and controlling the fight-or-flight response. Direct effects included deep relaxation, slowed heartbeat and breathing, reduced oxygen consumption and increased skin resistance.

This is something that you can do for yourself by following these steps:

- Sit quietly and comfortably.
- Close your eyes.
- Start by relaxing the muscles of your feet and work up your body relaxing muscles.
- Focus your attention on your breathing.
- Breathe in deeply and then let your breath out. Count your breaths, and say the number of the breath as you let it out.

Do this for ten or twenty minutes. An even more potent alternative approach is to follow these steps, but to use relaxation imagery instead of counting breaths in step 5. Again, you can prove to yourself that this works using the biofeedback equipment. "Deep Breathing," "Progressive Muscular Relaxation," and the steps leading to the "Relaxation Response" are three good techniques that can help you to relax your body and manage the symptoms of the fight-or-flight response. These are particularly helpful for both handling nerves prior to an important performance, and reducing stress generally. This is an excerpt from "Managing Stress for Career Success", the Mind Tools Stress Management Masterclass. These physical relaxation techniques are just some of the important skills that we explain. As well as explaining relaxation techniques, the Stress Management Masterclass shows you how to take action to tackle the root causes of job stress-a side-effect of this approach is that you become more effective and successful in your career.

2

Physical Education

Before embarking on an account of the nature of physical education, and its knowledge and values claims, it is necessary to first take a short detour and second, offer an apology. First, it is necessary–if we are to have a reflective view of the philosophical terrain in which sense can be made of the concept of physical education–to understand a little of the nature of philosophical thinking. Second, the account here is itself situated within a particular tradition of thought. It is not speak of continental philosophy where there might be rich seams indeed for philosophers of physical education to plough. In particular, the works of phenomenologists and hermeneuticists have tremendous potential to offer understandings of our experiences in the activities that comprise physical education. The manner in which this 'new' philosophy took a foothold in the UK and the USA–what came to be known as analytical philosophy of education–was nothing short of remarkable.

The classic UK texts of the 1960s and 1970s bear testimony to it: Dearden's Philosophy of Primary Education, Hirst and Peters' The Logic of Education and Peters' Ethics and Education are paradigmatic. A cursory glance at their contents pages indicates their subject matter. Each philosopher bore down on their subject matter with microscopic linguistic scrutiny; precisely what was meant by concepts so central to education as 'authority', 'democracy', 'discipline', 'initiation', 'knowledge', 'learning', and so on. No educational concepts escaped their analytical scrutiny. A very similar movement was carrying the day in the USA where philosophers of education centrally saw themselves engaged in the same enterprise–and with surprisingly similar results given the cultural and geographical distance that set them apart.

Despite the time that has elapsed since this highly original work, it is genuinely worthwhile to revisit their positions in order to better understand how to think philosophically about physical education as an educational enterprise. In the UK, Richard Peters developed the most powerful statement about the nature of education. In his inaugural lecture in 1965, he put forward a thesis that was to reach literally across the world through the old British Empire–many of whose educational lecturers were still taught in British universities–that education must be viewed, by all those who seriously investigate its nature, to comprise a certain logical geography.

Briefly, his thesis was that education, properly conceived, referred to the initiation of the unlearned into those intrinsically worthwhile forms of knowledge that were constitutive of rational mind. Its shorthand was that education referred essentially to the development of rationality. Despite the hugely influential educational effects of muscular Christianity, physical education enjoyed little more than a Cinderella existence, even in British education, throughout the twentieth century. And now, it was surely not to be invited to the ball. The hegemony of that great thesis cast physical education well and truly into the educational hinterland. we shall now consider that thesis in a little detail. The particular picture of education favoured by analytical philosophers of education, then, is that of the British philosopher of education Richard Peters and, to a lesser extent, his close colleague Paul Hirst. we shall refer to their theses collectively as the Petersian conception of education. It is familiar enough to anyone who read any English language philosophy of education from 1965 to 1985. For Peters, the many uses of the word 'education' might be reduced to the central case and the philosophical task was to tease out criteria implicit in that case.

This led Peters to develop his sophisticated account of education as the transmission of what was intrinsically worthwhile in order to open the eyes of initiates to a vaster and more variegated existence. That same worthwhile knowledge was continuous with the various forms of knowledge that Hirst had delineated by his own set of epistemological criteria. The Petersian thesis was summarised thus: _ 'education' implies the transmission of what is worth-while to those who become committed to it; _ 'education' must involve knowledge and understanding and some kind of cognitive perspective, which are not inert; and _ 'education' at least rules out some procedures of transmission, on the grounds that they lack wittingness and voluntariness on the part of the learner.

The first two conditions have been referred to as the axiological and epistemological conditions by two other philosophers, Andrew Reid and David Carr, both of who have sought to conceptualise physical education in similar ways, but who have come to rather different conclusions about its educational potential. The third criterion refers to the processes by which such transmission was ethically acceptable. It will comment on the analytical and epistemological dimension of Carr's and Reid's articles and then examine the axiological dimension of Reid's work which is the bedrock of his justification for the educational status of physical education. What is of significance in Reid is the idea that education, as conceived in the Petersian mould, is narrow and restrictive. Despite a lack of argumentation, he signifies a broader conception of education than is found in the accounts of philosophers such as Peters, Hirst, and Barrow or, for that matter, anyone housed within the liberal tradition. These philosophers of education conceive of education as the development of individual, rationally autonomous, learners. In their writings they sharply distinguish education from other learning-related concepts such as 'socialisation', 'training' and 'vocation' in terms of their content, scope, value and application.

Reid's conceptualisation of a broader account rests on the position of John White in his book Education and the Good Life where educators aim towards the development of personal well-being grounded in rationally informed desires of both a theoretical and practical kind. Education is thus subservient to, and continuous with, the kinds of development that enable an individual to choose activities, experiences and relationships that are affirmations of those informed choices. By contrast, Carr is more traditional in his account of education and therefore physical education. Like Barrow and Peters before him, he marks the education training distinction by a thesis about mind. For the earlier writers in liberal philosophy of education, all educational activities were broader and richer in scope than mere training which was a form of instruction with limited, focused ends. Education properly conceived, they argued, aimed at something much richer and more variegated. The educated mind did not focus on things limited in scope, such as training for the world of work, but rather helped learners to better understand their world, and their place within it.

As it was often said, education had no specific destination or goal as such; it was rather to travel with a new, enlarged view. Necessarily, this educated view was informed by an initiation into the forms of knowledge or rationality; aesthetic, mathematical, philosophical, scientific, religious, and so on. These were simply what being educated consisted in. Despite the fact that Carr recognises the value of practical as well as theoretical rationality, he undermines Reid's thesis about the importance of physical education conceived of as practical knowledge, and is driven back to the old liberal ground: The key idea here is the traditionalist one that certain forms of knowledge and understanding enter into the ecology of human development and formation–not as theories of a scientist or the skills of a golfer, but as the horizon of significance against which we are able to form some coherent picture of how the world is, our place in it and how it is appropriate for us to relate to others.

Strictly speaking, it matters not a hoot on the traditionalist picture whether such received wisdom is theoretical or practical or located at some point in between; what matters is that there should be–in the name of education–some substantial initiation into this realm of human significance alongside any training in vocational or domestic or merely recreational skills. This is not to deny any proper normative conception of the latter, or that any pursuit of such skills may involve considerable rational judgement and discrimination; it is rather to insist that the sort of rationality they do exhibit may not and need not have anything much to do with education. Very roughly, one might put the point of the liberal-traditionalist distinction between educational and non-educational knowledge by observing that the former is knowledge which informs rather than merely uses the mind.

Thus, Carr's account is little more than a brave leap back to the Petersian position. Now, as with all philosophical argument, one can dispute a position on its own terms, one can deny the presuppositions of those terms or one can either assert or argue for a counter position. The middle option can be seen in any of a legion of writers who attacked the liberal position for its normative presuppositions. Under the banner of ideological neutrality, it seemed to smuggle in an awful lot of values. Moreover, nearly every self-respecting sociologist of education cried that it entailed little more than a crystallisation of the kind of curriculum favoured by British grammar and public schools in the UK over the last 100 years or so. There is a point of considerable agreement between Carr and Reid that is typical of the liberal theory of education, and it is one that is typically used against the educational advocates of physical education. Both writers are keen to hold on to the liberal ideal that education has its own ends. This of course cuts across the grain of 'common sense' thinking that it is the job of education to effect socialisation, or produce a more efficient workforce, and so on. Reid says that a broader view of what education entails–the introduction to cultural resources–must not simply be thought of as the development of qualities of mind: The idea of introduction to cultural resources is to be taken here as an abbreviated way of referring to the complex and lengthy processes associated with the knowledge condition of education, with the teaching and learning which are required for effective appreciation and use of those resources. The sports and games which figure in physical education, then, are to be distinguished from work, the arts, intellectual illumination and so on in terms of their fundamentally hedonic orientation, but not in terms of their role as major cultural institutions, and thus in terms of their educational importance.

It seems, therefore, that Reid is not unhappy with the general model of education as initiation into sports and games as major cultural institutions. As we have seen, Carr parts company with Reid on epistemological issues to do with the development of rational mind, though not only there. Despite recognising the value of such initiation, Carr pejoratively refers to sports and games as merely a valuable part of one's schooling–but, note, not education. Carr's logical geography is restricted to the Petersian-liberal continent. Like many others before him, Reid wants to shift the ground of education away from the development of intellect as the sole basis and look also to a kind of 'pleasure principle'. Reid suggests that the nature and value of physical education is best characterised by a 'fundamentally hedonic orientation'. We shall consider these points in that order.

TERMINOLOGY OF PHYSICAL EDUCATION

The multiplicity of terms that are sometimes used synonymously for physical education makes it imperative that the meaning of these various terms be clarified.

Hygiene. Hygiene comes from the Greek word *hygieinos*, meaning healthful. This refers to the science of preserving one's health. It often refers to rules or principles which are prescribed for the purpose of developing and maintaining health. In past years many school physical education departments were known as departments of hygiene. A few still use this term. It appears that this term became popular as a result of legislation in various states which sought to have the effects of tobacco and alcohol brought to the attention of all students through a course which was often known by the name of hygiene. There are still many laws on the statute books prescribing such instruction. Since World War I, this term has become more or less obsolete. Newer terminology is being used, such as health education and personal and community health.

Physical Culture. The term "physical culture" is an obsolete term in education, having been used in the late nineteenth century to parallel the use of other names of courses which at times were called religious culture, social culture, and intellectual culture. This term is still used by some faddists in commercial ventures to popularize the beneficial effects of exercise. Such men as Bernarr Macfadden have, through their publications and business enterprises, done a great deal spread the use of such terminology. Physical culture has been used synonymously for physical training. It implies that through various physical activities health may be promoted. It is a term, however, which is not in use today in our institutions of learning.

Athletics. The term "athletics" refers to the games or sports which are usually engaged in the individuals who are strong, robust, skilled, and able to participate in vigorous exertions. The interest in athletics in the United States has been largely inherited from Great Britain. With the introduction of athletics into colleges and universities, there has been a rapid growth in all sports which were engaged in on an intercollegiate basis. The first intercollegiate meet was a boat race in 1852 between the crews of Harvard University and Yale University. The first intercollegiate football game is believed to have been played between Rutgers University and Princeton University in 1869. These rivalries still exist today.

Many lay persons frequently think of athletics and physical education as being similar in meaning. However, most physical education personnel think of athletics as one phase of a broad physical education programme, namely, that division of the programme which is concerned with interscholastic or intercollegiate sports competition. A director of athletics in a school has as his primary responsibility the direction of this competitive programme.

Gymnastics. The word "gymnastics" refers to exercises that are adaptable to or are performed in a gymnasium. It is the art of performing various types of physical exercises and feats of skill. The term has been and still is used extensively in the various physical education programmes on the continent of Europe. Anyone trained in physical education has heard mention of such programmes as the German and Swedish systems of gymnastics. Formal drills, such as calisthenics, were until recently utilized extensively in many physical education programmes in the United States. Today, when one thinks of gymnastics, formal drills come to mind which are conducted either with or without the use of apparatus. Americans do not use the term synonymously with physical education but, instead, with just that phase of the physical education programme which is concerned with formal drills. Physical express himself in various types of games rather than through formal drill. This is believed to be more in keeping with the democratic way of life.

Physical Training. The term "physical training" to many individuals has a military tinge. It is a term that has been used in school programmes of physical activity and also a term that has been used in the Armed Forces. Hetherington used the term to connote big-muscle activity in the school programme of physical education. On the other hand, both during World War II and at the present time its use refers to the entire programme of physical conditioning which the Armed Forces require men to go through as preparation for their rigorous duties. Most individuals agree that, because of the military connection, the term is used to imply training. This term has become rather outmoded for the modern-day physical education programmes that are found in the public schools. Today, physical education programmes realize outcomes other than just those that are concerned with the physical aspects. For example, there are sociological outcomes which result in an individual's better adaptation to group living. The term "physical education" also implies that physical activity serves the field of education in a much broader aspect than physical training does.

Physical Education. The term "physical education" is much broader and much more meaningful for day-to-day living than any of those terms that have been discussed previously. It is more closely allied to the larger area of education, of which it is a vital part. It implies that its programme consists of something other than mere exercises done at command. A physical education programme under qualified leadership aids in the enrichment of an individual's life.

Before formulating a definition of physical education, it is interesting to note how a few of the leaders in the field of physical education define this term.

Hetherington listed two things with which physical education is concerned. First, physical education is concerned with big-muscle activity and the benefits which may be derived therefrom and, second, with its contribution to the health and growth of the child so that he may realize as much as possible from the educational process without having growth handicaps.

Nash points out that physical education is one phase of the total education process, and that it utilizes activity drives that are inherent in each individual to develop a person organically, neuromuscularly, intellectually, and emotionally. These outcomes are realized whenever physical education activities are conducted in such places as the playground, gymnasium, and swimming pool.

Nixon and Cozens describe physical education as that phase of the total education process which pertains to vigorous activities involving the muscular system and the learnings that result from participation in these activities.

Sharman points out that physical education is that part of education which takes place through activities which involve the motor mechanism of the human body and which results in the individual's formulating behaviour patterns.

Williams, Brownell, and Vernier point to the fact that physical education implies selected physical activities which are conducted with reference to the benefits that may be derived from participation in these activities.

Voltmer and Esslinger refer to physical education as that phase of education which takes place through physical activity.

From these various definitions of physical education, it can be seen that any definition of the term should incorporate such concepts as selected physical activities and related learnings that are realized through participation in these activities and should show that it is a part of the educational process.

The author proposes the following as a definition of physical education:

Physical education, an integral part of the total education process, is a field of endeavor which has as its aim the development of physically, mentally, emotionally, and socially fit citizens through the medium of physical activities which have been selected with a view to realizing these outcomes.

In a larger sense, this definition of physical education means that the leadership in this field must develop a programme of activities in which participants will realize results beneficial to their growth and development; that they will develop, through participation, such physical characteristics as endurance, strength, and the ability to resist and recover from fatigue; that neuromuscular skill will become a part of their motor mechanism so that they may have proficiency in performing physical acts; that, socially, they will become educated to play an effective part in democratic group living; and that they will be better able to interpret new situations in a more meaningful and purposeful manner as a result of these physical education experiences.

MEANING

Physical Education is the process by which changes in the individual are bought about through movements' experiences. Physical Education aims not only at physical development but is also concerned with education of the whole person through physical activities. It would not be wrong if we say that Physical education is the play-way method of education.

Various Definitions of Physical Education are:

- Barrow defined Physical Education as an education of and through human movement where many of educational objectives are achieved by means of big muscle activities involving sports, games, gymnastic, dance and exercise.
- According to Webster's Dictionary Physical education is a part of education which gives instructions in the development and care of the body randing from simple callisthenic exercises to a course of study providing training in hygiene, gymnastics and the performance and management of athletics games.
- Jackson R. Sharman points out that physical education is that part of education which takes place through activities, which involves the motor mechanism of human body which results in an individual's formulating behaviour patterns.
- Charles A. Bucher defines physical education, an integral part of total education process, is a field of endeavour which has as its aim the development of physically, mentally, emotionally and socially fit citizens through the medium of physicalactivities which have been selected with a view to realizing these outcomes."
- Central Advisory Board of physical Education and Recreation defines Physical education as an education through physical activities for the development of total personality of the child to its fullness and perfection in body, mind and spirit.

AIMS AND OBJECTIVES OF PHYSICAL EDUCATION

AIMS

The ultimate goal is final end aim of physical education, it the way and achieved some certain objectives. The general education like aim of physical education is to develop human personality in its totality well planned activity programmes. In physical education aim at the all-round development of the personality of an individual or some development of human personality and it includes physical, mental, social, emotional and moral aspects to make an individual a good citizen. The physical education means at making an individual physical fit, mentally alert, emotionally balanced, socially well adjusted, morally true and spiritually uplifted.

The objectives of Physical Education are steps considered towards the attainment of the aim and the particular and precise means employed to realize an aim. An aim is achieved it become an objective in the action that goal on continuing. There some objectives of physical education are the objective of physical fitness is essential to leading a happy, vigorous and abundant life. The second is the objective of social efficiency is concerned with one proper adaptation to group living and all these qualities help a person to make him a good citizen. Another objectives of physical education is culture, it aims at developing an understanding and appreciation of one own local environment as well as the environment and a person understand the history, culture, religious practices etc and the aesthetic values associated with these activities. So find the best step the ultimate goal in aims of physical education.

OBJECTIVES

Physical education is an important part of every school curriculum and a class every pupil awaits. Physical education is that segment of the daily timetable that every student eagerly waits to attend, as it is the only official time when the students can be on the grounds, engaged in their favourite sports. One of the main objectives of physical education is to bring in this element of joy to the academic orientation of schools.

Physical education aims at dedicating a daily time for some physical activity for the students. The physical training class, as it is also called, involves sports, games, exercise and most importantly, a break from the sedentary learning indoors. One of the other important objectives of physical education is to instill in the students the values and skills of maintaining a healthy lifestyle. Daily physical activity promotes an awareness of health and well being among students. It boosts them to engage in physical activities on a daily basis.

It promotes them to lead a healthy life in adulthood. Physical education classes constitute programmes to promote physical fitness in students, train them in sports, help them understand rules and strategies in playing and teach them to work as a team. A very vital factor in physical education is to develop interpersonal skills in children. Sports aim at making them team players, developing a sportsman spirit in them and enhancing their competitive spirit. Sports that form a part of physical education classes help the students invest time in fruitful and competitive activities.

One of the other important objectives of physical education is to inculcate in the minds of the students, the importance of personal hygiene and cleanliness. Physical education classes aim at teaching the students, the habits of personal cleanliness and the importance of the maintenance of personal hygiene in life. Physical education classes also impart sex-education to the students, help them clarify their doubts and find answers to all the questions that occur to their minds. The sports, which are a part of the physical education class, help in developing motor skills in children. The ability to hold a racket or a bat, the ability to catch a ball and the ability to swing a bat are some examples of the motor abilities that can develop with the help of sports. The physical activity that is involved in physical education helps the students in bringing discipline to body posture and body movements. Hitting a ball with a bat or a shuttle with a racket as also aiming a ball for a goal or catching it to get the opponent team out, are some of the commonly observed actions in sports and are extremely beneficial in improving hand-eye coordination.

The very important objective of physical education is to encourage the upcoming sportsmen and women of the crowd. Physical education gives the budding sports people a platform to exhibit their talents. Those with a flair for sports get an opportunity to display their talent. Their small step on the school playground can eventually turn into a huge leap in the field of sports. Moreover, sports refresh the students' minds. Physical education class becomes enjoyable for the kids while proving helpful for their overall growth and development. Physical education is indeed one of the most fruitful activities of a school schedule.

PSYCHOLOGICAL ASPECTS OF PHYSICAL EDUCATION

DEFINITION OF PSYCHOLOGY AND SPORT PSYCHOLOGY

Psychology

Psychology is a broad discipline which seeks to analyze the human mind. Different disciplines within this field study why people behave, think, and feel the way they do. There are many different ways to approach psychology, from examining biology's role in mental health to the role of the environment on behaviour. Some psychologists focus only on how the mind develops, while others counsel patients to help improve their daily lives. The history of psychology dates back at least to 1879, when the German psychologist Wilhelm Wundt founded the first laboratory exclusively devoted to psychology. The most famous psychologist is perhaps Sigmund Freud, an Austrian who founded the field of psychoanalysis. Although Freud's theories had a huge impact on a wide variety of areas, including literature and film, many of his ideas are considered subjective from a modern perspective. There are dozens of different sub-disciplines of psychology, with each taking a somewhat different approach to understanding the mind. Some sub-disciplines include social psychology, clinical psychology, occupational health, and cognitive psychology. It is important to note that, even within a particular field, there may be different approaches.

Clinical psychology, for example, has four main schools: psychodynamic, humanistic, cognitive and behavioral, and systems therapy. The field of psychology is far wider than the image of a patient reclining on a couch, talking to his therapist or a researcher studying a rat running through a maze. A forensic psychologist may help legal professionals investigate allegations of child abuse or evaluate a suspect's competency. A legal psychologist might act as an adviser to a judge or as a trial consultant. An industrial-organizational psychologist might work with a company to help hire the best applicants or help improve workplace morale. A sports psychologist might work one-on-one with a player to help overcome a performance barrier, or work with an entire team to help improve group cohesion. Psychology should be further distinguished between *research psychology*, which seeks to establish facts about the mind by conducting experiments, and *applied psychology*, which seeks to help people with their problems. Some experiments have shown that the success rate for solving one's problems using only psychotherapy — talking to a professional psychologist — is the same as talking to a close friend, so the efficacy of applied or clinical psychology as a discipline can sometimes be difficult to quantify. Alternatively, many patients report that therapy has been greatly helpful in their lives. As compared to other hard sciences, like physics and biology, some critics argue the field suffers from a lack of scientific rigour. The objectivity of tools like surveys, through which data is collected in some cases, are often questioned. This is perhaps closely associated with the complexity of the mind which we haven't quite been able to fully or substantially understand. Psychological studies, however, continue to be held in order to try to get a better understanding of the mind and how it works. Fields like neuropsychology, which looks at how the structure of the brain affects mental health, use neuroimaging technology. Tools such as functional magnetic resonance imaging and positron emission tomography (PET) scans have assisted psychologists in making correlations between mental problems and biological states. For instance, in the 1980s, it was realized the schizophrenia was mainly caused by biological factors rather than maternal neglect or some other environmental explanation.

Etymology

The word psychology literally means, "study of the soul" and logia, translated as "study of" or "research". The Latin word psychologia was first used by the Croatian humanist and Latinist Marko Maruliæ in his book, Psichiologia de ratione animae humanae in the late 15th century or early 16th century. The earliest known reference to the word psychology in English was by Steven Blankaart in 1693 in The Physical Dictionary which refers to "Anatomy, which treats of the Body, and Psychology, which treats of the Soul."

History

The study of psychology in philosophical context dates back to the ancient civilizations of Egypt, Greece, China, India, and Persia. Historians point to the writings of ancient Greek philosophers, such as Thales, Plato, and Aristotle, as the first significant body of work in the West to be rich in psychological thought.

- Structuralism: German physician Wilhelm Wundt is credited with introducing psychological discovery into a laboratory setting. Known as the "father of experimental psychology", he founded the first psychological laboratory, at Leipzig University, in 1879. Wundt focused on breaking down mental processes into the most basic components, starting a school of psychology that is called structuralism. Edward Titchener was another major structuralist thinker.

- Functionalism: Functionalism formed as a reaction to the theories of the structuralist school of thought and was heavily influenced by the work of the American philosopher and psychologist William James. James felt that psychology should have practical value, and that psychologists should find out how the mind can function to a person's benefit. In his book, Principles of Psychology, published in 1890, he laid the foundations for many of the questions that psychologists would explore for years to come. Other major functionalist thinkers included John Dewey and Harvey Carr. Other 19th-century contributors to the field include the German psychologist Hermann Ebbinghaus, a pioneer in the experimental study of memory, who developed quantitative models of learning and forgetting at the University of Berlin; and the Russian-Soviet physiologist Ivan Pavlov, who discovered in dogs a learning process that was later termed "classical conditioning" and applied to human beings. Starting in the 1950s, the experimental techniques set forth by Wundt, James, Ebbinghaus, and others would be reiterated as experimental psychology became increasingly cognitive—concerned with information and its processing—and, eventually, constituted a part of the wider cognitive science. In its early years, this development had been seen as a "revolution", as it both responded to and reacted against strains of thought—including psychodynamics and behaviorism—that had developed in the meantime.
- Psychoanalysis: From the 1890s until his death in 1939, the Austrian physician Sigmund Freud developed psychoanalysis, a method of investigation of the mind and the way one thinks; a systematized set of theories about human behaviour; and a form of psychotherapy to treat psychological or emotional distress, especially unconscious conflict. Freud's psychoanalytic theory was largely based on interpretive methods, introspection and clinical observations. It became very well-known, largely because it tackled subjects such as sexuality, repression, and the unconscious mind as general aspects of psychological development. These were largely considered taboo subjects at the time, and Freud provided a catalyst for them to be openly discussed in polite society. Clinically, Freud helped to pioneer the method of free association and a therapeutic interest in dream interpretation. Freud had a significant influence on Swiss psychiatrist Carl Jung, whose analytical psychology became an alternative form of depth psychology. Other well-known psychoanalytic scholars of the mid-20th century included psychoanalysts, psychologists, psychiatrists, and philosophers. Among these thinkers were Erik Erickson, Melanie Klein, D. W. Winnicott, Karen Horney, Erich Fromm, John Bowlby and Sigmund Freud's daughter, Anna Freud. Throughout the 20th century, psychoanalysis evolved into diverse schools of thought, most of which may be classed as Neo-Freudian.[c] Psychoanalytic theory and therapy were criticized by psychologists and philosophers such as B. F.

Skinner, Hans Eysenck, and Karl Popper. Popper, a philosopher of science, argued that Freud's, as well as Alfred Adler's, psychoanalytic theories included enough ad hoc safeguards against empirical contradiction to keep the theories outside the realm of scientific enquiry. By contrast, Eysenck maintained that although Freudian ideas could be subjected to experimental science, they had not withstood experimental tests. By the 20th century, psychology departments in American universities had become experimentally oriented, marginalizing Freudian theory and regarding it as a "desiccated and dead" historical artifact. Meanwhile, however, researchers in the emerging field of neuro-psychoanalysis defended some of Freud's ideas on scientific grounds,[d] while scholars of the humanities maintained that Freud was not a "scientist at all, but... an interpreter."

- Behaviorism: In the United States, behaviorism became the dominant school of thought during the 1950s. Behaviorism was founded in the early 20th century by John B. Watson, and embraced and extended by Edward Thorndike, Clark L. Hull, Edward C. Tolman, and later B. F. Skinner. Theories of learning emphasized the ways in which people might be predisposed, or conditioned, by their environments to behave in certain ways. Classical conditioning was an early behaviorist model. It posited that behavioral tendencies are determined by immediate associations between various environmental stimuli and the degree of pleasure or pain that follows. Behavioral patterns, then, were understood to consist of organisms' conditioned responses to the stimuli in their environment. The stimuli were held to exert influence in proportion to their prior repetition or to the previous intensity of their associated pain or pleasure. Much research consisted of laboratory-based animal experimentation, which was increasing in popularity as physiology grew more sophisticated. Skinner's behaviorism shared with its predecessors a philosophical inclination towards positivism and determinism. He believed that the contents of the mind were not open to scientific scrutiny and that scientific psychology should emphasize the study of observable behaviour. He focused on behaviour–environment relations and analyzed overt and covert behaviour as a function of the organism interacting with its environment. Behaviorists usually rejected or deemphasized dualistic explanations such as "mind" or "consciousness"; and, in lieu of probing an "unconscious mind" that underlies unawareness, they spoke of the "contingency-shaped behaviors" in which unawareness becomes outwardly manifest. Among the behaviorists' most famous creations are John B. Watson's Little Albert experiment, which applied classical conditioning to the developing human child, and Skinner's notion of operant conditioning, which acknowledged that human agency could affect patterns and cycles of environmental stimuli and behavioral responses. Linguist Noam Chomsky's critique of the behaviorist model of language acquisition is widely regarded as a key factor in the decline of behaviorism's prominence. Martin Seligman and colleagues discovered that the conditioning of dogs led to outcomes that opposed the predictions of behaviorism. But Skinner's behaviorism did not die, perhaps in part because it generated successful practical applications. The fall of behaviorism as an overarching model in psychology, however, gave way to a new dominant paradigm: cognitive approaches.

- Humanism: Humanistic psychology was developed in the 1950s in reaction to both behaviorism and psychoanalysis. By using phenomenology, intersubjectivity and first-person categories, the humanistic approach sought to glimpse the whole person—not just the fragmented parts of the personality or cognitive functioning. Humanism focused on fundamentally and uniquely human issues, such as individual free will, personal growth, self-actualization, self-identity, death, aloneness, freedom, and meaning. The humanistic approach was distinguished by its emphasis on subjective meaning, rejection of determinism, and concern for positive growth rather than pathology. Some of the founders of the humanistic school of thought were American psychologists Abraham Maslow, who formulated a hierarchy of human needs, and Carl Rogers, who created and developed client-centered therapy. Later, positive psychology opened up humanistic themes to scientific modes of exploration.
- Gestalt: Wolfgang Kohler, Max Wertheimer and Kurt Koffka co-founded the school of Gestalt psychology. This approach is based upon the idea that individuals experience things as unified wholes. This approach to psychology began in Germany and Austria during the late 19th century in response to the molecular approach of structuralism. Rather than breaking down thoughts and behaviour to their smallest element, the Gestalt position maintains that the whole of experience is important, and the whole is different than the sum of its parts. Gestalt psychology should not be confused with the Gestalt therapy of Fritz Perls, which is only peripherally linked to Gestalt psychology.
- Existentialism: Influenced largely by the work of German philosopher Martin Heidegger and Danish philosopher Søren Kierkegaard, psychoanalytically trained American psychologist Rollo May pioneered an existential breed of psychology, which included existential therapy, in the 1950s and 1960s. Existential psychologists differed from others often classified as humanistic in their comparatively neutral view of human nature and in their relatively positive assessment of anxiety. Existential psychologists emphasized the humanistic themes of death, free will, and meaning, suggesting that meaning can be shaped by myths, or narrative patterns, and that it can be encouraged by an acceptance of the free will requisite to an authentic, albeit often anxious, regard for death and other future prospects. Austrian existential psychiatrist and Holocaust

survivor Viktor Frankl drew evidence of meaning's therapeutic power from reflections garnered from his own internment, and he created a variety of existential psychotherapy called logotherapy. In addition to May and Frankl, Swiss psychoanalyst Ludwig Binswanger and American psychologist George Kelly may be said to belong to the existential school.

- Cognitivism: Cognitive psychology is the branch of psychology that studies mental processes including how people think, perceive, remember, and learn. As part of the larger field of cognitive science, this branch of psychology is related to other disciplines including neuroscience, philosophy, and linguistics. Noam Chomsky helped to ignite a "cognitive revolution" in psychology when he criticized the behaviorists' notions of "stimulus", "response", and "reinforcement", arguing that such ideas—which Skinner had borrowed from animal experiments in the laboratory—could be applied to complex human behaviour, most notably language acquisition, in only a vague and superficial manner. The postulation that humans are born with the instinct or "innate facility" for acquiring language posed a challenge to the behaviorist position that all behaviour is contingent upon learning and reinforcement. Social learning theorists such as Albert Bandura argued that the child's environment could make contributions of its own to the behaviors of an observant subject. Meanwhile, accumulating technology helped to renew interest and belief in the mental states and representations—*i.e.,* the cognition—that had fallen out of favour with behaviorists. English neuroscientist Charles Sherrington and Canadian psychologist Donald O. Hebb used experimental methods to link psychological phenomena with the structure and function of the brain. With the rise of computer science and artificial intelligence, analogies were drawn between the processing of information by humans and information processing by machines. Research in cognition had proven practical since World War II, when it aided in the understanding of weapons operation. By the late 20th century, though, cognitivism had become the dominant paradigm of mainstream psychology, and cognitive psychology emerged as a popular branch. Assuming both that the covert mind should be studied and that the scientific method should be used to study it, cognitive psychologists set such concepts as "subliminal processing" and "implicit memory" in place of the psychoanalytic "unconscious mind" or the behavioristic "contingency-shaped behaviors". Elements of behaviorism and cognitive psychology were synthesized to form the basis of cognitive behavioral therapy, a form of psychotherapy modified from techniques developed by American psychologist Albert Ellis and American psychiatrist Aaron T. Beck. Cognitive psychology was subsumed along with other disciplines, such as philosophy of mind, computer science, and neuroscience, under the umbrella discipline of cognitive science.

Biopsychosocial Model

The biopsychosocial model is an integrated perspective towards understanding consciousness, behaviour, and social interaction. It assumes that any given behaviour or mental process affects and is affected by dynamically interrelated biological, psychological, and social factors. The psychological aspect refers to the role that cognition and emotions play in any given psychological phenomenon—for example, the effect of mood or beliefs and expectations on an individual's reactions to an event.

The biological aspect refers to the role of biological factors in psychological phenomena—for example, the effect of the prenatal environment on brain development and cognitive abilities, or the influence of genes on individual dispositions. The socio-cultural aspect refers to the role that social and cultural environments play in a given psychological phenomenon—for example, the role of parental or peer influence in the behaviors or characteristics of an individual.

Subfields

Psychology encompasses a vast domain, and includes many different approaches to the study of mental processes and behaviour.

- *Biological:* Biological psychology and a number of related fields study the biological substrates of behaviour and mental states. Behavioral neuroscience uses animal models to study the neural, genetic, cellular mechanisms of human behaviour and cognition, cognitive neuroscientists investigate the neural correlates of psychological processes in humans using neural imaging tools, and neuropsychologists conduct psychological assessments to determine, for instance, specific aspects and extent of cognitive deficit caused by brain damage or disease.
- *Clinical:* Clinical psychology includes the study and application of psychology for the purpose of understanding, preventing, and relieving psychologically based distress or dysfunction and to promote subjective well-being and personal development. Central to its practice are psychological assessment and psychotherapy, although clinical psychologists may also engage in research, teaching, consultation, forensic testimony, and programme development and administration. Some clinical psychologists may focus on the clinical management of patients with brain injury—this area is known as clinical neuropsychology. In many countries, clinical psychology is a regulated mental health profession. The work performed by clinical psychologists tends to be influenced by various therapeutic approaches, all of which involve a formal relationship between professional and client. The various therapeutic approaches and practices are associated with different theoretical perspectives and employ different procedures intended to form a therapeutic alliance, explore the nature of psychological problems, and encourage new ways of thinking, feeling, or

behaving. Four major theoretical perspectives are psychodynamic, cognitive behavioral, existential-humanistic, and systems or family therapy. There has been a growing movement to integrate the various therapeutic approaches, especially with an increased understanding of issues regarding culture, gender, spirituality, and sexual-orientation. With the advent of more robust research findings regarding psychotherapy, there is evidence that most of the major therapies are about of equal effectiveness, with the key common element being a strong therapeutic alliance. Because of this, more training programmes and psychologists are now adopting an eclectic therapeutic orientation.

- *Cognitive:* Cognitive psychology studies cognition, the mental processes underlying mental activity. Perception, learning, problem solving, reasoning, thinking, memory, attention, language and emotion are areas of research. Classical cognitive psychology is associated with a school of thought known as cognitivism, whose adherents argue for an information processing model of mental function, informed by functionalism and experimental psychology. On a broader level, cognitive science is an interdisciplinary enterprise of cognitive psychologists, cognitive neuroscientists, researchers in artificial intelligence, linguists, human–computer interaction, computational neuroscience, logicians and social scientists. Computational models are sometimes used to simulate phenomena of interest. Computational models provide a tool for studying the functional organization of the mind whereas neuroscience provides measures of brain activity.
- *Comparative:* Comparative psychology refers to the study of the behaviour and mental life of animals other than human beings. It is related to disciplines outside of psychology that study animal behaviour such as ethology. Although the field of psychology is primarily concerned with humans the behaviour and mental processes of animals is also an important part of psychological research. This being either as a subject in its own right or with strong emphasis about evolutionary links, and somewhat more controversially, as a way of gaining an insight into human psychology. This is achieved by means of comparison or via animal models of emotional and behaviour systems as seen in neuroscience of psychology.
- *Developmental:* Mainly focusing on the development of the human mind through the life span, developmental psychology seeks to understand how people come to perceive, understand, and act within the world and how these processes change as they age. This may focus on intellectual, cognitive, neural, social, or moral development. Researchers who study children use a number of unique research methods to make observations in natural settings or to engage them in experimental tasks. Such tasks often resemble specially designed games and activities that are both enjoyable for the child and scientifically useful, and

researchers have even devised clever methods to study the mental processes of small infants. In addition to studying children, developmental psychologists also study aging and processes throughout the life span, especially at other times of rapid change. Developmental psychologists draw on the full range of psychological theories to inform their research.

- *Educational and School:* Educational psychology is the study of how humans learn in educational settings, the effectiveness of educational interventions, the psychology of teaching, and the social psychology of schools as organizations. The work of child psychologists such as Lev Vygotsky, Jean Piaget, Bernard Luskin and Jerome Bruner has been influential in creating teaching methods and educational practices. Educational psychology is often included in teacher education programmes, in places such as North America, Australia, and New Zealand. School psychology combines principles from educational psychology and clinical psychology to understand and treat students with learning disabilities; to foster the intellectual growth of “gifted” students; to facilitate prosocial behaviors in adolescents; and otherwise to promote safe, supportive, and effective learning environments. School psychologists are trained in educational and behavioral assessment, intervention, prevention, and consultation, and many have extensive training in research.
- *Industrial-organizational:* Industrial and organizational psychology applies psychological concepts and methods to optimize human potential in the workplace. Personnel psychology, a subfield of I-O psychology, applies the methods and principles of psychology in selecting and evaluating workers. I-O psychology’s other subfield, organizational psychology, examines the effects of work environments and management styles on worker motivation, job satisfaction, and productivity.
- *Personality:* Personality psychology is concerned with enduring patterns of behaviour, thought, and emotion in individuals, commonly referred to as personality. Theories of personality vary across different psychological schools and orientations. They carry different assumptions about such issues as the role of the unconscious and the importance of childhood experience. According to Freud, personality is based on the dynamic interactions of the id, ego, and super-ego. Trait theorists, in contrast, attempt to analyze personality in terms of a discrete number of key traits by the statistical method of factor analysis. The number of proposed traits has varied widely. An early model proposed by Hans Eysenck suggested that there are three traits that comprise human personality: extraversion-introversion, neuroticism, and psychoticism. Raymond Cattell proposed a theory of 16 personality factors. The “Big Five”, or Five Factor Model, proposed by Lewis Goldberg, currently has strong support among trait theorists.

- *Social:* Social psychology is the study of how humans think about each other and how they relate to each other. Social psychologists study such topics as the influence of others on an individual's behaviour, and the formation of beliefs, attitudes, and stereotypes about other people. Social cognition fuses elements of social and cognitive psychology in order to understand how people process, remember, and distort social information. The study of group dynamics reveals information about the nature and potential optimization of leadership, communication, and other phenomena that emerge at least at the microsocial level. In recent years, many social psychologists have become increasingly interested in implicit measures, mediational models, and the interaction of both person and social variables in accounting for behaviour.

Research Methods

Psychology tends to be eclectic, drawing on knowledge from other fields to help explain and understand psychological phenomena. Additionally, psychologists make extensive use of the three modes of inference that were identified by C. S. Peirce: deduction, induction, and abduction.

While often employing deductive-nomological reasoning, they also rely on inductive reasoning to generate explanations. For example, evolutionary psychologists attempt to explain psychological traits—such as memory, perception, or language—as adaptations, that is, as the functional products of natural selection or sexual selection. Psychologists may conduct basic research aiming for further understanding in a particular area of interest in psychology, or conduct applied research to solve problems in the clinic, workplace or other areas. Masters level clinical programmes aim to train students in both research methods and evidence-based practice. Professional associations have established guidelines for ethics, training, research methodology and professional practice. In addition, depending on the country, state or region, psychological services and the title "psychologist" may be governed by statute and psychologists who offer services to the public are usually required to be licensed.

- *Qualitative and Quantitative Research:* Research in most areas of psychology is conducted in accord with the standards of the scientific method. Psychological researchers seek the emergence of theoretically interesting categories and hypotheses from data, using qualitative or quantitative methods (or both). Qualitative psychological research methods include interviews, first-hand observation, and participant observation. Qualitative researchers sometimes aim to enrich interpretations or critiques of symbols, subjective experiences, or social structures. Similar hermeneutic and critical aims have also been served by "quantitative methods", as in Erich Fromm's study of Nazi voting or Stanley Milgram's studies of obedience to authority. Quantitative psychological research lends itself to the statistical testing of hypotheses. Quantitatively oriented research designs include the experiment, quasi-experiment, cross-sectional study, case-control study, and longitudinal study. The measurement and operationalization of important constructs is an essential part of these research designs. Statistical methods include the Pearson product-moment correlation coefficient, the analysis of variance, multiple linear regression, logistic regression, structural equation modeling, and hierarchical linear modeling.

- *Controlled Experiments:* Experimental psychological research is conducted in a laboratory under controlled conditions. This method of research relies on the application of the scientific method to understand behaviour. Experimenters use several types of measurements, including rate of response, reaction time, and various psychometric measurements. Experiments are designed to test specific hypotheses (deductive approach) or evaluate functional relationships (inductive approach). A true experiment with random allocation of subjects to conditions allows researchers to infer causal relationships between different aspects of behaviour and the environment. In an experiment, one or more variables of interest are controlled by the experimenter (independent variable) and another variable is measured in response to different conditions (dependent variable). Experiments are one of the primary research methods in many areas of psychology, particularly cognitive/psychonomics, mathematical psychology, psychophysiology and biological psychology/cognitive neuroscience. Experiments on humans have been put under some controls, namely informed and voluntary consent. After World War II, the Nuremberg Code was established, because of Nazi abuses of experimental subjects. Later, most countries (and scientific journals) adopted the Declaration of Helsinki. In the US, the National Institutes of Health established the Institutional Review Board in 1966, and in 1974 adopted the National Research Act. All of these measures encouraged researchers to obtain informed consent from human participants in experimental studies. A number of influential studies led to the establishment of this rule; such studies included the MIT and Fernald School radioisotope studies, the Thalidomide tragedy, the Willowbrook hepatitis study, and Stanley Milgram's studies of obedience to authority.
- *Survey Questionnaires:* Statistical surveys are used in psychology for measuring attitudes and traits, monitoring changes in mood, checking the validity of experimental manipulations, and for a wide variety of other psychological topics. Most commonly, psychologists use paper-and-pencil surveys. However, surveys are also conducted over the phone or through e-mail. Increasingly, web-based surveys are being used in research. Similar methodology is also used in applied setting, such as clinical assessment and personnel assessment.
- *Longitudinal Studies*: Longitudinal studies are often used in psychology to study developmental trends across the life span, and in sociology to study life events throughout lifetimes or generations. The reason for this is that unlike cross-

sectional studies, longitudinal studies track the same people, and therefore the differences observed in those people are less likely to be the result of cultural differences across generations. Because of this benefit, longitudinal studies make observing changes more accurate and they are applied in various other fields. Because most longitudinal studies are observational, in the sense that they observe the state of the world without manipulating it, it has been argued that they may have less power to detect causal relationships than do experiments. They also suffer methodological limitations such as from selective attrition because people with similar characteristics maybe more likely to drop out of the study making it difficult to analyze. Some longitudinal studies are experiments, called repeated-measures experiments. Psychologists often use the crossover design to reduce the influence of confounding covariates and to reduce the number of subjects.

- *Observation in Natural Settings:* In the same way Jane Goodall studied the role of chimpanzee social and family life, psychologists conduct similar observational studies in human social, professional and family lives. Sometimes the participants are aware they are being observed and other times it is covert: the participants do not know they are being observed. Ethical guidelines need to be taken into consideration when covert observation is being carried out.
- *Qualitative and Descriptive Research:* Research designed to answer questions about the current state of affairs such as the thoughts, feelings and behaviors of individuals is known as descriptive research. Descriptive research can be qualitative or quantitative in orientation. Qualitative research is descriptive research that is focused on observing and describing events as they occur, with the goal of capturing all of the richness of everyday behaviour and with the hope of discovering and understanding phenomena that might have been missed if only more cursory examinations have been made.
- *Neuropsychological Methods:* Neuropsychology seeks to connect aspects of behaviour and mental activity with the structure and function of the brain. Cognitive neuropsychology and cognitive neuropsychiatry study neurological or mental impairment in an attempt to infer theories of normal mind and brain function. This typically involves looking for differences in patterns of remaining ability (known as 'functional disassociations') which can give clues as to whether abilities are composed of smaller functions, or are controlled by a single cognitive mechanism. In addition, experimental techniques are often used to study the neuropsychology of healthy individuals. These include behavioral experiments, brain-scanning or functional neuroimaging, used to examine the activity of the brain during task performance, and techniques such as transcranial magnetic stimulation, which can safely alter the function of small brain areas to reveal their importance in mental operations.

- *Computational Modeling:* Computational modeling is a tool often used in mathematical psychology and cognitive psychology to simulate a particular behaviour using a computer. This method has several advantages. Since modern computers process extremely quickly, many simulations can be run in a short time, allowing for a great deal of statistical power. Modeling also allows psychologists to visualize hypotheses about the functional organization of mental events that couldn't be directly observed in a human. Several different types of modeling are used to study behaviour. Connectionism uses neural networks to simulate the brain. Another method is symbolic modeling, which represents many different mental objects using variables and rules. Other types of modeling include dynamic systems and stochastic modeling.
- *Animal Studies:* Animal learning experiments aid in investigating the biological basis of teaching, memory and behaviour. In the 1890s, Russian physiologist Ivan Pavlov famously used dogs to demonstrate classical conditioning. Non-human primates, cats, dogs, rats and other rodents are often used in psychological experiments. Ideally, controlled experiments introduce only one independent variable at a time, in order to ascertain its unique effects upon dependent variables. These conditions are approximated best in laboratory settings. In contrast, human environments and genetic backgrounds vary so widely, and depend upon so many factors, that it is difficult to control important variables for human subjects. Of course, there are pitfalls in generalizing findings from animal studies to humans although animal models can be helpful in developing an understanding of human behaviour.

PERSONS, PRACTICES AND PHYSICAL EDUCATION

Parry captured the bigger picture with respect to the nature and values of physical education when he urged upon the profession a fundamental re-examination of the central concepts education, culture and personhood. He observed that any educational ideology could be challenged at a variety of levels: first, at the level of actual practices as legitimate expressions of the educational theory, or as efficient means to its goals; second, at the level of educational theory as a legitimate expression of the ideology; and, finally, at the level of ideology. The radical kind of conceptualisation was never fully taken up. But neither analytical philosophy of education nor the physical education profession at large is sympathetic to this dense continental philosophy that employs a language all of its own and seems antithetical to the common-sense strand of English-speaking analytical philosophy. Parry urged a less cognitivist conceptualisation of education and personhood. We think that the move of situating a less rationalistic view of persons necessarily opens up a proper consideration of the role of the emotions in our lives and especially in sports where they are channelled, frustrated, exposed, and potentially explored in self-critical ways. But this has rarely been addressed either theoretically or in the professional education of physical educators where spaces for non-applied, or immediately relevant, professional matters are rarely created. This very thin conceptualisation is preferable at two levels to the position made famous by Peters and retained by Carr. In the first instance, it is preferable on normative grounds in that it recognises a plurality of conceptions of education that emerge from some shared understanding of the need for societies to seek the grounds of their own continuance. How is this to be done other than by capturing the hearts and minds of its young?

On analytical grounds, so open a position does not prescribe in precise terms how this is to be done though one would want, as Peters did, to proscribe certain procedures on ethical grounds. To set out the traditional liberal distinctions as Carr does, and as Peters did before him, renders him open to the simple charge of ideology; no matter how internally coherent the thesis, he is always open to counter-ideological critique. It seems that Parry, Reid point in a different direction but none of us have travelled down it any distance here.

We have all signposted a less restricted account of education as the initiation into a range of cultural practices that have the capacity to open up the possibilities of living a full and worthwhile life. Reid has given us no clues as to a broader weltanshauung that informs the shared position against cognitive imperialism: he is merely at pains to stress the primacy of the hedonic.

Parry has suggested some liberalised Olympic thesis while intectual persons inclined towards a communitarian position central to which would be a stronger recognition for the dominant role that social practices like sport play in the formation of our identities and values. Furthermore, it will be clear from the position developed there that should be see such practices as one of the foundational bedrocks of character and identity formation which is one of the crucial tasks that fall predominantly, though not exclusively to formal education. At an analytical level, then, rather than arguing that X is education or not-education on the grounds of a pre-eminent criterion: cognitive depth and breadth recognition must be made for the fact that there are competing conceptions of education.

Instead, as a matter of conceptual necessity, it seems to me that despite the fact that these conceptions embody particular evaluative commitments regarding the nature of persons and society, they all share the formal notion that education is the development of persons towards the living of full and valuable lives. The next step of the argument is to develop an account of persons and the kinds of things that make their lives worthwhile over and above Peters' intellectual pursuits. Persons are beings who have the capacity to develop, evaluate, and live out life-plans based on a combination of projects, relationships and commitments. Among this combination of activities are a variety of practices which are valuable by virtue of their internal goods and their capacity to secure external goods in particular and unique ways.

The activities of physical education are exemplified by a certain range of sporting practices, which are taken as paradigmatic which can be characterised as mixed goods because they have the capacity to be valued not only for their internal goods but also for the particular manner in which they secure external goods.

Physical education can, therefore, contribute to the living of full valuable lives for persons and is thus of educational value. We think this is precisely the kind of rationale that can be either be read into the work of the American physical educator, Daryl Siedentop with his enormously influential model of sport education or simply might be explored as a philosophical justification for at least some portions of that model. This kind of argument, it might be said, holds true only for those practices we recognise as sporting games or athletic activities.

While the argument is long on initiation into those practices that are partly definitive of a culture and its identity it is short on the kinds of individualised, health-related activities.

Historically, there have been two strands in what is called physical education: sports and health. It seems clear to me that a different type of justificatory argument is required to support each. Maybe Carr is right here to classify the latter activities as valuable but not educationally valuable because of their lacking in what can be referred to as cultural significance or cultural capital. Time and space do not allow me to comment in any way here upon these other strands except to note the following. Those who look for conceptual unity are simply wasting their time. There is no meaningful essence to the concept in that way.

As Reid remarks, one must look rather to culture-specific, historical and political factors that have shaped the professions. Dance is a cultural practice that employs large motor-skilled activity like tennis or football. Some forms of gymnastics require interpretative movements and proceed with music like dance. Sculpting bodies, like training for rugby or netball, often requires the kinds of regimes and exercises that are common.

But these similarities are nothing more than that. If all that one can do is to point out commonalities then there is little that is philosophically interesting here for anyone attempting a conceptual analysis by necessary and sufficient conditions of linguistic usage. Reid's peroration towards value pluralism should extend so far as to recognise the inherent openness of the concept of physical education: pluralism in activities; pluralism in values.

No universal criterion of demarcation can be raised that will help physical educators to select activities is available, and so we should simply stop looking. Instead we should enquire as to the types and natures of rituals that sports instantiate in our modern world. And if, as Wollheim argues, traditions pass on what they possess, then we should see to it, as guardians of these great cultural rituals, that the values physical education has and gives are kept in good health.

VALUES OF PHYSICAL EDUCATION

The consequences of a quality programme presents the following benefits to those individuals involved Sensory stimulation through physical activity is essential for the optimal growth and development of the nervous system.

- Promotes cognitive function through imitation, symbolic play and the use of symbols.
- Assists in the development and refinement of perceptual abilities involving vision, balance, and tactile sensations.
- Enhances the function of the central nervous system through the promotion of a healthier neuronal network.
- Aids the development of cognition through opportunities to develop learning strategies, decision making acquiring, retrieving, and integrating information and solving problems.
- Fortifies the mineralization of the skeleton and promotes the maintenance of lean body tissue, while simultaneously reducing the deposition of fat.
- Leads to proficiency in the neuromuscular skills that are the basis for successful participation in games, dances, sports, and leisure activities.
- Is an important regulator of obesity because it increases energy expenditure, suppresses appetite, increases metabolic rate, and increases lean body mass.
- Improves aerobic fitness, muscle endurance, muscle power, and muscle strength.
- Is an effective deterrent to coronary heart disease due to its effects on blood lipids, blood pressure, obesity, and capacity for physical work?
- Improves cardiac function as indicated by an increased stroke volume, cardiac output, blood volume, and total hemoglobin.
- Is associated with a reduction in atherosclerotic disease.
- Promotes a more positive attitude towards physical activity and leads to a more active lifestyle during unscheduled leisure time.
- Enhances self-concept and self-esteem as indicated by increased confidence, assertiveness, emotional stability, independence and self-control.
- Is a major force in the socializing of individuals during adolescence?
- Is instrumental in the development and growth of moral reasoning, problem solving, creativity, and social competence
- Is an effective deterrent to mental illness and the alleviation of mental stress?
- Improves the psychosocial and physiological functions of mentally and physically handicapped individuals.
- Prevents the onset of some diseases and postpones the debilitated effects of old age.

COMPONENTS OF FITNESS

The components of fitness each work together to contribute to the ability of the body to handle physical demands. The more efficient the body functions, the higher the level of fitness. Optimal fitness is a combination of lifestyle, nutrition, habits, but it cannot be reached without an appropriate level of physical activity. Optimum physical performance is a combination of all the components of fitness; depending on the specific demands of the sport or activity, some components will require more attention than others, but each should be present as a part of an integrated training programme.

GENERAL COMPONENTS OF FITNESS

Endurance

Endurance is, simply put, the ability to *endure*, or an object or person's lasting quality. Thus, the longer a thing lasts, the greater the endurance. Endurance may refer to short-term—high intensity, anaerobic exercise such as sprinting—or long term, which may last hours or even days in duration, as in the case of marathons, triathlons, and ultramarathons.

In terms of fitness, endurance may be broken down into several types: aerobic endurance (cardiorespiratory endurance), anaerobic endurance, speed-endurance and strength-endurance. It is most commonly broken up into cardiorespiratory endurance and muscular endurance.

Well-trained endurance athletes are able to generate blood lactate levels that are 20-30% higher than those of untrained individuals under similar conditions. This produces significantly enhanced endurance as their muscles are better equipped to utilize it to fuel further muscular energy.

Cardiorespiratory Endurance

Cardiorespiratory endurance refers to the efficiency with which the body delivers oxygen and nutrients needed for muscular activity and transports waste products from the cells. It is also sometimes referred to as *aerobic endurance* or *aerobic fitness*. Improving aerobic endurance enables the heart, lungs, and muscles to do work over a longer period of time. Cardiorespiratory conditioning can decrease risk factors associated with heart disease, increase vitality, increase maximum oxygen uptake, and can aid weight loss or maintenance.

In addition to this, training cardiorespiratory endurance improves aerobic capacity caused by fibre adaptation, more specifically an increase in the size of mitochondria, which enhances the ability of the fibres to generate aerobic energy. It also facilitates an increase in capillary density, which enhances the fibres' capacity to transport oxygen, and thus to create energy. Finally, endurance training increases the number of enzymes relevant to theKrebs cycle, a chemical process within muscles that allows the regeneration of ATP under aerobic conditions. The enzymes involved in this process may actually increase by a factor of two to three after a sustained period of endurance training.

Flexibility

Flexibility refers to the movement in a joint or group of joints, during a passive movement (passive meaning no active muscle involvement is required to hold the stretch; instead gravity or a partner provides the force for the stretch). Flexibility is a general component of physical fitness. Additionally, good range of motion will allow the body to assume more natural positions to help maintain good posture. This component becomes more important as people age and their joints stiffen up, preventing them from doing everyday tasks. Stretching is therefore an important habit to start and continue as one ages. Flexibility of a joint depends on many factors, particularly the length and looseness of the muscles and ligaments due to normal human variation, and the shape of the bones and cartilage that make up the joint. The primary reasons for increasing flexibility are enhanced performance and reduced risk of injury. The rationale for this is that a limb can move further before an injury occurs.

Muscular Strength

Strength is recognised as the ability to exert force, typically measured in the amount of weight a person can lift or manipulate. There are five broad categories of strength, each with its own special training requirements:absolute, limit, speed, anaerobic, and aerobic. There are many factors that influence strength.

- Structural/Anatomical - muscle fibre arrangement, musculoskeletal leverage, ratio of fast vs. slow-twitch fibres, tissue leverage, scar tissue and adhesions (motion-limiting factors), elasticity, intramuscular/intermuscular friction, etc.
- Physiological/Biochemical - stretch reflex, Golgi tendon organ sensitivity, hormonal function, energy transfer systems efficiency, extent of hyperplasia, myofibrillar development, motor unit recruitment, cardiovascular and cardiorespiratory factors, etc.
- Psychoneural/Learned Responses - arousal level, pain tolerance, level of concentration, social learning,skill level, spiritual factors, etc.
- External/Environmental - equipment, weather, altitude, gravity, opposing/ assisting forces, etc.

Muscular strength is a general component of fitness. Strength level should only be to a point where the increased strength will not interfere with technique execution. When excessive amounts of strength are developed, range of motion, and speed of execution is decreased and coordination usually deteriorates. Thus, it becomes increasingly important to keep all of these factors well balanced.

BODY COMPOSITION

Components of Motor Fitness

Accuracy is the ability to control movement in a given direction or at a given intensity.

Agility

Agility is the ability to apply explosive movements to rapidly change directions.

Balance

Balance is the ability to exercise precise control over the body's position and movement.

Coordination

Motor coordination (sometimes called *hand-eye coordination*) is the coordinated functioning of muscles or groups of muscles in the execution of a complex task. Coordination itself, however, is a global system made up of several synergistic elements and not necessarily a singularly defined ability. Coordination is, in essence, the ability to integrate all the components of fitness so that effective movements are achieved.[8] Rhythm, spatial orientation and the ability to react to both auditory and visual stimulus have also been identified as elements of coordination. Motor coordination can be broken into two components: gross motor coordination and fine motor coordination. *Gross motor coordination* refers to gross motor skills, such as walking, running, climbing, jumping, etc. *Fine motor coordination* refers to fine motor skills, such as drawing, writing, typing, etc.. In reference to athletic performance, gross motor coordination may entail more complex movements than simply walking, jumping, or running. Athletic coordination is the ability to combine several distinct movement patterns into a singular distinct movement.

Power

Power, in physics, is the "rate at which work is performed," *i.e.* work is the "product of force and distance" (Work = Force x Distance). What this usually translates to is the ability to exert maximum muscular contractioninstantly in an explosive burst of movement, ie. *the ability of a muscular unit, or combination of muscular units, to apply maximum force in minimum time*. The two components of power are strength and speed, as with power exercises. (*e.g.* jumping or a sprint start, snatch, clean and jerk, etc.) Power is a vital component of motor fitness, and is applicable especially to a myriad of athletic activities, and therefore it should not be neglected. Despite the importance of power for athletics and function, the ability to produce powerful muscle contractions decreases with age, more so than other components, such as cardiorespiratory endurance. This decline also appears despite persistent training and otherwise good health.

Speed

Speed is the rate of motion, or equivalently the rate of change in position, often expressed as distance traveled per unit of time. A subcategory of speed is quickness, which is the ability of the central nervous system to contract, relax or control muscle function without involvement of any preliminary stretch.

Reaction Time

Reaction time is the interval time between the presentation of a stimulus and the initiation of the muscular response to that stimulus. A primary factor affecting a response is the number of possible stimuli, each requiring their own response, that are presented. Examples include how fast a sprinter can get off the blocks and react to a gun, how quickly a boxer can react to an opponent's punch, or how quickly a batter can recognise and respond to a pitch.

Quickness

A subcategory of speed is quickness, which is the ability of the central nervous system to contract, relax or control muscle function without involvement of any preliminary stretch. Quickness is measured as the time interval or reaction time between voluntary stimulation and the initiation of movement. This time should be distinguished from absolute movement speed, which is the interval from the beginning to the end of movement.

NEED AND IMPORTANCE OF PHYSICAL EDUCATION

In the Present World of Space age and automation era, all human beings appear to be living a more and more inactive life. They ride instead of walk, sit instead of stand and watches instead of participants. Such type of inactivity or sedentary life is detrimental to mental and physical health. Thus, there is great need for physical education as a part of balanced living. Physical education which is commonly a part of the curriculum at school level includes training in the development and care of the human body and maintaining physical fitness. Physical education is also about sharpening overall cognitive abilities and motor skills via athletics, exercise and various other physical activities like martial arts and dance. Here are some of the benefits that highlight the importance of physical education.

MAINTAINING SOUND PHYSICAL FITNESS

Physical fitness is one of the most important elements of leading a healthy lifestyle. Physical education promotes the importance of inclusion of a regular fitness activity in the routine. This helps the students to maintain their fitness, develop their muscular strength, increase their stamina and thus stretch their physical abilities to an optimum level.

Physical fitness helps to inculcate the importance of maintaining a healthy body, which in turn keeps them happy and energized. Sound physical fitness promotes, increased absorption of nutrients, better functioning of digestion and all other physiological processes and hence results in all round fitness.

OVERALL CONFIDENCE BOOSTER

Indulging in sports be it team sports or dual and individual sports, leads to a major boost in self-confidence. The ability to go on the field and perform instills a sense of self-confidence, which is very important for the development of a person's character. Every victory achieved on the field, helps to boost a person's self-confidence. Moreover, the ability to accept defeat on field and yet believe in your own capabilities brings a sense of positive attitude as well. Thus participation in sports, martial arts or even dance and aerobics, is always a positive influence on a student's overall personality and character and works wonders for his/her self-confidence.

AWARENESS ABOUT IMPORTANT HEALTH AND NUTRITION ISSUES

Physical education classes are about participating in the physical fitness and recreation activities, but they are also about gaining knowledge about the overall aspects of physical health. For example in today's world the problems of obesity, or anemia and bulimia are rampant amongst teenagers. Physical education provides an excellent opportunity for teachers to promote the benefits of healthy and nutritious food and cite the ill effects of junk food. Promoting sound eating practices and guidelines for nutrition are some of the very valuable lessons that can be taught through physical education classes at school level.

INCULCATING SPORTSMANSHIP AND TEAM SPIRIT

Participation in team sports, or even dual sports helps to imbibe a sense of team spirit amongst the students. While participating in team sports, the children have to function as an entire team, and hence they learn how to organize themselves and function together. This process of team building hones a person's overall communications skills and the ability to get along with different kind of people. Thus participating in team sports instills a sense of team spirit, which is a great value addition to anyone's personality and helps a lot in all the future endeavors.

DEVELOPMENT OF MOTOR SKILLS

The ability to concentrate, the ability to swing the racket just at the right time are some of the examples of development of motor skills in the physical education classes. Participation in sports and several physical education activities helps to sharpen the reflexes of the students. It also brings order and discipline to the body movements and helps in development of a sound body posture as well. The hand-eye co-ordination improves as well.

IMPORTANCE OF HYGIENE AND SEX EDUCATION

Physical education classes also include lessons about the importance of personal hygiene and importance of cleanliness. Thus the physical education classes help the students to know the important hygiene practices that must be practiced in order to maintain the health and well being throughout the life. In addition to this, the physical education classes also cover an important aspect that the children have to deal with at the age of puberty. Physical education classes also impart sex-education and hence help the students deal with their queries and doubts about the subject of sexuality.

ENHANCING OVERALL COGNITIVE ABILITIES

Physical education classes help to enhance the overall cognitive abilities of the students, since they get a lot of knowledge about the different kinds of sports and physical activities that they indulge in. For example a person who is participating in a specific type of martial arts class, will also gain knowledge about the origins of the martial art, and the other practices and historical significance associated with it. Thus physical education helps to enrich the knowledge bank of the students.

ENCOURAGING BUDDING SPORTSMEN

Physical education classes are an excellent opportunity for all the budding sportsmen and sportswomen who wish to make their mark in the world of sports. Physical education classes allow the budding sportsmen and sportswomen to explore and experiment with several areas until they find what interests them. After this, physical education classes also allow the students to indulge the sport of their choice and then go ahead to participate in several tournaments and competitions, which help to give the students an exposure to the competitive world of sports.

A STRESS BUSTER AND SOURCE OF ENJOYMENT

In addition to the health benefits and the knowledge benefits that the students get from the physical education classes, one important aspect of it remains to be recreation. Students, who are busy with their other subjects in the curriculum, often get exhausted with the listening, reading and writing pattern of studying and need a recreational activity as a source of recreation. Sports and other physical fitness activities offered in the physical education class are a welcome break for the students.

PROMOTING HEALTHY LIFESTYLE IN ADULTHOOD

Children, who learn the importance of health and hygiene in their early ages, tend to grow up to be responsible and healthy adults who are well aware of the benefits of a healthy lifestyle. Thus the overall physical education programme, that includes different types of physical activities and sports and also provides important information about hygiene and overall health, helps in creating well-informed pupils. A well-balanced and all-round physical education class helps to create responsible adults who know the importance of a healthy lifestyle.

3

Coaching: The Challenges and the Benefits

Coaching, is a teaching, training or development process via which an individual is supported while achieving a specific personal or professional result or goal. The individual receiving coaching may be referred to as the client or coaches. Occasionally, the term coaching may be applied to an informal relationship between two individuals where one has greater experience and expertise than the other and offers advice and guidance as the other goes through a learning process. This form of coaching is similar to mentoring.

The structures, models and methodologies of coaching are numerous, and may be designed to facilitate thinking or learning new behaviour for personal growth or professional advancement. There are also forms of coaching that help the coachee improve a physical skill, like in a sport or performing art form. Some coaches use a style in which they ask questions and offer opportunities that will challenge the coachee to find answers from within him/herself.

This facilitates the learner to discover answers and new ways of being based on their values, preferences and unique perspective. When coaching is aimed at facilitating psychological or emotional growth it should be differentiated from therapeutic and counseling disciplines, since clients of coaching, in most cases, are considered healthy (*i.e.* not sick). The purpose of the coaching is to help them move forward in whatever way they want to move, not to 'cure' them. In addition the therapist or counsellor may work from a position of authoritative doubt, but cannot claim the position of ignorance so vital for coaching, because of the assessment knowledge that underpins their work. The UK's Chartered Institute of Personnel Management reports that 51% of companies (sample of 500) 'consider coaching as a key part of learning development' and 'crucial to their strategy', with 90% reporting that they use coaching. More recent research in 2011 by Qa Research, an independent marketing research agency in the UK, found that 80% of organisations surveyed had used or are now using coaching, but also found that while 90% of organisations with over 2,000 employees had used coaching in the past five years, only 68% of companies with 230–500 employees had done the same. The basic skills of coaching are often developed by managers within organizations specifically to improve their managing and leadership abilities, rather than to apply in formal one-to-one coaching sessions. These skills can also be applied within team meetings and are then akin to the more traditional skills of group facilitation.

ORIGINS OF COACHING

Etymologically, the English term "coach" is derived from a medium of transport that traces its origins to the Hungarian word *kocsi* meaning "carriage" that was named after the village where it was first made. The first use of the term coaching to mean an instructor or trainer arose around 1830 in Oxford University slang for a tutor who "carries" a student through an exam. Coaching thus has been used in language to describe the process used to transport people from where they are, to where they want to be. The first use of the term in relation to sports came in 1831.

Historically the evolution of coaching has been influenced by many other fields of study including those of personal development, adult education, psychology (sports, clinical, developmental, organizational, social and industrial) and other organizational or leadership theories and practices. Since the mid-1990s, coaching has developed into a more independent discipline and professional associations such as the Association for Coaching, The International Coach Federation, and the European Coaching and Mentoring Council have helped develop a set of training standards. Janet Harvey, president of the International Coach Federation, was quoted in a New York Times article about the growing practice of Life Coaching, in which she traces the development of coaching to the early 1970s Human Potential Movement and credited the teachings of Werner Erhard's "EST Training," the popular self-motivation workshops he designed and led in the '70s and early '80s.

Thomas Leonard who founded "Coach U", "International Coach Federation", "Coachville" and "the International Association of Coaches" was an EST employee in the 1980 's. After having resigned from EST, Thomas Leonard continued being always concerned witheST or Landmark-Education as a business partner. He is always under the influence of Werner Erhard, and participated in foundation of "Coaches Training Institute". Naming of "coaching" is also Erhard 's influence clearly. Werner Erhard believed “Affirmations (New Age)” and took his original expressions in the seminars. For example, he often added “-ing” to behind the words which have both functions of a verb and the noun. The usage became like vogue words in the ’70s and early ’80s. The facilitative approach to coaching in sport was pioneered by Timothy Gallwey; before this, sports coaching was (and often remains) solely a skills-based learning experience from a master in the sport. Other contexts for coaching include executive coaching, life coaching, emotional intelligence coaching and wealth coaching.

APPLICATIONS

There are many definitions of coaching, mentoring and various styles of management and training. What follows are more succinct definitions of the various forms of helping. However, there may be overlap between many of these types of coaching activities. Managing is making sure people do what they know how to do. Training is teaching people to do what they don’t know how to do. Mentoring is showing people how the people who are really good at doing something do it. Counselling is helping people come to terms with issues they are facing. Coaching is none of these – it is helping to identify the skills and capabilities that are within the person, and enabling them to use them to the best of their ability. Professional coaching uses a range of communication skills (such as targeted restatements, listening, questioning, clarifying *etc.*) to help clients shift their perspectives and thereby discover different solutions to achieve their goals. These skills are used when coaching clients in any field. In this sense, coaching is a form of ‘meta-profession’ that can apply to supporting clients in any human endeavor, ranging from their concerns in personal, professional, sport, social, family, political, spiritual dimensions, *etc.*

Life Coaching

Life coaching draws upon a variety of tools and techniques from other disciplines such as sociology, psychology, positive adult development and career counseling with an aim towards helping people identify and achieve personal goals. Specialty life coaches may have degrees in psychological counseling, hypnosis, dream analysis, marketing and other areas relevant to providing guidance. However, they are not necessarily therapists or consultants; psychological intervention and business analysis may lie outside the scope of some coaches’ work. The phrase “life coach” dates from at least 1941.

Contemporary life coaching can be traced to the teachings of Benjamin Karter, a college football-coach turned motivational speaker of the late 1970s and early 1980s. Many life-coach training-schools and -programmes operate worldwide, providing options (classroom attendance or home study) for the individual who wants to gain a certificate or diploma and to take up paid work in the field of life coaching. Many certificates and a few diplomas are available within the profession. On August 14, 2012, the term *Life Coach* appeared for the first time in Merriam-Webster’s *Collegiate Dictionary*.

Regulation

Critics see life coaching as akin to psychotherapy but without restrictions, oversight, regulation, or established ethical policies. Regulators have addressed some of these concerns on a state-by-state basis. In 2009, the State of Tennessee issued a memorandum emphasizing that life coaches may be subject to discipline if they perform activities construable as personal, marital, or family counseling. Some other states have made no formal statement but have legal statutes that broadly define mental-health practice. Hawaii, for example, defines the practice of psychology as any effort aimed at behaviour change or to improve "interpersonal relationships, work and life adjustment, personal effectiveness, behavioral health, [or] mental health." Although such states usually provide some exclusions to licensure requirements (such as for ordained clergy), life coaches usually fall under such statutes. More favourably to life coaches, in 2004 the Colorado General Assembly specifically exempted trained life-coaches from licensure requirements that apply to other mental and behavioral health professionals in that state.

ADHD Coaching

ADHD coaching is a specialized type of life coaching that uses specific techniques designed for working with the unique brain wiring of individuals with attention-deficit hyperactivity disorder. Coaches work with clients to help them better manage time, organize, set goals and complete projects. In addition to helping clients understand the impact ADHD has had on their lives, coaches can help clients develop "work-around" strategies to deal with specific challenges, and determine and use individual strengths. Coaches also help clients get a better grasp of what reasonable expectations are for them as individuals, since people with ADHD "brain wiring" often seem to need external mirrors for accurate self-awareness about their potential despite their impairment.

Business Coaching

Business coaching is a type of personal or human resource development. It provides positive support, feedback and advice to an individual or group basis to improve their personal effectiveness in the business setting. Business coaching includes executive coaching, corporate coaching and leadership coaching. The Professional Business Coach Alliance, The International Coach Federation, the International Coaching Council and the Worldwide Association of Business Coaches provide a membership-based association for business coaching professionals.

These and other organizations train professionals to offer business coaching to business owners. According to a MarketData Report in 2007, an estimated 40,000 people in the US, work as business or life coaches, and the $2.4 billion industry is growing at rate of 18% per year.

According to the National Post, business coaching is one of the fastest growing industries in the world. There are almost as many different ways of delivering business coaching as there are business coaches. Some offer personal support and feedback, others combine a coaching approach with practical and structured business planning and bring a disciplined accountability to the relationship. Particularly in the small business market, business coaching is as much about driving profit as it is about developing the person.

Coaching is not a practice restricted to external experts or providers. Many organizations expect their senior leaders and middle managers to coach their team members to reach higher levels of performance, increased job satisfaction, personal growth, and career development. Business coaching is not the same as mentoring. Mentoring involves a developmental relationship between a more experienced "mentor" and a less experienced partner, and typically involves sharing of advice. A business coach can act as a mentor given that he or she has adequate expertise and experience. However, mentoring is not a form of business coaching.

DEFINITION OF COACHING

There are many definitions for what coaching is all about. The dictionary defintion of coaching* is: "A method of directing, instructing and training a person or group of people, with the aim to achieve some goal or develop specific skills. There are many ways to coach, types of coaching and methods to coaching. Direction may include motivational speaking and training may include seminars, workshops, and supervised practice."

*Definition taken from the Wikipedia, the free Encyclopaedia.

BENEFITS OF COACHING

The benefits of coaching is that the individual will be able to improve their work performance and skill set by receiving one-on-one training to develop career prospects. The majority of coaching is generally delivered within an organisation by an immediate supervisior or manager. However, many organisations these days employ professional external coaches to come into their organisation to provide this service.

Coaches should be willing to listen, observe and support the coachee's ability, knowledge and resourcefulness. External coaches are trained to deliver specific individual coaching sessions to meet the individual needs, following the methods of setting clear standards, goals, use of learning, feedback and evaluation.

DIFFERENCES BETWEEN COACHING AND MENTORING?

The differences between coaching and mentoring are quite clear:

Coaching

Generally follows the format of individual guidance that is focused on job performance and aimed at one person alone. The coach specificially advises the person on how to tackle and perform a particular task, they provide constructive feedback and delegate further similiar tasks, setting goals or higher-level tasks for the individual to complete. The coach in most instances will be an immediate supervisior or manager who will have overall resonsibility for the department's overall performance. Coaching is about having a positive relationship where the coachee respects, trusts and identifies with the coach.

Mentoring

Generally follows the format of generalised advice and guidance of career development. Mentoring is about developing a relationship between a more senior and experienced mentor and an inexperienced mentee to guide and develop the mentee's knowledge and career progression. The mentor generally will be someone who is not your immediate supervisior/manager or within your organisation can allow the luxury of talking to an independent impartial confidante who is not your manager, they will have the ability to listen to your issues, afford you the opportunity to vent unrestrictedly and support and assist you in achieving your goals. However saying this, some organisations who offer mentoring for their employees may allocate a mentor within the organisation who has similiar standing, status and power as your immediate supervisior.

COACHING PERFORMANCE

Achieving excellence through performance is accomplished in two major ways. The first way is taking a proactive stance by unearthing or preventing counter-productive methods. For example, you might implement diversity and sexual harassment training programmes before they become a problem within the organisation.

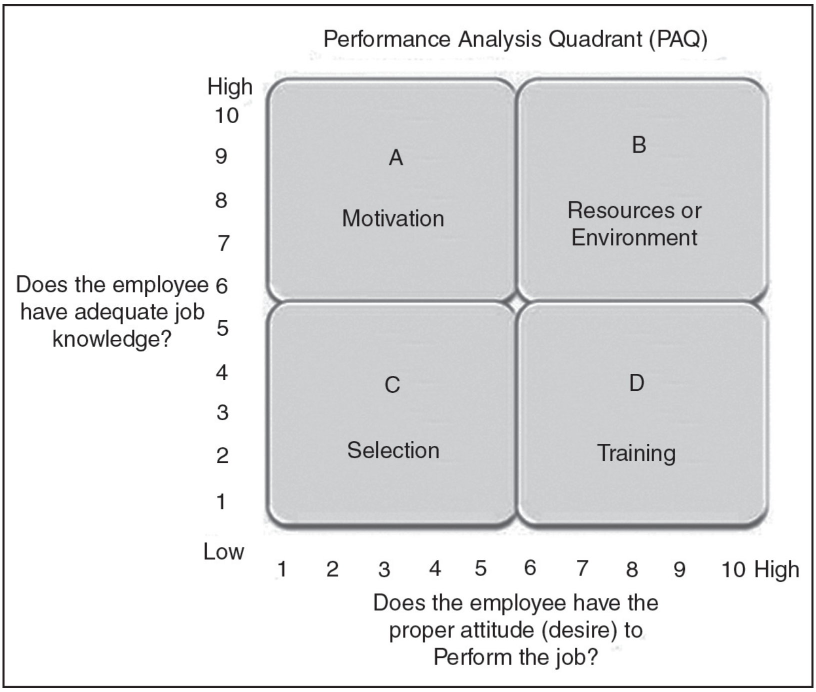
The second way is to correct performance problems as they arise within the organisation. This is accomplished by first, identifying the root cause and secondly, implementing a plan of action to correct the problem. Although people are our most important asset, it sometimes seems as if they are our biggest headache. Both training and coaching are are used to help fix performance problems when part of the solution requires learning, but each uses a different approach: 1) training uses a structured design to provide the employees with the knowledge and skills to perform a task, while 2) coaching is a process that helps the employees gain greater competence and overcome barriers to improve job performance on a as needed basis..

Coaching has been defined as 1) the processes of encouraging the individual to improve both job skills and knowledge, 2) to assist in problem solving or mastering new skills, 3) and the process of providing others with valuable information so that the organisation learns.

There are four major causes of performance problems:

- *Knowledge or Skills* - The employee does not know how to perform the process correctly - lack of skills, knowledge, or abilities.
- *Process* - The problem is not employee related, but is caused by working conditions, improper procedures, etc.
- *Resources* - Lack of resources or technology.
- *Motivation or Culture* - The employee knows how to perform, but does so incorrectly.

THE PERFORMANCE ANALYSIS QUADRANT (PAQ)



The Performance Analysis Quadrant (PAQ) is a tool to help in the identification. By asking two questions, "Does the employee have adequate job knowledge?" and "does the employee have the proper attitude (desire) to perform the job?" and assigning a numerical rating between 1 and 10 for each answer, will place the employee in 1 of 4 the performance quadrants:

1. *Quadrant A (Motivation):* If the employee has sufficient job knowledge but has an improper attitude, this may be classed as motivational problem. The consequences (rewards) of the person's behaviour will have to be adjusted. This is not always bad as the employee just might not realize the consequence of his or her actions.
2. *Quadrant B (Resource/Process/Environment)*: If the employee has both job knowledge and a favourable attitude, but performance is unsatisfactory, then

the problem may be out of control of the employee. *i.e.* lack of resources or time, task needs process improvement, the work station is not ergonomically designed, etc.

3. *Quadrant C (Selection):* If the employee lacks both job knowledge and a favourable attitude, that person may be improperly placed in the position. This may imply a problem with employee selection or promotion, and suggest that a transfer or discharge be considered.
4. *Quadrant D (Training and or Coaching)*: If the employee desires to perform, but lacks the requisite job knowledge or skills, then additional training or coaching may be the answer.

The solution does not have to be the same as the cause. For example, you can often fix a process problem with training or perhaps fix a motivation problem with attitude or (affective domain) coaching. While coaching may be used for all four problems depending on the circumstances, it is best used for Quadrant D—Training and/or Coaching. Large groups are generally dealt with training as it is more efficient, while coaching is used for small groups; however, coaching is normally required during training due to individual needs. Note: The four quadrants are based on Jones' (1993) description of the four factors that affects job performance.

Motivation

Often the employee knows how to perform the desired behaviour correctly, the process is good, and all resources are available, but for one reason or another, chooses not to do so. It now becomes a motivational issue. Motivation is the combination of a person's desire and energy directed at achieving a goal. It is the cause of action. Motivation can be intrinsic - satisfaction, feelings of achievement; or extrinsic - rewards, punishment, or goal obtainment. Not all people are motivated by the same thing, and over time their motivation may change.

Although most jobs have problems that are inherent to the position, it is the problems that are inherent to the person that cause us to loose focus from our main task of getting results. These motivational problems could arrive from family pressures, personality conflicts, a lack of understanding how the behaviour affects other people or process. When something breaks the psychological contract between the employee and the organisation, the leader must find out what the exact problem is by looking beyond the symptoms, find a solution, focus on the problem, and implement a plan of action. One of the worst situations that a leader can get into is to get all the facts wrong. Start by collecting and documenting what the employee is not doing or should be doing - tasks, special projects, reports, etc. Try to observe the employee performing the task. Also, do not make it a witch hunt, observe and record what the employee is not doing to standards. Check past performance appraisals, previous managers, or other leaders the employee might have worked with. Try to find out if it a pattern or something new.

Once you know the problem, then work with the employee by using coachingtechniques to solve it. Most employees want to do a good job, thus they will normally be happy to accept coaching to improve themselves. It is in your best interest to work with the employee as long as the business needs are met and it is within the bonds of the organisation to do so.

Process or Environmental Problems (Not Related to Employees)

Many performance problems are due to bad process, that is, the process does not support the desired behaviour. It has often been said that people account for 20percent of all problems while bad processes account for the rest. However, coaching can be used to get people to think about ways to solve the process or environmental problem.

Lacks the Skills, Knowledge, or Abilities to Perform

This problem generally arises when then is a new hire, new or revised process, change in standards, new equipment, new policies, promotion or transfer, or a new product. In this case, there is only one solution - training. The training may be formal classes, on-the-job, self-study, coaching, etc. To determine if training/coaching is needed, we only need to ask one question, “Does the employee know how to perform the task?” If the answer is yes, then training is not needed. If the answer is no, then training is required. However, this page will focus on coaching skills.

COACHING SKILLS

Some people tend to use the terms coaching, mentoring, and training interchangeably. However, there are differences. Mentoring is often thought of as the transfer of wisdom from a wise and trusted teacher. He or she helps to guide a person’s career, normally in the upper reaches of the organisation. However, this perception is starting to change as organisations are now implementing mentoring at all levels of the company’s structure. Training is about teaching or instructing a particular skill or knowledge and is normally given in a formal environment.

Coaching, on the other hand, is about increasing an individual's knowledge and thought processes with a particular task or process. It creates a supportive environment that develops critical thinking skills, ideas, and behaviours about a subject. Although it is closely tied to training, it is more personal and intimate in nature. The main difference between a coaching and a training is that the former is normally done in real time. That is, it is performed on the job. The coach uses real tasks and problems to help the learner increase his or her performance. While with training, learning is performed within the classroom.

Mentoring is more career developing in nature, while training and coaching are more task or process orientated. Also, mentoring relies on the mentor's specific knowledge and wisdom, while coaching and training relies on facilitation and developmental skills. Although there are these differences, you could say that the three are synergistic and complementary, rather than mutually exclusive as most people would agree that a good coach trains and mentors, a good trainer coaches and mentors, and a good mentor trains and coaches.

A performance coach is also a:

- Leader - who sets the example and becomes a role model.
- Facilitator - is able to instruct a wide verity of material.
- Team Builder - pulls people into a unified team.
- Peace Keeper - acts as a mediator.
- Pot Stirrer - brings controversy out in the open.
- Devil's Advocate - raises issues for better understanding.
- Cheerleader - praises people for doing great.
- Counselor - provides intimate feedback.

In order to coach, it helps to use a few facilitating techniques:

- Draws people out:
- "What do others think?" or "What do you think?"
- "I've heard from (name) so far... are there any other thoughts?"
- "And what else?"
- Silence (20-30 seconds) - gives the person a chance to think. Also, people tend to abhor silence, thus if you wait long enough, the person will usually speak up.
- "You look like you have something to say..."
- Interprets comments:
- Words verses tune or tone (many questions are not really questions but a need for self-assurance).
- Intent verses wording (learners often have a hard time wording new subject matters).
- Sees beyond the learners paradigms and filters.
- Clarifies thoughts or comments:

- Use models and experiences to bring life to the subject.
- Looks for multiple points to expound on the subject.
- Looking for similarities and differences.
- Senses group energy:
- Sparks up the group with various energizers.
- Takes breaks as needed.
- Has a sense of timing.
- Handling objections:
- Try not to personalize (the learner will become defensive).
- Reflect on the objection for a moment to ensure you understand the objection.
- Encourages conversation.
- Remembers to breath and relax.
- How we treat each other:
- Accepting each other into the group.
- Individual responsibility.
- Being right verses being successful.
- Influence verses dominance (pull rank).
- Confidentiality and trust.
- Supporting each other.
- Active listening.
- Conflict resolution.

RESOURCES

Just because the problem is caused by a lack of resources or technology, does not mean expenditures are needed. Remember, the solution does not always have to be the same as the cause. In this case you might be able to get your team to brainstormnew processes or procedures that will eliminate the need for new resources. In addition, coaching can be used to get people to try new methods that require less resources.

CAUSES OF PROBLEMS

Expectations or Requirements Have not Been Adequately Communicated

This motivational issue is not the fault of the employee. By providing feedback through coaching and ensuring the feedback is consistent, you provide the means for employees to motivate themselves to the desired behaviour. For example, inconsistent feedback would be for management to say it wants good safety practices, and then frowns on workers who slow down by complying with regulations. Or expressing that careful workmanship is needed, but reinforces only volume of production. Feedback must be provided on a continuous basis.

If you only provide it during an employee's performance rating period, then you are NOT doing your job. Also, ensure that there is not a difference in priorities. Employees with several tasks and projects on their plates must be clearly communicated as to what comes first when pressed for time.

With the ever increasing notion to do more with less, we must understand that not everything can get done at once. Employees often choose the task that they enjoy the most, rather than the task they dislike the most. And all too often that disliked task is what needs to get performed first.

Lack of Motivation

A lack of motivation could be caused by a number of problems, to include personal, family, financial, etc. Use coaching to help employees recognise and understand the negative consequences of their behaviour.

Shift in Focus

Today, it's a lucky employee (or unlucky if that employee thrives on change) that does not have her job restructured. Changing forces in the market forces changes in organisations.

When this happens, use coaching to ensure that every employee knows:

- How the job changed and what are the new responsibilities.
- Why the job was restructured.
- How their performance be evaluated and by whom.
- New skills they will need to learn.
- What responsibilities will be delegated.
- How their career will benefit from this transition.
- What new skills or training they need to perform successfully.
- How it will make them more marketable in the future.

By keeping them informed, you help to eliminate some of the fear and keep them focused on what must be performed.

PERFORMANCE FEEDBACK VERSES CRITICISM

In general, there are two different forms of information about performance — feedback and criticism. Feedback was originally an engineering term that refers to information (outcome) that is fed back into a process to indicate whether that process is operating within designated parameters. For example, the sensor in a car's radiator provides feedback about the engine temperature. If the temperature rises above a set point, then a secondary electrical fan kicks in.

When dealing with human performance, feedback refers to observable behaviours and effects that are objective and specific. This feedback needs to be emotionally neutral information that describes a perceived outcome in relation to an intended target. For example, during a meeting, you announce the tasks and how to perform them. A better way is to ask for input on how the tasks should be performed. This gives people the opportunity to take ownership of their work. People who receive feedback in this manner can use the data to compare the end results with their intentions. Their egos should be aroused, but not bruised.

Compare this to criticism that is emotional and subjective. For example, "You dominate the meetings and people do not like it!" The recipient has much more difficulty identifying a changeable behaviour other than to try to be less dominant. Also, the angry tone of the criticism triggers the ego's defensive layer and causes it to be confrontational or to take flight (fight or flee), thus strengthening the resistance to change; which is exactly the opposite of what needs to be done. Delivering effective performance feedback takes time, effort, and skill; thus people often fall back to criticism. Since we receive far more criticism than feedback, our egos have become accustomed to fighting it off. We have all seen people receive vital information, yet shrug it off through argument or denial, and then continue on the same blundering course.

Receiving Feedback

Being able to give good feedback should not be the only goal; we also must be aware of the need to receive and act upon feedback, even if it is delivered in a critical manner. That is, we need to develop skills that help us extract useful information, even if it is delivered in a critical tone.

Allowing attitudes of the criticizer to determine your response to information only weakens your chances for opportunity. Those who are able to glean information from any source are far more effective. Just because someone does not have the skills to give proper feedback does not mean you cannot use your skills to extract useful information for growth. When receiving information, rather it be feedback or criticism, think "How can I glean critical information from the message?" Concentrate on the underlying useful information, rather that the emotional tones. Also note what made you think it was criticism, rather than feedback? This will help you to provide others with feedback, rather than the same emotional criticism.

The Feedback Process

Giving feedback, instead of criticism, can best be accomplished by following two main avenues:

- Observing behaviour - Concentrate on the behaviour. Why is it wrong for the organisation, team, individuals, etc.; not why you personally dislike it. Your judgement needs to come from a professional opinion, not a personal one. Report exactly what is wrong with the performance and how it is detrimental to good performance.
- Concentrate on pointing out the exact cause of poor performance. If you cannot determine an exact cause, then it is probably a personal judgement that needs to be ignored.
- State how the performance affects the performance of others. Again, if it does not affect others, then it is probably a personal judgement.
- Do unto others, as you want them to do unto you - Before giving the feedback, frame the feedback within your mind.
- It might help to ask yourself, "how do I like to be informed when I'm doing something wrong?"
- What tones and gestures would best transfer your message? Remember, you want the recipient to seriously consider your message, not shrug it off or storm away.

SINGLE-LOOP AND DOUBLE-LOOP LEARNING

Coaching all too often focuses too narrowly on problem solving by identifying and correcting errors. Yet if the coaching is to transfer to the work environment, it must also look inwardly by allowing the learners to reflect critically on their own behaviour. Chris Argyris (1991) coined the terms *single-loop* and *double-loop* learning.

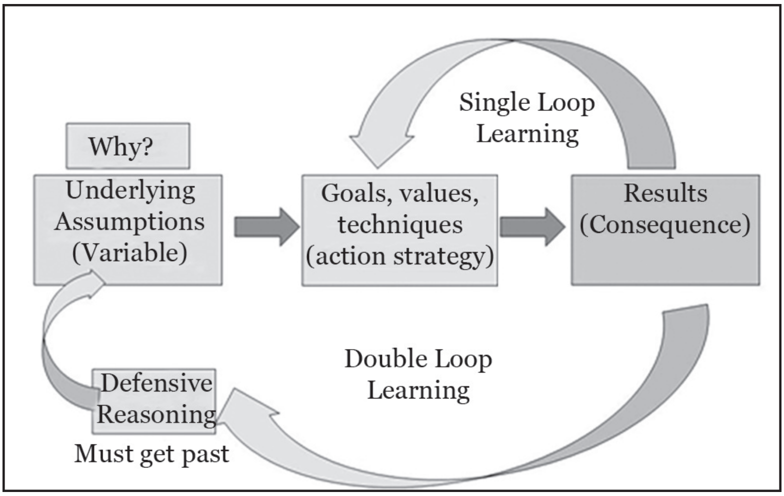


Fig. Double Loop Learning: Argyris and Schon

Single loop learning has often been compared to a thermostat in that it makes a decision to either turn on or off. Double loop learning is like a thermostat that asks "why" — Is this a good time to switch settings? Are there people in here? Are they in bed? Are they dressed for a colder setting? Thus it orientates itself to the present environment in order to make the wisest decision.

A person who is double loop learning is basically orientating herself to all possible solutions within her environment by asking a series of "whys" that is similar to Sakichi Toyoda's (the founder of Toyota) method who used a technique he called the *Five Whys* — when confronted with a problem you ask "why" five times. By the time the fifth why is answered, you should be at the root cause of the problem.

In Borat's cheesy video shown above, we notice that Borat is learning that all the packages are cheese in one form or another. He might even go on to an advanced cheese class and actually taste some cheeses and learn how to prepare a few dishes. And this is where a lot of coaching ends — at the single-loop learning stage in which we learn about something and how it can solve some of our problems.

Yet Borat still faces the daunting task of actually using his new skills and knowledge in the workplace. And when he meets the slightest form of resistance of putting his new found learnings into action, *e.g.*, "we only use block cheese," he is going to place the blame on the external environment. This allows any failings to fall upon external forces.

Thus his ability to learn has been shut down at the time that he really needs it the most. Rather than shifting to double-loop learning by asking deep reflective questions on why he cannot do it and then taking action to actually do it, he shifts to a defensive reasoning pattern that ensures any criticisms are shifted to others or to the environment. For coaching to be the most effective, then we need to allow the learners to look at their own reasoning patterns so that they can detect any inconsistencies between their espoused and actual theories of action.

Note that the supermarket manager is analogous to coaches who only teach "packaging" and pay no mind to the differences in substance. He presented every package as cheese even though the product might have been quite different from variety to variety. For example, Feta and Colby are both cheeses but if a recipe calls for one, they are not normally interchangeable. Distinctions often matter, even if it is only cheese.

Borat seems quite confused that the same product could be offered in so many different packages; he has not yet realized that the product within differs. A good coach allows the learner to go deeper into the process, rather than just gloss over the surface so that when he or she returns to work, they can use their new knowledge and skills in a variety of contexts. In compliance environments, employees are told what to do. Although you may turn them loose to perform their jobs, the goals and objectives come from upper-management. In commitment environments, employees are involved in determining the strategies, directions, and tasks needed to achieve the organisation's objective's.

This is accomplished by:

- Involve all essential people in developing action plans in areas that are critical to success.
- Identify critical success factors and formulate the plans necessary to achieve those objectives. Everyone in the department, from the front-line workers to managers are used in this process.
- Drive the methodology deeper into the organisation by cultivating an environment in which almost everything is linked to employee involvement. The heart of this strategy is by sharing information and involving people at all levels of the organisation. Also, hold regular team meetings in which everyone is encouraged to speak what is on their mind.
- Give workers direct access to top management. This keeps top-management in tune with the wants and needs of front-line employees.

By using various coaching techniques to bring them into the process, they understand the problems and have a say in the commitment. This engages their hearts, minds, and hands... the greatest motivators of all!

TRADITIONAL VERSUS GAMES APPROACH TO COACHING

As mentioned previously, transferring skills from practice to games can be difficult. A sound background of technical and tactical training prepares athletes for game situations. But you can surpass this level by incorporating gamelike situations into daily training, further enhancing the likelihood that players will transfer skills from practices to games. To understand how to accomplish this, you must be aware of two approaches to coaching: the traditional approach and the games approach.

TRADITIONAL APPROACH

Most coaches are comfortable with the traditional approach to coaching. This method often begins with a warm-up period followed by a set of drills, a scrimmage and finally a cool-down period. This approach can be useful in teaching the technical skills of softball, but unless coaches shape, focus and enhance the scrimmages or drills, the athletes may not successfully translate the skills to game situations, leaving coaches to ponder why their team practices better than it plays.

Games Approach

Using the tactical triangle in practice supplies athletes with the tools that they need to make appropriate and quick decisions. But unless they can employ these tools in game situations, they are of little value. You have surely seen players jump into the batting cage in practice and tear the cover off the ball on the tees or the pitching machine but then have trouble making good contact after the game begins. This type of hitter has learned the art of performing well in drills but has not learned how to transfer those technical skills to tactical situations that occur during a game. Some people call this choking, but a more accurate description would be failure to adapt. The same sort of thing happens to the player who can field every ground ball flawlessly in practice but bobbles easy grounders in a game or lets them go through her legs.

The best way to prevent this scenario is to use the games approach to coaching, which provides athletes with real-time, gamelike situations in training that allow them to practice and learn the skills at game speed. This philosophy stresses the importance of putting technical skills rehearsed in drills into use in practice. You can drill players in a skill like bunting until they are sore, but if they never get the opportunity to use the skill in a gamelike setting, they will not be able to perform when it really counts—in an actual game. When players make mistakes in game-speed situations, they learn. You have to provide gamelike opportunities in which players can feel secure about making mistakes so that they can file those mistakes in the "softball sense" parts of their brains. By doing so, the chances of their making the same mistakes in games will lessen.

The games approach emphasizes the use of games and minigames to provide athletes with situations that are as close to a real game as possible. This approach requires more than just putting the team on the field, throwing them a ball and letting them play. Rather, according to Launder, the games approach includes three components that make each minigame educational:

1. Shaping
2. Focusing
3. Enhancing

Shaping play means modifying the game in a way that is conducive to learning the skills that you want to teach in that particular setting. The games approach shapes play by modifying the rules, the environment (playing area), the objectives of the game and the number of players used. In a typical scrimmage situation, the stronger players dominate and the weaker players rarely get a chance to play an active role. When play is shaped, for example, by reducing the number of players—the weaker players are put into positions where they will have more opportunities to play active roles. But you cannot simply shape the play and expect miracles to happen. You need to focus your athletes' attention on the specific objectives that you are trying to achieve with the game. Young players are more apt to learn, or at least to reduce their reluctance to learn, if they know why you are asking them to grasp new tactical information.

Knowing how the tactic fits into the team's game plan or season plan also helps players buy into the tactic. You can assist your athletes with this phase by providing them with clear objectives and explaining how learning those objectives elevates their capability to play and helps their team win games. Shaping play and focusing players on objectives, however, cannot be successful unless you play an active role and work on enhancing their play. You can enhance minigames by adding challenges to make the contests between the sides equal. You can also enhance play by encouraging your players and give them confidence by frequently pointing out their progress.

Minigames also give you an opportunity to stop the game whenever you recognise an opportunity to teach something that will improve their play even further.

Most coaches have used aspects of the games approach one way or another in their training sessions. Although you may already have a basic understanding of how to use this approach, this book takes the concept further by presenting a games approach season plan as well as sample practices for you to use with your team.

Both the traditional and the games approach are sound coaching practices. Part IV examines both approaches to teaching the skills in softball. Although both approaches have value, the philosophy of this book slants towards the latter. Providing athletes with game-speed, real-time situations that have clear objectives creates a productive, fun-filled learning environment.

Athletes who have learned to think of training as a necessary evil will be more motivated to come to practice if they are engaged on a daily basis. More important, if they sense that they have ownership over what they learn in practice, they become more responsible team members. An added benefit is that softball players who learn through the games approach will be better prepared for competition because they have already faced stiff challenges in their everyday practice sessions.

Knowing how to teach the technical and tactical skills of softball is important, but you will never know how your players are performing unless you create good assessment systems. Next, you must learn how to evaluate players.

COACHING PROCESS

Coaching is best conducted if the coach is fully aware and knowledgeable of the theories and practices involved in the process. Aside from that, coaching is most effective if the coach can demonstrate a variety of skills, styles, and techniques suitable to the context in which coaching is conducted. The activity is dynamic and broad. In the workplace, a coach must know the theoretical concepts and must fully embrace their functions before progressing to the application.

THE VALUE OF COACHING PROCESS

Professional coaches follow a set of standards in their coaching activities. This includes utilizing a coaching model, a coaching flow, or simply coaching procedures. Following a coaching process serves as a guide for the coach to attain the main goal of coaching in an effective manner. Any incoherence in the coaching procedures will definitely not make the coaching a success. An effective implementation of coaching in workplace involves a constant adjustment to the ever-evolving systems and structures in an organisation. Hence, coaching processes must be modified regularly to suit the needs of the individuals. If a coach is stuck to a single approach in coaching, the outcome may not be effective. Individuals may see it as a routine practice in the office and will not recognise the effect of coaching on their careers. They will treat it more as a compliance to organisational practices rather than an opportunity to become better performers.

The coaching process allows a structured approach in its implementation but must not be restricted to one kind of approach. In coaching there must be variations in the use of styles and techniques.

The Stages in the Standard Coaching Process Model

A standard coaching process matrix has four stages.

Stage 1: Analyse for Awareness

Coaching is the solution when the learner realizes the need to develop performance or change certain ways in doing things. There has to be a willingness from the learner's end to undergo coaching, and the coach plays a vital role in making the learner realize this awareness.

Another way of determining the need for coaching is through a recommendation from the supervisor or team leader on the need to improve member's performance and enhance certain skills.

Stage 2: Plan for Responsibility

This stage gives the learner a chance to take on responsibility for developing performance. Although it is helpful to use learning programmes during coaching, this must not be strictly imposed on the person. Learners must also be actively participating in elaborating the learning style, in finding something that is conducive to their level and ability.

Stage 3: Implement Using Styles, Technique and Skill

After the planning stage in coaching, the next phase is to identify coaching styles and techniques that are deemed appropriate for the situation or the level of need of the learner. Moreover, this includes a test of the effective coaching skills of the coach to successfully conduct the coaching session. Coaching drills and activities geared towards developing the skill and performance of the individual are also used.

Stage 4: Evaluate Success

The last stage in the coaching process model is monitoring progress of the learner's performance after the coaching session. This has the role to check if the person has made any significant improvements or positive changes as a result of the coaching.

Four-Step Coaching Process

The four major steps in the process of coaching are:

1. *Observation:* A coach must have sufficient preparation before doing any coaching no matter it is an on-the-spot coaching or a schedule session. This would entail getting a good understanding of the learner's current performance as well as his strengths and weaknesses.
2. *Discussion:* During the more detailed preparation of the coverage for coaching the issues to be addressed must be discussed by the coach and the learner.
3. *Active Coaching:* This is where the actual coaching sessions occur. Feedback must be given and proper facilitation must be observed.
4. *Follow-up:* The last step is about keeping track of the learner's progress and performance trend. This is a chance for the coach to recognise any development and identify opportunities for continuing the coaching.

EXECUTIVE COACHING

Executive coaching is designed to help facilitate professional and personal development to the point of individual growth, improved performance and contentment. Most important, the coach attempts to stimulate the client's self-discovery by posing powerful questions and/or assigning homework that may take the form of "thought experiments" with written product or "field experiments" which are actions to try in the real world that may result in experiential learning and development of new approaches to situations. Coaches need to have a strong understanding of individual differences in a work place as well as the ability to adapt their coaching style or strategies.

It is suggested that those coaches who are unable to acknowledge these differences will do more harm than good. Many executive coaches have a specific area of expertise: sports; business or psychology. Regardless of specific area of focus, coaches still need to be aware of motivational needs and cultural differences.

Executive coaches work their clients towards specific professional goals. These include career transition, interpersonal and professional communication, performance management, organizational effectiveness, managing career and personal changes, developing executive presence, enhancing strategic thinking, dealing effectively with conflict, and building an effective team within an organization. An industrial organizational psychologist is one example of executive coaching.

CONTROL THEORY

Multiple factors affect coaching such as motivation, cultural differences, goals, and feedback. Control theory focuses on goals and feedback. The basic premise of control theory is that people attempt to control the state of some variable by regulating their own behaviour. With behavioral regulation, first compare the goal with feedback. After comparing the two you can now evaluate if there is a behaviour that can be changed to increase performance which will help reach your goal.

Expat and Global Executive Coaching

Expat and global Executive coaching deals specifically with the unique set of challenges created from crossing cultures following an international or domestic relocation. This niche of coaching tends to center around adapting to a new culture, identity issues created within relocating families, difficulties attaining professional goals amidst a changing political and social structure, and other social and personal hurdles unique to each individual. This method of coaching is either individual, or group-based and helps the client gain fulfillment, success and a sense of identity in the areas that are coached.

Career Coaching

Career coaching focuses on work and career or issues around careers. It is similar in nature to career counseling and traditional counseling. Career coaching is not to be confused with life coaching, which concentrates on personal development. Another common term for a career coach is career guide, although career guides typically use techniques drawn not only from coaching, but also mentoring, advising and consulting. For instance, skills coaching and holistic counseling are increasingly of equal importance to careers guidance in the UK.

Financial Coaching

Financial coaching is an emerging form of coaching that focuses on helping clients overcome their struggle to attain specific financial goals and aspirations they have set for themselves. At its most basic, financial coaching is a one-on-one relationship in which the coach works to provide encouragement and support aimed at facilitating attainment of the client's financial plans.

Recognizing the array of challenges inherent in behaviour change, including all too human tendencies to procrastinate and overemphasize short-term gains over long-term wellbeing, they monitor their clients' progress over time and hold the client accountable. This monitoring function is hypothesized to boost clients' self-control and willpower. Previous studies in psychology indicate that individuals are much more likely to follow through on tasks when they are monitored by others, rather than when they attempt to 'self-monitor'. Although early research links financial coaching to improvements in client outcomes, much more rigorous analysis is necessary before any causal linkages can be established. In contrast to financial counselors and educators, financial coaches do not need to be experts in personal finance because they do not focus on providing financial advice or information to clients.

Personal Coaching

Personal coaching is a process which is designed and defined in a relationship agreement between a client and a coach. It is based on the client's expressed interests, goals and objectives. A professional coach may use enquiry, reflection, requests and discussion to help clients identify personal and/or business and/or relationship goals, and develop action plans intended to achieve those goals. The client takes action, and the coach may assist, but never leads or does more than the client. Professional coaching is not counseling, therapy or consulting. These different skill sets and approaches to change may be adjunct skills and professions.

Systemic Coaching

Systemic coaching is a form of counseling that employs constructivistic conversation, aimed at human problem resolution. Systemic coaching recognizes that in order for two or more persons to interact effectively in a social system, any one individual or group of individuals within that system, each as an element of the whole, may require or benefit from coaching aimed at restoring equilibrium or creating a new alignments.

HEALTH COACHING

In the world of health and wellness, a health coach is an emerging new role. Health coaching is becoming recognized as a new way to help individuals "manage" their illnesses and conditions, especially those of a chronic nature. The coach will use special techniques, personal experience, expertise and encouragement to assist the coachee in bringing his/her behavioral changes about.

SPORTS COACHING

In sports, a coach is an individual that teaches and supervises, which involves giving directions, instruction and training of the on-field operations of an athletic team or of individual athletes. This type of coach gets involved in all the aspects of the sport, including physical and mental player development. Sports coaches train their athletes to become better at the physical components of the game, while others train athletes to become better at the mental components of the game. The coach is assumed to know more about the sport, and have more previous experience and knowledge. The coach's job is to transfer as much of this knowledge and experience to the players to develop the most skilled athletes. When coaching its entail to the application of sport tactics and strategies during the game or contests itself, and usually entails substitution of players and other such actions as needed. Many coaches work at setting their own rules and regulations. They are expected to provide and maintain a drug free environment, act as a role model both on and offof the fields and courts. Coaches must ensure that their players are safe and protected during games as well as during practices.

Dating Coaching

Dating coaches are coaches whose job is to direct and train people to improve their success in dating and relationships. A dating coach directs and trains his/her clients on various aspects of meeting and attracting long-term partners and meeting more compatible prospects. The focus of most programmes is on confident and congruent communication. Dating coaches may focus on topics important to the art of dating: interpersonal skills, flirting, psychology, sociology, compatibility, fashion and recreational activities. Neil Strauss in *The Game: Penetrating the Secret Society of Pickup Artists* also focuses on neuro-linguistic programming (NLP), theories of persuasion, history and evolutionary biology, body language, humor and street smarts.

Conflict Coaching

Conflict coaching may be used in an organizational context or in matrimonial and other relationship matters. Like many other techniques of this nature, it is premised on the view that conflict provides an opportunity to improve relationships, to create mutually satisfactory solutions and attain other positive outcomes when differences arise between and among people.

Victimization Coaching

Victimisation coaching is a type of life coaching that educates people who consider themselves as victims of crime or those who fear victimisation. Coaches work with groups of people to assist them on how to identify and approach potentially hazardous situations.

Christian Coaching

Christian coaching is becoming more common among religious organizations and churches. A Christian coach is not a pastor or counselor (although he may also be qualified in those disciplines), but rather someone who has been professionally trained to address specific coaching goals from a distinctively Christian or biblical perspective. Although training courses exist, there is no single regulatory body for Christian coaching. Some of these training programmes feature best-selling Christian authors, leaders, speakers or pastors. Several of these authors have developed their own coach training programmes, such as Henry Cloud and John Townsend, or John C. Maxwell.

COACHING ETHICS AND STANDARDS

One of the challenges in the field of coaching is upholding levels of professionalism, standards and ethics. To this end, many of the coaching bodies and organizations have codes of ethics and member standards and criteria according to which they hold their members accountable in order to protect coaching clients' interests.

VIABILITY AS A CAREER

Coaching as a career has become increasing popular over the past 15 years, fueled in part by the work of Thomas J. Leonard. However, the competitive marketplace has posed problems for some. Suzanne Falter-Barnes and David Wood demonstrated in a 2007 survey of 3,000 coaches that more than 50% are earning less than $10,000 a year. However, the survey canvassed a mixture of full-time and part-time coaches.

MANAGING CONFIDENTIALITY

The executive and other members of the organization must be able to open up and share information with the coach and one another without fear that the information will be passed on inappropriately or without their approval. Because each coaching situation is unique, it is important for all partners to develop a formal, written confidentiality agreement before the coaching begins.

This agreement specifies what information will and will not be shared, in which circumstances, with whom, and how. The agreement helps all coaching partners remain sensitive to confidentiality issues from each other's points of view. Coaching partners should communicate with other members of the partnership before sharing any information with anyone outside the partnership.

Coach's Commitments

- *Guidelines:* Work within the proprietary and confidentiality guidelines noted in the organization's financial, legal, and business contracts and documents.
- *Organizational information:* In general, within the boundaries of the law, keep all organizational information confidential unless it is otherwise available to the public. *Exception to this guideline:* You may be required to reveal to the appropriate representatives of the organization, and possibly to legal authorities, any information regarding illegal or unethical improprieties or circumstances that pose a physical or emotional threat to any individual, group, or organization.
- *Information about the executive:* Do not share with anyone except the executive himself any details regarding that executive unless members of the coaching partnership have agreed otherwise. *Exception to this guideline:* You are often obligated to provide the organization with a summary of your conclusions on the executive's current and potential ability to serve in his role. Share this summary with the executive and get his input as appropriate. Obtain a detailed agreement from all partners on what this summary will and will not include before the coaching begins.
- *Feedback from others about the executive:* You may often get feedback, usually under promise of anonymity, from members of the organization or other people familiar with the executive. Members of the coaching partnership should agree on the anonymity and confidentiality of such information before it is collected.

You should also obtain agreement, before coaching begins, on exactly how anonymously the feedback will be reported: no identification, identification by category of person (work group, level, *etc.*) or by specific name. You are obliged to the people from whom you obtain this feedback to be clear up front about the terms of this anonymity and confidentiality and to work strictly within these terms. Present any feedback to the executive in verbatim or summary form.

Commitments of the Executive and Other Partners

- Members of the organization who, as a result of coaching, learn confidential information about the executive, keep that information confidential unless otherwise agreed before the coaching begins.
- The executive responds to feedback from others in non-defensive ways, without second-guessing who might have said what or retaliating for feedback that is difficult to hear. This non-defensive response maximizes the trust the executive will share with others in the future.
- All members of the coaching partnership ensure that no confidential information coming out of the coaching process is shared electronically unless they can control access to that information.
- Before coaching begins, all partners consider how the confidentiality of each of the following types of information will be managed. They agree on what will or will not be shared, with whom, by whom, when, in what form, and under what circumstances:
  – Assessment results
  – Coaching goals
  – Job hunting and career aspirations
  – 360-degree feedback
  – Performance appraisals
  – Interpersonal conflicts.
  – Details of coaching discussions.
  – Proprietary or organizationally sensitive information
- The organization identifies an internal resource who can advise coaches and stakeholders on questions of confidentiality and other sensitive topics, and who can help resolve these issues.

PRE-COACHING ACTIVITIES

Certain activities can determine if coaching is appropriate in the first place, help select the most appropriate coach, and prepare both coach and executive for the process. This important set of behind-the-scenes activities, usually conducted by HR, includes sourcing, selecting, and orienting coaches, consulting withexecutives on their needs, matching coach to executive, and establishing standards for practice. The intent of these pre-coaching activities is to ensure the best possible experience and outcomes for the executive and the organization.

Executive's Commitments:

- Consult with appropriate stakeholders to determine if executive coaching is a viable option for you. Consider your organization's overall development focus, your specific learning needs, and the skills and experience of available coaches.
- Conduct exploratory interviews with several coaches before selecting the one who is best for you.
- Handle all business and financial contract requirements yourself, or make sure appropriate people in your organization handle them.
- Provide your coach with the necessary background information about your organization, specific business documents, and personal information.
- Begin the coaching process with a willingness to learn.

Coach's Commitments:

- Participate in the organization's process for selecting, matching, and orienting executive coaches.
- Provide the organization and the executive with requested background information about you and your practice, your rates, business practices, and references.
- Partner with the Human Resources staffand other stakeholders as needed.

Other Partners' Commitments:

- Establish business practices and standards for executive coaching.
- Develop a coach selection and orientation process.
- Apply criteria for analyzing coaching needs and matching the executive with the most appropriate coach.
- Provide feedback to the coaches you do not select.
- Consult with the executive to provide guidance and support in determining coaching needs, requirements, and desired outcomes.
- Partner with coaches to ensure their best fit with the organization and the executive.

CONTRACTING

The purpose of contracting in executive coaching is to ensure productive outcomes, clarify roles, prevent misunderstandings, establish learning goals, and define business and interpersonal practices. There are three major components of contracting: the Learning Contract, the Business/Legal/Financial Contracts, and the Personal Contract between the executive and the coach.

The Learning Contract includes:

- Purpose and objectives
- Timelines
- Scope and types of assessment

- Milestones
- Measures of success
- Identification and roles of stakeholders
- Confidentiality agreements
- Guidelines for the use of personal and coaching information
- Guidelines for the communication and distribution of information

Business/Legal/Financial Contracts include:

- Purpose and objectives
- Executive coaching standards and guidelines
- Organizationally sponsored proprietary and confidentiality statements
- Guidelines for relevant business practices
- Total costs of service
- Who is paying for coaching services
- Fee and payment schedules
- Guidelines for billing procedures
- Agreements on expense reimbursements
- Confirmation of the coach's professional liability insurance

Personal Contracts between the coach and the executive include:

- Guidelines on honesty, openness, and reliability between executive and coach
- Understanding of the coach's theoretical and practical approach and how coaching sessions will be structured
- Agreements on scheduling, punctuality, and cancellation of meetings
- Scoping of how much pre-work coach and executive will do before each coaching session
- Guidelines on giving and receiving feedback
- Understanding of when the coach will be available to the executive and vise versa, and how contact will be made
- Agreements on follow-up and documentation
- Confirmation of locations and times for meetings and phone calls

Executive's Commitments:

- Actively participate in establishing learning and personal contracts.
- As your organization deems appropriate, participate in establishing, monitoring, and administering business/legal/financial contract(s) with the coach.
- Adhere to the learning contract and use it to gauge progress and success.
- Adhere to the personal contract and hold the coach to it as well.

Coach's Commitments:

- Share your own standards and guidelines for contracting with the executive and organization while respecting and agreeing to use the organization's standards.
- Actively use the learning contract to plan and deliver coaching and to assess progress and results.

- Use the personal contract as the set of guidelines to follow in all interactions with the executive; hold the executive to the guidelines as well.
- Negotiate the terms of the contracts in good faith or have the appropriate representative(s) from your practice do so. Comply with the terms of the contract in full, or reestablish them as mutually agreeable between your practice and the executive's organization.

Other Partners' Commitments:

- Establish and disseminate standards for learning contracts in your organization.
- Actively participate in establishing and supporting the executive's learning contract.
- Respect the personal contract as established between the coach and executive.
- Ensure that the coach has and uses business/legal/financial contractual information.
- Expedite the contracting and payment process in your organization in support of the executive and the coach.

ASSESSMENT

The assessment phase of executive coaching provides both the coach and the executive with important information upon which to base a developmental action plan. The assessment is customized, taking into account the needs of the executive and the norms and culture of the organization.

The coach can select among a wide variety of assessment instruments, including personality, learning, interest, and leadership style indicators. Observing the executive in action in her usual work setting provides assessment data, as does interviewing her, her peers, direct reports, manager, and other stakeholders. In some cases, the coach administers a formal 360-degree assessment. There are times when an executive or her rganization chooses not to initiate a full executive coaching process. Sometimes, rather than providing full coaching, the executive participates in feedback debriefing/development planning. This process can be appropriate for gathering data, receiving feedback, and creating a development plan. It is often conducted without an executive coaching partnership as recommended in this Handbook. Without that partnership, however, it can be difficult for the executive to implement change in herself or in the system.

When separate assessment and development planning has been done and coaching is added after the fact, it may be necessary to include others in further data gathering, review, and goal setting. When assessment and planning are done without a formal coaching phase, some coaching should accompany the presentation and review of the results. This will help the executive not only understand the data and their implications, but also make the best use of the information to increase self-awareness and identify development areas with the greatest potential for success.

In addition to assessing the executive, it is also valuable to assess the team and organization with and within which the executive works. Such additional assessments are an important part of the systems perspective of executive coaching. By understanding the team and organizational environment, the executive and her coach can better determine what to change and how to achieve that change.

Additional assessments include such variables as the organizational culture, team communication, organizational trust, quality, employee satisfaction, efficiency, and profitability.

These systems factors may indicate how the organization operates, the results achieved, or predictive measures of likely success. They can be assessed through direct observation, questionnaires, focus groups, one-on-one interviews, and other methods. The data collected on the organizational system are often valuable to share with others besides the executive.

If the larger assessment is contemplated, the coaching partnership needs to decide ahead of time how to deal with the data and include these decisions in the learning contract.

Executive's Commitments:

- Maintain an open attitude towards feedback and other assessment results, considering all information as hypotheses to be proved or disproved.
- Invest the required time to expedite the assessment phase.
- Partner with the coach to identify situations, such as meetings and events, which might provide on you and your organization.
- Ask questions and digest feedback to make the best use of assessment information.

Coach's Commitments:

- Be knowledgeable in a broad range of assessment methodologies.
- Administer only those instruments for which you have been fully trained/certified or otherwise adequately prepared.
- Maintain the confidentiality of the executive by protecting the assessment data.
- Provide a safe, supportive environment in which to deliver assessment feedback. Deliver feedback in ways that encourage the executive to act upon her assessment.
- Offer a clear context for the strengths and limitations of the testing process.
- Help the executive use her assessment data to create a development action plan.

Other Partners' Commitments:

- Respect the agreed-upon level of confidentiality for executive coaching data.
- Provide information about the executive and the organization.
- Partner with the coach and executive to identify ways for the coach to directly observe the executive and the organization.

GOAL SETTING

Executive coaching is driven by specific goals agreed upon by all members of the coaching partnership. These goals focus on achievements and changes the executive can target, both for himself and for his organization. Initial goals are established when coaching begins and revised or refined as coaching progresses. Based on whether they should be achieved within weeks, months, or over a longer time period, goals can be divided into short-, mid-, and long-term targets.

Goals are based on valid and reliable data that exemplify how the executive should learn new skills, change his behaviour, work on organizational priorities, or achieve specific business results. After a specified time period, progress is measured against goals, and they are updated to adapt to the executive's changing capabilities and the organization's evolving priorities. Goal achievement is measured both quantitatively and qualitatively.

Executive's Commitments:

- Collaborate with and listen to your stakeholders to become aware of how others perceive your needs for change and development.
- Be honest about your own priorities for coaching.
- Clarify specifically what you will need to do so that others perceive you as achieving your goals.
- Invest time in the coaching and on the job based on the established goals.

Coach's Commitments:

- Facilitate collaboration between the executive and his stakeholders to identify and agree upon coaching goals.
- Accept responsibility only for coaching activities that are based on specific, measurable goals.
- Help members of the coaching partnership gather valid and reliable data as a basis for establishing goals.
- Document the coaching goals and communicate them to all partners.
- Assess coaching progress and adjust goals based on interim results and changing priorities.

Other Partners' Commitments:

- Be honest and direct about your goals for the coaching.
- Collaborate with the executive and other partners to agree on specific, measurable, achievable, challenging, time-bound, and practical goals.
- Base the goals on valid and reliable data about the executive's performance and organizational priorities.
- Provide ongoing feedback to bothexecutive and coach on the executive's progress towards his goals.
- Support the executive's efforts to achieve his goals.

- Allow the executive to take the agreed-upon time to achieve his goals before changing his responsibilities or the resources he needs.

TRANSITIONING TO LONG-TERM DEVELOPMENT

Upon completing the coaching sessions, the executive and his coach take whatever steps are necessary to ensure that the executive will be able to continue his development. Applying the results of the coaching within the context of the executive's long-term development is an important part of this process.

It usually includes the joint preparation of a long-term development plan identifying future areas of focus and action steps. The coach may also recommend a range of internal and external resources relevant to the executive's long-term development needs. In most cases, transitioning includes handing offthe development plan to the executive's manager or another stakeholder who agrees to monitor future progress in partnership with the executive.

The coach, executive, and other stakeholder incorporate into the long-term plan a regular review of progress towards objectives or goal reassessment. A successful executive coaching process serves as a catalyst for the executive's long-term development.

Executive's Commitments:

- When the coaching process is complete, discuss its results with your coach, including how successfully you feel your development needs have been addressed.
- Identify any areas where gaps might exist or further progress could be made.
- Identify any areas that may become more critical to address in your anticipated future roles.
- Participate in formulating a long-term development plan identifying specific areas of focus and action steps.
- Identify a manager or other organizational stakeholder who will take responsibility for monitoring your future development.
- Hold yourself accountable for adhering to your action plans, including a regular review of progress with your manager or other stakeholder.
- Provide feedback to your coach on performance, strengths, and development needs.
- Provide your organization with a forthright assessment of the coach's capabilities and organizational fit.

Coach's Commitments:

- Use your knowledge and expertise to guide the executive and other stakeholders in developing a long-range plan that targets areas of focus and action steps.
- Recommend internal and external means of development that best fit the needs of the executive and the organization.

- Communicate with the executive's manager or other stakeholders to ensure commitment to his future development, including regular progress reviews.
- After the coaching ends, make yourself available for questions and clarification.
- Check in with the executive occasionally, as appropriate, to maintain the relationship.

Other Partners' Commitments:

- Support the executive's future development, including a long-term development plan.
- Facilitate internal and external means of development for the executive including, but not limited to, rotational assignments, stretch assignments, mentoring opportunities, task force leadership or participation, and internal or external seminars or courses.
- Share constructive feedback about the executive's progress towards development objectives.
- Evaluate the effectiveness of the coach and the coaching process for future use in the organization.
- Provide feedback to the coach on performance, strengths, and development needs.

CORE COMPETENCIES OF THE EXECUTIVE COACH

WHY A COMPETENCY MODEL?

What are the essential competencies of the effective executive coach? This is a challenging and important question. A number of writers have addressed this question over the past decade.i Unfortunately, commercial concerns and a lack of research in the field make it difficult to find a reasonable answer. The danger for consumers is that, given the lack of standards, they may be vulnerable to fads and sales pitches without a clear sense of how to evaluate executive coaches and the coaches' ability to meet the consumers' needs.

Our goal in offering this work is to provide an initial framework of executive coaching to define the needs in specific situations and develop criteria for coach selection, not dissimilar to any good human resource selection process. In addition, we also hope to stimulate further research and dialogue on this critically important topic. The competencies we will describe here are based on our collective experience and judgement as executive coaches, executives, educators, and consumers of executive coaching services.

The competencies must be considered tentative – to be refined as further research becomes available.

Defining Executive Coaching

Our systems-oriented definition of executive coaching is:

- "Executive coaching is an experiential and individualized leader development process that builds a leader's capability to achieve short- and long-term organizational goals. It is conducted through one-on-one interactions, driven by data from multiple perspectives, and based on mutual trust and respect. The organization, an executive, and the executive coach work in partnership to achieve maximum impact."

A discussion of competencies has to begin with the question, "competent for what?" In this work we consider those competencies that we believe are necessary to effectively execute the tasks described in this "baseline" definition. Clearly such a set of tasks requires a broad set of competencies.ii Examination of the definition suggests that the competent executive coach will likely possess or display psychological knowledge, business acumen, organizational knowledge, knowledge about coaching, coaching skills needed to perform essential coaching tasks, and a set of personal attributes that serve as a foundation for these competencies and skills.

Construction of this Competency Description

Drawing on our own experiences as executive coaches, managers of coaching in organizations, and/or coaching researchers and educators we began a process of attempting to articulate these competencies, skills and attributes. Ultimately, we developed four competency areas including "psychological", "business", "organizational" and "coaching" competencies.iii We stress that it is quite likely that few coaches are highly effective across every single one of the competency anchors listed under each of these areas.

It is also the case that certain coaching engagements will draw more on one or more competencies or skill sets than other sets. However, in our experience, significant competence in each of the four areas is in all likelihood essential. While the need for some competencies may vary from situation to situation, we also articulate here a set of personal *attributes* that are likely to be important regardless of context.

Similar to the growing awareness on the part of leadership-development researchers and practitioners of the importance of emotional intelligence in that area, our experience suggests that emotional intelligence competencies are important to the effectiveness of executive coaches as well.iv Finally, the reader will note that we consider each competency or attribute from a basic or foundational level and from a more advanced perspective as well. In doing so, we are trying to capture the simple premise that individual coaches will demonstrate a varying level of effectiveness within each competency as a function of both their natural gifts and their current level of development and that competency at the basic level is essential while an advanced level can be helpful in many situations.

We realize that research on this area is continuing apace and that these competency descriptions are subject to further scrutiny and debate. Matching coach, executive, and organization still requires a bit of intuition on the part of those involved because of the state of research in the field and because of the ambiguous nature of many coaching assignments. In spite of these limitations, given the growth of executive coaching, we believe it is critically important to assist executives, organizations, coaches and coaching educators and researchers by providing focus and language and by so doing try to help move the dialogue forward.

Potential Uses of this Model

Executive coaching, by its very nature, must manage the expectations of a variety of stakeholders.

Consistent with that reality, we believe these competency descriptions should be of assistance to the following audiences:

- Organizations that contract withexecutive coaches to provide services to their employees. Typically, within an organization that utilizes the services of executive coaches, one or more individuals are responsible for sourcing coaches. This description of competencies should be useful as part of the application process, in the interviewing of potential coaches and as a strategy for organizing a database of coaching capabilities. The list should be useful in assessing coaches' references as well. Finally, those responsible for matching executives and coaches can use this model as an initial screening tool to assess the kind of issues with which the executive needs help and the skills and attributes required to offer that help.
- Executives who are choosing a coach. The executive who is considering coaching can be operating at something of a disadvantage without some sense of the universe of competencies from which he or she can draw. The executive using this list of competencies should consider her or his issues and needs and make sure that in the interview process she or he checks to see whether the coach appears to have the right competencies.
- Executive coaches planning their own development. Most executive coaches are lifelong learners, or should be. Those wishing to guide their development intentionally can consider their strengths and weaknesses by considering each of these competencies. Likewise, the executive coach can also use the competencies to consider the kinds of coaching cases to which he or she is best suited.
- Designers of training programmes and curricula for future executive. There are now a number of executive coaching training institutes and academic programmes and more are likely to start up over the next decade. Too often, training and education for coaches offers little more than a few days of workshops with some follow up coaching. As this list of competencies implies,

it makes little sense to expect that an individual can become an effective executive coach in one week, or one month for that matter. Those teaching executive coaches or running institutes or programmes should grapple with the serious challenge of helping their clients develop a broad and deep set of organizational, psychological, business, and coaching competencies.

COACHING AND MENTORING

The main purpose of investing in a coaching programme and holding coaching sessions is to help individuals empower their careers and increase their level of work performance. However the success of an organisational coaching programme is dependent on several factors. The effectiveness of the coaching process is an important factor. If coaches are not bound by coaching guidelines and a structured process, they may fail to meet the goals of coaching.

The skills and competencies of a coach also influence the success of the coaching programme. If the programme is well-structured with a set of guidelines and clear processes but the coach seems to lack the ability to implement and utilize these components, the coaching programme may turn out to be ineffective. This pitfall can be addressed by providing coaching tools and techniques to the coaches that they can use to carry out an effective coaching. Besides the coaching style, strategies in coaching can also prove to be helpful.

COACHING TOOLS AND TECHNIQUES DEFINED

The terms "coaching tools" and "coaching techniques" tend to be interchangeably used in context. Yet, these two terms have separate definitions. Coaching tools refer to the materials and instruments used during the coaching sessions. Examples of the common tools used in coaching are needs assessment, evaluation materials, interview tools, performance feedback tools, coaching drills and simulations, and other similar resources necessary for coaching.

On the other hand, coaching techniques pertain to the methods or approaches in applying and using the coaching tools. It is the art of implementing or using a tool. In order to ascertain a productive coaching outcome, the effectiveness of use of these coaching tools and techniques has to be evaluated. Coaches should be particular and selective in choosing only the best and most reliable coaching tools and techniques.

COACHING TOOL: THE GROW MODEL

A popularly used coaching tool that has been around for years now is the GROW model introduced by John Whitmore. This instrument is widely used and applied by many organisations and companies worldwide. The acronym GROW stands for the following:

G – Goal

Setting the goals and aims of the trainee

R – Reality

Assessing the current "reality" or the situation of the trainee before choosing a coaching style and deciding what techniques to use

O – Options

Outlining the suitable "options" or actions plans based on the needs of the trainee

W – Will

The agreed actions and solutions that will be taken by the trainee for the attainment of goalsOne important tip to consider when using the GROW model is to match it with the appropriate coaching techniques in order to promote an increased awareness and responsibility of the issues. An effective questioning technique may possibly work best along with the use of the model.

However, this coaching tool must only be employed when deemed appropriate to the session. Moreover, the model should not restrict the coaching sessions to arrive at premature conclusions. The coach must see to it that all possible options are explored and discussed. Lastly, an effective coach must not be limited to the use of this coaching tool in every session. He can use other tools and techniques for that matter. In order to make the process even more efficient, the coach must explore other approaches in coaching..

COACHING STRATEGIES AND TACTICS

In any type of coaching, the use of strategies and techniques makes the coaching sessions easier to conduct.

Here are three fundamental strategies and tactics that coaches can apply:

A. Build an atmosphere of involvement and ownership

- Make use of applicable knowledge, necessary tools and resources
- Share the coaching goals and objectives and make the trainee understand the current situation
- Establish clear standards and gauge for progress and coaching results
- Recognise progress in performance and positive changes after coaching

B. Provide challenging role, tasks and responsibilities

- Evaluate the skills and abilities of the trainee
- Create tasks that commensurate with their abilities
- Set expectations and provide clear directions
- Trust the trainee's ability to complete the task
- Acknowledge efforts and results

C. Provide ongoing coaching

- Constructive feedback should be given every time after coaching
- Document coaching sessions and performance trends of trainees
- Utilize applicable tools and resources
- Demonstrate effective listening and give of feedback

SPORTS (ATHLETIC) AND COACHING

Being a nation full of avid sports fans and players, interest in watching sports and participating recreationally continues to expand at a rapid rate. Some amateur athletes dream of becoming paid professionals, such as players, coaches, or sports officials; but very little actually succeed at making a full-time living as a professional athlete. Those who do beat the odds, discover short careers and insecure jobs. Disregarding the small chance of getting a professional job, there are several options for part-time coaching, instructing, refereeing, or umpiring in amateur athletics and in educational systems. Athletes and sports competitors entertain spectators by playing in officiated and controlled sports. Athletes must know the strategies of the game and abide by any rules and regulations as they play. Athletes can compete in a variety of team sports, such as baseball, basketball, football, hockey, and soccer; and they can compete in individual sports, such as golf, tennis, and bowling. Just as sports can be different, so can the degree of difficulty, varying from unpaid high school athletics to the professional level where the supreme athletes play on worldwide television.

On top of playing in games, athletes also spend numerous hours in hard practices every day, perfecting skills and learning teamwork under the direction of a coach or sports instructor. They also spend more time analyzing video tapes, so they may critique themselves on how they play and gain an advantage as they scrutinize their opponents' playing strategies and weaknesses. Some athletes receive strength training to build up muscle and endurance and prevent injuries. All levels of competition are very fierce and security in a job is always unstable. Consequently, several athletes must train consistently year round to keep exceptional form, technique, and good physical condition. There are very few breaks for professional athletes. In accordance to physical training programmes, athletes may also be required to go on strict diets during the prime playing season. Several athletes push themselves as hard as they can during practices and games, increasing their risk for injuries that could end their career. Minor injuries are detrimental as well because the athlete could be replaced.

Coaches instruct and arrange amateur as well as professional athletes in essentials of individual and team sports. In individual sports, instructors might often have these same responsibilities. Coaches prepare athletes for a competitive season through directing practices where athletes perform drills to improve their abilities and endurance. Using their knowledge in the sport, coaches help the athlete with proper form and technique in beginning higher level exercises trying to make the most of the players' physical potential. In addition to supervising the improvement of athletes, coaches also are accountable for running the team during while practicing and competing, and for instilling in their athletes good sportsmanship, the spirit of competition, and working together as a team. They may also choose, store, issue, and stock equipment and supplies. During competitions, for example, coaches decide who will play to optimize fluid team work. Additionally, coaches conduct team strategy and impose certain plays during games to surprise and overpower opponents. To select the optimum plays, coaches assess or "scout" the opposing team before the competition, permitting them to decide on game strategies and certain plays. Several high school coaches mainly academic teachers who complement their income by being a part time coach. College coaches treat their jobs as full time; they travel frequently looking for potential players.

Sports instructors instruct professional andnon-professional athletes individually. They arrange, teach, train, and direct all types of athletes in sports such as bowling, tennis, golf, and swimming. Because sports are so diverse (from weight lifting to gymnastics to scuba diving) and may involve self-defence sports such as karate, instructors usually specialize in one or two areas. In addition to coaches, sports instructors also may have daily practices and be in charge of equipment and supplies. Using their sporting, physiology, and corrective technique expertise, they establish what the kind of exercises to use and how hard the exercises should be. They also give the athlete certain drills while correcting bad technique. A few instructors also give directions on using training apparatus like trampolines or weights, correct athletes' weaknesses and enhance their training. Sports instructors use their best knowledge to evaluate athletes and their opponents to devise a game strategy.

Because of their different focuses, coaches and sports instructors often approach athletes in various ways. For instance, coaches work with the team during a game to optimize their winning chances. On the other hand, sports instructors, like those who instruct professional tennis players, usually are not allowed to coach their athletes during competition. Sports instructors spend a lot of time working one-on-one with athletes, which gives them time to plan custom training programmes for individuals. It is challenging for coaches and instructors to encourage players; however, this is vital an athletes success. Several coaches and instructors get great satisfaction doing their job, helping children or young adults socially and physically to improve and learn skills that will promote achievement in their sport.

Umpires, referees, and other sports officials officiate at competitive athletic and sporting events by examining the play, identifying infractions of rules, and imposing penalties according to the sports' rules and regulations. They predict the plays and then place themselves in the best spot to see the action where they evaluate the situation and decide any violations. A few sports officials may work by themselves, such as those in boxing. Others such as umpires work in groups, such as baseball umpires. Officials' jobs are highly stressful in all sports because they are often have to make a quick decision, sometimes causing strong disagreements among players, coaches, or those watching. Professional scouts assess the abilities of athletes, amateur and professional, to determine their talent and potential. The scout is a sports intelligence agent who primarily finds the best athletes that will qualify for his or her team and down the road, will bring success. Professional scouts usually work for scouting associations or do freelance work. As scouts search out new talent, they perform their work in secret so their opponents won't know of their interest in particular players. A college-level head scout is usually an assistant coach; however, freelance scouts may assist colleges as they provide coaches information about outstanding players. These scouts look for gifted high school athletes through reading the newspaper, talking to high school coaches and alumni, going to high school games, and reviewing videotapes of candidates' performances.

Athlete are known for having irregular hours. The coach, umpires, referees, and other sports officials also share these hours. Athletes, coaches, umpires, and related workers sometimes work weekends, evenings and even holidays. Through out most of the sports season and year, athletes and full-time coaches typically work more than 40 hours a week. A few coaches in educational systems, especially in high school, may coach several sports. Athletes, coaches, and sports officials may be outside in different weather conditions for a lot for outdoor sports; those coaching indoor have the luxury of climate-controlled facilitates, such as arenas, indoor stadiums, or gymnasiums. Athletes, coaches, and some sports officials travel regularly by bus, airplane, or car to scout out players at sporting events.

SPORTS (ATHLETIC) AND COACHING TRAINING AND JOB QUALIFICATIONS

According to various levels and types of sports, education and training qualifications for athletes, coaches, umpires, and other workers differ significantly. Every job requires a great deal of knowledge about the sport that is typically acquired after having years of experience at lower levels. Most athletes have played their whole life, starting in elementary school and continuing to play through high school and sometimes college. They compete in amateur competitions and on teams in high school and college, where professional scouts can observe them. The majority of schools mandate athletes to obtain specific academic requirements to maintain eligibility to play. It takes years of effort to become a professional athlete. Those wanting to play professionally must obtain amazing talent, desire and devotion to training.

High schools typically prefer to hire teachers to be coaches and instructors that will work part-time. If they can't find someone suitable, they hire an outside coach. A few entry-level positions for coaches or instructors mandate merely experience resulting from participation in the activity or sport. Several coaches start out as assistant coaches to acquire the necessary knowledge and experience required for advancing to head coach. Head coaches working at bigger schools who endeavor to contend at the uppermost levels of a sport require considerable experience coaching as a head coach of a different school or as an assistant coach. To achieve professional coach status, it typically takes several years of coaching and winning at the lower levels.

A bachelor's degree is required for all levels of public secondary school head coaches and sports instructors. (Information on teachers, including those focusing on physical education, can be found in the section on teachers—preschool, kindergarten, elementary, middle, and secondary are in another place in the Handbook.) It is required for those who are not teachers to meet specific State qualifications in order to certify to become a head coach. Private school coaches and sports instructors may not have to reach these same standards to certify. Course in degree programmes include the following: exercise and sports science, physiology, kinesiology, nutrition and fitness, physical education, and sports medicine.

For sports instructors wanting to instruct tennis, golf, karate, or anything else, certification is highly recommended. Sometimes, an individual must be CPR certified as well as 18 years old. Different certifying organisations particular to a sport have various requirements and training according to their expectations. One must participate in a clinic, camp, or school in order to certify. Those working part-time and in smaller places have a lower chance of needing formal education or training.

Umpires, referees, and other sports officials have certain requirements for each sport. They usually volunteer for intramural, neighbourhood, and recreational league competitions to start out their career. College referee candidates have to be certified through an officiating school and be assessed during a trial period. A few bigger college leagues mandate officials to be certified with further qualifications, such as being a resident in or near the league's boundaries and having prior experience that usually includes numerous years refereeing at high school and college level conference games.

Qualifications for professional sport officials are even stricter. For example, an umpire for professional baseball must have a high school diploma or an equivalent plus perfect vision and fast reflexes. However to certify for a professional position, it required that potential candidates go to professional umpire training school. At this time, the Professional Baseball Umpires Corporation (PBUC) has approved two schools curriculums for training. Leading graduates are chosen for additional assessment as they officiate in a rookie minor league.

Umpires, prior to being considered for major leagues, frequently need 8 to 10 years of experience in different minor leagues. In football candidates are required to obtain a minimum 10 years of officiating experience, with 5 of them at a collegiate varsity or minor professional level. Prospects for the NFL must be cross-examined by clinical psychologists to evaluate intelligence level and capability of handling tremendously stressful situations. Additionally, the NFL's security department carries out detailed background checks. Potential candidates are probably interviewed by a board from the

NFL officiating department and take a comprehensive test about the association's rules and regulations. A scout's requires experience playing professional or college sports. This knowledge allows them to recognise young players who have astonishing athletic talent and skill. The majority of starting scouts jobs include spotting talent in a certain area or region. Working hard and having success can lead to full-time jobs in charge of bigger regions. A few scouts proceed to scouting director positions or different sport executive positions. Athletes, coaches, umpires, and related workers possess developed skills in communication and leadership as well as being able to relate to others. To successfully instruct and encourage individuals or groups of athletes, coaches also must be creative and adaptable.

SPORTS AND COACHING JOB AND EMPLOYMENT OPPORTUNITIES

Through the year 2012, employment of athletes, coaches, umpires, and related workers is projected to rise about as fast as the average for all occupations. Employment will grow due to the general public persisting to participate more and more in formal sports for entertainment, recreation, and exercise. Job expansion will also be pushed by the growing numbers of baby boomers about to retire, during which they are projected take part in leisure-time activities, like tennis and golf, and need instruction. Active participants in high school athletics will be the posterity of the baby boomers, requiring more coaches and instructors.

Increasing opportunities are estimated for coaches and instructors because physical fitness is becoming more important in our society. All ages of Americans are participating in extra young and lower-leveled athletic competitions, are becoming members of athletic clubs, and are being motivated to engage in physical education. Coaches and instructor employment also will expand with increasing programmes for athletes in school and college and rising need for private sports training. Job expansion relating to sports among education will be determined by local school board decisions. Population expansion commands the building of more schools, especially in the developing suburbs. Nevertheless, when budgets become tight, financial support for athletic programmes is usually cut first. Because team sports are so popular, they are able to compensate by assistance from fundraisers, booster clubs, and parents.

State-certified academic and physical education teachers are likely to be optimum candidates for becoming coaches and instructors. Coaching opportunities will arise from the need to change several high school coaches.

Professional athlete jobs will persist to be highly competitive. Opportunities to have a career in professional sports, such as golf or tennis may increase as novel tournaments are founded and participant receive more prize money. The majority of professional athletes' careers are relatively short because of incapacitating injuries as well as age. As a result, a large fraction of the athletes in these jobs are changed each year, opening a few job opportunities. However, far more gifted young men and women have hopes of becoming a sports celebrity and will be competing for a restricted number of job openings. Persons seeking part-time umpire, referee, and other sports high school level official jobs will have the optimum job opportunities; however, competition is probable for college level jobs with higher salaries and additional competition professional level jobs. Competition is expected to be very intense for scouts, certainly for professional teams who have limited positions.

COACHING HAPPENS IN A CONTEXT OF TRUST

As Warren Bennis and others have suggested, coaching by its nature is very personal and is based on a unique and profound level of trust between the coach and those being coached. While coaching is not therapy, the quality of the relationship is similar and the consequences, whether positive or negative, can be just as profound. This is because the coaching relationship is based on two key elements. The first is that the coach is a competent observer and is responsible for generating a possibility with the coachee that is larger than what is available in the coachee's historical reality.

This necessitates a level of faith on the part of the coachee that the coach is acting solely in the coachee's interests. Second, the objective of the coaching relationship is to empower the coachee to take unprecedented action that more often than not is contrary and counterintuitive to the experience and historical competence of the person being coached. Consequently, the coachee is required to take significant personal risk to realize the benefits of the coaching relationship. If successful, the benefits are obvious and validate the coaching process. If unsuccessful, the coachee must be prepared to deal with and be responsible for potentially negative consequences.

When a group of seminar participants was asked to distinguish between morals, ethics and values, this was their most useful interpretation: *Morals are the rules of God, ethics are the rules of a community and values are our individual rules*. Ideally, these rules are aligned and coherent and become our framework for living. In most cases, however, we are faced with choices where the rules are not always clear or may seemingly be in conflict. Moreover, in many cases our actions are not so much a product of deliberate choices as natural or automatic responses to situations and circumstances in which the prevailing ethic is governed by conventional wisdom, expediency, habit or trial and error.

In the federal public service, most leaders are familiar with the Report of the Study Team on Public Service Values and Ethics, prepared with the guidance of the late John Tait. That report proposed four lenses for thinking about values:

- Democratic values
- Professional values
- Ethical values, and
- Personal values

This offers another way to look at the standards and expectations we hold of coaches and their professional conduct. In reading the remainder of this chapter, it may be useful to consider the connections to these categories and the implications for those coaching within a public sector context. We believe that for coaching to become a widely respected discipline or profession, there must be an agreed upon set of rules to govern our practice. In the absence of such rules, coaching may indeed become a passing fad and the potential of coaching as a new and empowering paradigm for organisations and management may be lost. We invite coaching colleagues and clients to engage this proposal as a starting point. This chapter offers a foundation that can be enriched or expanded based on whatever experience and commitment can be brought to bear. Here then are the ïrules' – the code of ethics – we propose:

A COACH IS CLEAR ABOUT THE LIMITS OF HIS/HER COMPETENCY

Competency is defined as the ability to make and keep promises. One of the paradoxes of coaching is that the coach does not need to know as much as the coachee in a particular field to be an effective coach nor does she need to be able to perform at the level of the coachee. This is obvious if we look at coaches of professional athletes or performing artists. Yet the coach does need to be able to promise a result that is beyond the capability of the coachee to accomplish by herself. For example, a person can be competent to coach people in such areas as their relationships with other people, their communication, how they observe themselves and their circumstances, and their relationship to their own commitments. Breakthroughs in these domains can be powerful for almost anyone in any field of work. That same coach may not be competent, however, to write computer programmes or undertake market research or prepare a balance sheet. The ethical issue is this. A coach needs to know enough about the areas being coached to listen effectively. At the same time, the coaches interventions should be focussed on the coachee's commitments and coaching requests.

A coach is always working with the phenomenon of "cognitive blindness," the boundary between '*what we know*' and '*what we don't know that we don't know.*' This is the area where coaching is powerful and valuable since none of us can observe ourselves in action nor can we observe our own world view. If a coach isn't vigilant and aware of her own blindness, the distinction between expertise and opinion can be lost. When this occurs, the coaching can be reduced to giving advice and there is a danger of the coachee acting on the view of someone less competent than himself.

One indicator of a coach's awareness and management of her own blindness is that she maintains a relationship with a coach herself. It is good to be suspicious of anyone purporting to be a coach who is not also a great coachee. We recommend that anyone hiring a coach ask explicitly in what areas is the person not competent. This does not restrict the coach from ever giving advice, but when aware of her limits of competency she is able to clearly announce that "this is not coaching" and in doing so manage the potential for confusion or unintended outcomes.

Professional and Clarifies the nature of the Coaching Process

A mark of professionals in any field is that they are very clear with their clients about what they do and do not offer. These promises may be recorded in writing and become a basis for later evaluation of results and value added. In a coaching relationship, there should also be a clear articulation of what is required of the coachee. Establishing expectations is important not only because it is a good professional practice, but also because the nature of the coaching relationship is inherently dynamic.

Neither the coach nor the coachee can predict nor should they proscribe what the process will be. It will change and evolve over time and in most cases involve creativity on the part of the coach. Consequently, clarifying the ground rules establishes a frame of reference in which the individuals can work. At the same time, the coaching client may well make new requests as he discovers the possibility of achieving results in areas other than those initially considered. Finally, the coach will be responsible for clarifying the nature of the coaching relationship, particularly with respect to the power of choice exercised by the coachee. In summary, the coach should be explicit about the coaching process, content, and relationship.

One reason that it is important to get the "agreement" clear at the beginning of a coaching relationship is that coaching can be extremely uncomfortable for the person being coached. Historical inertia and patterns of behaviour are often difficult to break and require exceptional clarity and commitment to do so. Coaching requires "unreasonable" actions on the part of the coachee, which can generate considerable resistance and rationalization that can either undermine the value of the coaching or produce classical "power and control" games between the coach and coachee. Part of any agreement should include a mutual understanding of where boundaries exist, what are the protocols and practices for dealing with expected resistance, and any other pitfalls that can be anticipated. Coaching works only when power and choice are vested in the coachee. A failure to understand and clarify this at the beginning of the relationship can result in the coach either assuming or being given a traditional authority-control role in the relationship. This will negate or obscure the value of the coaching and even if results are produced, the coachee may become co-dependent upon the coach for sustaining the results.

Coaches Recognize the Value of their Work and Maintain Professional Integrity

We believe that coaching is a privilege. When we coach others, they grant us permission to observe and intervene in their lives and in how they are 'being' in their work and in their world. The special trust required of them for the coach to make a contribution is exceptional and can only be regarded as a gift. The relationship is analogous to that typically found with doctors, priests, therapists, or very close friends. When someone relates to another this way, it is always possible to abuse or take advantage of the trust. This abuse can be sexual, political, economic, or personal domination. All such forms of abuse are inappropriate. One way to acknowledge and maintain this special relationship is to clarify exactly what the basis of compensation will be (if any) and rigorously reject any temptations to benefit beyond the pre-established terms. In an organisational context, this may be implicit in the managerial relationship, however, it is incumbent on the coach to make it explicit, just as he would if operating as an independent practitioner. We believe that coaching is a valuable service and we set daily or hourly fees based on what we and the client consider to be appropriate.

Once established and agreed upon, practices should be developed that keep the relationship on a professional level. When the coachee accomplishes breakthroughs, they can experience extraordinary emotion and gratitude. In some cases, these breakthroughs can be "priceless" or worth many times the cost of any fees. It is important to always acknowledge that the result belongs to the coachee and that the coach has already been compensated, whether formally or simply from the satisfaction of empowering others. In addition to establishing set fees to prevent inappropriate remuneration, there are other ethical issues which must be considered. One common abuse of coaching by professional consultants is to use coaching as a way to gain entry and preferred access to larger consulting contracts within the client's organisation. While many coaches are also qualified as trainers or management consultants, the relationship with the client is compromised if the coach uses it for such business development purposes. Another common example of abuse of the coaching relationship is the covert use of the coachee as a vehicle for promoting or marketing coaching to others. And since coaching involves a level of trust and openness, there is the possibility of sexual attraction. From an ethical perspective, encouraging or acting on such attraction would be another clear form of abuse.

A Coach has Clear and Observable Criteria

A common question is, "How can we measure the results of a coaching relationship?" This is difficult to answer because breakthroughs are inherently unpredictable. If a result can be predicted it isn't a breakthrough. Measurement is also difficult because the result of coaching occurs first and foremost in the coachee's '*way of being*,' which becomes the context for the coachee's actions and the coachee's results. It is holistic in nature. Our answer to the above question is always that the only place to look for the results of coaching is in the actions and the results of the coachee. The only criteria possible are the declared commitments of the coachee that, in his view, he could not have accomplished by himself.

An important aspect of most coaching relationships is that, while the commitments agreed upon at the beginning are a base line for assessing results, they are not necessarily the only results that will be produced. Because of the holistic nature of the process, coachees frequently report results in other domains. For example, although most of our coaching is in the context of work, results often occur in people's relationships with family or with themselves. We recommend a practice in which the coach and coachee frequently and rigorously assess outcomes in a variety of areas and document these over time. Such documentation is valuable because as breakthroughs occur there is a natural tendency to forget where one began, which increases the likelihood that the coaching relationship will not be correlated with the results. It is possible that the results from a coaching relationship may be acceptable or satisfactory and not be a breakthrough. These results should obviously be appreciated and acknowledged, but in our view they are also the kinds of results that can and should be available from other types of developmental relationships such as skills training, counseling, mentoring, normal supervision and even self-study. What distinguishes coaching as a discipline is its focus on achieving more than incremental improvement. The objectives of coaching are unprecedented actions and results on the part of those they coach.

Community and Empowers the Profession and Community

One-way to distinguish a profession from other kinds of work is that a profession is a community of practitioners that draw their identity and practices and to some degree their power from the community. They are not merely members of a club or association, but they relate to their colleagues as contributors to the field and as sources of new thinking and inspiration. While competition always exists within a profession, competition is not the primary factor governing strategy and action. There is a larger commitment to the clients and to the profession as a whole including general adherence to its traditions and practices. For example, a surgeon will not normally perform a procedure on a close relative, not because he or she isn't capable, but because the profession has declared that the risk of losing objectivity is inappropriate. This kind of solidarity is two-way in the sense that the individual derives power, knowledge, and value from the community and the community exists by virtue of each individual practitioner's responsibility and commitment to the community.

At the present time, there is no recognised group or organisation that represents those who are practicing coaching either professionally or as part of another profession or organisational role. Several associations exist and many coaches work with others in informal arrangements. At this point, there are no widely recognised licensing authorities or agreed standards for what constitutes the discipline of coaching.

The International Coach Federation is contributing to the development of such standards and has an accreditation process for Coach Training Programmes. This sort of formalisation of the profession is emerging but is not yet clearly endorsed by coaching practitioners. For this reason, it is important that anyone offering coaching be clear with those they coach about their qualifications, experience, references, and the underlying basis for their practice. Similarly, those people about to begin a coaching relationship will want to be clear about how prospective coaches' qualifications fit with their needs.

COACHEE ACCOMPLISHING BREAKTHROUGHS

When a coach can no longer be responsible for this possibility, he will withdraw from the coaching elationship. Another paradox of coaching is that breakthroughs are always occurring at the boundary between what is realistic and reasonable and what is unrealistic and unreasonable. Before Roger Bannister broke the 4-minute mile, there was no evidence that it was possible. Possibilities do not exist in reality, they only exist in the vision and commitment of human beings. If they existed in reality, they would be examples. One of the primary values of a coach is the ability to clarify and own the possibility of a coachee accomplishing the unprecedented in the face of a history in which there is no evidence that such a breakthrough could happen.

When we coach, we always deal with two people: first, the person with whatever history, talents, and knowledge she might possess, and second, the person as a possibility. A coach is always relating to the latter. If we stop relating to the person as a possibility, then we inevitably will fall into attempting to fix the person rather than empowering her to accomplish the results herself.

People often ask, "When do you give up coaching someone?" There may be many reasons to terminate a coaching relationship. One reason might be that the coachee is unwilling or unable to trust the coach. Another reason might be that the coachee's commitment to accomplishing a breakthrough result changes. Perhaps there is simply insufficient time, interest, or resources to do the necessary work. A coach will also give up when he can no longer see and be responsible for the possibility for which they are working. This is important because the coach is often the only one who sees the possibility when the conventional wisdom or the coachee does not. When the possibility disappears,then so does the reason for coaching. In all cases, if the coaching relationship is not terminated, it can devolve into a process of mutual rationalisation or co-dependence that is generally unhealthy and unproductive.

A Coach is Responsible

Like Harry Truman, a coach works in a context of "the buck stops here," not as a function of authority, but of responsibility. Another paradox of coaching is that, while the coach is 100 percent responsible, she must relate to the coachee or the team in a way that allows them to also be 100 percent responsible. When coaches deny their responsibility and blame winning or losing on the circumstances or on those they coach, they are undermining both themselves and the coachees and they will lose the context in which coaching is possible.

Responsibility is defined here as the "ability to respond," as distinct from taking on an assumption of blame or credit either before or after the fact. Being responsible means accepting, even owning, what happens in our world. This notion of responsibility opens a different relationship with the circumstances in which we find ourselves, including everything that has occurred, is occurring or might occur. As coaches, we promise breakthroughs. If they do not occur, then we are responsible for not keeping our promise. Successes and failures are relative to the observer and the time frame of the activity. Losing a game doesn't constitute a failure in the context of an entire season. It is always possible to criticize a strategy or process after the fact. A coach, however, is focused on action, outcomes, and the future, not on differing interpretations or stories of why something did or did not work out.

The point is that the ethical behaviour of a coach must be guided by her moment-to-moment judgements in the context of responsibility. This is a coach's way of being because the results of a coach are accomplished without control or force. This requires being fully ïpresent' and clear with respect to what is occurring in the conversations with the coachee and also present to the intended outcomes. If the coach attempts to control the actions of the coachee, then the coach's behaviour is a re-action to what is occurring. This is a sign that the coach has been co-opted by the circumstances or the coachee's story thus negating any possibilities for breakthroughs.

Success or failure is declared at the end of whatever time frames are established for the coaching. There are no clear rules for what the consequences should be if the results are declared a failure. In sports, failure may mean loss of the opportunity to continue coaching; in business, it may be payment of a penalty or return of some portion of fees. Failure may also be handled by a simple apology and a commitment to redesign the coaching strategy. The point is that coaches must be responsible or they will lose the relationship and creative context with those they coach. Coaching then becomes just another part of the game and loses the power to empower others.

Clear Distinction Between Coaching and Other Kinds of Helping Relationships

In our interpretation of coaching, one of the central ideas is that coaching changes the way people observe themselves in relation to their situation. This doesn't happen as a function of new knowledge or information, nor does it occur because they are given a new model. This kind of change is ontological in nature, a shift in a person's ground of being or context. This is also a way of describing what occurs when a person has a new opening for action. Historically, the subject of "being" has been addressed from many perspectives and generally regarded as psychological or spiritual in nature, which is why coaching is often regarded as an advisory role.

From our point of view, coaching has nothing to do with those disciplines or approaches that seek change through self-understanding or insights based on new information. Coaching is grounded in different paradigms and traditions and, while it may or may not have similar objectives to other disciplines, it is distinct. For example, counselors or therapists are often oriented towards fixing or revealing something related to a person's 'internal state.' Coaching is exclusively concerned with action and relegates issues like motive and internal state to the responsibility of the person being coached.

Given this view, it is dangerous when coaches attempt to resolve or promise to deal with human behaviour or other phenomena from any perspective other than coaching. A coach is not normally a therapist, although some therapists are capable of offering coaching as a distinct aspect of their practices. One way to manage this potential confusion between disciplines is by rigorously avoiding "why" questions with the coachee.

For example, when taking a risk, it is natural and common for a person to feel fear. Most people want to understand why they are afraid and have the coach help eliminate the fear. As a coach, it is important to have compassion and acknowledge the fear, but it is not the coach's job to make coachees feel unafraid. Rather, the coach's job is to empower the coachees to take action whether or not the coachees are afraid.

This distinction is particularly important when working with people in areas which are also the concern of other professions. Coaching and other disciplines deal with matters such as trust, being authentic, distinguishing recurring patterns of inaction or ineffectiveness and impacting upon cultural values such as responsibility, accountability, risk-taking, courage, and being able to complete and put the past in the past. Clarifying the distinction between coaching and other disciplines, including the theoretical foundations and practices upon which coaching is based, is important for preventing confusion. Avoiding any misrepresentation or confusion is the ethical responsibility of the coach.

A Coach is Committed to the Commitments of the Coachee

When we approach coaching, the power in a coaching relationship is always with the person or persons being coached. Thus, the senior context for coaching is to serve. A summary description of the coaching relationship is that the coach 1) listens for the commitments of the coachee, 2) observes action, and 3) interacts with the coachee until the actions and the commitments are in harmony. This harmony represents optimum performance. Beyond it, the coach works to assist the coachee to invent bigger possibilities, formulate new commitments, 'complete' past successes and failures, and sustain new competencies and results.

From our perspective, this is the ultimate ethic: *to be committed to the commitment of the other*. Many times this means being *more* committed to the other person's commitment than he is. It means being a support *for* the other person when they forget or are struggling with all the ways human beings have for giving up, or rationalizing away possibility, or avoiding being uncomfortable. It means giving the person absolute choice while at the same time having compassion for the difficulty of reinventing oneself consistent with a new commitment. A coach never needs to use force if she is clear within herself and with the coachee that choice and commitment rest with those being coached. It is the coach's authentic commitment to those she coaches that becomes the context for breakthroughs and unprecedented action.

No human being can observe himself or herself in action. This is why virtually all professional athletes and top performers always maintain relationships with a coach. In “Coaching and the Art of Management,” coaching is described as a dyad — it cannot occur except in relationship with another. It is in the relationship between the coach and the coachee that the results are created, even though the action that manifests the results of the coaching is the responsibility of the coachee. The coach is not a critic of the player; she observes what the player cannot observe for himself.

4

Basis of Sports Training

All activities which are part of human behaviour were subject to a long-term development. Let us take throwing, which is regarded a basic motor activity, as an example. In the deep past, throwing was necessary for feeding and defence. At present, throwing has lost its importance as one of the above mentioned activities but it is involved in different sports to a great extent (*e.g.* athletics, handball, baseball, etc.). The task of a prehistoric hunter was to hit the target precisely to get food. The aim of a present-day athlete is to throw the javelin as far as possible. The result of the activity in both examples can be considered a performance. Performance is understood as an extent to which motor task is accomplished.

With the prehistoric hunter, performance is evaluated dichotomically: hitting the target or missing and it is not restricted by any rules. In the case of the athlete, performance is evaluated following rules of the sports discipline which were set in advance, it is expressed by the length of the throw and is understood as a sports performance. An ability to achieve a given performance repeatedly is referred to as efficiency. The aim of sports training is to achieve maximum individual or team efficiency in a selected sports discipline limited by rules.

Reaching maximum efficiency in any activity is not possible over a day. Efficiency is conditioned by several interrelated areas. Sports training focuses on reaching maximum efficiency in motor abilities connected to a certain sports discipline. Supposed performance depends on motor ability and motor skill which are closely related to the sports discipline. Motor abilities can be described as relatively stable sets of inner genetic presuppositions needed to carry out locomotive activities. They include force, speed, endurance, coordination and flexibility. Motor abilities are manifested on the outside by sports skills. Sports skillsare presuppositions needed for implementing performance in a selected sports discipline which is limited by rules. Such presuppositions are gained through motor learning.

It, however, would not be possible to implement sports skills or develop locomotive abilities without motivation. Motivation is understood as an inner incentive to carry out certain activity. The final area needed for performance implementation is represented by tactical skills. Tactics means conducting a sports competition in a purposeful way.

The contents of sports training consists of individual key areas which are called components of sports training:

- Physical component is generally focused on developing motor abilities.
- Technical component focuses on acquiring sports skills through motor learning.
- Tactical component focuses on acquiring and further development of different ways to conduct sports contest on a purposeful basis.
- Psychological component is focused on improving the athlete's personality.

Example: Component of physical fitness means motor abilities which are a condition for maximum jump height. What is crucial in this matter is quick force. In volleyball, technical component is an acquired skill of offensive hit. Tactical component is represented by choosing direction and force of hit which depends on game situation analysis.

Psychological component is manifested on the outside by the player's reliance on him/herself to solve game situation successfully. Sports training is understood as a process of systematic development of each component in dependence on the duration of preparation which leads to achieving maximum efficiency in senior age within the selected sports discipline.

CHARACTERISTICS OF SPORTS TRAINING COMPONENTS

PHYSICAL COMPONENT

Physical component is primarily oriented towards systematic development of motor abilities and their manifestation through sports skills in a selected sports discipline.

Among the most important areas of motor abilities are the following:

- Force abilities
- Endurance abilities
- Speed abilities
- Coordinative abilities
- Flexibility.

Basic differentiation of motor abilities is not sufficient to describe the manifestation of individual abilities within the specific sports discipline. Physical requirements on the athlete during physical training are primarily related to the selected sports discipline. Some sports require carrying out motor activity with a high (*e.g.*, 400-m run) or low (*e.g.*, marathon run) intensity during the whole course of motor task.

Other sports, like soccer or basketball require the athlete to carry out different types of motor activity ranging from static positions to running with maximum speed, often accompanied by change of direction; and all that with a different intensity. Requirements of individual sports disciplines are related to physical capacity of the athlete and can be divided into following categories:

- The ability to develop a high power output in single action during competition such as kicking in soccer an jumping in basketball (force).
- The ability to perform prolonged exercise (endurance).
- The ability to sprint (speed).
- The ability to exercise at high intensity which are the basis on acceleration, maximum velocity and multidirectional change of movement (agility).

Well-designed training programmes are based on applying five principles during each stage of sports preparation. There are three basic principles: specificity, size of adaptation stimulus and progression.

Specificity

Sports preparation in a specific sport is characterised by specificity. The athlete improves his or her performance in specific activities which are the content of a specific sports discipline. For instance, take-off in attack strike in volleyball is characteristic for taking off from both feet, therefore while training quick force, specific exercise must be utilized which support the respective type of take-off.

Size of Adaptation Stimulus

Applying optimum and adaptation stimulus means applying smaller size during sports preparation than the one which the athlete is used to. However well the training programme may be designed, without applying optimum adaptation stimulus, it restricts the ability of the athlete to improve. Subliminal stimulus does not lead to desirable progressive changes in performance.

Progression

If systematic training is to lead to ever greater improvement, its volume and intensity must continuously increase. If the principle of progressive increase is applied properly, it lead to cumulative training effect (an example of this can be gradual increase in intensity of sports preparation by increasing the number of weekly trainings, increasing repetitions within each exercise, change of type or difficulty of exercise).

Technical Component

Technical training focuses on acquiring, keeping and transferring motor skills. Generally, from the point of view of sports training, motor skills are divided into two groups: Fundamental skills are based on natural ontogenetic development of a human. It includes gait, run, jump, climbing, and basic over arm throwing etc. Sports skills are based on contents of a specific sports discipline. In volleyball, the content of skills is for instance setting, reception, block, service etc. The aim of developing these skills is acquiring high level of automatisation. These skills accompany the athlete during the whole period of his sports career regardless of the performance level he or she is at.

The athlete keeps such skills for the whole of his sports career regardless of performance level. Acquiring these skills should be in compliance with long-term conception of sports training. According to this conception, training of a specific sports discipline must contain another large group of motor skills which do not form its contents but are important for reaching other aims of sports training. For example, they include gymnastic or athletic skills which are important for recovery, compensation and versatile development of an athlete. Movement skills can be classified according to three basic motor behaviour criteria.

General Versus Special Skills

General agility tasks targets the development of one or more basic coordinative abilities, whereas special tasks unify them in a skill specific manner. For example, standing on one foot represents an example of a general skill which develops static balance. On the other hand, standing on one foot on a balancing bar can be a part of a gymnastic set where it represents a special skill.

Closed Versus Open Skills

Closed agility skills have programmed assignments and predictable or stable environments. An example of a closed skill can be gymnastic routine or set in figure skating. Open skill have non programmed assignments and unpredictable or unstable environments. The context changes during performance, and the training objective is to rapidly respond and adapt to new or unforeseen stimuli and situation. An example of a open skill can be situation in games when a defence player must respond to unforeseen movement of the opponent.

TECHNIQUE AND ITS TRAINING

INTRODUCTION TO THE CONCEPT OF TECHNIQUE

Technique is a concept with varied applications and a great diversity of contents. In contrast with other human technical activities, in sport activities all human capacities are involved and evaluated immediately. For this reason, there is a need for a more specific approach to the concept of technique. Technique as the most rational and effective form to perform exercises. The ideal model of a movement relative to a specific sport activity. For team sports; Mechling (1983) defines technique as that movements or part of movements that permit to perform attack and defence actions with a concrete game purpose, and with a quite good execution quality (good with reference to an ideal model). It follows from these definitions that the sportsman, in order to obtain performance in his particular sport, must have learned a group of movements, following ideal models, result of different research, which will facilitate to execute precise actions to improve his motor skills.

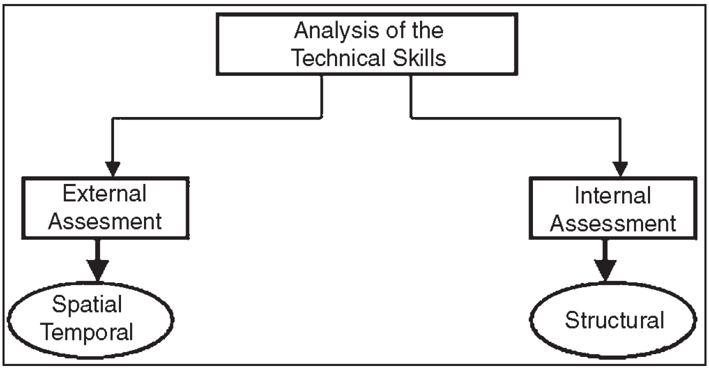
Therefore, when a person possess this or that movement within his motor repertory, it is said that he has this or that ability; that is why, the group of movements of different sport specialties are called sport technical abilities. Consequently, an sportsman will possess a good technical ability the best he adjust his movement to the ideal model, in addition to the best he can control it to obtain maximal performance during real competition. The aptitude of a person to acquire these abilities is understood as capacity. Thus, an individual with a better learning capacity, will have the possibility to have a major number of technical abilities to apply to sport executions. We should point out, in passing, that to apply correctly a technical ability during competition not only depends of this capacity, but also on tactical capacities which are not the motive of this article. But, how this ideal technical model is established? It is the product of research that proposes a concrete movement to be performed as the most effective. However, the ideal model changes, and the ideal now is not that ideal after a period of time, when science and experience of coaches improve. Therefore, we think that does not exist an ideal model, but the ideal model for each sportsman, which will depend upon his ability to perform during real competitive situations with reference to his possibilities to solve efficiently the situations proposed in that moment.

We neither agree with the term style as the personal interpretation that a person does of a determined ideal technical model, because his execution is a personal model acquired through practice and which is the ideal for him in that moment. Who can say that the ideal technical model of a person 15 years old must be the same than his ideal technical model when hi be 25 years old, when changes as corporal dimensions, weight, strength, concept of movement, contents, tactics, have occurred. The ideal model is transitory, thus the most important is to create a personal model that be adjusted to the game rules and facilitates maximal performance during competition at each stage of the sport life.

ANALYSIS OF THE TECHNICAL ABILITY

The need to examine the conditions in which the technique is executed. Each one proposes different options that are the base for the development of a particular analysis. All technical abilities are performed by precise segmentary and/or global movements.

Such precision makes the segmentary body parts to perform meticulous paths on the space. Also, that spatial trajectory must spend a certain period of time. All spatial and temporal aspects of a technical skill are easily observable and can be studied from different perspectives. They can be differenced from other called internal or structural aspects which will generate that external values manifested during sport practices of the athlete.



External factors are evident during the execution of each movement and they can even be observable by a non-specialist, while structural factors are of more complex assessment.

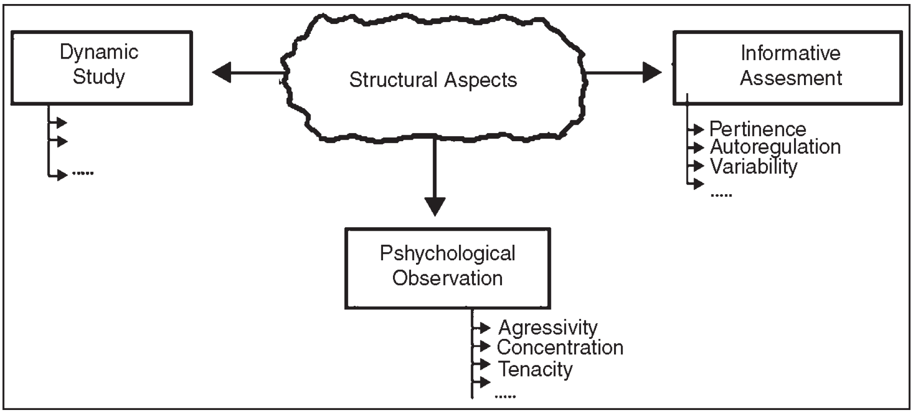


Fig. Description of the Spatial-Temporal Aspects.

The number of factors proposed for each analysis is obviously open for including new elements, depending on the knowledge of the specialist.

- Cinematic studies allow to know both segmentary and global speeds, accelerations, etc. External biomechanics is the science that develops these studies, which are often used for the description of technical skills.
- The medical-kinesiological observation is focused on the determination of technical conditions to prevent from deterioration of the human locomotor skeleton. A technical defect or a postural or muscular tone control defect, based on a kinesiological-postural criterion, might be very prejudicial for a future sport life.

- Technical skills may be externally assessed by a temporal criterion. Several qualitative factors (continuity, adequateness, anticipation, etc) are relevant for an effective execution of a technical movement.
- Many aesthetic factors of a technical skill could be assessed; as many as personal criteria of the aesthetic value of movement or things of nature (amplitude, symmetry, beauty, plasticity, expressiveness of skill, etc). We should not forget that the word technique comes from the Greek "tekhè" which most approximate meaning is "art". Unfortunately this value is marginalized by several coaches, except in those sport disciplines in which this concept is evaluated as part of performance.

In subsequent parts we will see how each of these factors is generated by one or several internal factors, from which they are their external manifestation. The best way of summing up this initial study is to state that all these spatial-temporal aspects (constituents of the external form of movement) are consequence of a correct coordination, and are the representation of an intention that generates movement.

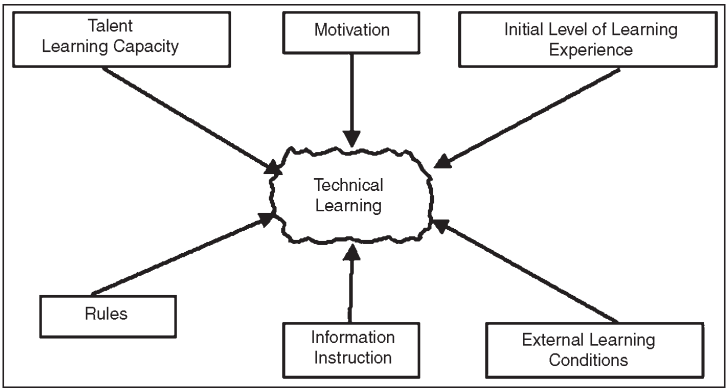


Fig. Description of the Structural Aspects.

Fundaments of the structuralism theory (Saussure, Köhler, Wertheimer, Kofka,...) assert that exists a close interaction among all internal factors influencing the execution of technical abilities. For instance, concentration allows to choose adequately the type of movement and the possibility of being executed under a determined strength condition (qualities from different categories). The interrelation is so that a single factor modify all others; and external aspects are the manifestation of all of them together.

- The dynamic study of a technique allows to know the muscular work conditions (forces, momentums,...) that bring about the observed movement. The "internal" biomechanics conducts these analyses and can explain many errors of the technical skills or their efficacy.
- There also exist the informative assessment of the conditions in which a specific sport technique is performed. For instance, pertinence to select a technique which is more or less compatible with the external conditions, progressive self-

regulation capacity to adjust movements to competition needs, or the variability of a technical model in order to adequate it to an opponent.
- Psychological observations of a technical skill offer us a knowledge of the psychological factors added to that technique. Concentration, agressivity, persistence,... are some features which can be observed during the execution of a technical skill and they produce typical personalities for each competitive situation.

As we mentioned, spatial-temporal aspects (external) are consequence of structural aspects (internal). It is evident that cinematic conditions are the result of force production (internal dynamic study). Temporal assessment depends a great deal on informative factors. Thus, aesthetic and medical-kinesiological factors depend in part on psychological contents of movement, but also on dynamic components. It seems clear that both external and internal assessments are necessary for the understanding of performance of a technical skill. Therefore, we can explain how various subjets who execute "the same" technical ability obtain very different values in some of these internal and external components and consequently perform, from a technical point of view, at a different level. For example, physical conditioning preparation, which is responsible of force application conditions (internal factor), can determine the outcome of a confrontation between two players of similar technical levels. Similarly, an appropriate tactical decision (that affects both internal and external factors) can be relevant in many cases. All described factors will have differential influence in each sport specialty. By analysing the technical abilities of a particular sport in the way it has been proposed, we will be able to define in a more precise way the technical training goals for this sport specialty and, in addition, to propose a more individualized technical learning.

THE MEANING OF TECHNIQUE IN DIFFERENT SPORTS

The concept "technique" does not have the same meaning in all sports. The factors previously studied play a different role for every sport.

In order to determine the different meaning of a technical concept, it is necessary to analyse each sport specialty independently.

Verjoshanskij (1985) develop different meanings of technique even within the same sport (athletics). For explosive strength specialties, he states that the technique must guarantee the capacity of producing a strong and concentrated impulse just at the determinant phase of execution.

Neumeier (1981) also states that at this determinant phase of execution the role of the technique is to reach maximal acceleration. We can see, then, that there is a clear concurrence of concepts (dynamic-cinematic).

While for endurance events the technique is efficacy or performing a skill economically (temporal and kinesiologic factors are the most relevant). Thus, their techniques must be learnt differently.

For sports practiced with an opponent, Gulinelli (1986) understands technique as the possibility to solve variable competitive situations. On the other, Djatschkov (1977) states that technique must develop precise strength and speed executions to the maximal extend. In this case internal-informative and external-temporal values are the most determinant. Therefore, such factors must be taught preferentially in order to develop technical skills adequated to the competitive needs.

Gulinelli (1986) holds that in artistics sports the technique have to increase the precision and expressivity of movement (external-aesthetic assessment, and internal-informative and psychological assessments are the most important). Evidently, it is necessary to understand the meaning of technique before the selection of a training model. The proposed analysis allows the coach to design very efficiently the learning objectives of a concrete technique, the first step of any learning plan.

TECHNICAL ASSESSMENT

The observation of a technique can be assessed from a formal and a real point of view. The formal assessment consist of an analysis of the similarity between a technical execution and its scientific ideal model. It is an objective assessment. In some sports (gymnastics, figure skating, etc) the athlete's performance is evaluated with such formal criterion. A judge, trained to observe, give an score that assesses the execution of this type of technical skill. In other words, there exist a direct concurrence technique-result; therefore, such formal assessment is equivalent to a real assessment of the result.

Real assessment is based on the result, independently from the similarity between a technical execution and its ideal model. An athlete can perform scoring many goals and having a poor model of the throwing technique. Thus, it is an indirect measurement or assessment of technique. On the other hand, in some athletics specialities to have a technique similar to a model is a guarantee to perform at high level. An athlete who does not know the jumping hurdles technique cannot obtain a good result in those specialities, although the technique is not evaluated, but the time (an indirect assessment of technique). Therefore, the indirect assessment conditions of a technique permit to classify sports, at least, in these two cited groups. What it is really relevant is to know how technique is evaluated in our sport specialty in order to plan learning and training goals depending on the real demands of that speciality. By doing that, different technical factors will be emphasized for each learning stage.

TECHNICAL TRAINING

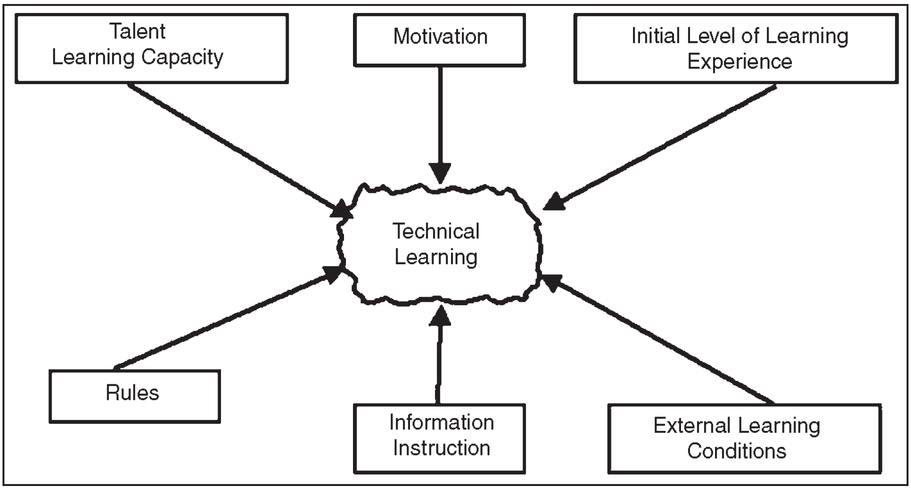
By the technical analysis we have the knowledge of all factors that can be taught in order to improve the technical skill of a particular athlete. Knowing the meaning of technique allow us to classify the importance of all these factors and, therefore, to clearly define the goals. Finally, with the evaluation of the technical skill in different sports, we can fix its relative importance and plan in sequence the learning goals during the athlete's sport life. It is now time to discuss how to improve a technical skill; however, it is impossible to describe all the training process of all different sports, due to the nature of this article. We are going to tackle the questions of the influential aspects in acquiring a sport technique and the proposal of a model based on learning stages.

To learn technical abilities, besides motor learning, requires also another learning that permits to adjust an athlete's motor activity to the predominant circumstances of a sport competition. Technical training requires a more complex improvement of various capacities. Thus, it is not only necessary to achieve a motor learning, but also a perceptual and decisional learning in order to achieve all learning needs which are required for a high level sport practice. These three type of learning must grow up homogeneously to perform in different sports; however, in each sport they will be emphasized particularly.

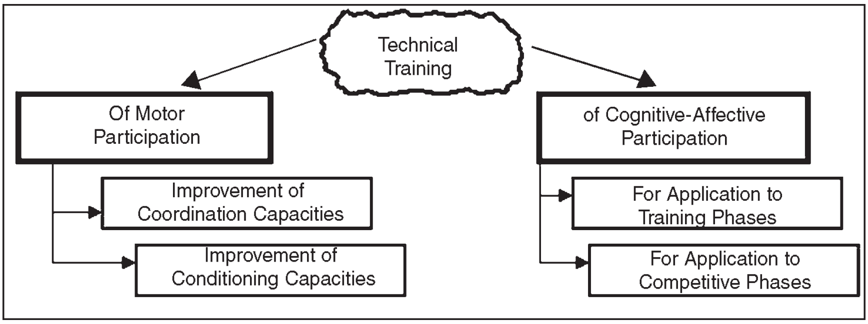
For instance, Mandoni (1985) proposes the following learning that must be reached in beginner basketball players:

- Learning of signals (stimulus-answer).
- Learning of connections (stimulus-answer, without linking)
- Linking of various stimulus-answers.
- Learning of concepts.
- Learning of rules.

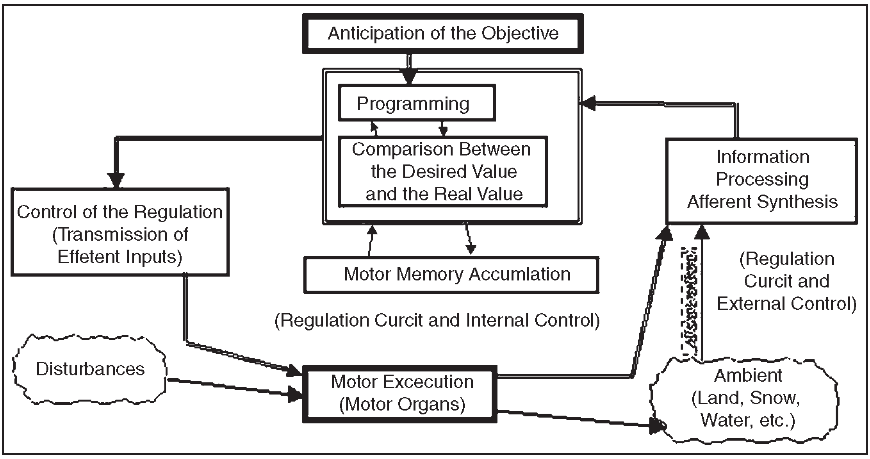
In this proposal perceptual and decisional learning are more emphasized than motor learning, while in other type of sports may happen the contrary. All in all, there is a group of factors that will influence when learning new sport techniques of any speciality:



On the other hand, technical training is a more complex process that affects the improvement of all technical factors, and it includes among them the motor learning as well as all other components set out. The training that is performed by execution of movements is the most meaningful in a training practice. This body motion can be aim towards the perfecting of movement, by improving its execution, or it also can be aim towards the acquisition of conditioning factors which improve the performance of a skill. In other words, to increase the value of both coordination and conditioning capacities of the athlete.

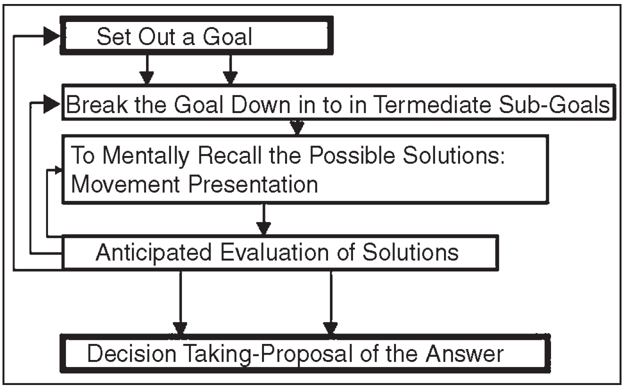


- In order to improve coordination capacities Manno (1984) proposes motor execution goals in conditions that assure the participation of all coordination capacities and specially the predominant coordination capacities in a concrete sport speciality. He makes a proposal, about the coordination of a desired motor act. A specific orientation of the objectives and the achievement of several trials under certain conditions, allow the learning of a particular movement, the learning of a concrete sport technique. Matveev (1982) asserts that "as soon as a movement is becoming usual, by a big increase in the amount of repetitions, this exercise no longer has any influence on the coordination capacities" and proposes a "systematic renewal of motor experiences" for constantly influence on coordination capacities. Added to that, motor learning is a part of training of the coordination capacities.

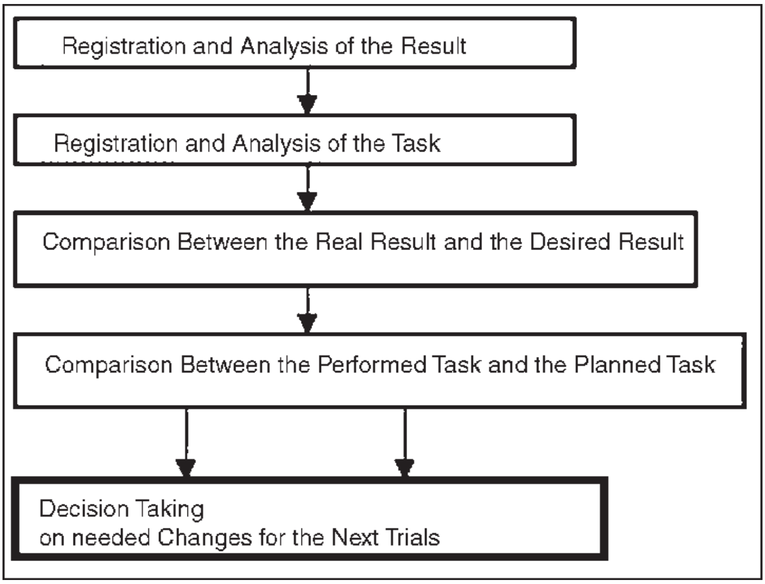


- The improvement of conditioning capacities is another facet of the motor participation training. It is done by the application of concrete systems which improve these factors. There exist well known systems to improve strength, speed or endurance, as well as flexibility and relaxation. We want to point out the importance of having an homogeneity in these two options of motor participation goals, which provides a balance of the technical behaviour and a continuous possibility of progress throughout the sport carrier of an athlete.

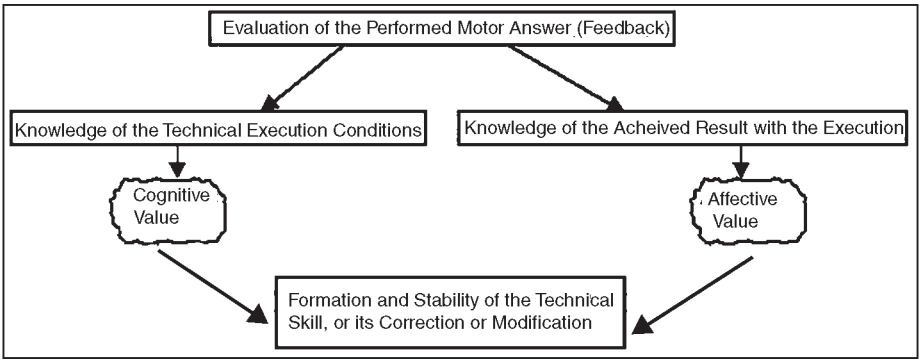
The technical training of cognitive-affective participation is commonly underestimated, supposing that the simple participation of such values in any physical practice means already its training. It is not really like that and Duran (1985) systematize all required operations to plan a motor answer that makes evident the need to improve these type of capacities in which it is not necessary a motor participation for its training.



After these "mental practices" the motor execution is performed, under conditions similar to the designed proposals in the preceding plan. Once the movement has been executed, it is necessary to evaluate it, taking again value cognitive-affective capacities for being able to perform the following operations proposed by Duran on the figure.

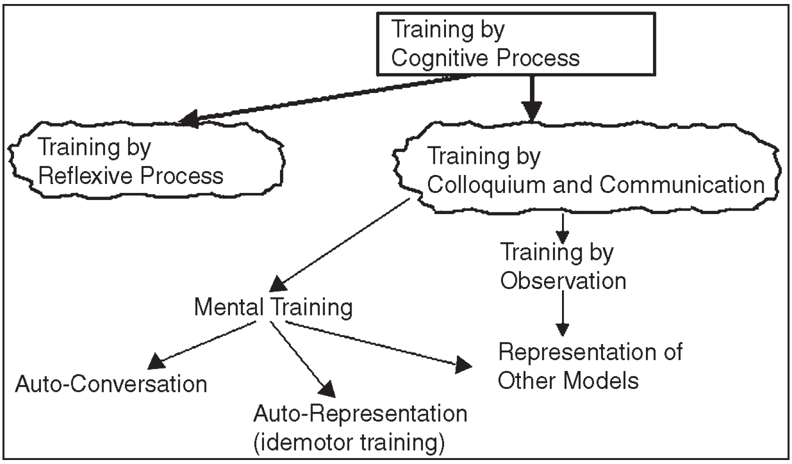


In such type of mental operations become apparent, both during training practices and during competitions, that the participations of cognitive capacities allows a technical execution be adjusted to the activity purpose of an athlete; and if the planning has bigger cognitive participation, the answer has both cognitive and affective values



Volpert (1976) proposes different possibilities of technical training without motor activity, therefore it permits to improve certain cognitive capacities. The representation of such training systems. Based on Volpert, Reider (1985) define the following cognitive objectives to practice:

- To improve the representation capacity of movement.
- To improve anticipation processes.
- Conscious perception of each information type (kinestesic, visual, tactile, acoustic).
- To improve observation capacities.
- To improve the capacity to compare the expected result with the real one.
- Motor schemes with attention focus.
- To improve the concentration capacity.



The achievement of such goals allows to develop more specific training methods so as to avoid execution mistakes which are based on the athlete's comprehension (cognition) of his own technical performance. For instance, Pöhlmann (1977), based on the called “principle of discrimination or clarification”, proposes attempts with artifacts of different weights in order to solve, by kinestesic differentiation, determined execution problems in relation, basically, to the application of forces.

There also exist options based on mistakes, proposed by Korengerg (1980), for knowing the real origin of errors during the execution of technical skills. Added to that, based on Meinel and Schnabel (1977), the cognitive processes attain great stability in the sportsman with reference to motor execution of technique for reaching his high performance in any circumstance (called “constancy of movement”). Cognitive processes have also an important participation in eliminating the called “speed’s barrier” or “speed’s block” (Ozolín, 1980 and Hollmann-Hettinger, 1980); in addition to other applications during training and competition.

Therefore, during the process of technical training is necessary to apply skills of motor participation as well as skill of cognitive-affective participation, which make its application more complex and extend it to, practically, a person’s sport life. It would be necessary to use different methods depending on what an athlete needs to improve his technique. Evidently, each of these training options are more or less effective to reach some or others technical goals that must be achieved during the course of different formation stages of the technical ability.

Stages of the Technical Learnig

This training process will allow the learning of technical abilities after years of uninterrupted experiences. On the one hand, Fitts (1964) proposes three stages for these acquisitions: cognitive stage, associative stage and autonomous stage. Gentile (1972) only define two stages, while Meinel (1977) develops also three levels of dominion: beginner, advanced and high dominion. Martin (1977) adds a fourth level. Both Ozolín (1970) and Matveev (1977) observe two moments: learning and advanced. More recently, Nadori (1985) define three levels under the following epigraphs: non-specific, semi-specific and specific stage for the technical ability acquisition. Most authors agree with the idea of defining a first moment in which it must be reached a basic “technical model”, understanding the conditions of the skill as well as recognising and interpreting environmental characteristics in which the technical movement must be performed and adapted. In a second technical stage, the sportsman must perfects his movements by eliminating unnecessary movements. He also must obtain an enough degree of stability and regularity, as well as perfect decisional processes, identifying the integrated environmental characteristics, predict events, and he also must rank answers that might be applied to determined competitive situations.

Some authors define, in a third level, phenomenon of constancy and availability that permits high levels of technical execution in any environmental situation. While others, which only propose two stages, include these achievements in the second like a final consequence of a well done training process. We believe that there is lack of a third or fourth stage of technical training. Stage in which the sportsman, owner of his technique, is able to create new ability forms, identifying them as something personal and, simultaneously, such new skills are as an additional stimulus to stay many years practicing sport. There exist sport specialities in which this process is more difficult, and it can even be interfered with the specific rules. Anyway, there cannot be any doubt that the person's sport life must be divided by stages in order to adjust the biological maturing processes with the development of technical abilities and others. Our proposal for team sports and sports of similar characteristics is:

(A) Stage of multipurpose general conditioning (8-10 year-old).

(B) Stage of oriented multilateral preparation (10-12 year-old).

(C) Stage of specific initiation (12-16 year-old).

(D) Stage of specialisation (16-20 year-old).

(E) Stage of perfecting (20-24 year-old).

(F) Stage of high stability of performance (more than 24 year-old).

For each stage we will define the differential criteria for the formation of the technical ability, as a consequence of an organised training under the parameters we have mentioned.

(A) At this stage the training of motor participation must develop all coordination capacities; with non-specific orientation, with constant modifications of environmental conditions of the practitioner. It is founded, therefore, on motor experimentation of all movement capacities. All areas of conditional capacities are developed in equilibrium to coordination capacities.

With reference to the training of cognitive participation, it should be focused on the understanding of movement as a human capacity, as well as the self-conscience of the individual during this movement. At this stage there is no difference among the training of different individuals; thus, all of them will after be able to practice all type of sports. Coaches should observe the areas of movements and cognitions in which each subject learn faster or is outstanding in comparison with his partners, in order to be able to develop next step correctly.

(B) When the athlete and coach are conscious of the best efficacy at executing a concrete spectrum of movements, this second stage of oriented multilateral preparation begins. The training of technical abilities is focused on the improvement of non-specific coordination of the group of movements in which the individual is talented, and in which they will be able to apply afterwards to some sport specialty. As much as the coach knows the selected sport he will propose an elementary orientation towards the practice of the technical actions of such specialty and to the complementary skills, based on his experience. At this moment it is of great importance to analyse the technique in order to propose the predominant space-temporal and structural aspects of that sport specialty that will be the fundamentals of posterior technical acquisitions in that specific ability.

At this stage begins the first differentiation of training of technical skills and the cognitive participation practices. This will allow, in addition to the observation of the biological development, to determine, with less errors, a future sport specialty which is initiated in a specific manner immediately.

(C) At this moment, all technical training is focused on reaching and developing the self-execution model of the specific movements of a concrete sport specialty. We could talk about an adaptation of abilities to specific skills with participation of cognitive processes so as they can be assimilated by the motor repertoire. These cognitive processes have now their main role during competition, place to test each technique achieved during training practices. The meaning of technique in the chosen sport specialty will determine the adjustment of achievements to the competition. The role of the coach is fundamental in identifying these meanings and presenting them to the athlete under simplified conditions during the practices.

Conditional capacities training is focused on that specific facets of the initiated specialty, but considering the degree of biological maturing of the subject. It is evident, at this stage, that the athlete is already a player or competitor of a determined athletic specialty.

(D) Talent personal qualities are basic at this specialisation stage. Morphological development, facility in technical learning and its fixing, efficacy in taking decisions, adaptability to competitive conditions, are among others, factors which will allow both the athlete and the coach to make sure that their choice was right. All external factors, space-temporal, must be solve with efficacy. Specific technical evaluation will define this efficacy and will permit to adjust structural factors progressively to needs required at this level. Methods of cognitive training must reach a profound knowledge of the sport and sport specialty, or of the specific position to which the athlete must adapt all the achieved skills. This stage is termed “stage of profound specialisation” by Tschiene (1985), with reference to athletics (track and field) in order to point out the importance of specific training methods.

The dynamic study and temporal assessment are the elements that allow the proposal of preferable goals for technical training at this level. Higgins (1977) proposes that the athlete must adapt and transfer his ability to the predominant conditions of environment. Differentially, in the next stage, the athlete will have to adapt his ability to unusual or varied situations.

(E) At this moment the sportsman has a dominion of all technical abilities and the purpose is its perfecting. That means that space-temporal and structural aspects must be in total concordance in order to obtain this perfection in competitive situations. The variability of these situations tests cognitive and specific conditional capacities which must be continuously trained to reach that perfection of the technical ability. It is fundamental to reproduce variable conditions of competition in a global form, even to increase them during training practices in order to never block the progress of the technical execution.

If these conditions of training are maintained, this perfection can be extended indefinitely; if also competitive conditions bring to the sportsman enough new stimulus that avoid periods of stagnation.

(F) The stage of high stability of performance. This is a consequence of the previous stage. The difference is that now, the possible motor solutions to competitive situations must be, in most of the cases, a product of personal elaboration of the athlete and his creative capacity. The athlete has a high level of technical stability, its technical capacity is consolidated in such a way that permits him to perform in all competitive conditions and, moreover, can create movement solutions which even allow him to be more effective. This success will permit him to stay during a longer period of time competing with a high level of motivation to do that. We believe this is possible through an appropriate distribution of the goals in the different stages proposed and by giving attention to the training of cognitive capacities over the different phases of technical training.

TRAINING PRECONDITIONS AND FRAMEWORK

Training courses for the socially disadvantaged are not significantly different from other training courses if modern didactic principles are taken into account. However, this target group requires greater didactic and educational expertise on the part of the trainer. Motivation, group dynamics, communication and cultural aspects assume greater prominence. In particular, the trainer needs to think about the perceived balance of power if he wishes to empower individuals who not infrequently carry the stigma of powerlessness. Three interacting components of the training course mutually influence one another: The trainer, the participants and the training course itself. In addition to these principal actors, the situational context should not be overlooked. We would like to address the above-mentioned components in more detail below.

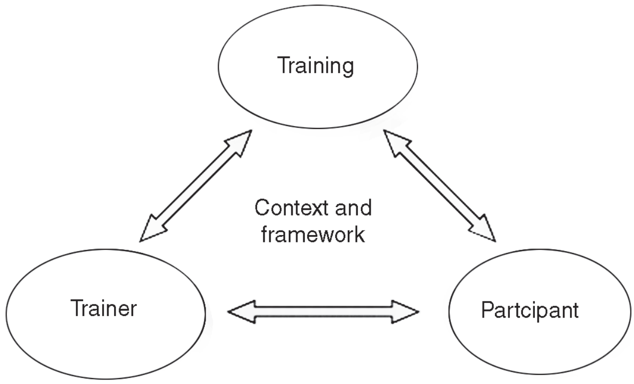


Fig. Interacting components of training.

THE TRAINER

In the following pages we would like to shed more light on the trainer's skills, characteristics and tasks which we consider to be necessary for positively and successfully implementing this training concept for the socially disadvantaged. We are aware that we are portraying an optimal picture and thereby setting high standards. It seems to us to be important to strike a balance between the maximum amount of available staffing resources that can be employed and the minimum number of personnel required to ensure the quality of the training course.

Nevertheless we would like to encourage small institutions as well to conduct a training course such as this. Before we turn our attention to the characteristics of the trainer, we would like to discuss the preferred number of trainers to implement the training course.

Individual Trainer or Team

Is it enough to have one single specialist available to implement the training course or is it necessary to work in a team? Our training concept aims to develop different skills areas using diverse methods which require the trainer to possess multiple skills. We would like to mention that we recommend training teams of at least two people for any type of training course.

There are many reasons for this:

- Implementing an interactive training course and coordinating the group requires a great deal of work, especially if the training course is to be designed to be interesting and maintain the participants' motivation. Nobody can deliver 100% performance throughout the entire day.
- It requires very different capabilities and skills to implement a training course and coordinate a group of participants. Nobody is perfect – a number of trainers can complement one another's strengths and specialist skills.
- To ensure that flexibility is designed into the training schedule and that the course is tailored to the participants' needs and resources, one needs to look at the entire sequence of events and processes as they relate to group dynamics.
- As with every human, trainers and participants as well are influenced by sympathy and antipathy. It would be regrettable if the training were to miss its mark because a participant could not get on with the trainer as an individual. Employing a training team increases the chance of each participant finding a contact person he can trust.

What appears to be optimal is:

- An interdisciplinary team in which different professional groups complement each other's skills and expertise;
- A team comprising different characters and both sexes;
- Including an individual who is familiar with the target group and its life situation.

It goes without saying that forming such an ideal team is not always easy, especially if the training course is being staged by a small institution. In this case it may help to cooperate with other institutions. When we talk about "the trainer" below, what we invariably mean by this is the option of a training team as well.

Skills and Knowledge

The trainer requires a number of fundamental skills and background knowledge enabling him to implement the training course effectively and successfully. A number of these skills are necessary for any training course but assume particular significance when dealing with the socially disadvantaged target group, especially if the latter are carrying the baggage of adverse previous experience with conventional education systems.

From our perspective, the following skills and knowledge are of particular importance for this training concept:

- Sensitivity to cultural aspects (in particular to the target group's social culture)
- Communication skills;
- Moderating and organisational skills;
- The ability to cooperate and work in a team;
- Background knowledge of group dynamics processes;
- The ability to excite and maintain the participants' motivation;
- Observation skills (observing group dynamics and the entire training process;
- The flexibility to adapt the training course to the relevant participating group and its needs;
- Expert knowledge of (interactive) educational approaches;
- Expert knowledge of lifelong learning;
- The ability to encourage participants to get involved and take an active part;
- Knowledge of and methods for planning learning processes;
- Practical experience of staging training courses;
- The awareness of one's own value system.

In summary, our conclusion is that the trainer should have completed basic pedagogical, social or psychological qualification.

Experiences with and Attitude Towards the Target Group

Sensitivity to cultural background is an additional important precondition the trainer should possess. To align the training course with the relevant target group, the trainer requires solid background knowledge of the target group's life situation, needs and resources. If he does not have any personal experience of dealing with the target group, it can be helpful to establish advance contact with someone representing the target group's interests or even to integrate them into the training team. Personal contact should be established with the participants a few days prior to the training course and an interview of sorts organised in which questions can be asked about the participants' specific expectations of the training course. Based on this, the trainer can develop the necessary empathy for the target group. In order to be able to work effectively with it, the trainer requires a certain degree of sympathy for and interest in the target group and should show tolerance for the participants' life experiences and values. Self-regulated learning in terms of empowerment can only be achieved if the trainer does not foist his own values and perspectives on the participants.

Instead, the participants should be engaged with on their own ground and be able to contribute their own experiences. The trainer can explain his own attitudes and offer new or complementary value concepts but should never present these as the "only true" point of view. The trainer supports the participants in devising new goals and skills by themselves; he is a sort of coach but not a leader. It seems to us to be very important that the "balance of power" between trainer and participants remains in equilibrium. What this requires is for the trainer constantly to reflect on his own influential position and the socially disadvantaged situation of the participants.

The trainer's Tasks

The trainer is responsible for the following general aspects of the training course:

- A clearly defined framework and clear and realistic objectives should be made transparent for the participants.
- The trainer should work with the participants' resources and align the training course with their experience and previous knowledge.
- He ought to observe and evaluate the training process and tailor it to the participants' needs if so required.
- The participants should be able to comprehend the training process so that they can appreciate their experiences within the group.
- The trainer should be aware of his role and in particular of his boundaries within the training course context (*e.g.,* education vs. therapy).
- Linked to this, he should collaborate with other organisations and, if necessary, make contact with institutions and experts capable of supporting participants with other goals and questions.

THE TRAINING COURSE

The training course can follow particular principles to meet participants' needs and to ensure a positive and productive atmosphere.

Trust

It is very i*mportant for the participants to feel accepted and to develop trust in the trainer and training. This can be fostered by the following aspects:*

- Training should always take place under the aegis of a known organisation or institution and not under the personal responsibility of an individual trainer.

- It can be helpful for participants to get to know the trainer and the premises before training commences. As a result they will be familiar with the setting and will already feel somewhat more confident when the training gets under way. Individuals who do not feel confident can cancel their participation in the training without losing face.
- The participants should receive detailed prior information about the content, techniques and rules of the training course.

Acceptance and Good Atmosphere

A good atmosphere within the group and mutual acceptance between participants are essential for staging a successful training course, especially when we are working with target groups who are familiar with rejection and social exclusion in their day-to-day lives.

This requires a number of basic rules:

- The group atmosphere has first priority. Conflicts between participants should be discussed immediately. Quite incidentally, these situations lend themselves to practicing social skills in a practical context.
- A number of basic rules should be agreed at the beginning of the training course, for example feedback rules, accepting different opinions and previous experiences *etc.* Depending on the group and timeframe, these rules can be developed together with the participants.
- Training should be fun. Sufficient time should be scheduled for breaks, relaxation or games and exercises to lighten the atmosphere.

Take Account of Participants' Needs

The training course should take account of the participants' needs and specific characteristics:

- When the training course gets underway, participants should have the opportunity to express their needs and expectations.
- Participants should be actively involved in designing the training course. In the process they can be helped to identify and select their learning goals for the training course themselves.
- The training course design and techniques should be consistent with the target group's cultural and social background. Participants should be able to introduce their cultural identity.
- The training course content and techniques should tie in with the target group's previous knowledge and everyday life.
- Participants need to be motivated and empowered to transfer their newly acquired skills to everyday life and try them out in "real" situations.

- The learning process should envisage small steps from the familiar to the unfamiliar and from simple to more challenging material. It is the trainer's responsibility to ensure that each participant is able to follow the learning process.
- The participants' questions and expectations should be answered.
- If possible, the participants should be able to continue their learning process after the training course. For example, this can be in the form of an ongoing training course or by group meetings for exchanging information.
- The results and learning successes should be acknowledged upon conclusion of the training course. Firstly, respect should be shown for the participants' commitment and successes. Secondly, the participants should receive a form of certificate or written reference that they can specifically use for their own purposes (*e.g.,* applications).

THE PARTICIPANTS

Not only the trainer and training course but the participants as well should satisfy certain fundamental criteria:

- So that the participants really can benefit from the training, they need to possess a modicum of capabilities enabling them to interact, comprehend and learn. This means for example that there has to be a certain command of the language in order to ensure communication.
- The participants must be motivated to take part in the training course and exercises. They should be open to learn new things and for the possibility of personal development or change.
- This presupposes that the participants' involvement in the training course is voluntary.
- One aspect which cannot categorically be answered is the question about the group's homogeneity or heterogeneity. A homogenous group may be easier to handle, the course content can be tailored to specific questions relevant to many or even all the participants. In a heterogeneous group, on the other hand, the participants can benefit from their differences and learn from other people's different experiences and knowledge. It seems advisable to maintain a balance between diversity and similarity and thereby take account both of the social learning objective while also enabling participants to feel quickly at home in the group. In the same manner, no one-size-fits-all statement can be made about the participants' familiarity and anonymity. Anonymity affords the participant the opportunity to speak and behave freely without the others having a preconceived opinion about him and without misgivings about how one's behaviour on the training course will affect existing relationships. On the other hand, depending on the target group, participants in a familiar peer group can

operate in a more free and less inhibited manner and provide one another mutual support during the training course. The trainer has to weigh up the influence of dependencies and strong social ties within the training group and select the set-up that seems to him to make the most sense in terms of promoting individual development.

- Thought should be given to exclusion criteria before and during training. Depending on what the training course purports to achieve, physical or mental illnesses, dependency, substance use etc may have a counterproductive effect on participation in the training course.

Contract between the Trainer and the Group

It can be helpful to conclude a contract between the trainer and the participants. This contract defines the training course's overriding goals, states what both parties are to contribute to the training and can lay down from the outset a number of fundamental rules and obligations (*e.g.,* regular participation, punctuality or the like). This type of agreement is an initial contribution to ensuring that participants themselves assume responsibility for the training course.

TRAINING CONTEXT

A number of other underlying conditions can influence the way a training course proceeds. The premises are of particular importance.

They should:

- Be readily accessible for the target group (travel costs and distance)
- Be known to the participants and accepted by them;
- Provide all the necessary resources for the training course;
- Provide for complimentary or low-priced meals and overnight accommodation if required;
- Be independent of institutions that the participants associate with other obligations (*e.g.,* therapeutic centre, job centre or police station).

CONTINUOUS VERSUS DISCRETE VERSUS SERIAL SKILLS

Continuous tasks have no identifiable start or finish. An example can be skills of cyclic character (cycling, skating, and rowing) Discrete tasks have a definite start and finish.

An example can be skills of acyclic character (throw, jump). Serial tasks are composed of discrete skills performed in sequence, with successful execution of each subtask determining the overall outcome. An example can be skills of a combined cyclic and acyclic character (javelin throwing, long jump).

TACTICAL COMPONENT

Tactical component of sports training focuses on different ways to conduct sports competition towards victory. Key terms of this component are strategy and tactics. Strategy means a plan which was created beforehand and is based on experience with a purposeful conduct of sports competition that has proved to lead to an expected result in a specific competition.

Tactics means practical execution of strategy in a specific race situation. Practical execution is based mainly on acquired possible solutions of specific race situation. Progress of acquiring possible solutions of race situations must be in compliance with the duration of sports training within the selected long-term conception of sports training.

Almost every volleyball team has some weakness which can be used as advantage for the opposing team. Let us suppose that the line-up of some anonymous team includes a player who is not so good at receiving of first hit on their half. At present, reception is a necessary basis of a good quality game in volleyball.

Let us further imagine that this team has got a very good setter. Strategy is then based on the fact that it is necessary to aim service at this player in the course of game and attempt to lead own offence over such part of the net that is defended by a player of a lower height.

Tactics is then based on practical solution of game situation when service is aimed in such a way so that the receive spiker view of the ball is made difficult. In such a situation he would have to make as long movement as possible towards the place of reception; offence is conducted according to the position of a specific defence player, etc. Another example of strategy could be summarised as follows: The basis of own good quality game is to make opponents argue with one another. Tactics is then to choose one of the opponents who is a bit choleric a talk to him at the right moment.

PSYCHOLOGICAL COMPONENT

Psychological component focuses on positive influence on the athlete's personality as far as fair play is concerned in dependence on the length of sports training with the aim to achieve maximum efficiency in senior age. There are no two exactly identical people in the world. Everyone is an original who acts as an individual on the outside. Personality of each individual is characterized by a number of factors. Among them, there are the following:

Temperament which is manifested on the outside through emotions and is related to the dynamics of mental processes. In practice, four basic types of temperament are distinguished: sanguine, choleric, phlegmatic, and melancholic.

Motivation is understood as an incentive which supports some kind of behaviour and is decisive in the kind and intensity of a person's acting. Acting can be described as an activity carried out to follow a clear-cut aim. Motivation is closely related to activation level. Activation level can be described as the level to which organism is activated. Relationship between activation level and sports performance has been proved to exist. The curve of dependence is in the shape of inverted U. The interpretation is that both very high and very low activation level is of a negative influence on sports performance.

Qualities of an individual are innate and can be divided into two positive (devotion, persistence) and two negative (dependence, selfishness) categories. Qualities of an individual are characterised with four dimensions: direction, intensity, scope and duration.

Attitudes are – as opposed to qualities – acquired and they are repeatedly manifested in given situations. Attitudes originate from echoing, maturing, rationality or on the basis of emotional reactions.

An example of temperament manifestation in sports can be a response of two different volleyball players to a game situation in which the referee makes an unintentional discriminating mistake against one of the teams. Each volleyball player knows that referees never change their statements. A phlegmatic player who does not get emotional easily and does not manifest his or her emotions much on the outside, will not comment on the referee's mistake much, rather he or she will get ready for the rest of the game. A choleric player, who gets emotional easily and manifests his or her emotions on the outside, will comment on the situation violently and will request a change of the referee's decision.

During my career as a coach, I encountered many cases of insufficient or, on the other hand, exaggerated motivation. Neither of these leads to a good performance in competition. It is necessary to keep activation level at an optimum level. It is upon the coach's feeling and experience to know how much motivation his trainees need to reach maximum sports performance. The following rule holds true: "neither too much, nor too little".

Both positive and negative qualities are manifested in any situation. In sports, these qualities are manifested much more in situations when the athlete or the team loses. Again, it is upon the coach's feeling and experience to be able to regulate these manifestations in the right way.

What can be used as an example from the area of sports is a team of athletes of a senior category of a high-league collective sport who considered training of physical fitness useless. It was very difficult to begin with fitness training. The athletes' attitudes towards fitness training changed with first successes. Now, the attitude of nearly all the athletes is totally different. For a well physically prepared athlete is able to resist fatigue more effectively and consequently manifest better performance.

Tasks of sports training focus on systematic development of the components of sports training. Development of individual components of training is influenced by the structure of sports performance.

STRUCTURE OF SPORTS PERFORMANCE

Sports performance is understood as an extent to which a motor task limited by rules of a given sports discipline is accomplished. Sports performance factors are understood as a relatively independent part of sports performance.

Traditionally recognized factors of sports training in any sports disciplines include:

- Somatic factors
- Fitness factors
- Technical factors
- Tactical factors
- Psychical factors

A common feature of the factors is that they can be affected by training (fitness, technical, tactical, and psychical) or they are taken into account in talent picking (somatic – *e.g.* selecting taller children for volleyball, basketball, or shorter for gymnastics respectively). Sports performance factors correspond to the above sports training components. Sports performance is influenced by a number of factors. The importance and hierarchy of the factors depends on a specific sports discipline.

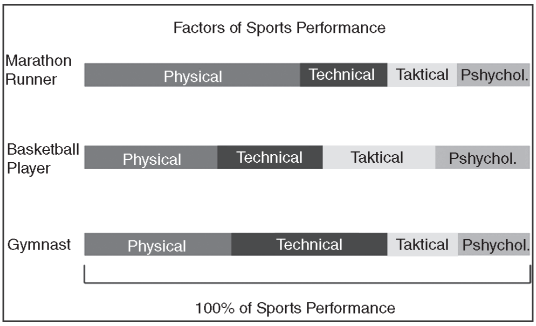


Fig. Example of sports performance factors in different sports discipline

For marathon runners, long-term endurance training is an important part of the year's micro cycle, while with sports gymnastics development of this kind of endurance is not a priority. The performance of an athlete in sport depends on the athlete's technical, tactical, physical, and psychological-social characteristic. These factors are linked with each other, *e.g.*, the technical skills can not be fully utilised without adequate levels of physical abilities. Conversely tactical component can not be fully utilised without adequate quality of technical skills. The physical demands in sport are related to the activities of the athlete.

The performance in selected sports discipline is based on the characteristic of the respiratory and cardiovascular systems as well as muscles, combined with the interplay of the nervous system. The muscular system is constituted by a multitude of components, which have important influence on the mechanical and metabolic behaviour of the muscle. Muscle morphology and architecture, and myosin isoform composition play a major role in the contractile strength characteristics of the muscle evaluated as maximal isometric, concentric, and eccentric contraction force, maximal rate of force development, and power generation. Glycolytic muscle enzyme levels and ionic transport systems are major determinants of anaerobic power and capacity. Likewise, mitochondrial enzyme levels and capillary density exert a strong influence on aerobic muscle performance in turn affecting the force development and the maximal power output of human skeletal muscle, while also influencing the endurance performance of the muscle fibres.

The respiratory, cardiovascular, and muscle characteristic are determined by genetic factors but they can also be developed by training. A number of environmental factors such as temperature and for outdoor sports, the weather and surface of competition ground also influence on the performance.

PRINCIPLES OF EXERCISE AND SPORT TRAINING

INDIVIDUALITY

Everyone is different and responds differently to training. Some people are able to handle higher volumes of training while others may respond better to higher intensities. This is based on a combination of factors like genetic ability, predominance of muscle fibre types, other factors in your life, chronological or athletic age, and mental state.

SPECIFICITY

Improving your ability in a sport is very specific. If you want to be a great pitcher, running laps will help your overall conditioning but won't develop your skills at throwing or the power and muscular endurance required to throw a fastball fifty times in a game. Swimming will help improve your aerobic endurance but won't develop tissue resiliency and muscular endurance for your running legs.

PROGRESSION

To reach the roof of your ability, you have to climb the first flight of stairs before you can exit the 20th floor and stare out over the landscape. You can view this from both a technical skills standpoint as well as from an effort/distance standpoint. In order to swim the 500 freestyle, you need to be able to maintain your body position and breathing pattern well enough to complete the distance. In order to swim the 500 freestyle, you also need to build your muscular endurance well enough to repeat the necessary motions enough times to finish.

OVERLOAD

To increase strength and endurance, you need to add new resistance or time/intensity to your efforts. This principle works in concert with progression. To run a 10-kilometer race, athletes need to build up distance over repeated sessions in a reasonable manner in order to improve muscle adaptation as well as improve soft tissue strength/resiliency. Any demanding exercise attempted too soon risks injury. The same principle holds true for strength and power exercises.

ADAPTATION

Over time the body becomes accustomed to exercising at a given level. This adaptation results in improved efficiency, less effort and less muscle breakdown at that level. That is why the first time you ran two miles you were sore after, but now it's just a warm up for your main workout.

This is why you need to change the stimulus via higher intensity or longer duration in order to continue improvements. The same holds true for adapting to lesser amounts of exercise.

RECOVERY

The body cannot repair itself without rest and time to recover. Both short periods like hours between multiple sessions in a day and longer periods like days or weeks to recover from a long season are necessary to ensure your body does not suffer from exhaustion or overuse injuries. Motivated athletes often neglect this. At the basic level, the more you train the more sleep your body needs; despite the adaptations you have made to said training.

REVERSIBILITY

If you discontinue application of a particular exercise like running five miles or bench pressing 150 pounds 10 times, you will lose the ability to successfully complete that exercise. Your muscles will atrophy and the cellular adaptations like increased capillaries (blood flow to the muscles) and mitochondria density will reverse. You can slow this rate of loss substantially by conducting maintenance/reduced programme of training during periods where life gets in the way, and is why just about all sports coaches ask their athletes to stay active in the offseason.

The principles of specificity, progression, overload, adaptation, and reversibility are why practicing frequently and consistently are so important if you want to improve your performance. Missed sessions cannot really be made up within the context of a single season. They are lost opportunities for improvement. Skipping your long ride on weekend A means you can't or shouldn't go as far as originally planned on weekend B (progression and overload). Skipping your Monday swim means your swimming skills and muscles won't be honed or stressed that day (specificity). Missing a week due to a vacation sets you back more than one week (adaptation and reversibility). Apply these principles to your training to get a better understanding of your body and how to achieve success.

THE FITT PRINCIPLE OF TRAINING

Think of The FITT principle as a set of rules that must be adhered to in order to benefit from any form of fitness training programme.

These rules relate to the Frequency, Intensity, Type and Time (FITT) of exercise...

These four principles of fitness training are applicable to individuals exercising at low to moderate training levels and may be used to establish guidelines for both cardio respiratory and resistance training. The FITT principle is used to guide the development of unique and bespoke fitness plans that cater for an individual's specific needs.

FREQUENCY

Following any form of fitness training, the body goes through a process of rebuild and repair to replenish its energy reserves consumed by the exercise.

The frequency of exercise is a fine balance between providing just enough stress for the body to adapt toand allowing enough time for healing and adaptation to occur...

1. CardioRespiratory Training The guidelines for cardiorespiratory training (also called aerobic conditioning) is a minimum of three sessions per week and ideally five or six sessions per week.

Experts suggest that little or no benefit is attained over and above this amount. Of course athletes often fall outside the suggested guidelines but even elite performers must give themselves time to rest.

2. Resistance Training The frequency of resistance training is dependent upon the particular individual and format of the programme. For example, a programme that works every body part every session should be completed 3-4 days a week with a day's rest between sessions.

On the other hand, aprogram that focuses on just one or two body parts per session, in theory you could be completed as frequently as six days per week. Many bodybuilders follow such a routine. Remember though, each time you complete a strenuous strength training session (regardless of the body part) you are taxing your body as a whole - including all the physiological systems and major organs.

INTENSITY

The second rule in the FITT principle relates to intensity. It defines the amount of effort that should be invested in a training programme or any one session. Like the first FITT principle - frequency - there must be a balance between finding enough intensity to overload the body (so it can adapt) but not so much that it causes overtraining.

Heart rate can be used to measure the intensity of cardiorespiratory training. Workload is used to define the intensity of resistance training.

Cardio Respiratory Training

Heart rate is the primary measure of intensity in aerobic endurance training. Ideally before you start an aerobic training programme a target heart rate zone should first be determined. The target heart rate zone is a function of both your fitness level and age. Here's a quick method for determining your target heart rate...

Heart Rate and Maximum Heart Rate

Heart rate is measured as beats per minute (bpm). Heart rate can be monitored and measured by taking your pulse at the wrist, arm or neck. An approximation of maximum heart rate (MHR) can also be calculated as follows: MHR = 220 - age.

Target Heart Rate

For beginners a target heart rate zone of 50-70 percent of their maximum of heart rate is a good place to start. So if, for example, you are 40 years old that gives you a predicted maximum heart rate of 180 (220 - 40). Multiply 180 by 50% and 70% and your reach a target zone of 90bpm - 126bpm. For fitter, more advanced individuals, a target heart rate zone of 70-85 percent of their maximum of heart rate may be more appropriate. Staying with the example above, that 40 year old now has a heart rate zone of 126bpm - 153bpm. There are limitations with heart rate and the heart rate reserve method, while no means flawless, may be a more accurate way to determine exercise intensity.

Resistance Training

For resistance training, workload is the primary measure of intensity.

Workload can have three components:

1. The amount of weight lifted during an exercise
2. The number of repetitions completed for a particular exercise
3. The length of time to complete all exercises in a set or total training session

So, you can increase workload by lifting heavier weights. Or you could increase the number of repetitions with the same weight. Finally, you could lift the same weight for the same number of repetitions but decrease the rest time between sets. However, only increase the intnesity using one of the above parameters. Do not increase weight and decrease rest time in the same session for example.

TYPE

The third component in the FITT principle dictates what type or kind of exercise you should choose to achieve the appropriate training response...

Cardio Respiratory Training

Using the FITT principle, the best type of exercise to tax or improve the cardiovascular system should be continuous in nature and make use of large muscle groups.

Examples include running, walking, swimming, dancing, cycling, aerobics classes, circuit training, cycling etc.

Resistance Training

This is fairly obvious too. The best form of exercise to stress the neuromuscular system is resistance training. But resistance training does not neccessarily mean lifting weights. Resistance bands could be used as an alternative or perhaps a circuit training session that only incorporates bodyweight exercises.

TIME

The final component in the FITT principle of training is time - or how long you should be exercising for. Is longer better?

Cardio Respiratory Training

Individuals with lower fitness levels should aim to maintain their heart rate within the target heart rate zone for a minimum of 20-30 minutes. This can increase to as much as 45-60 minutes as fitness levels increase. Beyond the 45-60 minute mark there are diminished returns. For all that extra effort, the associated benefits are minimal.

This also applies to many athletes. Beyond a certain point they run the risk of overtraining and injury. There are exceptions however - typically the ultra-long distance endurance athletes. In terms of the duration of the programme as a whole, research suggests a minimum of 6 weeks is required to see noticeable improvement and as much as a year or more before a peak in fitness is reached.

Resistance Training

The common consensus for the duration of resistance training session is no longer than 45-60 minutes. Again, intensity has a say and particularly grueling strength sessions may last as little as 20 - 30 minutes. Perhaps the most important principle of training (that ironically doesn't have it's own letter in the FITT principle) is rest. Exercising too frequently and too intensely hinders the body's ability to recover and adapt. As a rule of thumb, the harder you train, the more recovery you should allow for. Unfortunately many athletes don't have that luxury!

SPORTS TRAINING PRINCIPLES

The FITT principle is designed more for the general population than athletes.

Sport-specific training should be governed by a more in-depth set of principles. These include:

- Specificity
- Overload
- Adaptation

- Progression
- Reversibility
- Variation.

TRAINING PRINCIPLES TO IMPROVE ATHLETE PERFORMANCE

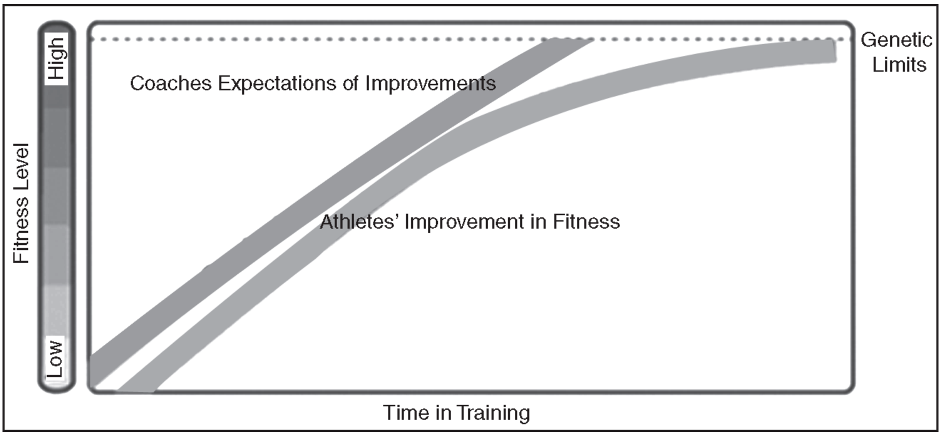
SPECIFICITY PRINCIPLE

The specificity principle asserts that the best way to develop physical fitness for your sport is to train the energy systems and muscles as closely as possible to the way they are used in your sport. Thus, the best way to train for running is to run, for swimming is to swim, and for weightlifting is to lift. In sports such as basketball, baseball, and soccer, the training programme should not only overload the energy systems and muscles used in that sport, but should also duplicate similar movement patterns. For example, in strengthening a quarterback's throwing arm, design the exercise to simulate the throwing movement. Warning: This principle can be taken too far. Ample evidence suggests that cross training, or doing another sport or activity, can help improve performance.

OVERLOAD PRINCIPLE

To improve their fitness levels, athletes must do more than what their bodies are used to doing. When more is demanded, within reason, the body adapts to the increased demand.

You can apply overload in duration, intensity, or both. If you increase a cross country runner's long-distance run by five minutes, you've added an overload of duration. If you instead ask the runner to run her normal distance but in a shorter amount of time, you've added an overload of intensity.



Progression Principle

To steadily improve the fitness levels of your athletes, you must continually increase the physical demands to overload their systems. If the training demand is increased too quickly, the athlete will be unable to adapt and may break down. If the demand is not adequate, the athlete will not achieve optimal fitness levels.

Diminishing Returns Principle

When unfit athletes begin a training regime, their fitness levels improve rapidly, but as they become fitter, the diminishing returns principle becomes law. That is, as athletes become fitter, the amount of improvement is less as they approach their genetic limits. A corollary to this principle is that as fitness levels increase, more work or training is needed to make the same gains. As you're designing training programmes, remember that fitness levels will not continue to improve at the same rate as athletes become fitter.

Variation Principle

This principle has several meanings. After your athletes have trained hard for several days, they should train lightly to give their bodies a chance to recover. Over the course of the year use training cycles (periodisation) to vary the intensity and volume of training to help your athletes achieve peak levels of fitness for competition. This principle also means that you should change the exercises or activities regularly so that you do not overstress a part of the body. Of course changing activities also maintains athletes' interest in training.

Perhaps you're thinking that the specificity principle and variation principle seem to be incompatible. The specificity principle states that the more specific the training to the demands of the sport, the better; and the variation principle seemingly asserts the opposite-train by using a variety of activities. The incompatibility is resolved by the degree to which each principle is followed. More specific training is better, but it can become exceedingly boring. Thus some variety that involves the same muscle groups is a useful change.

Reversibility Principle

We all know the following adage: Use it or lose it. When athletes stop training, their hard-won fitness gains disappear, usually faster than they were gained. The actual rate of decline depends on the length of the training period before detraining, the specific muscle group, and other factors. A person confined to complete bed rest is estimated to lose cardiovascular fitness at the rate of 10 percent a week. Smart coaches and athletes today recognise that maintaining a moderately high level of fitness year-round is easier than detraining at the end of the season and then retraining at the beginning of the next.

Individual Differences Principle

Every athlete is different and responds differently to the same training activities. The value of training depends in part on the athlete's maturation. Before puberty, training is less effective than after puberty. Other factors that affect how athletes respond to training include their pretraining condition; genetic predisposition; gender and race; diet and sleep; environmental factors such as heat, cold, and humidity; and of course motivation.

Moderation Principle

Here is another familiar adage: All things in moderation. Remember that training is a slow, gradual process. Give athletes time to progress. Your challenge as a physical conditioning coach is to design a training programme that progresses optimally using the principles just discussed. You want to gently coax your athletes' bodies into superior condition, not beat them up by overtraining. Make training fun. Design games and activities that challenge athletes to do the same work but without the drudgery of monotonous exercises. Be encouraging and promote a positive attitude about training. By all means don't use training activities as punishment for misbehaviors.

INSPIRATORY MUSCLE TRAINING: ERGOGENIC BENEFITS

Since the first reports of an ergogenic effect of specific inspiratory muscle training (IMT) in the mid-1990s, researchers have not only demonstrated its efficacy beyond reasonable doubt, but are also beginning to understand how it works. Alison McConnell takes a look at the latest thinking on IMT, its ergogenic benefits, and why serious athletes neglect it at their peril... For those readers who are unfamiliar with IMT, we should perhaps take a small step back in time to set the stage. In the early days of research in this area, those of us with an interest in ventilatory limitations to exercise performance were viewed with what might politely be called skepticism. The received wisdom has always been that there is no respiratory limitation to exercise performance; after all, maximal oxygen uptake is not limited by the transfer of oxygen across the lung, but by the ability to transport and utilize it. This being the case, what possible advantage could there be to increasing the ability of the respiratory pump muscles to ventilate the lungs? Furthermore, the respiratory muscles were thought to be 'super human', and immune to fatigue, by virtue of their continuous activity throughout life.

The first questions about these assumptions began to surface in the early-1990s, when compelling evidence emerged that the inspiratory muscles (specifically the diaphragm) exhibit fatigue in the same way that other skeletal muscles do. This was followed by evidence that the work and associated metabolic demands of the inspiratory muscles during intense exercise were far greater than anyone had anticipated. Not only that, but the inspiratory muscle would 'steal' blood (and oxygen) from the exercising limbs in order to meet their own metabolic demands. Any coach who was confronted with such evidence would instantly conclude that the respiratory pump muscles were a dangerous weak link and warranted specific training. However, the sport scientists were so bogged down by their preconceptions about there being no respiratory limitation to maximal oxygen uptake that it took a further five years, as well as overwhelming direct evidence, to persuade them that IMT is genuinely ergogenic.

HOW IMT WORKS

What is now emerging from the published literature is a clearer picture of just how IMT exerts its ergogenic effect. Well-conducted studies of IMT had demonstrated consistent changes (or lack of them) in a few key physiological outcomes after four to six weeks of IMT. These included no change in maximal oxygen uptake, but reductions in the intensity of breathing and whole-body effort sensations, as well as a lower *blood lactate* concentration and heart rate at equivalent intensities of exercise. These observations provided some vital clues about how IMT actually works and sparked a series of experiments to test the resulting hypotheses.

Changes in blood lactate concentration are reminiscent of the response to whole-body training, and lead to questions about an increase in the lactate threshold. We examined this question specifically in a carefully conducted study that used the most accurate and meaningful index of lactate turnover as the main outcome variable – the maximum lactate steady state (MLSS). If lactate concentration is lower after IMT because the exercise intensity corresponding to MLSS (or the lactate threshold) increases after IMT then we would expect to see a measurable change in exercise intensity at which

MLSS is achieved. In short, despite a significant reduction in the lactate concentration at the cycling power output that corresponded to MLSS, we did not observe any change in MLSS power. Our study was capable of detecting changes in MLSS power of as little as 2.5% (6 Watts). Thus, we concluded that IMT cannot operate via a mechanism linked to improvements in the lactate threshold.

Changes in heart rate and effort sensations hint that the cardiovascular strain of exercise may be reduced following IMT, but this is not because the oxygen cost of exercise is measurably lower, because this does not appear to change. However, the evidence that inspiratory muscles can 'steal' blood from locomotor muscles led some researchers to examine the interaction between inspiratory muscle work and cardiovascular *control* during exercise.

In an elegant series of experiments, researchers at the University of Wisconsin identified that when the inspiratory muscles are subjected to fatiguing bouts of work (breathing against an added external load), they provoke a reflex change in vasoconstrictor output to the limbs. In other words, in the face of fatigue, the inspiratory muscles signal the cardiovascular control centres to divert blood away from the working limbs. This has a two-fold benefit from the viewpoint of the inspiratory muscles; firstly, it ensures that they get more blood and oxygen (protecting their vital function); secondly, by restricting blood flow to the working limbs, the supply of oxygen and removal of metabolic by-products is impaired, which leads to an enforced decrease in exercise intensity, or even cessation (reducing the ventilatory demand).

The most recent study from this group went on to show that if the work of breathing is manipulated during very intense cycling exercise that the severity of leg fatigue is also changed – if inspiratory muscle work is increased, then leg fatigue is increased, and if respiratory work is reduced (by allowing a ventilator to 'breathe' for the subjects) leg fatigue is reduced. This is entirely consistent with the inspiratory muscles maintaining a high position in the 'pecking order' for the supply of blood flow. What we believe occurs after IMT is that the enhanced strength, power and fatigue resistance of the inspiratory muscles leads to an abolition or delay in the triggering of the reflex vasoconstriction. This mechanism is consistent with all of the physiological changes that arise after IMT.

PRACTICAL IMPLICATIONS

The first question that most coaches ask me is, 'How long does IMT take?' There are two answers to this: 1) the training itself requires about three minutes twice per day; 2) performance improvements are measurable within four weeks.

Until we fully understand the mechanisms (the explanation given above is our best guess at this moment in time) it will not be possible to offer specific guidance on training regimens for different sports. However, what has been identified in the laboratory is a regimen that works for time trials of between six and 60 minutes duration, in rowing, cycling and running – ie 30-repetition maximum load executed with maximal effort. The latter is important because it ensures that the muscle recruitment is maximal – *i.e.* as many muscle fibres as possible are forced to contract against the load.

Aside from the laboratory based research, my dealings with athletes in a range of sports have also led me to identify a number of postural challenges that appear to benefit from 'posture-specific' IMT. For example, the sport I have worked most extensively with is rowing, which is probably one of the toughest sports for the respiratory system.

Rowers normally inhale at two points in the stroke (just before the catch and just after the finish), with the largest breath being just after the finish. Both of these points in the stroke impose restrictions on breathing. At the finish, the hips are extended and the shoulders are behind the hips. This means that the muscles of the torso (including the inspiratory muscles) must work against gravity to prevent the rower from falling backwards. At the same time, the rower needs to take a large, fast breath, which means that the inspiratory muscles are subjected to competing demands for postural stability and breathing. Once the rower reaches the catch, he or she must take another breath, but in this position the movement of the diaphragm is impeded by the crouched body position. At the catch, the thighs push the liver, stomach and gut upwards against the diaphragm, compressing the abdomen. This compression makes it harder for the diaphragm to contract, flatten and move downwards, as it must do in order to inflate the lungs. Once again, looking at this from a pragmatic point of view, the obvious thing to do is to train the inspiratory muscles under the conditions where they experience the greatest challenge.

Summary

A very successful and knowledgeable coach friend of mine describes the inclusion of IMT in his athletes' training as a 'no-brainer'. In his view, there's nothing else that he can add to their training that requires so little time and provides such a large guaranteed benefit to their performance. He coaches indoor rowers, and has taken some of them to World Championship medals, so his opinion should count for something.

Scientists have seen IMT move from being the preserve of heretics (myself included), to a credible and intensely intriguing phenomenon that is now undergoing equally intense scrutiny. The next few years should see IMT really come of age, and as more coaches and athletes gain experience of manipulating the training to suit the demands of their own sports, the greater the benefits should be for sport as whole.

Finally, for the 20% or so of athletes who have asthma, it is worth mentioning that IMT is also becoming a well-established, drug-free method of managing asthma symptoms. Indeed, one IMT device has recently been made available on prescription, thanks to its proven efficacy in clinical trials. So, for athletes with asthma, or those with borderline symptoms that don't qualify for medication in competition, IMT could provide an even greater benefit.

TRAINING PLAN: MONITORING YOUR TRAINING INTENSITY

Measuring an athlete's training load through physiological monitoring provides the opportunity for the coach to build up a detailed training history, which should form the basis for any assessment of performance. For instance, using a training diary, it's possible to infer links between the volume of training and decrements in performance and make assumptions about the possible occurrence of the overtraining/overreaching syndrome.

Alternatively, the coach may wish to know what type, intensity, duration and frequency of training stimulus is optimal to develop maximal *aerobic capacity* (*VO2max*). Using a past history of training load and physiological test results, it's possible to determine how much training is required to improve VO2max. Successful monitoring and assessment of training may then lead to improvements in performance.

TRAINING LOAD

A number of methods to quantify 'training load' have been used for training and monitoring in different sports, including athlete self-assessment of training sessions, simple calculations of volume and heart rate (HR) response methods. The commonest method now used for monitoring training load is 'Training Impulse' or 'TRIMP', which was formulated by Bannister in 1975. Since Bannister first introduced the TRIMP it has been modified several times to refine its use of HR data.

Although HR monitoring is both valid and reliable, problems can occur when using this information to interpret physiological responses. For example, the relationship between HR and *blood lactate* is linear at low to moderate speeds/power outputs but as exercise intensity increases, HR increases linearly but blood lactate rises exponentially – ie the various physiological responses occur that are not necessarily reflected in HR data alone. To overcome this problem, Carl Foster introduced a weighting category based upon percentages of maximum heart rate to give a TRIMP score. He also used perceived rating of exertion (PRE) to provide an overall subjective assessment of a training session. In a further development, a research group from Spain gave weightings to three physiological stages based upon analysis of inspired and expired oxygen and CO_2.

Following an incremental exercise test to exhaustion, an analysis of oxygen and CO_2 *gas flows in the body can reveal two distinctive changes in ventilation, which represent different physiological occurrences in response to an increase in exercise intensity:*

1. The first identifiable point is the ventilatory threshold (VT) – it represents one of the body's initial responses to a change in homeostasis as a result of exercise;
2. The second identifiable point is the respiratory compensation point (RCP), which marks the exercise intensity at which heavy breathing is required to help remove metabolites from the body.

The Spanish researchers tested eight national and regional level male runners to determine their VT and RCP. They then used the modified TRIMP to provide the following weighting factors.

1. Light intensity, heart rates below the exercise intensity that elicits VT; 1 minute of exercise in zone 1 is given a score of 1;
2. Moderate inensity, heart rates between the exercise intensity that elicits VT and RCP; 1 minute of exercise in zone 2 is given a score of 2;
3. High-intensity heart rates above the RCP; 1 minute of exercise in zone 3 is given a score of 3.

After determining the training zones the researchers gave HR monitors to all participants and asked them to wear the monitor each time they trained. The purpose of this was to determine the relationship between training load and running performance during the most important competitions of the season (ie national cross country championships). Training load was monitored each session over a six-month period.

Results showed that the athletes in this study spent most of their training time (71%) at low intensities (zone 1). The percentage of time training at moderate (zone 2) and high (zone 3) intensities was 21% and 8% respectively. In this particular group of athletes, there was a significant correlation between the amount of time they spent training in zone 1 and performance in both the short and long cross country races; the athletes who spent more time training at a low intensity performed better in a high-intensity (30 minutes of continuous exercise in zone 3) race!

Five-zone Trimp

In another study, British scientists recognised that training at higher intensities should be awarded larger weighting factors because of the greater physiological demand imposed by high-intensity training. Using blood lactate and HR responses to a treadmill test and a concept known as the 'fractional elevation' in heart rate, they arrived at a five-zone system that more accurately reflects the increased physiological stress imposed by high-intensity activity.

Table. Training zones based upon the modified TRIMP (MHR= maximal heart rate

Zone	%MHR	Weighting	Training Type
5	95-100	5.91	Maximal training
4	90-94	4.01	OBLA traning
3	84-89	2.57	'Steady state' traning
2	77-83	1.65	Lactate threshold training
1	64-76	0.84	Moderate activity

Similar to previous TRIMP methods, the total time spent in each zone was multiplied by a weighting factor to calculate training load. The difference in this instance is that as exercise intensity increases, the weighting factor increases exponentially rather than linearly. In previous TRIMP methods low (zone 1) intensity exercise was given a weighting of 1 and high (zone 3) intensity a weighting of 3, a simple linear increase. The five-zone method gives high-intensity activity a weighting about seven times greater than low-intensity activity.

The researchers who developed this method gave eight male premier league hockey players a HR monitor and recorded their HR responses to training and competition from the start to the middle of the season. They found that those players who had a higher mean weekly TRIMP value had the largest change in VO2max and velocity at OBLA, suggesting that the greater the training volume, the greater the increase in VO2max and velocity at OBLA. The percentage changes in VO2max and velocity at OBLA were also correlated with the mean weekly time spent in high-intensity training (zones 4 and 5). This suggests that the more time spent training at a high intensity, the greater the improvements in VO2max and OBLA. Interestingly, the researchers calculated how much weekly TRIMP value players needed to accumulate to maintain VO2max and velocity at OBLA. They found that players needed a lower mean weekly TRIMP value to maintain VO2max than OBLA and needed to spend less time engaged in high-intensity training to maintain VO2max than OBLA.

These calculations are particularly useful to coaches when planning and monitoring training as they can ensure that each player is receiving a sufficient training stimulus to maintain or improve VO2max and OBLA throughout the season. The major limitation to this study is that it only took into account endurance training when hockey training most likely comprises of speed and *strength training* too.

Intermittent Sports Training

Most sports, especially team sports, comprise a mix of three basic physical qualities, speed, strength and endurance. However, previous TRIMP systems have been limited to identifying the training load of endurance training and have not been able to account for other training loads using one single TRIMP method.

French researchers attempted to quantify the training loads of endurance, sprint and strength training using one single TRIMP method for use with intermittent sports training. The Work Endurance *Recovery* (WER) method created by this team is based upon the theory that training-induced physical stress can by quantified in relation to an athlete's maximal work capacity. So rather than training loads being quantified in relation to physiological responses (*e.g.* blood lactate, HR or VO2max and CO2), it is based on 'fatigue occurrence', making it possible to monitor different training load types using a single equation. The researchers above compared the WER method with TRIMP methods to determine how accurately they measured training loads by asking participants to exercise until exhaustion in three separate sessions – one endurance, one sprint and one strength. They then assumed that each session provided a similar training load – *i.e.* around 33% of total volume each and because the WER equation calculates training load in relation to fatigue; the total work capacity of 100% divided by three sessions equates to about 33%. They discovered that there was no difference between the calculated training loads of sprint or strength training regardless of which TRIMP method used. However, large differences in training loads were observed between endurance training and sprint/strength training; endurance session training loads were 2.5-2.8 times higher than sprint training and 5.1-5.5 times higher than the strength training. This confirmed that the TRIMP methods are incompatible when comparing training loads of different exercise modes. By contrast, when they used WER, they found that each session represented around 31 -35% of the cumulated training load indicating that there was no difference between the training sessions and confirming the group's hypothesis.

SPORTS TRAINING IN THE PREPARATION OF A BOXER

As far as boxing is concerned it has also experienced many changes, and the most important ones are those applied to the rules in order to protect the boxer and, more specifically, to the subject in question: reducing the amount of rounds for world belts and for being in the 10 first boxers in world ranking. Initially, it descends from the round limitless fights and/or due to abandoning them until making 15 and then 12 rounds –mid 1980's- for world fights and 10 rounds for scoring in world rankings.

On this sense, reducing the amount of rounds is a factor that forces to achieve what was achieved before, only that in little time: the same or superior amount of punches, the increase of movements of the torso and of the movements around the ring; all being made in the same or less time unit (3 minutes per round or about 50 minutes per belt and near to 40 by scoring in world rankings). This reducing trend forced the introduction of changes in preparation systems at the same time, and more specifically, in the quasi-training as the structural unit of the entire preparation process of the boxer.

Given the fact that boxing presents its own problems concerning the adjustment of changes to new times and that also allows the solution of such problems, the author had as a basic objective the proposal of a new technical training approach for the boxer. This approach is based on the integral and systemic consideration of all kinds of preparation (physical, psychic, tactical and integral) with the meaningful predominance of the development of each and every physical unit (strength, speed, resistance, flexibility and coordination).

SPORT TRAINING

Since sport training is a source and important part of sports preparation, it can also be considered as the structural unit of the entire preparation process. In other words, sports training is to the boxer's preparation process what the filament is to the muscle, or what the neuron is to the nerve system. There are many definitions of sports training (ST); however, in this case we will consider it a specialized pedagogical process aimed to integrally educate and also to develop the motor qualities of the trainee; to assimilate and improve the technical and basic habits and to move the psychic and functional reserves of the athlete. All this will be achieved through the use of several physical exercises and/or motor actions that will have the adaptation of the organism to extreme or necessary efforts as an objective. It will also seek to reach the highest technical and sports results in the chosen sport.

Sports training can be observed and considered from many other points of view: As a source and important part of the periodicity of the preparation process, according to Matveev (1991) and Platonov (2004); to form microstructures (from 2 to 14 days), meso structures (from 15 days to 6 weeks) and macrostructures (from 7 weeks to 4 years). In other words, sport training is the key element to compose and generate preparation plans and/or programmes for the boxer or for any other sportsman.

As a preparation process that takes many years to complete, a process in which the athlete goes from one technical level to another, since he is a rookie until reaching and stabilizing high sports achievements. We do this from the moment in which the person is a rookie until reaching and keeping high sports achievements; the second one, divides such process into five (5) stages, predominantly pointing at cyclic sports (swimming, cycling, rowing, canoeing, and others); the third and last one, divides it into four (4) stages for the complex coordination disciplines concerning their structures or high technical skills. These were divided in fewer stages, perhaps because of the fact that they were characterised as early specialisation sports.

In spite of the justified differences presented in the table, the fact that sports training is universally divided into stages and that it is a process that lasts many years unifies the criteria given by those specialists as a common denominator.

As a complex and dynamic system from a cybernetic position, meaning, from the theory of information (information exchange) as a functional closed circuit with its direct and reverse links.

From the cybernetic position, the training process is a complex and dynamic system (CDS) in which the role of system's director corresponds to the teacher- coach, and the role of being conducted goes to the athlete.

Due to its nature, the direction of the CDS is the passing process from one stage to the other through the variable system's influences. In order to achieve this, the system director must count on a) the object's model (athlete) in his current status; b) the final model of the status that is necessary to reach (high sport scores); c) the model of the principles, the methods and ways of influence and changes that would be produced on the subject; d) the ways of reception and analysis of the results given by the preparation process (receiver and informative systems).

In this case, the direction of the ST is one of the most important premises that guarantee the keeping of the ideal structure and the making of programmes, plans and objectives of the problem.

This direction is led by the teacher-coach with an active participation of the athlete and establishes three (3) operation groups:

1. Collecting information about the state of athletes including physical, technical-tactical and psychic preparation standards. The reaction of the different functional systems (nerve, cardio vascular, breathing, osteo-muscular, among others) to the competition and training loads.
2. The analysis of that information, by comparing the obtained parameters with the wanted parameters in order to correct, to plan and to achieve the goal.
3. Taking and making decisions by preparing and presenting objectives and tasks, plans, programmes, methods and ways to guarantee the achievement of the goal.

Kinds of Training

There are many and diverse kinds of training, therefore, naming them all is really difficult. Nevertheless, those could be classified in groups by considering their trends, according to the kind of preparation, the intensity or magnitude of the efforts and even the quality of training, among others.

Some of these kinds will be shown next without untying their structures and contents; otherwise, not only would it be another topic, but it would also be extended and broad.

According to the kinds of Preparation, there are Several Kinds of Training:

Physical training can be general, special and complementary and at the same time, be subdivided into other kinds of training in order to develop force (static and dynamic, among others), speed (frequency of movement, time of movement, anticipation), power (with predominance of speed and force), resistance (aerobic, non lactic and lactic acid, anaerobic resistance), flexibility (passive, active, elongation).

Flexibility (passive, active, elongation, joint mobility); coordinative quality (spatial orientation, still- dynamic balance, rhythm, precision, coordination of movement, differentiation of muscular efforts); Basic and/or special technical training that can be subdivided into other trainings for initial learning and notion creation, for a deep learning to assimilate the skill and for perfecting and stabilizing habit. Tactical training of a basic and/or special kind is subdivided into trainings for the attack, defence, counter attack, persuasion and/or deceit, among others. Psychic training, that can be also subdivided into autogenous and ideomotion training, hypnosis and others

There are kinds of training that are voluminous and intensive, according to the magnitude of the efforts that are made. They can be subdivided into trainings of moderate, medium, maximal and sub maximal magnitude. With that magnitude trend, the kinds of training can also be determined for both sexes, masculine and feminine or for different ages (children, teenagers, youngsters, adult and mature people).

According to the quality of the training, which encloses all kinds of training, that quality can be also subdivided into excellent, good, satisfactory, average, bad, dangerous and fatal. Knowing these and other kinds of training along with their structures and contents is very important and fundamental for the teacher-coach because the quality of the entire preparation process and the achievement of high world class scoring results depend on that knowledge; just as the current boxing sport requires it.

TACTICS AND TACTICAL PREPARATION IN ARCHERY “

In going in for sports, which have preparation for participation in the sports competition as the nearest goal on the level of sports exercises technique, teaching sports tactics takes place. Archery is not an exception. A word of tactics came into sport from the military terminology (from the Greek word of taktika - art of constructing troops), component of military art, including theory and practice of preparation and conducting the fight. There are several popular definitions of the notion - SPORTS TACTICS. Sports tactics is an art of competing with an «opponent». Its main task is the most expedient use of forces and possibilities for victory.

The basic tool of tactics is the sports technique, applied in the permanent and changing conditions of external environment, according to the plan and tasks, which arise. Sports tactics is purposeful methods of the use of technical principles in tactical actions in order to solve the tasks of the competition taking into account the rules, positive and negative descriptions of the preparation, as well as environmental conditions. Tactics in sport is an expedient and conscious change of conduct, actions of a sportsman or of a team depending on conditions and situation that is developing (or which can be foreseen) in order to achieve success in the sports competition.

Now technical, physical and psychological preparation of archers is on high enough, approximately identical level. Therefore other conditions being equal the level of tactical art of archers very often determines the victory at the biggest international competitions. Level of tactical readiness depends on mastering tools of sports tactics by an archer (*i.e.* technical and tactical actions), as well as mastering its forms (individual, team). Structure of tactical readiness follows from the character of strategic tasks, which form the basic directions of sports contest.

These tasks can be linked with the participation of an archer in the series of competitions and with the goal of his/her preparation for the main competition of season and, thus, be of a perspective nature. They can be local as well, *i.e.* related to participation in a separate competition, to shooting on a separate distance, or in a separate one sparring meeting. The basis of tactical readiness of separate sportsmen and teams is the mastering of modern tools, forms and kinds of tactics by them; accordance of the tactics to the level of development of archery; accordance of the tactical plan to the peculiarities of the competition (condition of the place of the competition, peculiarities of regulation of the competition, weather conditions, readiness of participants for the competition; way of judging; fans' behaviour, etc.); binding of the other sides of readiness, such as technical, psychical, and physical, with the level of perfection as well as with the quality and level of financial side. While developing tactical plan tactical actions experience of the best sportsmen, main competitors, their technique and physical abilities, psychical readiness; technique, tactical and functional abilities of partners, variations of tactics in different conditions, depending on the character of technique and tactical actions of competitors and partners, run of the competition should be taken into account.

Peculiarity of the archery is a deciding factor, which determines the structure of tactical readiness of an archer. For example, one of the basic components of tactical readiness in archery is the choice of a reasonable tactical plan and its use in the competition. Peculiarity of tactical preparation in archery is also the fact that an archer uses complex technical equipment such as bows, arrows, plungers, sights, stabilizers, etc in his sports activity. Knowledge of principles of action of equipment, which is used, its advantages and disadvantages considerably enrich an arsenal of tactical abilities of an archer.

Tactical actions of an archer at competitions are directed on the achievement of optimum success. They should be built according to the quality of equipment, which he/she uses, to the tactical knowledge, to the level of his/her technical, physical and psychical readiness, to the level of development of physical abilities, volitional qualities, speed of reactions and other components.

Concrete tasks, which are set by a sportsman for himself/herself in the competition, also refer to the questions of tactics. And it is a very important part of tactical preparation as it affects eventual result substantially. Range of possible tasks is very large. Choice of the tasks, which are set by a sportsman independently or which are recommended by the coach, should be made very attentively and with great responsibility. Tasks can be applied both to the participation in the competition on the whole and to its definite stages.

Let us try to examine some of the possible variants of this kind of tasks:

- To show the highest possible result
- To show the set result
- To win, indifferently with what result
- To show a result or take a place, which will give a right to continue the competition (in 1/32, 1/16 finale, etc)
- To show a record result on a definite distance or in an exercise
- To improve one's effectiveness gradually as the competition proceeds
- To take a definite place, for example, to take the first place or to be one of the best three, six, thirty two and etc participants
- To overcome a definite competitor

- To test new equipment (bow, arrows, stabilizers, etc)
- At participation in the competition to try to pay above all attention to the quality of one's technical actions, and not to the hit of the target or to the result
- During participation in the competition not to make some (concrete) technical mistake
- At shooting at the definite distances a task is set for the archer for the predominant hit of arrows in the definite circle of the target or in other words, to "agree" to optimum and real hits at the moment. For example, while shooting for 90 meters a task is set not to go out of the size "7", at 70 and 50 meters not to go out outside the size "8", and at 30 meters to "agree" to the hit in the size "9"
- To focus main attention on participation in the "Team Olympic round", but not on the individual shooting
- To check the quality of adjustment of his/her own bow and arrows
- To try to adhere to the optimum rhythm of shooting, or shoot quicker than usual etc.

Tactics of an archer in the great deal depends on the tasks, which follow the regulation and the rules of the competition. For example, the introduction of new rules of conducting the competitions in archery changed the requirements to the tactical preparation of an archer a lot.

That is why not all are called to this; favourites of archery were able to adjust to new tactical tasks and were even forced to halt active training in archery. While solving tactical tasks great amount of possible conditions, on which their implementation depends, should be taken into account.

As an example we can consider the case of necessity of participation of an archer in the competition on condition of deterioration of his/her special endurance (conditionally, because of illness or protracted business trip, he/she had an unforeseen large recess in the regular trainings on the eve of the competition). When constructing tactical charts and actions such archer should better give up participation in the competition because of the threat of causing harm to his/her motive habits.

But if because of definite circumstances there is no such possibility, this archer can be recommended to use an easier bow, to facilitate force of limbs of the bow or aim him/her at shooting at a more rapid rate with the purpose of economy of forces and energy used for treatment of separate shots, etc.

Tactical skills of an archer are interlinked with his/her psychological preparation. There are cases, when it is very difficultly to conduct a border between the tactical actions of an archer and his/her actions, relating to his/her psychological preparation, to be more concrete to his/her own psycho adjusting. In the psychological aspect tactical actions are the product of difficult psycho motoring processes, which flow consistently and conjointly.

Psycho motoring processes of a tactical action take place in three main phases:

- Assistance and analysis of competition situation;
- Mental decision of the special tactical task;
- The impellent decision of a tactical task.

These three phases create a successive decision rank of tactical tasks and are in the close interaction. Moreover, archer's memory plays here a deciding role. Tactical actions in the cybernetic plan are a searching system, directed at the target, which among the possible targets not only elects the most favourable one, but its decision will be also improved in the course of action. In the course of growth of effectiveness in sport as well as in archery, number of questions, which relate to the tactical preparation of archers, is constantly multiplied. To the modern questions of tactics it is possible to refer not only actions of an archer executed by him/her directly at competitions, but also those executed after their limits. So, for example, rituals of assembling and preparation to the future competition, feed on the eve of the competition, equipment of an archer, sequence of choice of options for the training, etc has a direct attitude towards tactical preparation of archers.

Activity of tactical actions is an important index of sports skills. A sportsman of high qualification should be able to impose his/her will to the competitor, to make permanent psychical pressure at him/her by the variety and efficiency of his/her actions, self-control, will for victory, faith in success. Efficiency of tactical activity in archery is determined by the capability of an archer to anticipate spatially and temporally that is to guess ways of the development of the situation at the competition. The exact choice of an archer of the aiming point in difficult weather conditions depends in a great deal on this ability.

METHODS OF TACTICAL PREPARATION

Tactical preparation is a process, directed on study of essence and basic theoretical-methodological principles of sports tactics; mastering of the basic elements, methods, variants of tactical actions; improvement of tactical way of thinking; study of the information, necessary for the practical realisation of tactical readiness; practical realisation of tactical readiness. Improvement of tactical archers' skills is as a rule carried out in the process of tactical training. Definition: Tactical Training is a type of training, which has the knowledge and the selective application, by the sportsman or team, of the most proper technical proceedings in order to solve certain partial or final tasks of the contest for an object. But in reality, the tactical skills of an archer is improved practically constantly, both during other types of training, and at the time spare from the basic types of training.

This peculiarity is explained by the fact that in the tactical preparation of an archer both accomplished domain by the tactical methods and develop an archer's habit of tactically correct thinking are important in the equal measure. Definition: Tactical thinking in sports is a sportsman's ability to estimate quickly and correctly the conditions of a sports struggle, which change constantly, and according to this to realize his/her own tactical plans or make changes in them effectively in connection with the circumstances. Among the basic directions of tactical preparation in archery we should mark out the following: study of essence and basic theoretical methodical principles of sports tactics; collection and analysis of the information, necessary for the practical realisation of tactical readiness; master the basic elements, principles, variants of tactical actions; improvement of tactical way of thinking; practical realisation of tactical readiness.

The study of essence and basic theoretical methodical principles of sports tactics consists in the theoretical mastering of general principles of sports tactics, tactics of archery, rules of judging and principles that concern the concrete competitions, peculiarities of tactics in family types of sport, tactical experience of the best sportsmen, methods of development of the tactical conception, etc.

A study of essence of tactics is a necessary pre-condition for mastering tactical actions, development of tactical abilities and skills, forming of tactical way of thinking. The knowledge of theoretical methodical principles of sports tactics helps to estimate a competition situation precisely, to pick up means and methods of competition activity adequately taking into account the individual features, qualification, level of readiness of a competitor and partners. An archer acquires the tactical knowledge during his/her whole sports life (career), multiplying it volume according to the growth of his/her skills and accumulation of experience. The objectivity and expedience of his/her tactical conceptions, plans and projects in great deal depends on the plenitude of this knowledge.

The whole complex of verbal and visual methods helps an archer to gain the knowledge of the theory of tactics. Special literature, lectures, conversations, explanations, viewing competitions, films and videotape recordings, their analysis, etc are the sources of this kind of knowledge. However, it should be remembered that this knowledge alone not supported by the personal motive experience of an archer can not affect sports results positively. For the practical mastering of the concrete principles of tactical skills of an archer we can recommend to conduct special lessons (trainings).

The main aim of this kind of lessons should be gaining of concrete tactical knowledge, principles or skills. These can be special lessons dealing with improvement of team rounds shooting, lessons imitating shooting in the conditions of time «trouble» - (shooting of definite quantity of arrows in the limited period of time, for example, high-quality release of three arrows for 30 seconds), special lessons for verification and improvement of «sense of time», etc. As an example it is suggested to conduct lessons putting an accent on acquisition of knowledge and abilities in the sphere of tactical preparation: As it is generally known, there is a wide difference in the «cost of amendments of breech-sight» on different distances which depends on the individual features of archers, on the large variety of technical descriptions of materiel, which they use. For the armament of archers with the practical knowledge about the «cost of amendments of their own bows» it is possible from time to time to suggest executing similarity of laboratory works at the training. For example, at 30 meters distance it is suggested to the archers to execute a few series with the combination of center of grouped of arrows with the center of target. After this it is recommended to make alterations in breech-sight on the concrete size (for example: to displace foresight for 5 mm to the right and also for 5 mm downwards).

Then the archers are recommended to execute a series of shots and to define the center of new grouped exactly. Simple arithmetic division of distances on horizon and on the vertical line from the center of the target to a new center of grouped for the quantity of the amendments (in this case 5) borne in breech-sight will be a cost of amendment in 1 mm at 30 meters distance. On all remaining distances 'costs of amendments' are determined similarly. It is possible to pay attention to the small distinction in the cost of 'amendments' in sight of the horizontal and vertical planes. This distinction can be explained by the fact that there is a small distinction in their characters. When we make amendments in the vertical plane, length of the aiming line forms distance from eye to foresight, and when we make amendments in the horizontal plane we take the distance from bowstring to foresight into account. But because of comparatively small difference in these distances, it is possible to ignore such sizes.

Collection and analysis of the information which is necessary for the practical realisation of tactical readiness. Foremost, this concerns information about the economic feasibilities and features of materiel part that is used, possible competitors and partners in the team, environment and terms of conducting of future competitions.

The most essential information about competitors is the information about their physical readiness, their technical tactical manner to conduct competition fight, peculiarities of their conduct in different (favourable and unfavourable) conditions of competitions, characteristics of personality, volitional and psychical qualities. Collection of information about environment and terms of future competitions are necessary for the design of such training terms, which would be adequate terms of future competitions, so that sportsmen would gradually adopt themselves to the specific conditions of competitions and prepare their equipment, clothes and shoes to the concrete terms of the competitions. Here it is necessary to take into account terms, place and time of conducting of competitions, climatic conditions (temperature, humidity of air), quantitative and qualitative entry list of the competitors, members and qualification of judges, state of the sports buildings (quality of shields for catching of arrows must be of the special interest), etc.

Improvement of tactical way of thinking. In the process of preparation to competitions it is impossible to foresee all possible situations of future competition fight. Therefore one of the basic tasks of tactical preparation of a sportsman is to improve his/her tactical way of thinking. Thus it is necessary to develop the following abilities:- to perceive quickly, to realize and analyse competition situations adequately;- to estimate a situation quickly and precisely and to make decisions according to the circumstances and level of one's own readiness;- to foresee actions of a competitor (team partner); - reflection to display one's actions according to the goal of competitions and tasks of a concrete competition situation.

It is important for an archer to develop his/her spatial and temporary foreseeing of situations before they are going to happen. Together with the growth of sports qualification an archer acquire capacity of the exact temporal and spatial extrapolation of technical-tactical actions. Speaking about tactics we should also put an accent on the ability of a sportsman to operate mental material, which consists of knowledge, verbal instructions (of the coach), picture of motions, competition situations.

The tactical thought is developed by exercises, which consist of observation and search for the tactical essence when weather conditions change during the competitions, at the change of one's own state (appearance of fatigue, discovering of one's own technical errors), breakages and damages of equipment, gestures, motions, actions, intentions, and states of competitors. Concentration of attention and consciousness of an archer on the search of effective methods of fight for victory helps to improve tactical thought. Tasks related with the improvement of tactical thought should induce to the analysis of possible aspects of competition situations in the fight for victory. A sportsman must remember about the results of actions in similar situations (his/her own and other sportsmen's), to make decision in the limited spans of time.

Method of training with competitor and method of training with the supposed competitor are the basic specific methods of improvement of tactical thought. Exercises done on the special devices, training devices, individual lessons with the coach and tests are also used with this purpose. Among the methods of improvement of tactical skills participation in great number of the sports competitions, different in terms, entry list and scale is especially valuable. Exactly in competitions valuable experience is gained, which helps to find a reasonable form of acting in the process of sports fight. Analysis of tactical activity in the conditions of training and competitions is also very important.

Archers should tell the coach about things they succeeded to find out in competitions, what their actions were caused by, and what intentions they had, what prevented to fulfill the plan of conduct of the competitions. They must be able to consider and analyse their tactical actions on training and competitions in details. At this time coach together with them analyses their senses, determines how quickly and correctly archers perceived the conditions of competitions and reacted on them, whether they were attentive and observant, what prevented them from executing the tasks, how physical and moral-volitional qualities were showed up in competitions, how technical skills were realized.

Mastering tactical actions. Tactical essence of actions is as a rule mastered practically together with the mastering technical principles (archer's technique). Exactly in the process of practical mastering of technical principles archers are conscious, that their technical actions must be connected with realisation of possible tactical variants. The task of tactical essence and its possible efficiency means that an archer will be able to do the optimum choice of action and successfully solve a tactical task in the definite situations operatively. Mastering of essence of different tactical situations foresees the study of typical situations and peculiarities of behaviour in any of them, as well as preparation to the conduct of competition fight with different competitors, in different conditions.

In teaching tactics of archery it is possible to use all aggregate of verbal, visual and practical tools and methods of preparation. However, the most complete practical tools and methods are used here. The principle of modeling of a sportsman's activity in competitions is the basis of practical methods of tactical preparation. With this aim artificial conditions or situations that model specific activity of a sportsman are introduced into the training of an archer. For example, during implementation of a control or shooting "on the account" a coach can artificially model such situation when a bowstring is torn or a stabilizer breaks off, etc at the competition.

Practically, this can be realized in the request of the coach to change the bow-strings on spare and continue shooting if possible without the decline of its quality (imitation of breaking-off of the bow-string), or to remove stabilizers (imitation of breaking-off of stabilizers) and also continue the successful conduct of shooting, or continue shooting using the spare bows. Except for acquisition of tactical skills this kind of exercises is also of purely psychological use.

The point is that this kind of exercises diminishes a degree (level) of anxiety in real circumstances of competitions and archers at the subconscious level stop to be fearful of possible unexpected cases and are not already lost in the cases if similar unforeseen situations happen during competitions in reality. Method of training with a competitor is used for the detailed working out of variants of tactical actions, tactical improvement taking into account the individual features of archers; improvement of volitional qualities; bringing up of the ability to use one's possibilities in different tactical situations which are created in the course of competitions.

The exercises with the conditioned situations, where a sportsman that acts as a competitor, operates within bounds of the tasks expressed definitely by the coach and models the fragments of separate competition situations are also used. On different stages of long-term preparation and in different periods of training macro cycle of tactical skills improvement is paid different attention to. The greatest attention is paid to the tactical preparation on the stage of maximal realisation of individual possibilities, when a sportsman gets ready to the highest achievements. On the stage of the specialized basic preparation main key components of tactical skills will be improved. On the first and the second stages of long-term preparation the tactical improvement deals, mainly, with the theoretical and practical preparation. The biggest volume of tactical preparation in macro cycle is at the end of preparatory period and during the period of competitions. On the first stage of preparatory period separate components of tactics will improved. The work considerably increases on the stage of direct preparation to the basic competition. The level of technical skills, physical and psychical readiness, which was formed before this stage, gives a possibility to proceed to the improvement of tactics in its closest approaching to the terms of future competition activity.

The choice of this or another tactical variant, its improvement and its realisation in the competition activity depends on the level of tactical skills of a sportsman, development of his/her motive qualities and possibilities of the major functional systems, moral-volitional and psychical readiness. In fact it is possible to consider process of tactical preparation as an original uniting factor in relation to other component parts of sporting skills. Practical realisation of tactical readiness. This is a synthesizing direction of tactical preparation, its purpose is to solve tactical tasks, such as creation of integral notion of the course of competitions; formation of individual style of conduct of competition fight; decisive and opportune embodiment of the accepted decisions through the rational principles and actions taking into account the peculiarities of weather conditions, of competitor's actions, of the environment, of judging, of the competition situation, etc.

The integral notion of tactics of competition activity is formed and varies during the sports activity. The most noticeable overvalues and changes in the sportsmen's notion take place after their participation in the main competitions, which more than any others test their sports skills, make them notice all "pros and cons" states of their readiness, compare new information with the notions that were acquired earlier.

It is Possible to Make A List of Minimum of Knowledge, Abilities and Skills, That an Archer Must Have in The Field of Tactical Preparation:

- To be able to estimate and to memorize size of the shift of the middle point of hitting (MPH) of the arrow on all distances of shooting on condition of moving pin of sight for 1 mm both in vertical, and in the horizontal plain.
- To be able to calculate the necessary amendments and bring them into sight.
- To know the size of shift of MPH of his/her own arrows at shooting on different distances at different speed directions of wind.
- To be able to shoot with the bearing-out of point of aiming in different areas of the target and beyond it.
- To be able to determine force and direction of wind by the external signs.
- To have optimum variants of warming up exercises and to be able to make use of them before training and competitions.
- To know an optimum psychological microclimate for himself/herself and to be able to create it successfully at competitions.

- To have a clear picture of rational nourishment (taking into account the individual features of his/her organism) and strictly adhere to him in the period of participation in the competitions.
- To be able to use the pauses of rest between shots, between series and between distances during competitions rationally.
- To be able to have under the control the state of his/her equipment constantly.
- To have comfortable sports clothes at competitions for different weather conditions that have been fitted and tested preliminary in the conditions of shooting.
- To be able to adjust shooting correctly during ranging series and to provide the constant control above position of MPH. To react to displacement of MPH in time.
- To feel well the motion of time given for implementation of series of shots, to be able to use it. To not be afraid of time troubles. To be able to shoot quickly with high quality, after the protracted expectation of an opportunity, because of the gusts of wind being in pose of readiness to the shot as well.
- To be able to shoot in different temporal modes. At necessity to be able to shoot with high quality without the use of clicker, if that is required by circumstances that have turned up.
- To have affected variants of conduct and actions in cases of breakage or problems with clothes or equipment of an archer.
- To have the list of all the necessary things that must be in the case for a bow and arrows on inner side of it. To work out a habit to check up on this list presence of all necessary things before going for training and competitions.

At request it is possible to continue this list further, as in the tactical preparation of an archer there is practically no borders. There are cases when an archer makes a tactical decision successfully and in future no one ever repeats this. Important components of forming of integral notion about the tactical preparation are: - awareness of the sportsman of his/her own technique-tactical arming, features of individual manner, advantages and lacks of preparation; - understanding of intercommunication between the preparatory actions and basic ways of conduct of competition fight;- understanding of character of initiative and value of such tactical elements as suddenness, timeliness and etc.; - understanding of necessity of self-control and conscious risk, knowledge of variants of conduct in different moments of competitions, ability to conduct warming up exercises and regulate one's own mental condition independently; - mastering of power of counteract with competitors different in style of shooting and in force; - understanding of psycho- tactical specifics of competition fight; - clear picture of purpose of preparation, participation in the separate competitions and separate meetings, about the paths of attaining a set purpose and solving separate tasks.

An ultimate goal of tactical preparation is the formation of a definite style of conduct of competition fight. Style (manner) of conduct of tactical fight must include general principles of tactic of archery and take into account individual features of a sportsman, including his/her lacks. The practice of sport for a long time has marked out the odd cases of loss of strong sportsmen and strong teams to the weak enough competitors. This is the result not to so big extent of a bad mood, but to a bigger extent of inadequate reflection of a model of the competitor. Most frequently this happens, if an experienced sportsman underestimates possibilities of his/her opponent or over-estimates his/her own.

5

Educational Psychology

Educational Psychology by its very nature is treated as a science. Like science it employs different objective methods of data collection, experiments and drawing inferences. It deals with the study of an organised and systematic body of knowledge concerning human behaviour of all kinds. It seeks to make its subject matter scientific and exact by utilising special means and appliances. It also aims at understanding, explaining, predicating and controlling facts and phenomena relating to human behaviour.

IT IS A SOCIAL SCIENCE

Educational psychology is a social science. Like any other social science it studies human beings relating to education.

Like the economist, political scientist, sociologist or anthropologist, the educational psychologist conducts investigations and experiments, collects data and statistics, and makes interpretations and inferences. He does not work in an ivory tower nor makes any arm- chair speculations.

IT IS AN APPLIED SCIENCE

Educational psychology is an applied science. It applies the psychological principles and methods in the field of education. The knowledge about the growth and development of the child, learning conditions and theories belonging to the field of psychology is used for methods of teaching and school organisation.

It is a Positive Science

Educational psychology is a positive science which deals with things as they are. It is not a normative science which deals with things as should be. It studies the child's growth and development as they are. It never deals with what the child should do or should not do. This task is taken care of by ethics and philosophy which are normative sciences.

SCOPE OF EDUCATIONAL PSYCHOLOGY

Educational psychology is psychology in relation to education. It deals with the behaviour of the individual in various educational environments. Psychology studies the behaviour of the individual in different conditions and situations. Therefore, scope of general psychology is broader than that of educational psychology, which is comparatively limited. Educational psychology deals with the child as a whole-his physical, mental, emotional and social development at various stages. Such development is resulted by heredity and environment, different biological, social and cultural factors. Hence all these aspect and factors come under the purview of educational, psychology.

Learning is the key-concept in educational psychology. It is so important and broad-based that educational psychology is also called learning psychology. The learning process, learning and maturation, nature and conditions of learning, factors influencing;, learning, motivation, attention and interest, various kinds of learning and laws of learning come under the scope of educational psychology.

The psychological principles underlying various methods of teaching, educational innovations and experiments, educational objectives, mental health and hygiene, special provisions for slow- learners, gifted, handicapped and deprived children are usefully applied to educational development. These are, therefore, part and parcel of educational psychology. The psychological tools and techniques, methods and approaches are profitably used by the educational psychologists for undertaking research studies and experiments in various fields. New methods and techniques are also developed by them for collection, interpretation and analysis of data. All these constitute the scope of educational psychology.

AIMS OF EDUCATIONAL PSYCHOLOGY

Educational psychology aims at the harmonious growth and rightful conduct of the children in too. It aims at the development of 'wholesome personality' and 'continuous growth'. It aims at helping the teacher in providing facts and generalisation in his task of assisting the child to develop the harmonious personality.

A few of the specific aims and objectives are discussed here with:

(1) Educational psychology aims at developing right attitudes in the teacher about the educational problems. An effective management of learning is a key

problem in educational psychology. From educational psychology the teacher knows the proper method of imparting instruction. He also knows the amount of learning which can be acquired by the child.

(2) Educational psychology aims at assisting the teacher to organise the material which to be taught to the child. The teacher studies the whole child and his mental make-up and chooses and organises the subject-matter properly.

(3) Educational psychology aims at studying heredity growth and maturation, environmental influences, language, thinking, the development of language and the process of socialisation in relation to their effect on the child as a learner. It assists the teacher to set up appropriate educational situations in order to bring about desirable change.

(4) It aims at assisting the teacher in treating their pupils with sympathy and understanding. It also aims at creating positive attitude towards learning.

(5) It aims at helping the teacher to understand his own task. The teacher faces many problems in the classroom as teaching situation. Educational psychology develops in the teacher a scientific attitude to solve different problems of education faced by him.

(6) Educational psychology aims at teaching, teachers how best to help their pupils to learn more effectively both in and out of class.

(7) It aims at providing the teacher with the proper method of teaching. The various teaching procedure are put in practice in teaching the child. Those which are psychologically sound are recommended for the teacher's use.

(8) Conducting research is another important objective of educational psychology. Knowledge in this field keeps on developing from this practice of research methods.

(9) Educational psychology aims at the application of research findings in the learning situation itself.

(10) It also aims at developing sound methods of measuring and evaluating the achievements of the pupils objectively with pure objectively.

(11) Another significant aim of educational psychology is to guide the administrators in the organisation and administration of the educational instruction and to provide a scientific basis for the supervision of instruction.

(12) Educational psychology also aims at helping the teacher to provide proper guidance programme in the school having thorough knowledge of individual difference.

RELATIONSHIP BETWEEN PSYCHOLOGY AND EDUCATION

- According to Aristotle, "Education is the creation of a sound round in a sound body."
- According to Pestology, "Education is the natural, harmonious and progressive development of man's innate powers."

- According to Gandhi ji, “By education I mean around drawing out of the best child and man, body, mind and spirit.”

The relation between psychology and education is very intimate. Psychology has been defined as the science of behaviour. It seeks to understand and explain behaviour in terms of mental and bodily activities. Its chief problem is how and why we behave, how we Think, know, feel and act and why we think, know, feel and act in the way in which we do.

It tries to understand the conditions from which acts of behaviour arise and to arrive at general principles which govern behaviour so as to interpret, control and predict it. Education, as we have seen above, is an attempt to mould and shape behaviour. It tries to help young people to grow and develop along certain lines, to acquire knowledge and skill and to learn certain ways of thought and feeling so that they may be absorbed in adult social life.

The science of psychology must be basic to such an attempt, for any influence on behaviour, to be effective, must be planned and worked according to the principles of psychology. Education, therefore, must be based on psychology and from the very first step which he takes to educate the child, the educator must depend upon psychological knowledge. Education deals with young people and the conditions that promote or retard growth and development; it selects and strengthens those influences which promote healthy growth and tries to eliminate and weaken those which retard it.

As a result of this study it formulates certain principles on which organisation and administration in schools should be based; it has to study the needs and interests of children and provide for their healthy satisfaction and expression; it has to devise effective methods of teaching so that children may learn more quickly and better.

All this is not possible without a knowledge of psychology which explains how young people grow, what dominant interests mark the several stages of their growth, how they differ from one another and grow at different rates, how they learn new skills or acquire new knowledge, how they react to the influence of teachers and class-mates.

Psychology is expanding rapidly and our growing knowledge of the minds and behaviour of young people promises to be an effective guide in the solution of our educational problems. The Education and Psychology are complementary to each other. Psychology is an essential element to education. Without its help problems of education cannot be solved. Both education and psychology are concerned with behaviour.

Modem education is based and founded on psychology. The child is imparted education only after making a thorough study of his interests, aptitudes, intelligence and personality both is two distinct branches of knowledge but they are closely related.

- According to Jha, “The process of education is entirely at the mercy of psychology.”

- According to Davis, “Psychology has made a distinct contribution to education through its analysis of pupils’ potentialities and differences.”
- According to Skinner, “The entire range of behaviour and personality is related to educational psychology.”

Psychology is the science of behaviour and education in its narrower sense is the modification of behaviour. The modification in behaviour cannot be easily brought unless we know the science of behaviour. The teacher must know about the developmental stages, personality development and emotions of the students in order to be successful teacher. Unless the teacher is fully aware of psychological characteristics of the child, he may not be successful in bringing the desirable changes in the behaviour of students. This leads us to believe that education and psychology are intimately related to each other.

Below are some points which show how far education and psychology are related to each other:

(1) Education is concerned with aims, ideals and standards of life and psychology determines whether these aims are attainable or not.

(2) Education demands the teacher to know the child as well as the subject matter of instruction, where the psychology helps to know about the child.

(3) Psychology also helps the teacher to teach effectively undertaking effective and appropriate teaching techniques.

On the basis of the following points the relationship between Psychology and education can easily be seen:

PSYCHOLOGY AND AIMS OF EDUCATION

Psychology helps the educator in the realisation of educational aims by helping him to bring out improvement in the quality of instruction by providing him ability and insight into the child’s attitudes, ideas, aptitudes, interests and emotions, etc.

Psychology and Teacher

Psychology helps the teacher to understand the learner, learning process and the learning situations. Psychology states that teacher should have sympathetic and affectionate attitude towards the learner. They should have genuine interest in the teaching profession.

Psychology and Curriculum

Psychology suggests that the curriculum should be integrated, flexible, co-related and child-centred. There should be different co-curricular activities in the school. Co- curricular activities are considered as an important part of education because they are important media for sublimation of instincts and for the development of personality.

Psychology and Methods of Teaching

Various methods of teaching like Project method, Heuristic method, Montessori Method, Play-way method are based on sound psychological principles.

Psychology and Text Books

Psychology tells the teachers and the students that text books should be attractive, well illustrated and according to the mental level of the pupils. These may act as good aids to the learners.

Psychology and Innovations

Psychology has made significant contribution by introducing innovative ideas for improving the process of teaching and learning such as-Activity-centred teaching, Micro-teaching, Programmed instruction, Interaction analysis.

Psychology and Audio-visual Aids

Psychology states that to develop interest among students, teacher should properly use audio-visual aids. Use of audio-visual aids makes the learning easy, interesting and effective.

Psychology and Time Table

Time table is prepared according to the psychological principles. While preparing it, the teacher should keep in mind the relative importance of different subjects and their toughness and the fatigue of students.

Psychology and School Administration

Psychology helps in solving problems of administration by mutual discussion among various agencies of school. It provides a scientific basis for the supervision of instruction.

Psychology and Discipline

Psychology tells us the ways of dealing with problems of delinquent, backward, handicapped and gifted children and helps in maintaining discipline. It states that discipline should be self-discipline, dynamic and constructive through participation in purposeful activity.

Psychology and Evaluation

Psychological tools help the teacher to evaluate the achievement of the pupils *and suggests improvements in examination. Teacher can control, direct and predict the behaviour of students on the basis of research studies in class-room teaching. Thus, education and psychology are closely and intimately related.

METHODOLOGY IN EDUCATIONAL PSYCHOLOGY

The research methods used in educational psychology tend to be drawn from psychology and other social sciences. There is also a history of significant methodological innovation by educational psychologists, and psychologists investigating educational problems. Research methods address problems in both research design and data analysis. Research design informs the planning of experiments and observational studies to ensure that their results have internal, external and ecological validity. Data analysis encompasses methods for processing both quantitive (numerical) and qualitative (non-numerical) research data. Although, historically, the use of quantitative methods was often considered an essential mark of scholarship, modern educational psychology research uses both quantitative and qualitative methods.

QUANTITATIVE METHODS

Perhaps first among the important methodological innovations of educational psychology was the development and application of factor analysis by Charles Spearman. Factor analysis is mentioned here as one example of the many multivariate statistical methods used by educational psychologists. Factor analysis is used to summarize relationships among a large set of variables or test questions, develop theories about mental constructs such as self-efficacy or anxiety, and assess the reliability and validity of test scores. Over 100 years after its introduction by Spearman, factor analysis has become a research staple figuring prominently in educational psychology journals.

Because educational assessment is fundamental to most quantitative research in the field, educational psychologists have made significant contributions to the field of psychometrics. For example, alpha, the widely used measure of test reliability was developed by educational psychologist Lee Cronbach. The reliability of assessments are routinely reported in quantitative educational research. Although, originally, educational measurement methods were built on classical test theory, item response theory and Rasch models are now used extensively in educational measurement worldwide. These models afford advantages over classical test theory, including the capacity to produce standard errors of measurement for each score or pattern of scores on assessments and the capacity to handle missing responses.

Meta-analysis, the combination of individual research results to produce a quantitative literature review, is another methodological innovation with a close association to educational psychology. In a meta-analysis, effect sizes that represent, for example, the differences between treatment groups in a set of similar experiments, are averaged to obtain a single aggregate value representing the best estimate of the effect of treatment. Several decades after Pearson's work with early versions of meta-analysis, Glass published the first application of modern meta-analytic techniques and triggered their broad application across the social and biomedical sciences. Today, meta-analysis is among the most common types of literature review found in educational psychology research. Other quantitative research issues associated with educational psychology include the use of nested research designs (*e.g.*, a student nested within a classroom, which is nested within a school, which is nested within a district, etc.) and the use of longitudinal statistical models to measure change.

QUALITATIVE METHODS

Qualitative methods are used in educational studies whose purpose is to describe events, processes and situations of theoretical significance. The qualitative methods used in educational psychology often derive from anthropology, sociology or sociolinguistics. For example, the anthropological method of ethnography has been used to describe teaching and learning in classrooms. In studies of this type, the researcher may gather detailed field notes as a participant observer or passive observer. Later, the notes and other data may be categorized and interpreted by methods such as grounded theory. Triangulation, the practice of cross-checking findings with multiple data sources, is highly valued in qualitative research.

Case studies are forms of qualitative research focusing on a single person, organization, event, or other entity. In one case study, researchers conducted a 150-minute, semi-structured interview with a 20-year-old woman who had a history of suicidal thinking between the ages of 14 to 18. They analyzed an audio-recording of the interview to understand the roles of cognitive development, identity formation and social attachment in ending her suicidal thinking.

Qualitative analysis is most often applied to verbal data from sources such as conversations, interviews, focus groups, and personal journals. Qualitative methods are thus, typically, approaches to gathering, processing and reporting verbal data. One of the most commonly used methods for qualitative research in educational psychology is protocol analysis. In this method the research participant is asked to think aloud while performing a task, such as solving a math problem. In protocol analysis the verbal data is thought to indicate which information the subject is attending to, but is explicitly not interpreted as an explanation or justification for behaviour. In contrast, the method of verbal analysis does admit learners' explanations as a way to reveal their mental model or misconceptions (*e.g.*, of the laws of motion).

The most fundamental operations in both protocol and verbal analysis are segmenting (isolating) and categorizing sections of verbal data. Conversation analysis and discourse analysis, sociolinguistic methods that focus more specifically on the structure of conversational interchange (*e.g.*, between a teacher and student), have been used to assess the process of conceptual change in science learning. Qualitative methods are also used to analyze information in a variety of media, such as students' drawings and concept maps, video-recorded interactions, and computer log records.

EXPERIMENTATION IN EDUCATIONAL PSYCHOLOGY

In every science, as far as possible, knowledge is obtained first hand. This knowledge is obtained systematised and organised. How is this knowledge obtained? It is done through observation and experiment.

To observe is to find a fact; to experiment is to make a fact. Experiment is observation under known and pre-arranged conditions. Observation is the general thing of which experiment is the special form. When observation is controlled and ceases to be general, we have an experiment.

The tremendous growth of Psychology during the last fifty years is due mostly to experimental method. Until the late 19th century Psychology was not considered as a Science. It was regarded as a branch of Philosophy and the methods employed for Psychological investigations were those of Philosophy such as observation and introspection.

The first attempt of making experiments in Psychology was made by Wundt in 1879. He established the first laboratory at Leipazig in Germany. Experiments were made chiefly in the field of sensation, perception, association and memory.

Later on, Germany and America took the lead and by 1894, a score of research laboratories started working.

The last two decades have brought in important developments in technique, methods and trends in Psychology. In America and England there is a psychologist for every school. His task is to take care of the mental health of the children and this experiment is proving very successful.

It consists in the recognition of the inter-connection of the human organism and the external world. The essential principle of the method is to place a human individual in a certain situation and to note how he reacts therein. It is on the basis of his reaction in relation to the situation in which has been placed that the laws of objective Psychology have been derived.

Woodworth expresses it as a stimulus- Response chain. (SR) ‘S’ is the stimulus presented to the individual or the situation in which he is placed and ‘R’ is the individual’s reaction to the stimulus or the way in which he behaves in the situation. The essential task of Psychology is to take the course from ‘S’ to ‘R’ and to state the routes from ‘S’ to ‘R’ under various circumstances.

Experiment in Psychology is performed on a subject who is a living being. The living organism has got certain characteristics and the experimenter must take these into account both in the conduct of the experiment as well as in the interpretation of the results. The experimental psychologist has to depend on the co-operation of his subject, whether an adult or a child or an animal. He cannot lose sight of the time-honoured saying, “one man can take a horse to water but

CRITICISM

1. *Further:* Experimental method has serious limitations when it is applied to Educational Psychology. While dealing with human beings it cannot be as objective as it is claimed by its advocates, because human behaviour cannot be controlled to that extent as is possible in the case of physical and chemical phenomena.
2. *Secondly:* human behaviour under controlled conditions is different from spontaneous behaviour. Therefore, the experimental findings should be cautiously applied to the spontaneous behaviour of human beings under normal and free conditions.

CONCLUDING REMARKS

It is not always possible for every teacher to experiment scientifically because of no special training in doing so and of lack of facilities for experimentation in schools.

Yet he can adopt new teaching methods with better results.

1. New teaching methods suggested by Psychology should be tried.
2. Experimental Psychology helps the teacher in dealing satisfactorily with the backward, the deficient, the emotionally deranged and the delinquent child.
3. Experiments can also be carried on in the development of character and leadership. Schemes of self-government and of giving responsibility to students can be tactfully introduced in schools.

LONGITUDINAL STUDY

A longitudinal study is a correlational research study that involves repeated observations of the same variables over long periods of time — often many decades. It is a type of observational study. Longitudinal studies are often used in psychology to study developmental trends across the life span, and in sociology to study life events throughout lifetimes or generations. The reason for this is that unlike cross-sectional studies, in which different individuals with same characteristics are compared, longitudinal studies track the same people, and therefore the differences observed in those people are less likely to be the result of cultural differences across generations. Because of this benefit, longitudinal studies make observing changes more accurate, and they are applied in various other fields. In medicine, the design is used to uncover predictors of certain diseases. In advertising, the design is used to identify the changes that advertising has produced in the attitudes and behaviours of those within the target audience who have seen the advertising campaign.

Because most longitudinal studies are observational, in the sense that they observe the state of the world without manipulating it, it has been argued that they may have less power to detect causal relationships than do experiments. But because of the repeated observation at the individual level, they have more power than cross-sectional observational studies, by virtue of being able to exclude time-invariant unobserved individual differences, and by virtue of observing the temporal order of events. Some of the disadvantages of longitudinal study include the fact that it takes a lot of time and is very expensive. Therefore, it is not very convenient. Longitudinal studies allow social scientists to distinguish short from long-term phenomena, such as poverty. If the poverty rate is 10 per cent at a point in time, this may mean that 10 per cent of the population are always poor, or that the whole population experiences poverty for 10 per cent of the time. It is not possible to conclude which of these possibilities is the case using one-off cross-sectional studies. Types of longitudinal studies include cohort studies and panel studies. Cohort studies sample a cohort, defined as a group experiencing some event (typically birth) in a selected time period, and studying them at intervals through time. Panel studies sample a cross-section, and survey it at (usually regular) intervals. A retrospective study is a longitudinal study that looks back in time. For instance, a researcher may look up the medical records of previous years to look for a trend.

CROSS-SECTIONAL

When psychologists want to investigate age differences between groups of children, they frequently use a cross-sectional design. In a cross-sectional design different children at different ages are assessed at the same time. For instance, if one were interested in the development of arithmetic abilities, different groups of children at ages 4, 5, 6, and 7 could be given tests that assess addition and the strategies children use to arrive at their answers. In a very brief time the test giver would have an idea of how this important skill changes with age. (If the researcher used a longitudinal design, in which the same children were tested repeatedly over time, it would take them four years to get the same information.) The intent of a cross-sectional design is to allow psychologists to efficiently describe change over time and to identify the various mechanisms associated with those changes. Cognitive strategies—the deliberate plans children use to organise their problem solving—have been studied extensively in developmental psychology using cross-sectional designs. For example, Jane Gaultney (1998) investigated the use of memory strategies in third-, fourth-, and fifth-grade students with and without learning disabilities. Specifically, she examined the degree to which children organised their recall of related items (*e.g.*, sorting and later recalling together items from the same semantic category) and the extent to which such strategy use benefited their memory performance.

She found that typically developing children benefited more from the use of strategies than children with learning disabilities. This difference increased with age and was due to the lack of progress by older children with learning disabilities. Her research also showed that children with learning disabilities experienced more benefit when they used strategies to study items for later recall, whereas typically developing children benefited most from strategies when actually recalling the items. Such an approach identifies potentially important age differences in children quickly and provides some insights into the specific problems children of different ages and abilities have.

Conducting Research from a Developmental Perspective

The use of cross-sectional designs in education implicitly involves a concern for age differences, and this awareness implies an interest in development. A developmental-psychological perspective is concerned with age-related changes and the factors associated with those changes. The deeper issues of how, when, and ultimately why psychological change takes place shape the formulation of research questions, which in turn influence the methods chosen to answer these questions.

While the importance of measuring age-related change should not be underestimated, it is often the normative, or perhaps even more critical, non-normative influences associated with change that most interest developmental and educational psychologists. A broad holistic view of development leads to questions about children's thinking in both social and academic environments that can lead to techniques for enhancing education.

Factors Involved in Conducting Cross-Sectional Studies

Several factors must be taken into account in conducting a study using a cross-sectional design. For instance, if researchers are interested in the development of scientific reasoning in school-age children, they can design a study in which groups of children are given problems to solve requiring reasoning associated with

a) Developing an hypothesis,
b) Deciding how to test the hypothesis,
c) Collecting data, and
d) Evaluating the hypothesis.

Questions immediately surface regarding how they might go about conducting such a study and what pitfalls they must avoid in order to make sure the results of their study are interpretable.

In such a study researchers must ensure that each age and experimental group is balanced in terms of important demographic characteristics. For example, in the study of scientific reasoning, they may give children in grades 3 through 8 different types of science instruction or different textbooks to read. They would need to balance the age and instruction groups for gender. They would need to consider if children from different age groups are from different schools, and if so if the schools differ in any important way, such as socio-economic status (SES) or quality of instruction.

Also, they would consider if the children in the various grades or instruction groups differ in academic abilities, motivation, or previous knowledge of the subject matter in any systematic way. Academic and developmental differences are often associated with gender, SES, motivation, and a host of other factors. Researchers may be interested in how these factors affect scientific reasoning, but in order to interpret unambiguously any age or instruction differences they may find in their study, they have to insure that their groups are balanced, as much as reasonably possible, in terms of demographic and related factors. Another consideration when doing cross-sectional research is the tasks children in each group perform. These tasks should be age-appropriate, so they are not too difficult for the youngest children or too easy for the oldest. Researchers must check that the wording of the problems is comprehensible to children of all ages to be tested. Once they have decided what tasks they are going to give children, they have to decide how to administer them. Children in each group should be tested in a similar fashion. However, because of differences in the social, emotional, and particularly cognitive abilities of children of different ages, it may not be possible to use the identical procedures with all children. What is important is that children of all ages understand the task and what they are supposed to do.

As a result, testing procedures may have to be varied to ensure that both younger and older children understand the tasks, but not varied so much as to make the tasks qualitatively different for children of different ages. For example, it is possible to read or rephrase directions, or even use pictorial representations of the directions, to help younger children understand the expectations of the task. Robust results from research are predicated on ensuring that attention is given not only to the design of the study, but also to the manner in which the research design is implemented.

Using Cross-Sectional Designs and Other Research Methods

Cross-sectional designs can be used in conjunction with both experimental and correlational studies. Experimental research is the gold standard for answering questions about cause-and-effect relationships. Experimental studies involve the manipulation of one or more factors, or variables, and observation of how these manipulations change the behaviour under investigation. For example, Schwenck, Bjorklund, and Schneider (2007) were interested in factors that influence memory strategy development in groups of first- and third-grade children. To assess these factors, they gave some children instruction in strategy use on some trials, whereas others received no such instruction.

They also varied the type of materials that children were asked to remember, including items that were either typical of their category (*e.g.*, shirt, dress, pants for clothes) or atypical of their category (*e.g.*, socks, tie, belt for clothes). They found age differences not only in how well children of different ages remember, but also in children's abilities to benefit from strategy training. For instance, older compared to younger children required less explicit prompting to use a strategy, were more likely to use a strategy when atypical category items served as stimuli, and generalised a strategy to new sets of words. They also identified differences in children's tendencies to use a strategy but not to experience any benefit in recall, termed a utilisation deficiency, something important to educators concerned with teaching strategies in the classroom.

In correlational studies researchers make comparisons between two or more variables. Keep in mind that identifying a relationship between two variables does not imply causation, but if a relationship does exist, further understanding may be gained by pinpointing the nature of the relationship and then conducting experimental studies. Cross-sectional designs have been used in correlational studies leading to understanding about different developmental trends in the use of reading strategies for children with different levels or styles of learning aptitudes.

For example, in one study (Siegel and Ryan, 1988), researchers compared the phonological processing (specifically, reading pronounceable pseudowords such as blurt) of 7- to 14-year-old children with and without reading disabilities. Both groups of children showed marked improvement in pseudoword reading with age, although at each age tested, children with reading disabilities performed significantly worse than typically developing children. In fact, the pseudoword reading of 13- to 14-year-old reading-disabled children was comparable to that of their 7- to 8-year-old typically developing peers.

Contributions to Teacher Practice Using Cross-Sectional Designs

A prominent myth exists that practitioners, most notably teachers, have no need for an understanding of research. The poor transfer of scientific research findings to application in the classroom creates a void between empirical enquiry and classroom-based practices. In short, teachers must be able to understand the meaning of research findings and how to incorporate the results into pedagogical practices. This transfer of knowledge poses a challenge for many teachers who may need more than simple exposure to the research literature in order to make fundamental change. The challenge to teach essential knowledge and foster skill development in children is the primary responsibility of the classroom teacher.

It is critical that teachers be able to pose and ultimately answer questions about important aspects of the teaching and learning process. Evaluating the outcomes of schooling based solely on students' abilities to engage in particular skills is clearly an inadequate approach to gain an understanding of the entire schooling process. There is no paucity of variables one must take into account when investigating why and how children make academic and social progress, and equally as important are the factors that account for failure to make adequate progress. A holistic view of the full teaching and learning process is necessary to account for the mechanisms underlying learning, and this often involves an ability to interpret the findings of developmental/educational research and sometimes to perform simple studies.

Research is not the enemy of teachers, and it is possible to structure research that can help answer some of the most salient questions about teaching and learning. There is common ground between researchers and teachers. Each group desires to answer questions about children's ability to learn content and develop skills. For many teachers, the daily grind of classroom instruction is reason enough to shy away from any sort of formal enquiry into the teaching and learning process, but it is exactly these daily routines that provide a rich source of potential issues from which to gain further insight.

The cross-sectional design is a relatively accessible approach that can be used in the classroom to answer questions about how children change with age in regard to a particular ability or process and how children of different ages respond to different types of instruction. The following paragraphs provide an example of a cross-sectional study that a classroom teacher could conceivably implement, beginning with the formulation of a research question to the execution of the project.

Example of a Cross-Sectional Study

Formal enquiry begins with the generation of questions about instruction and student learning. From here, the challenge is to isolate a particular aspect of the process that warrants attention. Research is most easily facilitated with a clear focus. For example, a teacher may wonder about how students benefit from the use of reading-comprehension strategies to recall information from a science textbook. Perhaps there are age differences in children's ability to benefit from using reading-comprehension strategies that can be traced using a cross-sectional design.

After formulating the research question, it is then necessary to identify groups of children to participate in the study and to determine what, if any, experimental groups teachers want to include. They may decide based on the science curriculum at their school that children much below grade 3 would be too young and children much beyond middle school would be too old for the questions they wish to ask. In the present example, they may include at each grade tested an experimental group that gets special reading-comprehension strategy instruction, as well as a control group that gets standard classroom instruction or perhaps some extra time with the teacher to balance for the amount of teacher-time children in both groups receive. (They may also want to test special groups of children, such as those with learning disabilities, slow readers, or those from less-advantaged homes, and compare them to typically developing children, but the design can be kept simple for this example.)

When identifying groups of participants, it is important to recruit approximately equal numbers of boys and girls and to ensure that the different ages and instructional groups are balanced for other factors that may influence children's performance, such as academic achievement and SES. The next step is to determine the measurement for the study. In this case, the teachers would need to develop or identify a reading-comprehension strategy that they could teach to their participants (likely obtained from the research literature) and a measure to identify whether using such a strategy works, that is, enhances recall of science-text content.

They may be able to use tasks already used in the classroom, for instance, questions taken from the science text or teacher-made questions. They want to make sure that the questions are age-appropriate and that the testing methods are similar across ages. Based on such an experiment, they may find that whereas third-grade children can implement the reading-comprehension strategy, the students tend not to generalise it to new texts and perhaps experience only minimal improvement in recall from its use. In contrast, older children not only use the strategy but also generalise it to new texts and experience substantial recall benefits from its use.

Such findings would help in the development of different types of instruction for children of different ages and perhaps cause teachers to ask more questions about what it is about the reading-comprehension instruction that works (or does not work) for children of different ages. Of course, there are a number of possible outcomes from such a study, but the point here is that one is able to get an idea of how strategy use is developed and then make a determination of which strategies to use and when.

Cross-sectional designs permit researchers or teachers to collect quickly information about age changes in some ability. However, in order to assess true developmental change, longitudinal approaches, which test the same children over time, are necessary. Such studies are not out of the question for classroom teachers.

For example, the assessment of reading-comprehension strategy instruction on children's recall of science information could be performed once in November and again in May to see if individual children make improvements over the course of the school year.

Research has not always been held in high regard or maintained a prominent place in the typical classroom. This does not have to be the case if research studies are designed to be easily understood and implemented in the classroom. Teachers typically reflect on the teaching and learning process, and the use of research designs simply allows these questions to be tested and better understood.

IMPORTANCE OF EDUCATIONAL PSYCHOLOGY IN CHILD DEVELOPMENT

Early childhood is considered to be "an ideal period for learning." Watson thinks, 'the scope and intensity of learning during this period exceeds that of any other period of development." The following points should be kept in view by the educator while planning the education of the child at this stage.

PROPER ATMOSPHERE

A healthy, peaceful and secure atmosphere should be provided for the child. This is necessary both at home and the school.

Proper Treatment

The child has to depend on others for the satisfaction of his requirements. Hence he should always receive an affectionate, sympathetic, and courteous treatment from others an atmosphere of fear and repression will not allow the child to develop properly.

Developing the Endowments of the Child

A child takes birth with certain natural endowments. The educator should try to locate them and plan for their maximum development.

Satisfaction of Curiosity

Every child is curious to know the details of his social environment. The parents and the teachers should try to pacify this need of the child by answering the questions asked by him.

Developing Self-Sufficiency

The infant depends on others for the satisfaction of his needs. The child has to do everything himself. The educator should try to develop a spirit of self-sufficiency in the child and should provide him opportunities so that the child is able to do most of the things himself.

Developing a Social Sense

Efforts should be made by the teacher to develop a rational social séances in the child. The Child starts developing a social sense by the end of this stage.

Encouraging the Instinctive Development of the Child

The child at this stage is guided by his instincts. The teacher should not try to suppress the instinctive development of the child.

It should be sublimated in the best interests of the personality development of the child.

Formation of Good Habits

This is considered to be the most flexible age of the child. Steps should be taken towards the formation of good habits by the child.

Learning by Doing

The child is active by nature. This ought to be satisfied through the use of activity method which advocates learning by doing.

Use of the Play-Way

Play is natural to the child. Let him play and learn. This will make learning easier and acceptable to the child.

Sensory Training

There are five senses of the child. These should be properly developed at this stage. This sensory training was also emphasized by Madam Montessori.

Use of Stories and Pictures

Children at this stage are interested in stories and multi-coloured pictures. They should be made to learn by using story telling method and with the help of multi-coloured pictures and charts.

APPROACH OF EDUCATIONAL PSYCHOLOGY

Recently the approach of educational psychology has changed significantly in the United Kingdom, as has its contribution to teacher education. In part these changes reflect political decisions to alter the pattern of teacher training: based on the belief that theory is not useful, and that "hands-on" training is preferable. However, discipline and teacher education have each been changing of their own accord. Moving away from the emphasis on all-encompassing theories, such as those of Jean Piaget, Sigmund Freud, and B. F. Skinner, the concerns of educational psychologists have shifted to practical issues and problems faced by the learner or the teacher.

Consequently, rather than, for example, impart to teachers in training Skinner's theory of operant conditioning, and then seek ways of their applying it in classrooms, educational psychologists have tended to begin with the practical issues—how to teach reading; how to differentiate a curriculum (a planned course of teaching and learning) across a range of children with differing levels of achievement and needs; and how to manage discipline in classrooms. Theory-driven research increasingly suggested that more elaborated conceptions of development were required. For example, the earlier work on intelligence by Binet, Burt, and Lewis Madison Terman focused on the assessment of general intelligence, while recognizing that intellectual activity included verbal reasoning skills, general knowledge, and non-verbal abilities such as pattern recognition.

More recently the emphasis has shifted to accentuate the differing profiles of abilities, or "multiple intelligences" as proposed by the American psychologist Howard Gardner, who argues that there is good evidence for at least seven, possibly more, intelligences including kinaesthetic and musical as well as the more traditionally valued linguistic and logico-mathematical types of intelligence.

There has also been a shift in emphasis from the student as an individual to the student in a social context, at all levels from specific cognitive (thinking and reasoning) abilities to general behaviour. For example, practical intelligence and its links with "common sense" have been addressed and investigations made into how individuals may have relatively low intelligence as measured by conventional intelligence tests, yet be seen to be highly intelligent in everyday tasks and "real-life" settings. Recognition of the impact of the environment on a child's general development has been informed by research on the effects of poverty, socio-economic status, gender, and cultural diversity, together with the effects of schooling itself. Also, the emphasis has changed from one of regarding differences in their performance on specific tasks as deficits compared with some norm, to an appreciation that deficits in performance may reflect unequal opportunity, or that differences may even reflect a positive diversity. It is now apparent that there are also important biological factors determined by a child's genetic make-up and its prenatal existence, as well as social factors concerned with the family, school, and general social environment. Because these various factors all interact uniquely in the development of an individual, consequently there are limitations in the possible applicability of any one theory in educational psychology.

SCOPE OF EDUCATIONAL PSYCHOLOGY

In this age of science and technology, psychology has been considered as one of the youngest, yet one of the most influential sciences. It has influenced education in many different ways and has give a new turn; a psychological turn to the human mind. For a skilful teacher in this day and age, a great deal of knowledge of educational psychology is highly indispensable.

The subject psychology has two aspects pure and applied. Pure psychology formulates techniques for the study of human behaviour, which finds the practical shape in its applied aspects, *i.e.* branches of applied psychology like clinical psychology, crime psychology, industrial psychology, occupational psychology and educational psychology. Educational Psychology as a Branch of Applied Psychology. As discussed above educational psychology is nothing but one of the branches of applied psychology. It is an attempt to apply knowledge of pure psychology to the field of education. It consists of application of psychological principles and techniques to human behaviour in educational situations. In other words, Educational Psychology is a study of the experience and behaviour of the learner in relation to educational environment.

In order to develop a clear understanding of the term educational psychology it is necessary to understand the meaning of psychology and education separately.

MEANING OF PSYCHOLOGY

Curiosity in man has led to know his surroundings which mainly conclude nature and other fellow men. There is always a desire to 'know' what 'one' is, what is his background, what is it made of, what are the associated factors and in what way can it be useful to one self? Each question will lead to more questions, Psychology forces to answer many of the questions we have about ourselves, other people and the nature of human life; why do we feel lonely? Why do we forget? How people learn? What makes someone creative? Why do we take drugs? What makes some one help others? and so on. Psychology had its formal beginning when Wilhelm Wondt established his psychological laboratory in Leipzig Germany in 1879. But in real sense, interest in psychology as a discipline dates back to the work of Plato, Aristotle and other philosophers.

Psychology – The Science of Soul

The Greek Philosophers conceived psychology as a science of soul, as early as 400 B.C. In fact, the term psychology literally means the science of soul. Etymologically, it is composed of two Greek words "Psyche" and Logos means soul and science respectively. Goeckel named it as psychologia. Soul is a being which dwells on the body and with the end of life it leaves the body. Soul is a metaphysical idea. It can neither be perceived on imagined nor its nature and function can be studied by scientific methods of observation, experiment etc. Therefore, definition of psychology as the science of soul has been discarded by the modern psychologists.

Psychology – The Science of Mind

Some regard psychology as the science of mind. Historically the French philosophers like Descartes (1596-1650) and the Britisher philosophers like Locke considered psychology as the science of mind. Descartes tried to understand body mind relationship in terms of their interaction, the study of nervous system, and interest of references as innate actions, etc. The definition of psychology as the science of mind is not acceptable at present. Mind is an ambiguous a concept as the soul. It is not at all possible to carry on scientific observation and experimentation on mind. This definition also does not include the overt behaviour of man and animal which are also important subject matter of psychology. Therefore, the definition of psychology as the science of mind has been discarded.

Psychology – The Science of Consciousness

Psychology has also been defined as the science of consciousness. Historically such a definition has been propounded by the Leipzing school of psychologists led by Withelm Wundt (1832-1920). Wundt defined psychology as the science of immediate experience with consciousness being the main subject matter. He postulated that conscious experience can be reduced in to elements and the primary aim psychology is the analysis of conscious experience in to its elements. But the definition of psychology as the science of consciousness is not acceptable. That is because mental life does not consist only of consciousness. There are unconscious and subconscious mental process which influence our behaviour in various ways without our knowledge.

Psychology – The Science of Experience

Titchner (1867 – 1927), the leader of the structuralists defines psychology as the science of conscious experience which is dependent upon the experiencing person. To give an example the physicit and the psychologist may be investigating about sound. But whole the former investigates the phenomena as such, the latter is interested as to how it is perceived by the observer. The mind is nothing but the sum total of the conscious experiences as perceived by a person. The subject matter of psychology is the study of such conscious experience which constitutes mind. The method of study of conscious experience is through the introspection of a trained observer.

Psychology – Study of Behaviour

Watson (1878-1958), an American brought about a revolution in psychology called behaviourism. He argued that psychology is to be regarded as a science and as a science it is to limit itself to the study and analysis of publicly observable events such as the behaviour of the subject rather than subjective matters like his private mental states. He defined psychology as “the science of behaviour.”

Meaning of Science and Behaviour

Science has been defined as “Systematic study of knowledge” concerning the relationship between the cause and effect of a particular phenomenon. In order to collect the scientific data and systematised material, science employs various kinds of methods of enquiry such as observation classification formulation of hypothesis, analysis of evidence etc. It also organises and develops our knowledge of the word, we live on, Psychology too aims at same thing. It uses scientific methods to study human behaviour. It also helps us to understand control and predict human behaviour.

What is Social Learning Theory?

The social learning theory proposed by Albert Bandura has become perhaps the most influential theory of learning and development. While rooted in many of the basic concepts of traditional learning theory, Bandura believed that direct reinforcement could not account for all types of learning. His theory added a social element, arguing that people can learn new information and behaviours by watching other people. Known as observational learning (or modelling), this type of learning can be used to explain a wide variety of behaviours.

IMPORTANCE OF EDUCATIONAL PSYCHOLOGY FOR TEACHER

The contribution of educational psychology to the theory and practice of education is rich and varied. The knowledge of educational psychology is important as it provides teachers with some basic skills and guidelines to solve the problems of teaching- learning process. According to John Adams, “Teacher should know John as well as Latin”. It means teacher should know child and subject- matter. A teacher should know the nature, capacities, likings and aptitudes and attitudes of the child. Child is like a book, teacher should know each and every page of it.”

- Skinner’s View, “The teacher needs psychology to bridge the lives of the young and the aims of education in our democratic society.”

- Kuppuswamy's View, "Psychology contributes to the development of the teacher by providing him with a set of concepts and principles."

Teaching is an art. Knowledge of educational psychology is very useful and indispensable for the teacher because it gives knowledge to the teacher.

KNOWLEDGE OF INNATE NATURE

The child has got natural urges instincts, potentialities and propensities. These innate qualities are the "Prime movers" of his behaviour. The teacher who knows psychology can make his teaching very successful while keeping in view innate nature of the child.

Knowledge of Behaviour

Educational psychology assists the teacher in knowing the behaviour of the child at different stages of development. It also helps the teacher in understanding the physiological and psychological basis of behaviour, *i.e.*, nervous system, glands, instincts, emotions, sentiments, motives, play, intelligence, heredity and environment, etc.

Knowledge of Guidance

It helps the teacher in giving guidance to the pupils by having an understanding of interests, abilities, aptitudes, achievements, problems, educational and vocational plans of the pupils.

Knowledge of Unconscious Mind

It helps the teacher in knowing the unconscious mind of the students and plays very important role in the development of the personality of the individual.

Knowledge About Himself

It helps the teacher to know about himself. He learns the psychology of being a teacher and acquaints himself with the traits of a successful teacher.

Understand Development Characteristics

Children pass through different stages of development as infancy, childhood and adolescence. These developmental stages have their own characteristics. If the prospective teacher knows the characteristics emerging at different stages of development, he can utilise these characteristics in imparting instruction and moulding their behaviour according to the specified goals of education.

Understand The Individual Differences

No two individuals are alike. The teacher with the knowledge of the kind of individual differences may adjust his teaching to the needs and requirements of the class and thus may be helpful in creating conducive environment in the schools where the students can develop their inherent potentialities to the maximum. Understand the Nature of Classroom Learning. The knowledge of educational psychology provides a teacher the knowledge of learning process in general and problems of classroom learning in particular. The teacher by the knowledge of educational psychology can understand the principles of learning and various approaches to learning process, problems of learning and their remedial measures and also about factors affecting and guidance for effective learning.

Understand Effective Teaching Methods

Educational psychology gives us the knowledge of appropriate teaching methods. It helps in developing new strategies of teaching. It also provides us with the knowledge of different approaches evolved to tackle the problems of teaching at different age levels.

Understand the Problems of Children

By studying educational psychology a teacher may understand the causes of the Problems of the children which occur at different age levels and can successfully solve them.

Knowledge of Mental Health

By studying educational Psychology teacher can know various factors which are responsible for mental ill health and maladjustment and can successfully help in Central hygiene.

Measurement of Learning Outcome

Psychological tools help the teacher to assess the learning outcome of the students and also to evaluate his teaching methods for required modification.

Curriculum Construction

Psychological principles are also used in formulating curriculum for different stages. Needs of the students, their developmental characteristics, learning patterns and needs of the society all are to be included in curriculum construction.

Research

Educational psychology helps in developing tools and devices for the measurement of various variables which influence the behaviour and performance of students.

Helps to Develop Positive Attitude

The teacher training programme aims to develop positive attitude towards teaching profession and provides the prospective teachers with the necessary competencies to meet the classroom challenges. Training colleges provide the knowledge of organising the subject matter in a sequential order to suit the needs of the class. The trainees are also acquainted with the techniques of motivating children for learning.

Understanding of Group Dynamics

Educational psychology helps the teacher to recognise the importance of social behaviour and group dynamics in classroom teaching learning.

PROBLEM OF DISCIPLINE

With the knowledge of educational psychology teacher utilises the importance of indirect discipline rather than corporal punishment. It tells the teacher that discipline should be self-discipline, dynamic, positive and constructive through participation in purposeful activity. Pleasure and pain, reward and punishment, praise, etc., should be judiciously used. If the teacher is unaware of the principles of educational psychology he may be unable to solve the problems of his students and thereby fail to induce order and discipline among them.

SCHOOL AND CLASS ADMINISTRATION

Former autocratic method of administration in school and class has been charged by democratic way of life wherein the teachers and administrators are more democratic, co-operative and sympathetic and problems of administration solved by mutual discussion.

Use of Audio-Visual Aids

Educational psychology has helped the teachers to make use of various types as audio-visual aids in classroom teaching so as to make the concept more clear, definite and learning to last longer.

Time-Table

The knowledge of psychology is helpful to the teacher in preparing time-table. He should keep in main the relative importance and toughness of different subjects and level and index of fatigue of the students.

Use of Innovations

Activity-centred teaching, discussion method, micro-teaching, etc., are some innovative ideas adopted to improve the teaching learning process.

Co-curricular Activities

Activities like debate, drama, games are given due importance along with theoritical subjects for the harmonious development of the personality of children.

Production of Text Books

Educational psychology has helped in planning of text books according to the intellectual development of children, their needs and interests at different age levels. Undoubtedly the study of educational psychology may be very helpful to equip our prospective teachers with necessary skills to deal with classroom teaching learning problems.

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Personality and Sport

Despite popular opinion, no distinguishable “athletic personality” has been shown to exist. That is, no consistent research findings show that athletes possess a general personality type distinct from the personality of non-athletes. Also, no research has shown consistent personality differences between athletic subgroups (, *e.g.,* team athletes vs. individual sport athletes, contact sport athletes vs. non-contact sport athletes).

Research has identified several differences in personality characteristics between successful and unsuccessful athletes. These differences, however, are not based on innate, deeply ingrained personality traits but rather result from more effective thinking and responding in relation to sport challenges as well as higher levels of motivation. Specifically, successful athletes, compared with less successful athletes, are

- More self-confident,
- Better able to cope well with stress and distractions,
- Better able to control emotions and remain appropriately activated,
- Better at attention focusing and refocusing,
- Better able to view anxiety as beneficial, and
- More highly determined and committed to excellence in their sport.

Olympic and World champion athletes have defined mental toughness as the natural or developed psychological edge that enables you to cope with competitive demands and remain determined, focused, confident, and in control under pressure. These athletes identify the following as critical personality responses that represent mental toughness: loving the pressure of competition, adapting to and coping with distractions and sudden changes, channeling anxiety, not being fazed by mistakes in the process, being acutely aware of any inappropriate thoughts and feelings and changing them immediately to perform optimally when needed, using failure to drive yourself, learning from failure, and knowing how to rationally handle success—an impressive list of qualities that we all would like to have as part of a mentally tough personality.

Although most sport personality research has focused on the influence of personality on sport behaviour, research has also examined the effects of sport participation on personality development and change. A belief commonly held in American society is that sport builds character or that sport participation may develop socially valued personality attributes.

Research shows, however, that competition reduces prosocial behaviours such as helping and sharing and that losing magnifies this effect. Sport participation has been shown to increase rivalrous, antisocial behaviour and aggression and has been linked to lower levels of moral reasoning.

Nevertheless, the sport story has a positive side. Research in a variety of field settings has demonstrated that children's moral development and prosocial behaviours (cooperation, acceptance, sharing) can be enhanced in sport settings when adult leaders structure situations to foster these positive behaviours. Interventions with children were successful in building character when naturally occurring conflicts arose and were discussed with the children to enhance their reasoning and values about sport and life events. The moral of the story is this: Sport doesn't build character, people do.

PERSONALITY AND EXERCISE

As in sport, researchers have found no "exercise personality" or set of personality characteristics that predict exercise adherence. Exercisers cannot be differentiated from non-exercisers based on an overall personality type. Two personality characteristics, however, are strong predictors of exercise behaviour. Individuals who are more confident in their physical abilities tend to exercise more than those who are less physically confident. A second important predictor of exercise behaviour, obviously, is self-motivation, with self-motivated individuals beginning and continuing exercise programmes and less motivated individuals dropping out or never starting at all.

A personality type termed "obligatory exercisers" describes individuals who participate in exercise at excessive and even harmful levels. For these individuals, exercise becomes the central focus of life, and their behaviour becomes pathological in terms of their need to control themselves and their environment. Clinical evidence demonstrates a similar link between anorexia nervosa, a psychopathological eating disorder, and compulsive exercise. Specialists in exercise psychology attempt to help individuals plan and engage in exercise behaviour that is healthy and non-controlling to enhance total well-being.

Echoing the idea that sport builds character, exercise or fitness training has also popularly been associated with positive personality change and mental health.

The personality characteristic that researchers have most frequently examined in this area is self-esteem. Self-esteem is our perception of personal worthiness and the emotions associated with that perception. Think of self-esteem as how much we like ourselves. Research has generally confirmed that fitness training improves self-esteem in children, adolescents, and adults. Research has also shown that exercise positively influences perceptions of physical capabilities, or self-confidence. Interestingly, the research indicates that these changes in self-esteem and self-confidence may result from perceived, as opposed to actual, changes in physical fitness. In addition, many aspects of intellectual performance have been related to physical activity, suggesting that cognitive functions respond positively to increased levels of physical activity.

Many people also associate exercise with changes in mood and anxiety. Most individuals say that they “feel better” or “feel good” after vigorous exercise, which emphasizes the important link between physical activity and psychological well-being. In addition, research documents that anxiety and tension decline following acute physical activity. The greatest reductions in anxiety occur in exercise programmes that continue for more than 15 weeks. Much research has been conducted to determine whether exercise or fitness reduces people’s susceptibility to stress, and the generally accepted conclusion is that aerobically fit individuals demonstrate a reduced psychosocial stress response. A tentative explanation for this finding is that exercise either acts as a coping strategy that reduces the physiological response to stress or serves as an “inoculator” to foster a more effective response to psychosocial stress.

Prolonged physical activity is also associated with decreases in depression and a lessening of depressive symptoms in individuals who are clinically depressed at the outset of the exercise treatment. Explanations for these changes range from the distraction hypothesis, which maintains that exercise distracts attention from stress, to other explanations that focus on the physiological and biochemical changes in the body after exercise.

PERSONALITY WITHIN SPORTS

The mind is one of the most important things to an athlete, as it is one of the greatest strengths they have and it can dramatically affect their performance; this is why personality is one of the biggest influences in sport.

Your personality is your attitude, feelings and how you react to various situations within life; this gives you your unique perception of the world around you, what you do and how you react to this. Your personality is build up of your psychological core, which gives you your basic and simple aspects of personality, so your likes and dislike; these cannot be rewritten, but through life it does grow. You then have your typical response which is partially rewritable and evolves; this is your attitude, so how you generally react based upon your beliefs and opinions concentrated within your psychological core. Finally there is the role related behaviour which is your specified behaviour based on your suited role; this means it is the controlled and restricted personality of the individual based upon the role they are taking. It is these three factors which make up the personality of a person.

There are two main personality types which roughly defines everyone, this affects various aspects of their personality, but this is especially within athletes. The two extremes are introverts and extroverts.

Introverts are commonly the people who rely on themselves and do individual sports, or have a position which reflects this personality type. Introverts are generally independent, more intelligent, timid, practical, self-assured and serious as they are self dependent.

Extroverts are typically more relaxed, trusting, group dependent and less intelligent; this is because they rely on others during their sport, or in their everyday life.

Although these are two extremes of the types of personalities, the majority of people have aspects of both types, but will generally sway one way. An example of this is that introverts can also take part in some team games; however, they will almost always be put in a position that suits this personality, *i.e.,* a position that does not involve as much team work.

As personality is important within sport to aid the motivation and ability of the athlete, it is important to understand not just what a personality is, but also how it is developed.

Personalities are believed to be developed in various ways as there are many theories. However, there are three theories which stand out from the others, and most of these other theories are derived from these three.

One of these theories is the Trait theory. The Trait theory predicts that the personality of one's self is not influenced by their surroundings, but instead by the inherited personality of both parents, within the passing on of genes. The theory states that someone's personality cannot be changed and that people are fixed in their ways; this means that our personalities are determined simply by what your parents have each inherited from their parents (, etc.,).

According to this theory it means that experience will not change our personalities; this means that someone will repeat their behaviour in the same situation. Due to this it means that someone's behaviour/attitude and personality becomes predictable as the role-related behaviour, typical behaviour and psychological core stay the same throughout a person's life. The problem with this theory is that people do in fact adapt to their surroundings and situations, as we make mistakes and learn from them; however, some people do find it harder or near impossible to change or adapt their personality. The theory also suggests that we merely inherit the personalities of our parents; although this still makes us unique it makes our humanity very limited as we are not as adaptable as people think. The theory is also criticised as children are taught to behave and they adapt to that as it is what they are taught, similar to how athletes are trained through discipline, etc.,

Another theory discussing the origin of our personalities is the Social Learning theory; this theory hypothesises that we learn through interacting with the people around us by analysing their personality at a subconscious level then seeing how they react to stimulus's/situations and copying them to develop our own unique personality. According to this theory it means that the psychological core develops and that then affects the typical and role-related behaviour as they are constantly adapting to develop their personality to their situations. By having the personality adaptable, the person is therefore able to adapt to the same situation or something new by using previous experiences to help them make a personally rational decision.

Although this theory seems more accurate, as we can learn from our mistakes, people sometimes do appear to inherit some traits or how their mind analysis and interprets things from their parents, as they have a similar understanding of things. However this could be said that it is because of the upbringing and the duplication of aspects of the parent's personalities through socially learning, but it could also be through inheritance of the genes as genes do play a part.

The final theory of personality development is the Interactional Approach theory; this theory is a concoction of the previous theories. The mixture states that you inherit characteristics and then your personality develops over time through socialisation; this means that people's personality will change and develop depending on how they have lived, for example they will learn from their mistakes. However, they are still influenced by the mixture of their parent's genes, hence, inheriting parts of their personality. Although this theory has the strengths of the other two theories, it also has its weaknesses. Despite being proved that we do inherit our parent's genes and that we do learn from our mistakes/situations, it contradicts the social learning theory that says that we have a blank canvas when we are born; which some people believe is true.

From these theories and from reading various articles on the subject I have developed my own theory. Despite being similar to the Interactional Approach Theory, it is more in depth and its development is better explained.

My theory states that there is a blank canvas for the personality, this canvas is then developed by socially learning. The learning that we take part in is natural as our brain absorbs information detailing and acknowledging our surroundings and the behaviour of others, most of which will come from the people you see the most or have the most influence over you (*, i.e.,* your parents, and then in later years, your role models). However, this is defined by a primitive/animalistic core which causes curiosity and a very minor behavioural personality; this is the blank canvas, as the canvas has to be made of something which will then affect the final portrait.

The basic core gives an infant something that resembles a personality, but a very basic one as they do not know any better; it is simply our brains trying to interpret the data it is taking in while the natural instinct within this core controls its main functions. As the infant grows and the brain develops, it becomes more and more influenced by the people surrounding them, this means they then socially learn which causes the individual to develop their psychological core, and then, as they grow, the rest of the layers of their personality.

As much as socially learning is the main source of developing a personality, the various genes which influence the primitive behavioural instincts do vary; this is because these are passed on genetically as they are the basic and most simple instructions given by the brain, *i.e.,* to eat, drink. excrete and to interact to survive. Although the parents mixed genes will create a unique individual, this is only a temporary core and simple instinct (a demonstration and basis for them to create their own); this is before their minds start to develop their own psychological core.

So this primitive core is not their personality per say, it is just a natural reaction based upon the basic instructions of the brain of how to react to stimulus's, taken from the genes of their parents. However, this primitive core does not completely disappear, after all they are the building blocks and the centre of their entire consciousness. As the brain develops these genes make their brain more susceptible to certain areas when analysing their experiences and developing their own personality, so they are more likely to notice, remember or copy specific behaviour as they have already had that in the example from their parents.

According to this theory, it means that a person's personality is developed due to their socialisation, but that personality is aided and defined further, especially at first, as they start to create the psychological core. Due to the development of this new core it will overlap the primitive core and move them further away from their natural origins/primitive behaviour, as this is then a much stronger influence to their decisions and reactions to stimulus's/situations throughout their life.

Although having these genes do mean that some people may be more susceptible to develop certain traits within their personality, this is a very minor factor and would not cause a great change, as the person's experiences affect them more due to the natural ability for humans to analyse and adapt. These theories, along with everything psychology related, can all be applied to sport. For example the decisions made within sports by their coaches, *i.e.,* positions played, how to communicate with the athlete and how to work efficiently with them. However, this is dependent upon which theory the coach or player most likely favours/believes in.

If a coach was selecting a team from a group of players, if they believed in the trait theory then that would affect their decision as the coach may know the players parents so they would judge their personalities as they would think that the player would inherit those traits or not progress any further than they have already. However, if the coach believed in either the social learning theory or the interactional approach theory, then they would look at the individual players and judge them as they can teach them how to behave appropriately (, etc.,) and see who has the most potential and who he/she can manage. Although the individuals personality does matter, due to the need for respect and honouring the rules, personality also matters when choosing a position or the sport they want to participate in. When the coach chooses a position in a sport, they will place somebody in that position based not only on their performance and skill, but their personality as well, *i.e.,* whether they are an introvert or an extrovert.

As described previously, an introvert is more independent and adapted to relying on themselves. Although introverts are commonly known to do solo sports, they can do team sports as well. In the cases where they do team sports they are more likely to play a position that is reliant upon themselves, or where they do not have to work with others as much. For example, an introvert would be placed in goal, as they are away from the main game and the responsibility is solely on them; this means that it suits their personality and they will feel more comfortable playing in that position.

An introvert is unlike an extrovert as the extrovert would rather rely on others and work with people more; this means that they would be part of the team, so they would be strikers in football, or part of the pack in rugby during the scrum.

PSYCHOANALYTIC THEORY

Psychoanalytic theory refers to the definition and dynamics of personality development which underlie and guide psychoanalytic and psychodynamic psychotherapy. First laid out by Sigmund Freud, psychoanalytic theory has undergone many refinements since his work.

HISTORY

Sigmund Freud, Melanie Klein, and Jacques Lacan are often treated as canonical thinkers by Lacanian psychoanalysts, although there are considerable objections to their authority, particularly from other psychoanalytical schools and feminism. Precisely in the interest of a theoretical approach to psychoanalysis, Lacan read Freud with G. W. F. Hegel's *Phenomenology of Spirit*. Major thinkers within psychoanalytic theory in France include Nicholas Abraham, Serge Leclaire, Julia Kristeva, Jacques Derrida, Jean Laplanche, Rene Major, Luce Irigaray, and Jacques-Alain Miller; their work is anything but unitary — Derrida, for example, has remarked that virtually the entirety of Freud's meta-psychology, while possessing some strategic value previously necessary to the elaboration of psychoanalysis, ought to be discarded at this point, whereas Miller is sometimes taken as heir apparent to Lacan because of his editorship of Lacan's seminars, his interest in analysis is even more philosophical than clinical, whereas Major has questioned the complicity of clinical psychoanalysis with various forms of totalitarian government. Some of the theoretical orientation of psychoanalysis in both German and French and, later, American contexts results in part from its separation from psychiatry and institutionalisation closer to departments of philosophy and literature (or American cultural studies programmes). Psychoanalytic theory heavily influenced the work of Frantz Fanon, Herbert Marcuse, Louis Althusser, and Cathy Caruth, among others. The implications for the work of these thinkers are varied; Fanon's interests were in racial and colonial identity, whereas Marcuse and represent distinct Marxist positions that, among other things, attempt to use psychoanalysis in the study of ideology, whereas Caruth, coming from a background in de Manian de-construction and working in comparative literature, has articulated notions of trauma through literary studies informed by philosophy, psychology, neurology, and Freudian and Lacanian theory. Theory can be so expansive a container as to include the work of Gilles Deleuze and Felix Guattari, who believed psychoanalysis ultimately radically reductionist and strongly opposed the psychiatric institutions of their time.

Psychoanalytic theory sometimes heavily informs gender studies and queer theory. Psychoanalytic theory proposes that every individual personality is the result of childhood conflict.

Id, Ggo, and Super-ego

Id, ego, and super-ego are the three parts of the psychic apparatus defined in Sigmund Freud's structural model of the psyche; they are the three theoretical constructs in terms of whose activity and interaction mental life is described. According to this model, the uncoordinated instinctual trends are the "id"; the organised realistic part of the psyche is the "ego," and the critical and moralising function the "super-ego."

Even though the model is "structural" and makes reference to an "apparatus", the id, ego, and super-ego are functions of the mind rather than parts of the brain and do not necessarily correspond one-to-one with actual somatic structures of the kind dealt with by neuroscience.

The concepts themselves arose at a late stage in the development of Freud's thought: the structural model was first discussed in his 1920 essay "Beyond the Pleasure Principle" and was formalised and elaborated upon three years later in his "The Ego and the Id." Freud's proposal was influenced by the ambiguity of the term "unconscious" and its many conflicting uses.

The terms "id," "ego," and "super-ego" are not Freud's own. They are latinisations by his translator James Strachey. Freud himself wrote of "das Es," "das Ich," and "das Uber-Ich"—respectively, "the It," "the I," and the "Over-I" (or "Upper-I"); thus to the German reader, Freud's original terms are more or less self-explanatory. Freud borrowed the term "das Es" from Georg Groddeck, a German physician to whose unconventional ideas Freud was much attracted (Groddeck's translators render the term in English as "the It").

Id

The id comprises the unorganised part of the personality structure that contains the basic drives. The id acts as according to the "pleasure principle", seeking to avoid pain or unpleasure aroused by increases in instinctual tension.

The id is unconscious by definition. In Freud's formulation, It is the dark, inaccessible part of our personality, what little we know of it we have learnt from our study of the dream-work and of the construction of neurotic symptoms, and most of this is of a negative character and can be described only as a contrast to the ego. We all approach the id with analogies: we call it a chaos, a cauldron full of seething excitations... It is filled with energy reaching it from the instincts, but it has no organisation, produces no collective will, but only a striving to bring about the satisfaction of the instinctual needs subject to the observance of the pleasure principle.

Developmentally, the id is anterior to the ego; *i.e.* the psychic apparatus begins, at birth, as an undifferentiated id, part of which then develops into a structured ego. Thus, the id: contains everything that is inherited, that is present at birth, is laid down in the constitution — above all, therefore, the instincts, which originate from the somatic organisation, and which find a first psychical expression here (in the id) in forms unknown to us".

The mind of a newborn child is regarded as completely "id-ridden", in the sense that it is a mass of instinctive drives and impulses, and needs immediate satisfaction. This view equates a newborn child with an id-ridden individual—often humorously—with this analogy: an alimentary tract with no sense of responsibility at either end. The id is responsible for our basic drives such as food, water, sex, and basic impulses. It is amoral and selfish, ruled by the pleasure–pain principle; it is without a sense of time, completely illogical, primarily sexual, infantile in its emotional development, and is not able to take "no" for an answer.

It is regarded as the reservoir of the libido or "instinctive drive to create". Freud divided the id's drives and instincts into two categories: life and death instincts—the latter not so usually regarded because Freud thought of it later in his lifetime. Life instincts (Eros) are those that are crucial to pleasurable survival, such as eating and copulation. Death instincts, (Thanatos) as stated by Freud, is our unconscious wish to die, as death puts an end to the everyday struggles for happiness and survival. Freud noticed the death instinct in our desire for peace and attempts to escape reality through fiction, media, and drugs. It also indirectly represents itself through aggression.

Ego

The Ego acts according to the reality principle; *i.e.* it seeks to please the id's drive in realistic ways that will benefit in the long term rather than bringing grief. The Ego comprises that organised part of the personality structure that includes defensive, perceptual, intellectual-cognitive, and executive functions. Conscious awareness resides in the ego, although not all of the operations of the ego are conscious. The ego separates what is real. It helps us to organise our thoughts and make sense of them and the world around us.

According to Freud, "The ego is that part of the id which has been modified by the direct influence of the external world... The ego represents what may be called reason and common sense, in contrast to the id, which contains the passions... in its relation to the id it is like a man on horseback, who has to hold in check the superior strength of the horse; with this difference, that the rider tries to do so with his own strength, while the ego uses borrowed forces Freud, *The Ego and the Id* (1923)."

In Freud's theory, the ego mediates among the id, the super-ego and the external world. Its task is to find a balance between primitive drives and reality (the Ego devoid of morality at this level) while satisfying the id and super-ego. Its main concern is with the individual's safety and allows some of the id's desires to be expressed, but only when consequences of these actions are marginal. Ego defence mechanisms are often used by the ego when id behaviour conflicts with reality and either society's morals, norms, and taboos or the individual's expectations as a result of the internalisation of these morals, norms, and their taboos.

The word *ego* is taken directly from Latin, where it is the nominative of the first person singular personal pronoun and is translated as "I myself" to express emphasis. The Latin term *ego* is used in English to translate Freud's German term *Das Ich*, which literally means "the I".

Ego development is known as the development of multiple processes, cognitive function, defences, and interpersonal skills or to early adolescence when ego processes are emerged.

In modern English, ego has many meanings. It could mean one's self-esteem, an inflated sense of self-worth, or in philosophical terms, one's self. However, according to Freud, the ego is the part of the mind that contains the consciousness. Originally, Freud used the word ego to mean a sense of self, but later revised it to mean a set of psychic functions such as judgement, tolerance, reality-testing, control, planning, defence, synthesis of information, intellectual functioning, and memory.

In a diagram of the Structural and Topographical Models of Mind, the ego is depicted to be half in the consciousness, while a quarter is in the preconscious and the other quarter lies in the unconscious.

When the ego is personified, it is like a slave to three harsh masters: the id, the super-ego, and the external world. It has to do its best to suit all three, thus is constantly feeling hemmed by the danger of causing discontent on two other sides. It is said, however, that the ego seems to be more loyal to the id, preferring to gloss over the finer details of reality to minimize conflicts while pretending to have a regard for reality. But the super-ego is constantly watching every one of the ego's moves and punishes it with feelings of guilt, anxiety, and inferiority. To overcome this the ego employs defence mechanisms. The defence mechanisms are not done so directly or consciously. They lessen the tension by covering up our impulses that are threatening. Denial, displacement, intellectualisation, fantasy, compensation, projection, rationalisation, reaction formation, regression, repression, and sublimation were the defence mechanisms Freud identified. However, his daughter Anna Freud clarified and identified the concepts of undoing, suppression, dissociation, idealisation, identification, introjection, inversion, somatisation, splitting, and substitution.

Super-ego

The Super-ego aims for perfection· It comprises that organised part of the personality structure, mainly but not entirely unconscious, that includes the individual's ego ideals, spiritual goals, and the psychic agency (commonly called "conscience") that criticises and prohibits his or her drives, fantasies, feelings, and actions.

"The Super-ego can be thought of as a type of conscience that punishes misbehaviour with feelings of guilt. For example: having extra-marital affairs." The Super-ego works in contradiction to the id. The Super-ego strives to act in a socially appropriate manner, whereas the id just wants instant self-gratification. The Super-ego controls our sense of right and wrong and guilt. It helps us fit into society by getting us to act in socially acceptable ways.

The Super-ego's demands oppose the id's, so the ego has a hard time in reconciling the two. Freud's theory implies that the super-ego is a symbolic internalisation of the father figure and cultural regulations. The super-ego tends to stand in opposition to the desires of the id because of their conflicting objectives, and its aggressiveness towards the ego. The super-ego acts as the conscience, maintaining our sense of morality and proscription from taboos.

The super-ego and the ego are the product of two key factors: the state of helplessness of the child and the Oedipus complex. Its formation takes place during the dissolution of the Oedipus complex and is formed by an identification with and internalisation of the father figure after the little boy cannot successfully hold the mother as a love-object out of fear of castration.

The super-ego retains the character of the father, while the more powerful the Oedipus complex was and the more rapidly it succumbed to repression (under the influence of authority, religious teaching, schooling and reading), the stricter will be the domination of the super-ego over the ego later on — in the form of conscience or perhaps of an unconscious sense of guilt (*The Ego and the Id*, 1923).

In Sigmund Freud's work *Civilization and Its Dis-contents* (1930) he also discusses the concept of a "cultural super-ego". The concept of super-ego and the Oedipus complex is subject to criticism for its perceived sexism. Women, who are considered to be already castrated, do not identify with the father, and therefore form a weak super-ego, leaving them susceptible to immorality and sexual identity complications.

Advantages of the Structural Model

The partition of the psyche defined in the structural model is one that "cuts across" the topographical model's partition of "conscious vs. unconscious". Its value lies in the increased degree of diversification: although the Id is unconscious by definition, the Ego and the Super-ego are both partly conscious and partly unconscious. What is more, with this new model Freud achieved a more systematic classification of mental disorder than had been available previously:-"Transference neuroses correspond to a conflict between the ego and the id; narcissistic neuroses, to a conflict between the ego and the superego; and psychoses, to one between the ego and the external world"-Freud, "Neurosis and Psychosis" (1923).

PSYCHOANALYTICAL THEORIES

Although Freudian theory was a brilliant breakthrough for the field of psychology by focusing on psychoanalysis as the area of concern, Freudian theory itself was engulfed in controversy almost from the date of publication. As all watershed discoveries go, Freud's theories have found themselves active pro and anti schools of thought and spin-offs from Freudian theory continue to rain hard and fast amongst the various psychoanalytical camps of thinkers. Some key names associated with Freud today are those of Carl Jung, Alfred Adler, Karen Horney and Erik Erikson. Freud's ideas about normal masculine and feminine personality were by no means universally accepted. Many of his own students objected to his biological approach to gender identity and gender typing and came up with their own numerous reinterpretations. Alfred Adler (1954), for one, believed humans to be primarily social beings rather than sexual ones. "Masculine dominance is not a natural thing," said Adler, arguing that it was actually a logical outcome of the conflicts of primitive peoples where the males emerged in the role of warriors owing to greater physical endurance. Thus boys and men were the privileged and preferred members of the family with greater social value, as girls soon learnt from observation. The emphasis of Adler's theory has been that a girl is born into a biased society "that robs her of her belief in her own value" thus undermining her self-confidence.

Taking exception to Freud's idea of female development, Karen Horney (1926, 1966) did not see penis envy and castration complex as important parts of female personality development as there is no reason to believe girls see themselves as lacking anything even if boys might perceive little girls as having 'lost' their penises. Horney in fact went a step ahead by postulating that womb envy was as pertinently applicable to men as penis envy could be to girls, speculating that the male obsession with achievement is really an over-compensation for an inferiority complex that stems from being unable to compete in the face of female reproductive potential-i. e. womb envy leads to a femininity complex. Horney's position quite clearly has been that there are no differences in the psychological make-up of men and women that can be accounted for strictly on a biological basis. Amongst the reformulators of Freud's theories, Erik Erikson (1963, 1964) emphasised the concept of 'inner space' believing that a woman's psychological development is governed by the fact that "her somatic design harbours an 'inner space' designed to bear the offspring of chosen men and, with it, a biological, psychological and ethical commitment to take care of human infancy." (1964) Erikson based his ideas of inner space affecting gender identity on his observations of the play behaviour of ten to twelve year-old children. The typical girl's play construction was an interior scene with people and furniture, thus being peaceful and static while the boy's were action-oriented. Erikson saw this as parallel to anatomical detailing of the sexes-the female organs are internal and nurturing as opposed to the external, active organs of males. Lastly, Carl Jung (1971) stressed the integration of the animus and the anima (the masculine and feminine components of personality) in his psychoanalytic conception of androgyny. Today, though Freud's theories have attracted many more students, followers and critics, and though there are many derivatives and adaptations of the psychoanalytic personality theory, few social scientists give much weight to concepts like penis envy-although some psychoanalysts still use them to interpret behaviour.

A FEMINIST PERSPECTIVE

Nancy Chodorow (1978, 1979) in her book The Reproduction of Mothering put forth the thesis that masculine-feminine identities and roles are not biologically determined but are 'reproduced' in every generation by social arrangements. Accepting Freud's starting thesis that early life events and unconscious motivation determine personality, Chodorow proposes that the process of gender typing and the development of gender identity happens as early as within the first two years of life.

According to Chodorow, a newborn infant has no sense of 'self'-it cannot differentiate between itself and its caretaker (usually the mother). Because the infant is absolutely dependent on its caretaker for the gratification of all its needs, it is psychologically merged with the mother. But as the mother leaves and returns, meets or fails to meet the infant's needs, responds to others and asserts her own needs, the baby starts perceiving her as separate and begins to make a 'not me' distinction. This process of differentiation during which infants begin to perceive the boundaries between themselves and their primary love object is therefore gradual and requires cognitive and psychological maturity as it is not an automatic process. Most importantly-the infant's sense of self develops in relation with the primary caretaker.

Because most child rearing is done by women, girl infants experience a caretaker who is like them in a very fundamental way, so their movement from "I and you are one" to "I am like you" is fluid and overlapping, based as it is on a sense of oneness with the mother. Because of this feeling of continuity that girls have with their mothers, they grow up with a sense of connection with the rest of society in general.

Boys, on the other hand, have a much more complex and difficult gender identity lesson to learn-despite the first love object being the mother, they have to make the 'not-female' 'not-mother' distinction for themselves and fathers being less available in general makes this an even more challenging task. Boys therefore have a more precarious gender identity and are therefore also more concerned with knowing what is 'masculine' from 'feminine' and drawing rigid boundaries between the two.

These early differences in the formation of gender identity have important consequences for the personality development of the future and the adult role playing behaviours. Boys who have defined masculinity as the opposite of femininity grow into men who believe in the superiority of whatever qualities they hold as masculine and inevitably devalue women. They repress and deny the 'feminine needs' for closeness and connection with others which in turn reduces their ability to be warm and loving fathers and hinders their experience of closeness within other relationships.

Contrariwise, women tend to define themselves in terms of their relationships with others and feel a need for connectedness which men find difficult to satisfy; so-claims Chodorow-women have babies, satisfying their need for connection with the formation of the mother-infant bond. The cycle is thus endlessly repetitive-and new generations of boys and girls form new generations of males and females, fathers and mothers, and in the consequent differences is social role playing that follow, opportunities for women in the wider world will continue to be curtailed as long as they are responsible for children.

This research study focused on the psychological aspects of the entrepreneurial intention of small-scale women entrepreneurs in the manufacturing, trading and service sectors. The variables studied were life situation antecedence, personality, entrepreneurial motivation and business related variables.

The tools used for the study were life situation antecedence scale, personality questionnaire and entrepreneurial motivation scale.

Three hundred women entrepreneurs from manufacturing, trading and service sectors and two hundred non-entrepreneurs from supervisory and clerical cadres from India participated in this study. Univariate and multivariate analyses were done to process the data. The results reveal that there are significant contributions of life situation antecedence, personality and business related variables in contributing to entrepreneurial intention in small-scale women entrepreneurs. The women entrepreneurs have been found to have lower psychological support, poorer work condition and lesser competence compared to women non-entrepreneurs in life situation antecedence. Certain life situation antecedence variables, personality variables and motivational factors were found to explain differences in the entrepreneurial intention of women entrepreneurs in the manufacturing, trading and service sectors.

Women perform an important role in building the backbone of the nation's economy, especially the small and medium-scale enterprises, as well as the cottage industries

(Epstein, 1983). It has been recognized that women have an important role to play in synthesizing social progress with economic growth of developing countries. With the socio-psycho-cultural and economic changes taking place in India, women are slowly entering the field of entrepreneurship.

It is increasingly recognized that women have a vast entrepreneurial talents that could be harnessed. In India, women entrepreneurship is slowly gaining credibility as an important activity in contributing to national economy, helping to foster economic independence of women by letting them hold the reins of their destiny, thus leading to empowerment. Small and medium sized entrepreneurs play a very important role in social and economic development (UNIFEM, 1995).

Running a business is never easy especially for women in a society like India where definitions are not always consistent with the notions of women having economic roles involving risk-taking, initiating, planning and coordinating market-oriented activities. Words like capability, credibility and confidence are the terms used while evaluating women entrepreneurs and which pose problems for them (Cannon, etal 1988).

A number of research observations have been made by Indian researchers that throw light on the psychological implications of women entrepreneurs in India. Azad (1989) stated 'The education and socialization of girls tend to inhibit entrepreneurship in two ways. Firstly an ideal of femininity is encouraged in girls, the values of which are contrary to those qualities needed for enterpreneurship. Women internalize those values and limit their aspirations accordingly. Secondly, those who resist this socialization, with the exception of a few, are punished for their temerity and independence. Girls are taught not to take initiative, be assertive or be independent. When they show these qualities, which underlie successful employment, they receive fewer rewards than boys or gain no approval'.

Mohuiddin (1987) discussed that 'Women entrepreneurs as an opportunity of productive work for women is not merely a means of higher income, but as a means of self respect, to the development of their personality and to a sense of participation in the common purposes of the society. Women Entrepreneurs in India represent a small group who have broken away from the beaten track and are exploring new vistas of economic participation and achievement satisfaction.

They have long stories of trials and hardships; their task has been full of challenges. They have even encountered public prejudice and criticism. Family opposition and social constraints have to be overcome before establishing themselves as independent entrepreneurs. The most serious barrier to women entrepreneurs undoubtedly continues to be the persistence of the belief held by both men and women, that entrepreneurship is a male domain. The resistance, apathy, shyness, inhibition, conservatism, poor response are all governed and generated by cultural traditions, value systems and social sanctions.

Rathore and Chabra (1991) also describe that the psychosocial barriers of Indian women entrepreneurship are poor self-image of women, discriminating treatment, faulty socialization, role conflict and cultural values. Iyer (1991) points out that the women's own personality traits such as shyness, lack of articulation, inability to communicate and attitude towards money matters inhibit and limit their growth as entrepreneurs. Shalini (1992) stresses that many women entrepreneurs feel that they have to overcome the belief that women are not as serious in business as men are, lack of respect for women and lack of confidence to start a business.

One striking fact that can be noted from all these observations is that the women who take up entrepreneurship, in spite of the reported difficulties and psychological barriers, are the ones who have the courage to transcend the limited boundaries, the culture generally prescribes for them. Women also run a lot of risks in the process of venturing into business. The question now arises "If there is so much of a price to pay in the process and with a number of other career options available to them, why are they taking up entrepreneurship". This query is one of the motivating factors in venturing into this research study.

ANALYSIS OF CULTURE AND PERSONALITY

An individual’s personality is the complex of mental characteristics that makes them unique from other people. It includes all of the patterns of thought and emotions that cause us to do and say things in particular ways. At a basic level, personality is expressed through our temperament or emotional tone. However, personality also colours our values, beliefs, and expectations. There are many potential factors that are involved in shaping a personality. These factors are usually seen as coming from heredity and the environment. Research by psychologists over the last several decades has increasingly pointed to hereditary factors being more important, especially for basic personality traits such as emotional tone. However, the acquisition of values, beliefs, and expectations seem to be due more to socialization and unique experiences, especially during childhood.

Some hereditary factors that contribute to personality development do so as a result of interactions with the particular social environment in which people live. For instance, your genetically inherited physical and mental capabilities have an impact on how others see you and, subsequently, how you see yourself. If you have poor motor skills that prevent you from throwing a ball straight and if you regularly get bad grades in school, you will very likely be labelled by your teachers, friends, and relatives as someone who is inadequate or a failure to some degree.

This can become a self-fulfilling prophesy as you increasingly perceive yourself in this way and become more pessimistic about your capabilities and your future. Likewise, your health and physical appearance are likely to be very important in your personality development. You may be frail or robust. You may have a learning disability. You may be slender in a culture that considers obesity attractive or vice versa. These largely hereditary factors are likely to cause you to feel that you are nice-looking, ugly, or just adequate. Likewise, skin colour, gender, and sexual orientation are likely to have a major impact on how you perceive yourself. Whether you are accepted by others as being normal or abnormal can lead you to think and act in a socially acceptable or marginal and even deviant way.

There are many potential environmental influences that help to shape personality. Child rearing practices are especially critical. In the dominant culture of North America, children are usually raised in ways that encourage them to become self-reliant and independent. Children are often allowed to act somewhat like equals to their parents. For instance, they are included in making decisions about what type of food and entertainment the family will have on a night out. Children are given allowances and small jobs around the house to teach them how to be responsible for themselves. In contrast, children in China are usually encouraged to think and act as a member of their family and to suppress their own wishes when they are in conflict with the needs of the family. Independence and self-reliance are viewed as an indication of family failure and are discouraged. It is not surprising that Chinese children traditionally have not been allowed to act as equals to their parents.

Despite significant differences in child rearing practices around the world, there are some similarities. Boys and girls are socialized differently to some extent in all societies. They receive different messages from their parents and other adults as to what is appropriate for them to do in life. They are encouraged to prepare for their future in jobs fitting their gender. Boys are more often allowed freedom to experiment and to participate in physically risky activities. Girls are encouraged to learn how to do domestic tasks and to participate in child rearing by baby-sitting. If children do not follow these traditional paths, they are often labelled as marginal or even deviant. Girls may be called "tomboys" and boys may be ridiculed for not being sufficiently masculine.

There are always unique situations and interpersonal events that help to shape our personalities. Such things as having alcoholic parents, being seriously injured in a car accident, or being raped can leave mental scars that make us fearful and less trusting. If you are an only child, you don't have to learn how to compromise as much as children who have several siblings. Chance meetings and actions may have a major impact on the rest of our lives and affect our personalities.

For instance, being accepted for admission to a prestigious university or being in the right place at the right time to meet the person who will become your spouse or life partner can significantly alter the course of the rest of your life. Similarly, being drafted into the military during wartime, learning that you were adopted, or personally witnessing a tragic event, such as the destruction of the World Trade Centre towers in New York, can change your basic perspective.

ARE THERE PERSONALITY TYPES?

We often share personality traits with others, especially members of our own family and community. This is probably due largely to being socialized in much the same way. It is normal for us to acquire personality traits as a result of enculturation. Most people adopt the traditions, rules, manners, and biases of their culture. Given this fact, it is not surprising that some researchers have claimed that there are common national personality types, especially in the more culturally homogenous societies. During the 1940's, a number of leading anthropologists and psychologists argued that there are distinct Japanese and German personalities that led these two nations to view other countries as trying to destroy them. The concept of national personality types primarily had its origins in anthropology with the research of Ruth Benedict beginning in the 1920's. She believed that personality was almost entirely learned. She said that normal people acquire a distinct *ethos*, or culturally specific personality pattern, during the process of being enculturated as children. Benedict went on to say that our cultural personality patterns are assumed to be "natural" by us and other personality patterns are viewed as being "unnatural" and deviant. She said that such feelings are characteristic of all people in all cultures because we are ethnocentric. Benedict compared the typical personalities of the 19th century North American Plains Indians with those of the farming Pueblo Indians of the Southwest. She said that the bison hunting Plains Indians had personalities that could be typified as being aggressive, prone to violence, and seeking extreme emotional states. In contrast, she said that the typical Pueblo Indian was just the opposite—peaceful, non-aggressive, and sober in personality.

Benedict's views were especially popular in the 1930's among early feminists such as her student Margaret Mead. This was because if personality is entirely learned, it means that feminine and masculine personality traits are not biologically hard-wired in. In other words, culture rather than genes, makes women nurturing towards children and passive in response to men. Likewise, culture makes men aggressive and domineering. If this is true, these stereotypical behaviours can be altered and even reversed. Mead carried out ethnographic field work among the Polynesian and Melanesian peoples of the South Pacific to find examples of societies in which femininity and masculinity have very different and even opposite characteristics from those found in the Western World. She began her research in Samoa in 1925 where she discovered a relaxed adolescence in which sex is talked about freely by boys and girls rather than hidden or suppressed.

Note: In 1983, J. Derick Freeman argued that Margaret Mead was wrong in her assertion about a relaxed Samoan adolescence in regards to sexuality. He described Samoan society as being comparatively puritanical as a result of Christian missionary influences. Other researchers have countered by saying that Freeman did most of his fieldwork a generation after Mead and that Samoan society may have changed in that time.

Most anthropologists today believe that Benedict and her students went too far in their assertions about the influence of culture on personality formation and in discounting heredity. They also tended to over simplify by defining people who did not share all of the traits of the "national personality type" as being deviants. It is more accurate to see the members of a society as having a range of personality types. What Benedict was describing was actually the modal personality. This is the most common personality type within a society. In reality, there is usually a range of normal personality types within each society.

In the early 1950's, David Riesman proposed that there are three common types of modal personality that occur around the world. He called them tradition oriented, inner-directed, and other directed personalities. The tradition-oriented personality is one that places a strong emphasis on doing things the same way that they have always been done. Individuals with this sort of personality are less likely to try new things and to seek new experiences. Those who have inner-directed personalities are guilt oriented. That is to say, their behaviour is strongly controlled by their conscience. As a result, there is little need for police to make sure that they obey the law. These individuals monitor themselves. If they break the law, they are likely to turn themselves in for punishment. In contrast, people with other-directed personalities have more ambiguous feelings about right and wrong. When they deviate from a societal norm, they usually don't feel guilty. However, if they are caught in the act or exposed publicly, they are likely to feel shame. Advocates of Riesman's concept of three modal personalities suggest that the tradition-oriented personality is most common in small-scale societies and in some sub-cultures of large-scale ones. Inner-directed personalities are said to be more common in some large-scale societies, especially ones that are culturally homogenous.

In contrast, the other-directed personality is likely to be found in culturally diverse large-scale societies in which there is not a uniformity in socialization processes and there is considerable anonymity for city dwellers. While Riesman's analysis of personalities was insightful, critics have pointed out that individuals may have characteristics of all three of his identified modal types.

For instance, most North Americans probably do not feel guilty about exceeding speed limits when they are driving on freeways, however, they would feel very guilty hitting someone with their car and would likely call the police. In other words, for some infractions of the law they are other-directed (or shame-controlled), and for others they are inner-directed (or guilt-controlled). Likewise, many people like to do some things in the same way every day but seek new experiences in other areas of their lives. You may like to wear the same style of clothes and spend your leisure time at the same place with your friends most days. However, you may easily get bored eating the same kinds of food every day and regularly try new restaurants when you go out to eat. In other words, you are tradition-oriented for some things but not others.

THE RIDDLE OF PERSONALITY AND SPORT

Most of us at one point or another have played or participated in a sport, whether it is volleyball, tennis, karate or pole-vaulting. Have you ever sat back and wondered why you chose that particular sport to play besides the simple fact that you love participating in it? Recent studies have shown that the complex of multiple personality traits that composes each individual may be a significant factor in which sport you prefer to play.

Traits can be described as people's characteristic behaviours and conscious motives. The broadest category of personality traits involves extraversion and introversion. People reflecting traits of extraversion tend to be excitable, outgoing, lively, sociable and impulsive.

They love the lime-light, work well in groups, and tend to dislike being alone for long periods of time. People reflecting traits of introversion tend to be reserved, reclusive, thoughtful, calm, and rational. They are more interested in their own mental self, work better alone, and are controlled in social situations, preferring closer, more personal relationship. Although traits of introversion and extroversion are reflective of personality, that doesn't mean that everyone is classified as one or the other, many people have traits associated with both extraversion and introversion.

In a study done by Urska Dobersek and Cart Bartling (2007), athletes from four different sports, three individual sports and one team sport, and non-athletes were given standard personality tests including the Eysenck Personality Questionnaire which measured emotionality and tough-mindedness and the Global 5 survey, which measured extraversion, introversion, emotional stability, orderliness, accommodations and intellect. Each subject's personality traits were viewed in association with the sport they preferred and conclusions were drawn between personality traits and were linked to the type of sport preferred.

The study showed significant differences in individuals who played team sports, like volleyball, and people who played individual sports, like tennis, track and golf. Participants on the volleyball team, a team sport, tended to display more traits associated with introversion such as being reliable and thoughtful. Learning to cooperate with other players and sharing the recognition for a win with other people tend to require being less bold and outgoing, and instead, being calmer, rational, and aware of surroundings. Participants of individual sports, where the pressure is all on you to perform reflected traits of extraversion such as being outgoing, energetic, spontaneous and to some extent egotistical.

These findings were interesting because many researchers and scientists would say the opposite is true. Many would agree that individual sports athletes would show traits of introversion versus team sports participants who would display traits of extraversion.

They argue this view because individual sports require a high level of thinking and being aware of self, which are characteristic traits of introversion, and team sports require sociability, and therefore openness, skills which are characteristic traits of extroversion. In a study using the Eysenck Personality Inventory done by Eagleton and his colleagues (2007), the researchers studied 90 undergraduate team sport participants, individual sport participants and non-participants. They found that team sport participants scored higher on traits associated with extraversion, like liveliness, responsiveness and being outgoing, compared to individual sport participants and non-participants, who displayed traits of introversion, such as being reserved, passive and controlled.

The book "Sport Psychology," by Matt Jarvis (1999), he upholds this view that team sport players are more extroverted than individual sport players who exert telic dominance, a motivational mode where individuals become cautious and serious-minded and form well thought out plans which are characteristics of introverts. A study does by Schurr and her colleagues studied team and individual sport athletes versus non-athletes. It revealed that team players were more anxious, aggressive and excitable, all traits of extraversion, as compared to individual sports athletes, who were more submissive and controlled. The book "Angles on Applied Psychology," by Julia Russell (2003), also agrees with this theory.

Her book mentions one study that compared the personalities of Irish female students who participated on the university hockey club to students who were on the tennis team using the Eysenck personality Questionnaire. Participants on the hockey team reflected more traits of extroversion than did the participants on the tennis team. These studies reveal some of the difficulties in predicting behaviours based on personality traits. There is no one perfect personality mold that fits each individual sport, but it is possible that research on links between personality traits and sport preference in the future could reveal the secret as to why one person chooses soccer over badminton and possibly determine success rates of individuals in certain sports.

PERSONALITY CHARACTERISTICS AND SPORTING BEHAVIOUR

CHARACTERISTICS OF PERSONALITY

Personality is fundamental to the study of psychology. The major systems evolved by psychiatrists and psychologists since, Sigmund Freud to explain human mental and behavioural processes can be considered theories of personality. These theories generally provide ways of describing personal characteristics and behaviour, establish an overall framework for organising a wide range of information, and address such issues as individual differences, personality development from birth through adulthood, and the causes, nature, and treatment of psychological disorders.

Type Theory of Personality

Perhaps the earliest known theory of personality is that of the Greek physician Hippocrates (c. 400 B.C.), who characterised human behaviour in terms of four temperaments, each associated with a different bodily fluid, or "humor." The sanguine, or optimistic, type was associated with blood; the phlegmatic type (slow and lethargic) with phlegm; the melancholic type (sad, depressed) with black bile; and the choleric (angry) type with yellow bile. Individual personality was determined by the amount of each of the four humors.

Hippocrates' system remained influential in Western Europe throughout the medieval and Renaissance periods. Abundant references to the four humors can be found in the plays of Shakespeare, and the terms with which Hippocrates labeled the four personality types are still in common use today. The theory of temperaments is among a variety of systems that deal with human personality by dividing it into types. A widely popularized (but scientifically dubious) modern typology of personality was developed in the 1940s by William Sheldon, an American psychologist. Sheldon classified personality into three categories based on body types: the endomorph (heavy and easy-going), mesomorph (muscular and aggressive), and ectomorph (thin and intellectual or artistic).

Trait Theory of Personality

A major weakness of Sheldon's morphological classification system and other type theories in general is the element of oversimplification inherent in placing individuals into a single category, which ignores the fact that every personality represents a unique combination of qualities. Systems that address personality as a combination of qualities or dimensions are called trait theories. Well-known trait theorist Gordon Allport (1897-1967) extensively investigated the ways in which traits combine to form normal personalities, cataloguing over 18,000 separate traits over a period of 30 years.

He proposed that each person has about seven central traits that dominate his or her behaviour. Allport's attempt to make trait analysis more manageable and useful by simplifying it was expanded by subsequent researchers, who found ways to group traits into clusters through a process known as factor analysis. Raymond B. Cattell reduced Allport's extensive list to 16 fundamental groups of inter-related characteristics, and Hans Eysenck claimed that personality could be described based on three fundamental factors: psychoticism (such antisocial traits as cruelty and rejection of social customs), introversion-extroversion, and emotionality-stability (also called neuroticism). Eysenck also formulated a quadrant based on intersecting emotional-stable and introverted-extroverted axes.

Psychodynamic Theory of Personality

Twentieth-century views on personality have been heavily influenced by the psychodynamic approach of Sigmund Freud. Freud proposed a three-part personality structure consisting of the id (concerned with the gratification of basic instincts), the ego (which mediates between the demands of the id and the constraints of society), and the superego (through which parental and social values are internalized).

In contrast to type or trait theories of personality, the dynamic model proposed by Freud involved an ongoing element of conflict, and it was these conflicts that Freud saw as the primary determinant of personality. His psychoanalytic method was designed to help patients resolve their conflicts by exploring unconscious thoughts, motivations, and conflicts through the use of free association and other techniques. Another distinctive feature of Freudian psychoanalysis is its emphasis on the importance of childhood experiences in personality formation. Other psychodynamic models were later developed by colleagues and followers of Freud, including Carl Jung, Alfred Adler, and Otto Rank (1884-1939), as well as other neo-Freudians such as Erich Fromm, Karen Horney, Harry Stack Sullivan (1892-1949), and Erik Erikson.

Phenomenological Theory of Personality

Another major view of personality developed during the twentieth century is the phenomenological approach, which emphasizes people's self-perceptions and their drive for self-actualization as determinants of personality. This optimistic orientation holds that people are innately inclined towards goodness, love, and creativity and that the primary natural motivation is the drive to fulfill one's potential. Carl Rogers, the figure whose name is most closely associated with phenomenological theories of personality, viewed authentic experience of one's self as the basic component of growth and wellbeing.

This experience together with one's self-concept can become distorted when other people make the positive regard we need dependent on conditions that require the suppression of our true feelings. The client-centered therapy developed by Rogers relies on the therapist's continuous demonstration of empathy and unconditional positive regard to give clients the self-confidence to express and act on their true feelings and beliefs. Another prominent exponent of the phenomenological approach was Abraham Maslow, who placed self-actualization at the top of his hierarchy of human needs. Maslow focused on the need to replace a deficiency orientation, which consists of focusing on what one does not have, with a growth orientation based on satisfaction with one's identity and capabilities.

Behavioural Theory of Personality

The behaviourist approach views personality as a pattern of learned behaviours acquired through either classical (Pavlovian) or operant (Skinnerian) conditioning and shaped by reinforcement in the form of rewards or punishment. A relatively recent extension of behaviourism, the cognitive-behavioural approach emphasizes the role cognition plays in the learning process. Cognitive and social learning theorists focus not only on the outward behaviours people demonstrate but also on their expectations and their thoughts about others, themselves, and their own behaviour. For example, one variable in the general theory of personality developed by social learning theorist Julian B. Rotter is internal-external orientation. “Internals” think of themselves as controlling events, while “externals” view events as largely outside their control. Like phenomenological theorists, those who take a social learning approach also emphasize people’s perceptions of themselves and their abilities (a concept called “self-efficacy” by Albert Bandura).

Another characteristic that sets the cognitive-behavioural approach apart from traditional forms of behaviourism is its focus on learning that takes place in social situations through observation and reinforcement, which contrasts with the dependence of classical and operant conditioning models on laboratory research.

Aside from theories about personality structure and dynamics, a major area of investigation in the study of personality is how it develops in the course of a person’s lifetime. The Freudian approach includes an extensive description of psychosexual development from birth up to adulthood. Erik Erikson outlined eight stages of development spanning the entire human lifetime, from birth to death. In contrast, various other approaches, such as those of Jung, Adler, and Rogers, have rejected the notion of separate developmental stages. An area of increasing interest is the study of how personality varies across cultures.

In order to know whether observations about personality structure and formation reflect universal truths or merely cultural influences, it is necessary to study and compare personality characteristics in different societies. For example, significant differences have been found between personality development in the individualistic cultures of the West and in collectivist societies such as Japan, where children are taught from a young age that fitting in with the group takes precedence over the recognition of individual achievement. Cross-cultural differences may also be observed within a given society by studying the contrasts between its dominant culture and its subcultures (usually ethnic, racial, or religious groups).

DEPENDENT PERSONALITY DISORDER

Personality traits are enduring patterns of perceiving, relating to and thinking about one’s environment and oneself that are exhibited in a wide range of social and personal contexts. Only when personality traits are inflexible, maladaptive and cause significant functional impairment or subjective distress are they considered personality disorders. The essential feature of a personality disorder is a continuing pattern of inner experience and behaviour that deviates noticeably from the expectations of the individual’s culture and is manifested in at least two of the following areas: cognition/ thinking, affectivity/emotional expression, interpersonal functioning or impulse control. This persistent pattern is inflexible and pervasive across a broad range of personal and social situations, and leads to clinically significant distress or impairment in social, occupational or other important areas of functioning. The pattern is stable and of long duration, which means its onset can be traced back to at least adolescence or early adulthood. This pattern is not better accounted for as a manifestation or consequence of another mental disorder and is not due to the direct physiological effects of a substance (such as drug abuse, medication, exposure to a toxin) or a general medical condition (such as head trauma).

Dependent personality disorder is described as a pervasive and excessive need to be taken care of that leads to a submissive and clinging behaviour as well as fears of separation. This pattern begins by early adulthood and is present in a variety of contexts. The dependent and submissive behaviours are designed to elicit caregiving and arise from a self-perception of being unable to function adequately without the help of others. Individuals with dependent personality disorder have great difficulty making everyday decisions (such as what shirt to wear or whether to carry an umbrella) without an excessive amount of advice and reassurance from others.

These individuals tend to be passive and allow other people (often a single other person) to take the initiative and assume responsibility for most major areas of their lives. Adults with this disorder typically depend on a parent or spouse to decide where they should live, what kind of job they should have and which neighbours to befriend. Adolescents with this disorder may allow their parent(s) to decide what they should wear, with whom they should associate, how they should spend their free time and what school or college they should attend.

This need for others to assume responsibility goes beyond age-appropriate and situation-appropriate requests for assistance from others (such as the specific needs of children, elderly persons and handicapped persons). Because they fear losing support or approval, individuals with dependent personality disorder often have difficulty expressing disagreement with other people, especially those on whom they are dependent. These individuals feel so unable to function alone that they will agree with things that they feel are wrong rather than risk losing the help of those to whom they look for guidance. Individuals with this disorder have difficulty initiating projects or doing things independently.

They may go to extreme lengths to obtain nurturance and support from others, even to the point of volunteering for unpleasant tasks if such behaviour will bring the care that they need. Individuals with this disorder feel uncomfortable or helpless when alone, because of their exaggerated fears of being unable to care for themselves. When a close relationship ends (such as a breakup with a lover or the death of a caregiver), individuals with Dependent Personality disorder may urgently seek another relationship to provide the care and support they need. They are often preoccupied with fears of being left to care for themselves.

SOCIAL LEARNING THEORY

This theory suggests that our personality is not a stable characteristic. And it can constantly change due to the people we are around and socialize with. Our personality also changes due to the changes id social situations. It also makes the point that we are highly unlikely to behave in the same way when we are in a sporting situation and in a non-sporting situation. It also suggests that we learn in sporting situations through two different ways which are modelling and reinforcement. Modelling means that and individual is likely to model themselves on people they feel they can relate to, such as individuals in the same sport or gender. It basically says that we look up to someone and copy there actions. Reinforcement-this is important because if an individuals behaviour is reinforced or rewarded it is likely that the behaviour will be repeated. You have to have high attention to retain the skill. Motor responses and motivation skills have to be high. The differences between the trait theory and this theory is that the trait theory suggest that your personality is stable and nothing can change were as this theory suggest that your personality is not stable and can change due to many things.

The Interactional Approach and Martens Schematic View

This theory is the only theory which is widely accepted by most sport psychologists. It tells us that if we are going to accurately predict behaviour in a sporting setting its important that you consider the situation the individual is in and the individual’s characteristics. This theory is basically a mix between the social learning theory and marten’s schematic view. Marten’s schematic view is almost the same the only differences is that it says there are three different distinctive levels that relate to each other which are the physiological core, typical responses and role related behaviour.

The physiological core is often referred to the real you/what you believe in what your interests are and your attitude towards work and play.

Typical responses are seen as the usual way you respond to any given situation and are also a good indicator of your psychological core. Role related behaviour determined by circumstances you are in which are ever changing especially within a sporting environment for example in football you wont be the same all the way through the game you may get frustrated at some points. Role related behaviour is seen as the changeable aspect of personality. Martens schematic and the interactional approach are very similar in the way that they both agree that your personality can be changed due to the situations you face in sporting and none sporting environments.

AN ANALYSIS INTO THE RELATIONSHIP BETWEEN PERSONALITY AND BEHAVIOUR IN SPORT

Personality is defined as “The sum of the characteristics that make a person Unique” There are many ways to define a personality; personality research has led to numerous theories that help us to explain how and why certain traits develop. Not all of our behaviour can be put down to conscious control sometimes it is our unconscious that controls our actions. To assess and define the personality of an athlete we can use certain theories to look in to the way athletes behave in a sporting environment and social environments, we can relate to the personality in 3 ways; role related behaviour (social situation), Core (stable aspects of the personality) and the outer layer (typical responses). Believed that the personality is made up of three phases; the psychological core, typical responses and role related behaviour. This can be illustrated by three concentric circles that represent the structure of the personality. The psychological core (inner core) being the hardest to reach, typical responses represent the usual manner in which a response to social and environmental situations and role related behaviour is the surface of the personality that responds to our environmental perception at any given time. These three layers can be measured with the following theories Trait, Situational and interactional approaches. Weinberg and Gould (1995) Measuring the personality with these three approaches, each having its own attributes can help us to define a personality to a certain extent.

Trait approach: Assumes that fundamental units of our personality are relatively stable, that is that our personality traits are enduring and consistent across a variety of situations, being one of the earliest theories to be used, the trait approach bases its self on a position that a person is predisposed to act a certain way. The traits as relative, the two most significant traits ranging on a continuum from introversion to extroversion and from stability to emotionality. Weinberg and Gould (1995) Behaviour generally resides within the person and that the role of the situation or environmental factors is minimal. Traits are considered to predispose a person to act a certain way, regardless of the situation or circumstances. For example if a football player is competitive, then he will be predisposed to play hard to win. A predisposition does not mean that the football player will always play hard to win; it simply means he might or is likely to.

Situation approach: This approach is largely determined by the situation or environment, it draws itself from social learning theory, Albert Bandura formulated a social learning theory, through which he conceptualized self-regulatory mechanisms explains behaviour in terms of observational learning and social reinforcement, using four categories,

a. Attention
b. Retention
c. Motor production
d. Motivational responses.

But what it does not account for is that some situations can influence some people's behaviour but others it may not. You can influence behaviour in sport and physical education by changing reinforcements in the environment, still the situation approach, like the trait approach cannot truly predict behaviour. Situations can influence some people's behaviour but other people will not be swayed by the same situation.

Interactional approach: Weinberg and Gould (1995) Considers the situation and person as co determinants of behaviour that is as variables that together determine behaviour. Interactional approach looks at the individual's psychological traits and the particular situation to help understand behaviour. found that the interactions between persons and situations could explain twice as many behaviours as traits or situations alone. Interactional approach is the only approach that takes time to study both traits and situations over a period of time. You can only truly assess a person over times as people react individually in sports and other situations.

This makes the interactional approach more favoured as you can look at an athlete in various sporting and social environments. If we look at the personality traits of Sir Bobby Robson we could say he was an extrovert–stable using The Eysenck Personality Inventory (EPI), if we look at some of his traits and what was his social position and environment, we can pick certain traits out to back up the extrovert–stable personality. Robson (1990).

For instance he was a kind man, well mannered and thought full to others. Yet in his professional life he was demanding, out spoken, successful and a leader. Most of what the public saw of Sir Bobby Robson was his outer layer (role related behaviour) as a manager of England and various other teams including Newcastle United his outer layer was foremost, projecting a Psychological image to those around him of an imposing father figure with authority. The extrovert side of his personality enabled him to rise to the top of his profession as a player and a manger, using such traits as ambitious, animated, charismatic, energetic and hardy.

We have looked at what a personality is and defined it; we have looked at the different ways in which we can assess a personality using three approaches:

a. Trait
b. Situational
c. Interactional.

We know the personality has three specific layers being the Inner Core, Typical Responses and the Outer Layer. These layers can be measured by one of the three approaches, but to truly assess a personality takes time. Using traits of Sir Bobby Robson we devised that his personality was extrovert–stable, using the Eysenck personality inventory in theory. But does it truly allow us to see his inner core. Using the interactional approach would allow us to look at it over time in various social environments and sporting environments allowing us to look a different responses to different situation that occur. So to truly know a person's personality takes time and endurance. There for the interactional approach would be more advantageous to truly get to know a person or an athlete.

COACH'S CODE OF BEHAVIOUR

The coach's code of behaviour is a positive document for all coaches. It affirms a coach's support for the concepts of responsibility, trust, competence, respect, safety, honesty, professionalism, equity and sportsmanship. The code also provides a reference point for clubs, parents, athletes, schools and employers to expect that a coach will demonstrate appropriate standards of behaviour.

Coach Code of Behaviour

Safety and Health of Participants:
- Place the safety and welfare of the participants above all else.
- Be aware of and support the sport's injury management plans and return to play guidelines.

Coaching excellence:
- Help each person (athlete, official, etc) to reach their potential. Respect the talent, developmental stage and goals of each person and encourage them with positive and constructive feedback.
- Encourage and support opportunities for people to learn appropriate behaviours and skills.
- Support opportunities for participation in all aspects of the sport.
- Treat each participant as an individual.
- Obtain appropriate qualifications and keep up-to-date with the latest coaching practices and the principles of growth and development of participants.

Honour the sport:
- Act within the rules and spirit of your sport.
- Promote fair play over winning at any cost.
- Respect the decisions of officials, coaches and administrators.
- Show respect and courtesy to all involved with the sport.
- Display responsible behaviour in relation to alcohol and other drugs.

Integrity:
- Act with integrity and objectivity, and accept responsibility for your decisions and actions.
- Ensure your decisions and actions contribute to a harassment-free environment.
- Wherever practical, avoid unaccompanied and unobserved one-on-one activity (when in a supervisory capacity or where a power imbalance exists) with people under the age of 18.
- Ensure that any physical contact with another person is appropriate to the situation and necessary for the person's skill development.
- Be honest and do not allow your qualifications or coaching experience to be misrepresented.
- Never advocate or condone the use of illicit drugs or other banned performance enhancing substances or methods.
- Never participate in or advocate practices that involve match fixing.

Respect:

- Respect the rights and worth of every person, regardless of their age, race, gender, ability, cultural background, sexuality or religion.
- Do not tolerate abusive, bullying or threatening behaviour.

Code of Behaviour Agreement form

All coaches wishing to become registered with the NCAS are required to sign an individualised coach's code of behaviour agreement form.

This form requires coaches to:

- Agree to abide by the code of behaviour of the relevant national sporting organisation and/or training provider. The Australian Sports Commission has developed a coach's code of behaviour. National sporting organisations and training providers must use this code unless they develop their own.
- Acknowledge that the national sporting organisation and/or training provider may take disciplinary action against them, if they breach the code of behaviour (national sporting organisation and training providers are required to implement a complaints handling procedure in accordance with the principles of natural justice, in the event of an allegation).
- Acknowledge that disciplinary action against them may include de-accreditation from the NCAS.

7

Motivation in Sports

Motivation is an internal energy force that determines all aspects of our behaviour; it also impacts on how we think, feel and interact with others. In sport, high motivation is widely accepted as an essential prerequisite in getting athletes to fulfil their potential. However, given its inherently abstract nature, it is a force that is often difficult to exploit fully.

Some coaches, like Portugal manager Luiz Felipe ‘Big Phil’ Scolari, appear to have a ‘magic touch’, being able to get a great deal more out of a team than the sum of its individual parts; others find motivation to be an elusive concept they are forever struggling to master.

What is it that makes individuals like the 45-year-old sprinter Merlene Ottey, who competed in her seventh Olympics in Athens 2004, churn out outstanding performances year in, year out? Elite athletes such as Ottey have developed an ability to channel their energies extremely effectively. Indeed, motivation is essentially about the direction of effort over a prolonged period of time.

There are numerous approaches to the study of motivation. Some are based on schedules of positive and negative reinforcement (eg BF Skinner’s and Ivan Pavlov’s behaviourism) while others focus on an individual’s sense of mastery over a set of circumstances (eg Albert Bandura’s self-efficacy theory). This article explores the constituents of motivation using a contemporary approach, popularised by Americans Edward Deci and Richard Ryan, known as self-determination theory, which emphasises the role of individual choice.

The key findings from recent literature and provide four evidence-based techniques relating to the enhancement of motivation. You will be able to tailor the motivational techniques to enhance your participation in sport or the performance of others. You will learn that motivation is a dynamic and multifaceted phenomenon that can be manipulated, to some degree at least, in the pursuit of superior sporting performance.

MOTIVATION

Motivation is the activation or energization of goal-oriented behaviour. Motivation is said to be intrinsic or extrinsic. The term is generally used for humans but, theoretically, it can also be used to describe the causes for animal behaviour as well. This stage refers to human motivation. According to various theories, motivation may be rooted in the basic need to minimize physical pain and maximize pleasure, or it may include specific needs such as eating and resting, or a desired object, hobby, goal, state of being, ideal, or it may be attributed to less-apparent reasons such as altruism, selfishness, morality, or avoiding mortality. Conceptually, motivation should not be confused with either volition or optimism.

MOTIVATION CONCEPTS

Intrinsic and Extrinsic Motivation

Intrinsic Motivation

Intrinsic motivation comes from rewards inherent to a task or activity itself-the enjoyment of a puzzle or the love of playing. This form of motivation has been studied by social and educational psychologists since, the early 1970s. Research has found that it is usually associated with high educational achievement and enjoyment by students. Intrinsic motivation has been explained by Fritz Heider's attribution theory, Bandura's work on self-efficacy, and Ryan and Deci's cognitive evaluation theory.

Students are likely to be intrinsically motivated if they:

- Attribute their educational results to internal factors that they can control,
- Believe they can be effective agents in reaching desired goals,
- Are interested in mastering a topic, rather than just rote-learning to achieve good grades.

Extrinsic Motivation

Extrinsic motivation comes from outside of the performer. Money is the most obvious example, but coercion and threat of punishment are also common extrinsic motivations. While competing, the crowd may cheer on the performer, which may motivate him or her to do well. Trophies are also extrinsic incentives. Competition is in general extrinsic because it encourages the performer to win and beat others, not to enjoy the intrinsic rewards of the activity. Social psychological research has indicated that extrinsic rewards can lead to overjustification and a subsequent reduction in intrinsic motivation. In one study demonstrating this effect, children who expected to be rewarded with a ribbon and a gold star for drawing pictures spent less time playing with the drawing materials in subsequent observations than children who were assigned to an unexpected reward condition and to children who received no extrinsic reward

Self-control

The self-control of motivation is increasingly understood as a subset of emotional intelligence; a person may be highly intelligent according to a more conservative definition, yet unmotivated to dedicate this intelligence to certain tasks. Yale School of Management professor Victor Vroom's "expectancy theory" provides an account of when people will decide whether to exert self control to pursue a particular goal. Drives and desires can be described as *a deficiency or need that activates behaviour that is aimed at a goal or an incentive.*

These are thought to originate within the individual and may not require external stimuli to encourage the behaviour. Basic drives could be sparked by deficiencies such as hunger, which motivates a person to seek food; whereas more subtle drives might be the desire for praise and approval, which motivates a person to behave in a manner pleasing to others. By contrast, the role of extrinsic rewards and stimuli can be seen in the example of training animals by giving them treats when they perform a trick correctly. The treat motivates the animals to perform the trick consistently, even later when the treat is removed from the process.

MOTIVATIONAL THEORIES

The Incentive Theory of Motivation

A reward, tangible or intangible, is presented after the occurrence of an action with the intent to cause the behaviour to occur again. This is done by associating positive meaning to the behaviour. Studies show that if the person receives the reward immediately, the effect would be greater, and decreases as duration lengthens. Repetitive action-reward combination can cause the action to become habit. Motivation comes from two sources: oneself, and other people. These two sources are called intrinsic motivation and extrinsic motivation, respectively. Applying proper motivational techniques can be much harder than it seems. Steven Kerr notes that when creating a reward system, it can be easy to reward A, while hoping for B, and in the process, reap harmful effects that can jeopardize your goals. A reinforcer is different from reward, in that reinforcement is intended to create a measured increase in the rate of a desirable behaviour following the addition of something to the environment.

Drive-reduction Theories

There are a number of drive theories. The Drive Reduction Theory grows out of the concept that we have certain biological drives, such as hunger. As time passes the strength of the drive increases if it is not satisfied. Upon satisfying a drive the drive's strength is reduced. The theory is based on diverse ideas from the theories of Freud to the ideas of feedback control systems, such as a thermostat. Drive theory has some intuitive or folk validity. For instance when preparing food, the drive model appears to be compatible with sensations of rising hunger as the food is prepared, and, after the food has been consumed, a decrease in subjective hunger. There are several problems, however, that leave the validity of drive reduction open for debate. The first problem is that it does not explain how secondary reinforcers reduce drive. For example, money satisfies no biological or psychological needs, but a pay check appears to reduce drive through second-order conditioning. Secondly, a drive, such as hunger, is viewed as having a "desire" to eat, making the drive a homuncular being-a feature criticized as simply moving the fundamental problem behind this "small man" and his desires. In addition, it is clear that drive reduction theory cannot be a complete theory of behaviour, or a hungry human could not prepare a meal without eating the food before he finished cooking it. The ability of drive theory to cope with all kinds of behaviour, from not satisfying a drive, or adding additional drives for "tasty" food, which combine with drives for "food" in order to explain cooking render it hard to test.

Cognitive Dissonance Theory

Suggested by Leon Festinger, this occurs when an individual experiences some degree of discomfort resulting from an incompatibility between two cognitions. For example, a consumer may seek to reassure himself regarding a purchase, feeling, in retrospect, that another decision may have been preferable. Another example of cognitive dissonance is when a belief and a behaviour are in conflict. A person may wish to be healthy, believes smoking is bad for one's health, and yet continues to smoke.

Need Theories

Need Hierarchy Theory

Abraham Maslow's theory is one of the most widely discussed theories of motivation.

The theory can be summarized as follows:

- Human beings have wants and desires which influence their behaviour. Only unsatisfied needs influence behaviour, satisfied needs do not.
- Since, needs are many, they are arranged in order of importance, from the basic to the complex.
- The person advances to the next level of needs only after the lower level need is at least minimally satisfied.

- The further the progress up the hierarchy, the more individuality, humanness and psychological health a person will show.

The needs, listed from basic (lowest-earliest) to most complex (highest-latest) are as follows:

- Physiology (hunger, thirst, sleep, etc.,)
- Safety/Security/Shelter/Health
- Belongingness/Love/Friendship
- Self-esteem/Recognition/Achievement
- Self actualization

Herzberg's Two-factor Theory

Frederick Herzberg's two-factor theory, AKA intrinsic/extrinsic motivation, concludes that certain factors in the workplace result in job satisfaction, but if absent, they dont lead to dissatisfaction but no satisfaction. The factors that motivate people can change over their lifetime, but "respect for me as a person" is one of the top motivating factors at any stage of life.

He distinguished between:

- Motivators; which give positive satisfaction, *and*
- Hygiene factors; that do not motivate if present, but, if absent, result in demotivation.

The name Hygiene factors is used because, like hygiene, the presence will not make you healthier, but absence can cause health deterioration. The theory is sometimes called the "Motivator-Hygiene Theory" and/or "The Dual Structure Theory." Herzberg's theory has found application in such occupational fields as information systems and in studies of user satisfaction.

Alderfer's ERG Theory

Clayton Alderfer, expanding on Maslow's hierarchy of needs, created the ERG theory. Physiological and safety, The lower order needs, are placed in the existence category, while love and self esteem needs are placed in the relatedness category. The growth category contains our self-actualization and self-esteem needs.

Self-determination Theory

Self-determination theory, developed by Edward Deci and Richard Ryan, focuses on the importance of intrinsic motivation in driving human behaviour. Like Maslow's hierarchical theory and others that built on it, SDT posits a natural tendency towards growth and development. Unlike these other theories, however, SDT does not include any sort of "autopilot" for achievement, but instead requires active encouragement from the environment. The primary factors that encourage motivation and development are autonomy, competence feedback, and relatedness.

Broad Theories

The latest approach in Achievement Motivation is an integrative perspective as lined out in the "Onion-Ring-Model of Achievement Motivation" by Heinz Schuler, George C. Thornton III, Andreas Frintrup and Rose Mueller-Hanson. It is based on the premise that performance motivation results from the way broad components of personality are directed towards performance. As a result, it includes a range of dimensions that are relevant to success at work but which are not conventionally regarded as being part of performance motivation. Especially it integrates formerly separated approaches as Need for Achievement with, *e.g.,* social motives like Dominance. The Achievement Motivation Inventory (AMI) is based on this theory and assesses three factors relevant to vocational and professional success.

Cognitive Theories

Goal-setting Theory

Goal-setting theory is based on the notion that individuals sometimes have a drive to reach a clearly defined end state. Often, this end state is a reward in itself. A goal's efficiency is affected by three features: proximity, difficulty and specificity. An ideal goal should present a situation where the time between the initiation of behaviour and the end state is close. This explains why some children are more motivated to learn how to ride a bike than mastering algebra. A goal should be moderate, not too hard or too easy to complete. In both cases, most people are not optimally motivated, as many want a challenge (which assumes some kind of insecurity of success). At the same time people want to feel that there is a substantial probability that they will succeed. Specificity concerns the description of the goal in their class. The goal should be objectively defined and intelligible for the individual. A classic example of a poorly specified goal is to get the highest possible grade. Most children have no idea how much effort they need to reach that goal.

Models of Behaviour Change

Social-cognitive models of behaviour change include the constructs of motivation and volition. Motivation is seen as a process that leads to the forming of behavioral intentions. Volition is seen as a process that leads from intention to actual behaviour. In other words, motivation and volition refer to goal setting and goal pursuit, respectively. Both processes require self-regulatory efforts.

Several self-regulatory constructs are needed to operate in orchestration to attain goals. An example of such a motivational and volitional construct is perceived self-efficacy. Self-efficacy is supposed to facilitate the forming of behavioral intentions, the development of action plans, and the initiation of action. It can support the translation of intentions into action.

Unconscious Motivation

Some psychologists believe that a significant portion of human behaviour is energized and directed by unconscious motives. According to Maslow, "Psychoanalysis has often demonstrated that the relationship between a conscious desire and the ultimate unconscious aim that underlies it need not be at all direct. "In other words, stated motives do not always match those inferred by skilled observers. For example, it is possible that a person can be accident-prone because he has an unconscious desire to hurt himself and not because he is careless or ignorant of the safety rules. Similarly, some overweight people are not hungry at all for food but for fighting and kissing. Eating is merely a defensive reaction to lack of attention. Some workers damage more equipment than others do because they harbour unconscious feelings of aggression towards authority figures. Psychotherapists point out that some behaviour is so automatic that the reasons for it are not available in the individual's conscious mind. Compulsive cigarette smoking is an example.

Sometimes maintaining self-esteem is so important and the motive for an activity is so threatening that it is simply not recognized and, in fact, may be disguised or repressed. Rationalization, or "explaining away", is one such disguise, or Defence mechanism, as it is called. Another is projecting or attributing one's own faults to others. "I feel I am to blame", becomes "It is her fault; she is selfish". Repression of powerful but socially unacceptable motives may result in outward behaviour that is the opposite of the repressed tendencies. An example of this would be the employee who hates his boss but overworks himself on the job to show that he holds him in high regard.

Unconscious motives add to the hazards of interpreting human behaviour and, to the extent that they are present, complicate the life of the administrator. On the other hand, knowledge that unconscious motives exist can lead to a more careful assessment of behavioral problems. Although few contemporary psychologists deny the existence of unconscious factors, many do believe that these are activated only in times of anxiety and stress, and that in the ordinary course of events, human behaviour—from the subject's point of view—is rationally purposeful.

Intrinsic Motivation and the 16 Basic Desires Theory

Starting from studies involving more than 6,000 people, Professor Steven Reiss has proposed a theory that find 16 basic desires that guide nearly all human behaviour.

The desires are:

- Acceptance, the need for approval
- Curiosity, the need to think
- Eating, the need for food
- Family, the need to raise children
- Honour, the need to be loyal to the traditional values of one's clan/ethnic group
- Idealism, the need for social justice

- Independence, the need for individuality
- Order, the need for organized, stable, predictable environments
- Physical Activity, the need for exercise
- Power, the need for influence of will
- Romance, the need for sex
- Saving, the need to collect
- Social Contact, the need for friends (peer relationships)
- Status, the need for social standing/importance
- Tranquility, the need to be safe
- Vengeance, the need to strike back

In this model, people differ in these basic desires. These basic desires represent intrinsic desires that directly motivate a person's behaviour, and not aimed at indirectly satisfying other desires. People may also be motivated by non-basic desires, but in this case this does not relate to deep motivation, or only as a means to achieve other basic desires.

TYPES OF MOTIVATION

- *Achievement Motivation*: It is the drive to pursue and attain goals. An individual with achievement motivation wishes to achieve objectives and advance up on the ladder of success. Here, accomplishment is important for its own shake and not for the rewards that accompany it. It is similar to 'Kaizen' approach of Japanese Management.
- *Affiliation Motivation*: It is a drive to relate to people on a social basis. Persons with affiliation motivation perform work better when they are complimented for their favourable attitudes and co-operation.
- *Competence Motivation*: It is the drive to be good at something, allowing the individual to perform high quality work. Competence motivated people seek job mastery, take pride in developing and using their problem-solving skills and strive to be creative when confronted with obstacles. They learn from their experience.
- *Power Motivation*: It is the drive to influence people and change situations. Power motivated people wish to create an impact on their organization and are willing to take risks to do so.
- *Attitude Motivation*: Attitude motivation is how people think and feel. It is their self confidence, their belief in themselves, their attitude to life. It is how they feel about the future and how they react to the past.
- *Incentive Motivation*: It is where a person or a team reaps a reward from an activity. It is "You do this and you get that", attitude. It is the types of awards and prizes that drive people to work a little harder.
- *Fear Motivation*: Fear motivation coercions a person to act against will. It is instantaneous and gets the job done quickly. It is helpful in the short run.

REQUISITES TO MOTIVATE

- We have to be Motivated to Motivate
- Motivation requires a goal
- Motivation once established, does not last if not repeated
- Motivation requires Recognition
- Participation has motivating effect
- Seeing ourselves progressing Motivates us
- Challenge only motivates if you can win
- Everybody has a motivational fuse, *i.e.,* everybody can be motivated
- Group belonging motivates

Motivating Different People in Different Ways

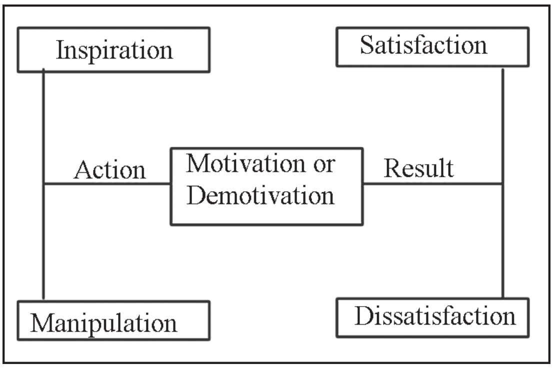
Motivation is not only in a single direction, *i.e.,* downwards. In the present scenario, where the workforce is more informed, more aware, more educated and more goal oriented, the role of motivation has left the boundries of the hierarchy of management.

Apart from superior motivating a subordinate, encouragement and support to colleague as well as helpful suggestions on the right time, even to the superior, brings about a rapport at various work levels. Besides, where workforce is self motivated, just the acknowledgement of the same makes people feel important and wanted.

Difference between Motivation, Satisfaction, Inspiration and Manipulation

Motivation refers to the drive and efforts to satisfy a want or goal, whereas satisfaction refers to the contentment experienced when a want is satisfied. In contrast, inspiration is bringing about a change in the thinking pattern.

On the other hand Manipulation is getting the things done from others in a predetermined manner.



Hence, manipulation or external stimulus as well as inspiration or internal stimulus acts as carriers of either demotivation or motivation which in turn either results into dissatisfaction or satisfaction depending upon.

THE MOTIVATIONAL DYNAMICS OF SPORT

Motivation is an internal energy force that determines all aspects of our behaviour; it also impacts on how we think, feel and interact with others. In sport, high motivation is widely accepted as an essential prerequisite in getting athletes to fulfil their potential. However, given its inherently abstract nature, it is a force that is often difficult to exploit fully. Some coaches, like Portugal manager Luiz Felipe 'Big Phil' Scolari, appear to have a 'magic touch', being able to get a great deal more out of a team than the sum of its individual parts; others find motivation to be an elusive concept they are forever struggling to master.

What is it that makes individuals like the 45-year-old sprinter Merlene Ottey, who competed in her seventh Olympics in Athens 2004, churn out outstanding performances year in, year out? Elite athletes such as Ottey have developed an ability to channel their energies extremely effectively. Indeed, motivation is essentially about the direction of effort over a prolonged period of time.

There are numerous approaches to the study of motivation. Some are based on schedules of positive and negative reinforcement while others focus on an individual's sense of mastery over a set of circumstances. Some of the key findings from recent literature and provide four evidence-based techniques relating to the enhancement of motivation. You will be able to tailor the motivational techniques to enhance your participation in sport or the performance of others. You will learn that motivation is a dynamic and multifaceted phenomenon that can be manipulated, to some degree at least, in the pursuit of superior sporting performance.

DIFFERENT TYPES OF MOTIVATION

One of the most popular and widely tested approaches to motivation in sport and other achievement domains is self-determination theory. This theory is based on a number of motives or regulations, which vary in terms of the degree of self-determination they reflect. Self-determination has to do with the degree to which your behaviours are chosen and self-initiated. The behavioural regulations can be placed on a self-determination continuum. From the least to the most self-determined they are amotivation, external regulation, introjected regulation, identified regulation, integrated regulation and intrinsic motivation.

Amotivation represents a lack of intention to engage in a behaviour. It is accompanied by feelings of incompetence and a lack of connection between one's behaviour and the expected outcome. For example, an amotivated athlete might be heard saying, 'I can't see the point in training any more – it just tires me out' or 'I just don't get any buzz out of competition whatsoever'. Such athletes exhibit a sense of helplessness and often require counselling, as they are highly prone to dropping out. External and introjected regulations represent non-self-determined or controlling types of extrinsic motivation because athletes do not sense that their behaviour is choiceful and, as a consequence, they experience psychological pressure. Participating in sport to receive prize money, win a trophy or a gold medal typifies external regulation. Participating to avoid punishment or negative evaluation is also external. Introjection is an internal pressure under which athletes might participate out of feelings of guilt or to achieve recognition.

Identified and integrated regulations represent self-determined types of extrinsic motivation because behaviour is initiated out of choice, although it is not necessarily perceived to be enjoyable.

These types of regulation account for why some athletes devote hundreds of hours to repeating mundane drills; they realise that such activity will ultimately help them to improve. Identified regulation represents engagement in a behaviour because it is highly valued, whereas when a behaviour becomes integrated it is in harmony with one's sense of self and almost entirely self-determined. Completing daily flexibility exercises because you realise they are part of an overarching goal of enhanced performance might be an example of integrated regulation. Intrinsic motivation comes from within, is fully self-determined and characterised by interest in, and enjoyment derived from, sports participation.

There are three types of intrinsic motivation, namely intrinsic motivation to know, intrinsic motivation to accomplish and intrinsic motivation to experience stimulation. Intrinsic motivation is considered to be the healthiest type of motivation and reflects an athlete's motivation to perform an activity simply for the reward inherent in their participation.

Flow: The Ultimate Motivational State

Mihalyi Csikszentmihalyi, the highest level of intrinsic motivation is flow state. Flow is characterised by complete immersion in an activity, to the degree that nothing else matters. Central to the attainment of flow is a situation in which there is a perfect match between the perceived demands of an activity and an athlete's perceived ability or skills.

During flow, self-consciousness is lost and athletes become one with the activity. For example, a World champion canoeist I work with often describes how the paddle feels like an extension of her arms while she is in flow.

An overbearing or unrealistic challenge can cause excess anxiety, which means that coaches need to ensure that athletes set realistic goals. Conversely, if athletes bring a high level of skill to an activity and the challenge that it provides is relatively low, such as Barcelona and Brazil's Ronaldinho playing in a minor football league, this can result in boredom. To promote flow, it is important to find challenges that are going to stretch athletes just a touch further than they have been stretched before.

Recent Motivation Research based on SDT

A study examining the relationship between athletes' goal orientations and their levels of intrinsic and extrinsic motivation indicated that British collegiate athletes with task-related or personal mastery goals were far more likely to report high self-determination than athletes with ego-orientated or social comparison-type goals. The study provided tentative support for the proposition that focusing on personal mastery and self-referenced goals promotes intrinsic motivation to a greater degree than focusing on winning and demonstrating superiority over others.

This has important implications for practitioners who work with children, given the wealth of evidence that suggests that a focus on personal mastery and intrinsic motivation brings the most positive motivation outcomes. A very recent study showed that during competition deemed to be important, intrinsically motivated athletes developed task-oriented coping strategies. Conversely, extrinsically motivated athletes tended to avoid dealing with key issues and were far less likely to achieve their goals. In another study, researchers adopted a qualitative approach to answer the question 'why does the "fire" of elite athletes burn so brightly?'. They sought to demystify the differences between high achievers and also-rans in the world of sport.

Their interviews with 10 elite Australian track and field athletes revealed three overarching themes:

1. Elite athletes set personal goals that were based on both self-determined and extrinsic motives;
2. They had a high self-belief in their ability to succeed;
3. Track and field was central to their lives – everything rotated around their involvement in the sport.

Using a statistical procedure known as 'cluster analysis', colleagues and I have identified two types of 'motivation profile'. The first was characterised by high levels of both controlling and self-determined types of behavioural regulations and the second by high self-determined and low controlling motivation. A comparison of the two profiles on the motivation outcomes of enjoyment, effort, positive and negative affect, attitude towards sport, strength and the quality of behavioural intentions, satisfaction, and frequency of attendance showed that participants in the first profile reported higher levels on all eight positive consequences when compared to those in the second profile. This finding suggests that the simultaneous presence of high extrinsic and high intrinsic motivation is likely to yield the most positive benefits for adult athletes.

However, it is critical that extrinsic motives are nurtured on a firm foundation of high intrinsic motivation. Without high intrinsic motivation, athletes are likely to drop out when they encounter problems such as injury, non-selection or demotion. We conducted a follow-up study confirming the profiles identified in 2000 and came up with a similar solution using a new sample of adult athletes. Importantly, we found that participants in cluster 1 also reported better concentration on the task at hand.

ENHANCING MOTIVATION IN SPORT

With Torino Olympic Games underway, the sports world will be in the spotlight for much of this month. Psychologists have been supporting the U.S. Olympic mission formally since the 1980s and a team of sport psychologists will be on-hand in Italy this month to continue the work they have been doing for the past 4 years with some of the country's finest athletes. As important as performance enhancement can be for elite athletes, it is only a small piece of how psychologists are contributing to the world of sport. Millions of youth participate in organised sports annually in the United States and another interesting line of enquiry in sport psychology focuses on how organised sport experiences can be used to foster optimal motivation. Enhancing motivation can lead to the sustained, high-quality engagement in sport that is required for the development of Olympic-level expertise and it may also contribute to healthy youth development which will be the focus of this essay.

THE VALUE OF YOUTH SPORT PARTICIPATION

One of the most powerful rationales for promoting youth sport participation draws from the documented benefits of physical activity. The United States Surgeon General and the American College of Sports Medicine (2000) endorse regular physical activity to reduce long-term risk for disease (*e.g.*, diabetes, cardiovascular disease, some forms of cancer). Strikingly, the prevalence of diseases such as type-II diabetes recently increased dramatically in children and youth. This increase is widely attributed to concurrent increases in childhood obesity. Overweight status among children and adolescents in the United States has more than tripled in the past 25 years.

Given the nature of energy balance (*i.e.*, caloric intake vs. energy expenditure), increasing youth physical activity will surely be one part of the solution to the current childhood obesity crisis. Unfortunately, daily physical activity is being cut out of school curricula across the country. The greatest single source of organised youth sport participation appears to be recreational sport programmes, such as those sponsored by community recreation departments; but it is well-established that youth sport participation rates experience a steady decline starting between ages 10-13 years. Getting and keeping youth involved in organised sport programmes outside of school is a motivation problem of great importance for public health. Of course, physical activity and its benefits for physical health represent only one class of youth sport outcomes. Sport is also a powerful context for youth psychosocial development. Youths' subjective experience during organised sports and other structured voluntary activities is unique because they report greater concentration than they do when playing with friends in unstructured settings, and greater enjoyment than they do in structured activities such as school. These conditions are ideally-suited for social learning and internalizing environmental characteristics.

Notwithstanding a few undesirable correlates, the available evidence suggests a generally positive profile of correlates associated with youth sport participation. Compared to non-athletes, high school athletes report greater liking of school, are less likely to dropout, have higher grade point averages, are more likely to attend college, are less socially-isolated, attain greater occupational success, and have greater increases in self-esteem through high school.

On balance, youth sport participation seems to be a positive developmental experience; however, it seems apparent that not all youth sport programmes are equal with respect to their developmental yield for youth. Many factors are likely to play a role in determining the quality of a youth sport experience. My colleagues and I are among a group of scientists who focus on the role that coaches play in determining the developmental yield of youth sport participation.

THE IMPORTANCE OF YOUTH SPORT COACHES

Behaviour observation research has provided compelling evidence that coaching behaviours influence the quality of youth sport experiences. In one study, youth reported greater liking for basketball when their coaches exhibited high levels of mistake-contingent technical instruction, and low levels of keeping control and general encouragement. Similarly, youth evaluated their coaches more positively when the coaches exhibited high levels of instructive (*e.g.*, general and mistake-contingent technical instruction) and supportive (*e.g.*, reinforcement, mistake-contingent encouragement) behaviours, and low levels of punishment. Interestingly, Smith and Smoll (1990) also found that youth self-esteem at the beginning of the season moderated the effects of coach behaviour on youth evaluations – low self-esteem athletes' evaluations of coaches seem to be especially influenced by the coaches use of the desirable coaching behaviours. Clearly, what coaches do impacts how youth evaluate those coaches and the activities that are organised by those coaches.

Beyond a specific behavioural repertoire, coaches are able to create motivational climates by the way they choose to structure the setting. To illustrate the role of coaching climates on young athletes' sport experience, consider a recent study of female and male recreational swim league participants aged 8 – 18 years. We were interested in whether and how the perceived coaching climate predicted changes in youths' reasons for swimming. Youth completed measures of their situational motivation (*i.e.*, their reasons for swimming) at the beginning, middle, and end of the season. At the beginning and end of season, youth also rated their achievement goals. Achievement goals represent the purpose or aim of their achievement behaviour. We employed Elliot's (1999) 2×2 model of achievement goals that distinguishes four goals based on their *definition of competence* (*i.e.*, task- or self-referenced criteria vs. normatively-referenced criteria) and the *valence of the goal* (*i.e.*, approaching competence vs. avoiding incompetence). At the end of the swim season, youth rated their perceptions of the coaching climate –

that is, the degree to which youth perceived the coaches as emphasizing each of the four achievement goals when evaluating the youths' competence. Results indicated that youth perceptions of avoidance coaching climates positively predicted approximately 40 per cent of the change in youths' corresponding avoidance achievement goals during the season. Additionally, to the extent that youth increased their focus on avoiding self-referenced incompetence (*e.g.*, not performing worse than they previously performed), they described their reasons for swimming as being more externally regulated (*i.e.*, done to satisfy external demands, such as parents' directives) and more amotivated (*i.e.*, done without a clear purpose in mind). Thus, avoidance coaching climates in swimming appear to be linked with deterioration in the self-determination of young swimmers' motivation.

The emerging conclusion from this literature – coaches may influence youth motivation both through their observed behaviours and the motivational climate they create. Similar to the literature on developmental correlates of youth sport participation, evidence for coaching effects on youth sport motivation is based largely on non-experimental research that does not permit strong causal inferences. For this reason, a number of researchers in this area have turned to experimental designs to test their hypotheses about the critical factors for optimizing youth sport experiences.

The seminal coach training efficacy trials involved Coach Effectiveness Training. This programme focused on teaching coaches a behavioural repertoire and philosophy of winning based on some of the behavioural research. This behavioural repertoire is designed to enhance youth perceptions and recall of coaches, and ultimately youth evaluative reactions in the sport setting. To accommodate recent theoretical developments and emerging research findings, my collaborator and I have posited that coach training programmes may influence youth motivation (and ultimately some important indicators of youth development) via a sequence of cascading changes in (a) coaches observed behaviours and activity structures, (b) youth perceptions of coaches behaviours and the coaching climate, and (c) youth self-perceptions.

This model provides a framework for evaluating the experimental coach training literature (unless otherwise specified, this brief review includes studies employing various psychosocially-based coach training programmes). First, it appears that some coach behaviours may be modified by brief training programmes. Specifically, coaches' use of reinforcement following desirable behaviours appears to be the behaviour most amenable to change following training. Other theoretically-important behaviours may be sufficiently well-engrained that they are resistant to modification or infrequent enough to escape detection of modest changes in their base rates. Second, athletes evaluate CET-trained coaches more positively than non-CET-trained coaches. These findings are based on post-training differences in youth perceptions of coach behaviours; it will be important to determine whether randomly-assigned coach training programmes can account for changes in youth perceptions of coaches.

SATISFACTION AND MOTIVATION IN SPORT

Self-determination theory examines the effects of the social context on motivation and individual behaviours. Grounded in this framework, numerous studies in the last 20 years have investigated individuals' motivation in a variety of settings and activities. According to Vallerand and Fortier (1998), self-determination theory is especially helpful in studying individual patterns of sport participation. People may engage in activities for different reasons. Athletes are intrinsically motivated when they engage in an activity for the pleasure and satisfaction derived from the activity itself, whereas extrinsic motivation describes behaviours performed to attain material or social rewards. The understanding of human behaviour also involves considering a motivation. When people are amotivated, they do not perceive a relationship between their actions and the resulting outcomes. Consequently, they no longer identify any good reasons for practicing their sport.

INTRINSIC AND EXTRINSIC MOTIVATION

According to self-determination theory, intrinsically motivated behaviour is associated with satisfaction of three psychological needs. The need for autonomy reflects the need to perceive behaviour as freely chosen. The need for competence refers to the urge to effectively interact with the social environment. The need for relatedness pertains to the desire to feel connected with other individuals. Vallerand and his colleagues suggested that three dimensions of intrinsic motivation exist: intrinsic motivation to know, intrinsic motivation to accomplish things, and intrinsic motivation to experience stimulation.

Firstly, intrinsic motivation to know involves engaging in sport for pleasure and satisfaction experienced while one is learning and exploring something new.

Secondly, intrinsic motivation to accomplish things operates when one is engaged in an activity for the pleasure derived from trying to surpass oneself or to improve skills.

Thirdly, intrinsic motivation to experience stimulation refers to engaging in sport in order to experience the pleasant sensations derived from the activity itself. In the sport domain, many studies have corroborated this tripartite conceptualization of intrinsic motivation.

Extrinsic motivation is also considered to be a multidimensional construct. Deci and Ryan (1985) have identified fours forms of extrinsic motivation that can be classified on the self-determination continuum from high to low levels of self-determination. These different types of extrinsic motivation (*, e.g.,* integrated regulation, identified regulation, introjected regulation, and external regulation) occupy the continuum between intrinsic motivation and a motivation. Research in sport has supported the presence of this continuum. Firstly, integrated regulation deals with behaviours that are so integrated in one's life that they are part of the individual's self and value system. Secondly, identified regulation reflects participation in an activity because one holds outcomes of the behaviour to be personally significant, although one may not enjoy the activity itself. Thirdly, introjected regulation refers to behaviours that are partly internalized by the individual but that remain non-self-determined because contingencies from external control sources have been internalized without having been endorsed by the person. Finally, external regulation characterises behaviours that are controlled by external sources such as rewards.

DETERMINANTS OF MOTIVATION

Cognitive evaluation theory, a sub-theory within the larger self-determination theory, details the environmental factors that promote and undermine the development of intrinsic motivation trough their impacts on perceptions of competence and autonomy. According to Deci and Ryan (1985), when there is a shift from an internal to an external perceived locus of causality, one's feelings of self-determination and consequently, one's intrinsic motivation towards the activity decreases. The concept of perceived locus of causality refers to the degree to which individuals feel that they are the origin of their own behaviour. On the one hand, when athletes experience an internal locus of causality, their actions are self-determined and volitional. On the other hand, when they experience an external locus of causality, the initiation of behaviour is influenced by external factors.

In the sport domain, many environmental and interpersonal factors (*, e.g.,* rewards, coaches' behaviours) may affect athletes' feelings of competence, autonomy, and relatedness, and, in turn, intrinsic motivation. For instance, scholarship athletes exhibited higher levels of perceived competence and intrinsic motivation than non-scolarship athletes. Sport competition also constitutes a social factor that may affect feelings of autonomy and intrinsic motivation because emphasizing winning at all costs may lead individuals to adopt an external perceived locus of causality. Vallerand, Gauvin and Halliwell have conducted a research to examine the effects of competition on intrinsic motivation.

Twenty-three subjects aged between 10 and 12 years were randomly assigned to one of two conditions: competition or intrinsic-mastery orientation. The task was a stabilometer motor task. In the competition condition, subjects were told that they were competing against other participants. In the intrinsic-mastery condition, subjects were encouraged to perform as well as they could. The amount of time spent on the stabilometer during a free-choice period served as the dependent measure of intrinsic motivation. Results revealed that participants in the intrinsic mastery condition were more intrinsically motivated than those in the competition condition.

Fortier, Vallerand, Brière and Provencher (1995) have examined the relationships between competitive and recreational sport structures, gender, and athletes' sport motivation. Participants were 399 athletes aged between 17 and 25 years and involved in four different sport activities (soccer, basketball, volleyball, and badminton). Results revealed that recreational athletes demonstrated more intrinsic motivation to accomplish things and to experience stimulation than competitive athletes, while exhibiting less identified regulation and less a motivation than this group. Many other studies have clearly revealed that competition undermined individuals' intrinsic motivation. However, when competition represents an interesting challenge, one's intrinsic motivation may be boosted. For instance, results from four studies conducted by Tauer and Harackiewicz (2004) with boys from a youth basketball camp showed that intergroup competition was consistently conducive to intrinsic motivation. According to Vallerand (2007), measurement and methodological issues could explain the inconsistent findings in the literature. Deci and Ryan (1985) also suggested that competition may enhance or reduce intrinsic motivation functions of individual perceptions of the competitive situation. As a consequence, Vallerand (2007) considers that future studies are required to better understand and analyse the effects of competition on intrinsic motivation.

In accordance with Vallerand's (1997) postulates, we consider that researchers need to take into consideration perceptions of competence, autonomy and relatedness in order to understand and explain the links between competition and motivation in the sport context. However, a few studies in sport have included both environmental factors and basic need satisfaction as predictors of motivation. In addition, all of these motivational studies have used scales developed and validated in other contexts in order to measure these feelings because there was not a scale for measuring perceptions of autonomy, competence and relatedness in the sport domain.

For instance, Sarrazin and his colleagues (2002) have used four items adapted from the Perceived Competence in Life Domains Scale to assess feelings of competence. To assess perceived autonomy, three items adapted from the Perceived Autonomy towards Life Domains Scale were used. To assess perceptions of relatedness, four items adapted from the Feelings of Relatedness Scale were used. Recently, Gillet, Rosnet and Vallerand (under revision) have developed and validated the Basic Need Satisfaction in Sport Scale to measure perceptions of competence, autonomy and relatedness in the sport context. Although other studies are needed to support the structure, the reliability and the construct validity of this scale, this questionnaire should allow researchers to test postulates from self-determination theory in the sport domain.

The Present Study

The first purpose of the present investigation was to analyse the relationships between competitive and recreational sport structures, perceptions of autonomy, competence and relatedness, and the different forms of motivation. We consider that it was necessary to go beyond the Fortier and her colleagues' (1995) work by including athletes' perceptions of autonomy, competence, and relatedness in order to examine the relationships between competition and motivation in sport. According to cognitive evaluation theory and past studies, it was hypothesized that competitive athletes would perceive themselves less autonomous and would display a less self-determined motivational profile than recreational athletes. The second purpose of this research was to identify motivational differences across gender because, according to Fortier and her colleagues (1995), it seems important to consider gender differences in motivational regulations in sport. These researchers have shown that male athletes reported lower levels of intrinsic motivation to accomplish things and identified regulation but higher levels of external regulation and a motivation than females. More generally, some research has provided data suggesting that male athletes exhibited a less self-determined motivation than female athletes. In the same way, the results of Hollembeak and Amorose's (2005) study showed that compared to males, females reported higher scores on autonomy, relatedness, and intrinsic motivation. In line with past investigations, it was hypothesized that female athletes would display a higher self-determined profile than males. It was also hypothesized that they would perceive themselves more autonomous and connected to each other, than male athletes.

Contrary to Fortier and her colleagues' (1995) research, the present study is not restricted to the relationships between competitive and recreational sport structures, gender, and individuals' motivation in order to provide a more complete understanding of the motivational processes involved in athletes' behaviour. Indeed, we consider that motivational differences could be due to the nature of the sport activities (*, e.g.,* individual versus collective sports) and the level of competition (*, e.g.,* district, regional, or national levels). Thus, the third purpose of our investigation was to analyse the relationships between perceptions of autonomy, competence and relatedness, and the different types of motivation for individual and team sport athletes. A tennis player must choose and individually register for the competitions he wants to participate. Conversely, a basket-ball player is not concerned with this problem because schedule of play is determined by the Basket-Ball Federation. In consequence, participation in individual sports should enhance feelings of autonomy. Thus, we hypothesized that athletes involved in team sports to be less autonomous than the ones in individual sports.

Finally, the fourth purpose of the present study was to examine the relationships between levels of competition (*, i.e.,* district, regional, or national levels), perceptions of autonomy, competence and relatedness, and the different forms of motivation.

Does an athlete engaged in national competitions play a match for the same reasons as a competitive athlete who plays at a regional level? Chantal and his colleagues (1996), in a sample of Bulgarian athletes, have reported that higher performing athletes (*, i.e.,* title and medal holders in national and international events) exhibited significantly higher levels of external regulation than those who did not win titles or medals in national and international competitions. Based on these findings, it was hypothesized that athletes at the lowest level of competition (*, i.e.,* district level) would display less external regulation than the ones involved in sport activities at higher levels (*, i.e.,* regional and national levels).

METHOD

Participants

Two hundred and eighty-eight athletes (83 females and 205 males; *M* age = 19.4 years, *SD* = 1.47) participated in this study. Participants were enrolled at two French universities. One hundred and fifty students were involved in team sports and 138 in individual sports. The sample was made up of 232 competitive athletes, including 52 females and 180 males, and 56 recreational athletes, including 31 females and 25 males. Sixty-six competitive athletes were ranked at the district level, 114 were at the regional level, and 52 were at the national level. Athletes at district level trained on average 4.22 hours (*SD* = 1.72) per week, athletes at regional level trained on average 5.35 hours (*SD* = 2.54) per week, and athletes at national level trained on average 8.53 hours (*SD* = 4.53) per week.

Measures

Athletes completed the French version of the Sport Motivation Scale. This questionnaire measures seven types of motivation, namely intrinsic motivation to know, intrinsic motivation to accomplish things, intrinsic motivation to experience stimulation, identified regulation, introjected regulation, external regulation, and a motivation.

The SMS does not include the construct of integrated regulation even though Pelletier and Kabush (2005, cited in Pelletier and Sarrazin, 2007) have attempted to develop an integrated regulation subscale. Items are scored on a seven-point Likert scale, ranging from 1 (does not correspond at all) to 7 (corresponds exactly). Previous research in sport has provided support for the factorial structure and the reliability of this scale. In the present study, Cronbach alpha coefficients for the seven subscales were calculated, and all internal consistencies ranged from.68 to.89. Because, there were no specific hypotheses about the intrinsic regulations, the three intrinsic components (*, i.e.,* intrinsic motivation to know, intrinsic motivation to accomplish things, and intrinsic motivation to experience stimulation) were combined into a single intrinsic motivation score (a =.90).

The Basic Need Satisfaction in Sport Scale (Gillet *et al.,,* under revision) was used to measure athletes' feelings of autonomy, competence, and relatedness. This instrument is composed of three subscales assessing these three perceptions. There are five items per subscale (*, i.e.,* a total of fifteen items). The response scale has a Likert format ranging from 1 (not at all true) to 7 (very true).

Recently, Gillet and his colleagues (under revision) have provided evidence for the factorial structure, the construct validity and the internal consistency of this questionnaire. In the present investigation, all Cronbach alpha coefficients were above the minimum criterion of.70. The internal consistency values were.71 for competence.82 for autonomy, and.81 for relatedness.

Procedure

Participation in this investigation was voluntary. After obtaining informed consent, all of the athletes completed a series of questionnaires individually at the beginning of a training session. The athletes were informed that there were no right or wrong replies. They were also assured that their answers would remain anonymous and confidential.

RESULTS

A 2 × 8 multivariate analysis of variance (MANOVA) was conducted to determine significant differences between competitive and recreational athletes on eight dependant variables (perceptions of competence, autonomy and relatedness, as well as intrinsic motivation, identified regulation, introjected regulation, external regulation and a motivation). Results revealed significant differences between competitive and recreational athletes, $F(8, 276) = 4.30$, $p < .001$.

Univariate F values indicated that competitive and recreational athletes differed on external regulation, $FF(1, 283) = 11.65$, $p < .001$, and perceived autonomy, $F(1, 283) = 17.75$, $p < .001$. Specifically, competitive athletes felt less autonomous and demonstrated more external regulation than recreational athletes. Means and standard deviations for the different subscales.

Table: Motivation and Basic Needs Subscales by Level of Competition.

Subscales Athletes	Competitive (N=232)		Recreative (N=56)		Total (N=288)	
Sample	SD	M	SD	M	SD	M
Intrinsic Motivation	5.26	0.96	5.36	0.95	5.31	0.96
Identified Regulation	4.37	1.11	4.22	1.10	4.34	1.11
Introjected Regulation	5.22	1.25	4.90	1.17	5.16	1.24
External Regulation*	3.25	1.31	2.58	1.19	3.12	1.32
Amotivation	1.43	0.81	1.53	0.78	1.45	0.81
Competetence	5.24	0.86	5.03	0.95	5.20	0.88
Autonomy*	4.47	1.33	5.30	1.38	4.36	1.37
Relatedness	5.63	0.82	5.64	0.90	5.63	0.83
Note: *p<.001						

A 2 × 8 MANOVA was conducted with the three basic needs (, *i.e.,* autonomy, competence and relatedness) and the five types of motivation (, *i.e.,* intrinsic motivation, identified regulation, introjected regulation, external regulation and a motivation) as dependent variables, and gender as the independent variable. Results indicated significant differences between males and females, $F(8, 276) = 4.98$, $p < .001$. Univariate F values indicated that male and female athletes differed on intrinsic motivation, $F(1, 283) = 4.02$, $p < .05$, external regulation, $F(1, 283) = 16.39$, $p < .001$, and perceived competence, $F(1, 283) = 7.19$, $p < .01$. Specifically, male athletes felt more competent and exhibited more external regulation than females, while demonstrating less intrinsic motivation than this group. Means and standard deviations for the different subscales.

Table: Motivation and Basic Needs Subscales by Gender.

Subscales	Males (N=205)		Females (*N=83)*	
	M	SD	M	SD
Intrinsic Motivation**	5.24	0.97	5.49	0.89
Identified Regulation	4.35	1.13	4.34	1.05
Introjected Regulation	5.13	1.22	5.24	1.30
External Regulation*	3.31	1.28	2.63	1.29
Amotivation	1.48	0.84	1.39	0.73
Competetence**	5.29	0.84	4.99	0.95
Autonomy*	4.55	1.39	4.83	1.31
Relatedness	5.62	0.85	5.65	0.81
Note: *p<.001, **p<0.5				

A 2 × 8 MANOVA was conducted with the three needs (*, i.e.,* autonomy, competence and relatedness) and the five forms of motivation (*, i.e.,* intrinsic motivation, identified regulation, introjected regulation, external regulation and a motivation) as dependent variables, and the nature of the sport activities (*, e.g.,* individual versus collective sports) as the independent variable.

Results indicated significant differences between athletes in individual and team sports, $F(8,\ 276) = 12.31$, $p < .001$. Univariate F values indicated that athletes in individual and team sports differed on perceived autonomy, $F(1, 283) = 92.03$, $p < .001$. Specifically, athletes involved in team sports felt less autonomous than the ones in individual sports. Means and standard deviations for the different subscales.

Table: Motivation and Basic Needs Subscales by Team or Individual Sport.

Subscales	Team Sports (N=150)		Individual Sports (N=138)	
	M	SD	SD	M
Intrinsic Motivation	5.26	0.96	5.36	0.95
Identified Regulation	4.41	1.03	4.27	1.18
Introjected Regulation	5.14	1.30	5.19	1.18
External Regulation	3.24	1.28	2.98	1.35
Amotivation	1.49	0.89	1.41	0.70
Competetence	5.17	0.89	5.24	0.87
Autonomy*	3.98	1.17	5.35	1.22
Relatedness	5.54	0.88	5.72	0.77
Note: *p<.001				

A 2 × 8 MANOVA was conducted with the three needs (*, i.e.,* autonomy, competence and relatedness) and the five types of motivation (*, i.e.,* intrinsic motivation, identified regulation, introjected regulation, external regulation and a motivation) as dependent variables, and the level of competition (*, e.g.,* district, regional, or national levels) as the independent variable.

Results indicated significant differences between athletes at district, regional and national levels, $F(16,\ 436) = 2.16$, $p < .01$. Univariate F values indicated that athletes differed on intrinsic motivation, $F(2,\ 225) = 5.90$, $p < .01$, introjected regulation, $F(2,\ 225) = 5.13$, $p < .01$, and external regulation, $F(2, 225) = 3.87$, $p < .05$. Newman-Keuls post hoc comparisons were conducted to determine which of the three level groups differed on these variables. Firstly, no significant differences between the regional and national levels emerged among the study variables. Secondly, athletes at regional level displayed more intrinsic motivation and more external regulation than the ones at district level. Finally, athletes at district level exhibited less intrinsic motivation, less introjected regulation, and less external regulation than the ones at national level.

Table: Motivation and Basic Needs Subscales by Level of Competition.

Subscales	District Level (N=66)		Regional Level (N=114)		National Level (N=52)	
	M	SD	M	SD	M	SD
Intrinsic Motivation*	5.00	0.96	5.43	0.92	5.50	0.84
Identified Regulation	4.21	1.04	4.42	1.19	4.49	1.02
Introjected Regulation*	4.89	1.36	5.23	1.20	5.62	1.13
External Regulation**	2.89	1.11	3.33	1.37	3.54	1.36
Amotivation	1.44	0.85	1.40	0.79	1.51	0.83
Competetence	5.23	0.80	5.33	0.85	5.06	0.95
Autonomy*	4.27	1.19	4.45	1.30	4.78	1.53
Relatedness	5.59	0.79	5.60	0.81	5.76	0.86
Note: *p<.001, **p<0.5						

Four factorial MANOVAs were conducted to test sport structure (*, i.e.,* competitive versus recreational) × gender, nature of activities x gender, nature of activities × level of competition and gender × level of competition interactions on eight dependant variables (*, i.e.,* perceived autonomy, perceived competence, perceived relatedness, intrinsic motivation, identified regulation, introjected regulation, external regulation, and a motivation). No significant differences emerged. A 2 × 2 × 8 MANOVA was also conducted to determine significant differences between competitive and recreational athletes, as well as between athletes in individual and team sports on the three needs (*, i.e.,* autonomy, competence and relatedness) and the five types of motivation (*, i.e.,* intrinsic motivation, identified regulation, introjected regulation, external regulation and a motivation).

Results revealed significant differences, $F(8, 274) = 2.79$, $p <.01$. Univariate F values indicated that athletes differed on autonomy, $F(1, 281) = 7.97$, $p <.01$. Tukey's HSD post hoc procedure was used for pair wise comparisons between the different groups. Firstly, no significant differences emerged between competitive athletes engaged in individual sports and recreational athletes involved in team sports. Secondly, recreational athletes engaged in individual and team sports and competitive athletes involved in individual sports felt more autonomous than competitive athletes engaged in team sports.

DISCUSSION

The first purpose of this investigation was to examine the relationships between competitive and recreational sport structures, and athletes' perceptions of autonomy, competence and relatedness, as well as motivation. In line with the first hypothesis, results revealed that recreational athletes felt more autonomous and exhibited less external regulation than competitive athletes.

The present results showed that competitive athlete's behaviours were regulated by external factors such as material or financial rewards. Our results also provided support for the cognitive evaluation theory's predictions about competitive structures. Indeed, according to Deci and Ryan (1985), when there is a shift from an internal to an external perceived locus of causality, athletes' perceptions of autonomy decline.

Results of the present research suggest that it is highly possible that competitive structures lead to a change in competitors' perceived locus of causality from internal to external. Contrary to Fortier and her colleagues' (1995) study, our research took into consideration perceptions of autonomy and showed that competitive athletes perceived themselves less autonomous than recreational athletes.

Thus, it is highly probable that sport competition leads to a decrease in individuals' feelings of autonomy and leads them to focus on the external aspects of the activity. However, contrary to our hypotheses and past studies, we did not find any significant differences between competitive and recreational athletes on intrinsic motivation. It is somewhat surprising that no differences in intrinsic motivation emerged between groups because recreational athletes exhibited higher scores on perceived autonomy than competitive athletes and intrinsically motivated behaviour is positively related to autonomy need satisfaction. Consequently, further research with an experimental design is required to confirm the possible harmful effect of competition on intrinsic motivation in the sport setting.

The second purpose of this study was to examine plausible gender differences in athlete's motivational profiles. Firstly, results revealed that male athletes perceived themselves as being more competent than female athletes. This is not surprising given male athletes typically display higher self-confidence than female athletes.

One explanation suggested for gender differences in perceived competence may be that males are boastful and think they will do better than they do. In line with our hypotheses, the present results also showed that male athletes exhibited more external regulation and less intrinsic motivation than female athletes. In other words, females appeared to take part in sport activities for the pleasure derived from the activity itself more than for extrinsic motives.

These results were in line with past studies in the sport context and confirmed that gender differences should be taken into consideration in the sport domain. However, contrary to our hypotheses and a study conducted by Hollembeak and Amorose's (2005), the present findings did not confirm that females reported higher scores on perceived autonomy and perceived relatedness than males. Consequently, future investigations should continue to explore gender differences in basic need satisfaction in order to gain a better understanding of the motivational processes underpinning sport participation.

The third purpose of the present investigation was to study motivational differences among athletes involved in individual and team sports. Results were in line with our predictions. Firstly, our results revealed that athletes in individual sports felt more autonomous than athletes in team sports. Secondly, recreational athletes in individual and team sports and competitive athletes in individual sports perceived themselves more autonomous than competitive athletes involved in team sport activities. With regard to these results, we believe that recreational and individual sport activities allow individuals to satisfy their need of autonomy. Nevertheless, it is inappropriate to make causal inferences due to the cross-sectional design of the present investigation. Therefore, this assumption should be examined in future studies using experimental designs.

A fourth purpose of the present research was to identify motivational differences between athletes at district, regional and national levels. In line with our hypotheses, results showed that athletes at regional and national levels displayed more external regulation than the ones at district level, while athletes at national level exhibited more introjected regulation than the ones at district level. The present results also revealed that athletes at district level displayed less intrinsic motivation than those at regional and national levels. Thus, our results provide only partial support for Chantal and his colleagues' (1996) findings. These dissimilar results could be due to the specific cultural context which prevailed in Bulgaria at the time of the study. Indeed, Chantal and his colleagues (1996) postulated that the social context which prevailed in Bulgaria under the communist regime and the pressures associated with highly competitive structures could have led to increase non self-determined forms of motivation. Consequently, future research would do well to examine differences on basic need satisfaction and motivation according to the level of sport competition.

More generally, results from the present investigation showed that the best athletes (*, i.e.,* those ranked at national level) displayed both high levels of intrinsic motivation and external regulation. Firstly, they suggested that high-level sport performers' behaviours were not solely intrinsically motivated. These results are comparable to those of Mallet and Hanrahan (2004) who also reported that elite male and female track and field athletes had several participation motives (*, e.g.,* excitement, enjoyment, money, social recognition). Secondly, the present findings suggested that sport performance could be positively related to both self-determined and non-self-determined motivation. Therefore, future research with sport performers should examine the relationships between multidimensional motivational regulations and sport performance which is an outcome variable of central interest.

The present study had two major limitations. Firstly, the present investigation did not use an experimental design. Thus, we cannot be assured of the direction of the relationship between sport structures, nature of sport activities and motivation. Is it athletes' sport motivation that determines their choices to engage in competitive or recreational structures, or is it sport structure that impacts athletes' motivational orientations? Is it individual sport that enhances athletes' feelings of autonomy, or is it individuals' perceptions of autonomy that determines their choices to participate in individual or team sport activities?

Having recourse to longitudinal or experimental designs should allow researchers to accurately examine the relationships between sport structures, nature of sport activities, perceptions of autonomy, competence and relatedness, and sport motivation. A second limitation pertained to the nature of sport activities examined in this study. The participants were distinguished only in terms of participation in individual or teams sports. No significant interaction effect between gender and nature of sport activities has emerged in the present investigation. Due to restricted number of participants, it was not possible to compare, for example the motivation of basketball and tennis players. Future research should investigate this potential interaction effect between nature of sport activities and gender with a larger sample of athletes. In sum, results from the present investigation showed that competitive athletes exhibited lower level of perceived autonomy and higher level of external regulation than recreational athletes. They also revealed that female athletes displayed a higher self-determined profile than males, and that athletes involved in individual sports and recreational team sport athletes felt more autonomous than competitive athletes in team sports. Finally, athletes at the district level had lower scores on intrinsic motivation and external regulation than the athletes involved in sport activities at regional and national levels. The present findings underscore the importance of considering motivational differences in the sport domain as a function of sport structures, gender, nature of sport activities, and level of competition. They also offer several directions for future research to expand our knowledge of the motivational processes in sport.

INTEGRATED THEORY OF MOTIVATION

The integrated theory of motivation includes the notions of intrinsic and extrinsic motivation. The theory is based upon the centerpiece of self-determination. Self determination is the unifying that brings meaning to the overall concept of motivation. Social factor and psychological mediator are seen as determinants of motivation that lead to certain consequences.

SOCIAL FACTORS

Social factors facilitate or cause feelings of competence, autonomy, and relatedness. The specific social identified in the model include experiences o success and failure, experiences with competition and cooperation, and coaches' behaviours.

Success and Failure

As athletes participate in a sport they have many opportunities to experience, *i.e.,* to say they may have to face failure or negative feedback or successor positive feedback. Successful experiences on one hand leads to the belief that one is competent and efficacious relative to skills being learned and performed. Whereas, failure feedback on the other hand leads to a reduction in the belief that one is competent and efficacious.

Competition and Cooperation

Achievement situations tend to focus upon either competition or cooperation. Putting the emphasis upon defeating the opponent is an ego or competitive goal orientation that is associated with a loss in intrinsic motivation. Also, consistent with self determination theory is the observation that competition reduces feelings of autonomy, as the focus is external and not internal. Just as competition relates to any goal orientation, cooperation relates to a task or mastery goal orientation.

Coaches' Behaviour

The third social factor that can influence an athletes' perception of competence, autonomy and relatedness is the Coaches' behaviour. The coach could insists on being incomplete control and determining, even down to the last detail, exactly what transpires on the playing field or practice field. Or the coach be more democratic in nature and is willing to share the perception of control with athletes. The controlling coach risks destroying the intrinsic motivation of the athlete for the sake of personal control or even perhaps more wins.

PSYCHOLOGICAL MEDIATORS

There are three psychological mediators which determine motivation, they are:

Competence

Self determination theory values competence as a prerequisite for motivation, but by itself it is not a sufficient condition for its development. It is clear that competence (self confidence) is critical to the development of intrinsic motivation, but without autonomy you do not have self determination, and without self determination you do not have intrinsic motivation. Competence without autonomy gives rise to the efficacious pawn. In the efficacious pawn you have an individual who is confident that he can successfully perform a task, but who is doing it for an external reason. When the external reason is removed, he will no longer be motivated to perform the task, although he may do it without enthusiasm or real motivation.

Autonomy

The concept of autonomy is central to self determination theory. You cannot exhibit self-determination without autonomy. According to self-determination theory, every individual has the basic innate need to be an "origin" and not a "pawn".

Relatedness

The third innate psychological need is the need for relatedness. Relatedness is necessary for a person to be self actualized, or to realize his full potential as an athlete and as a human being. Relatedness has to do with the basic need to relate to other people, to care for others and have others care for you. As all human beings are social animals and for that they have to interact with other people similarly it is very interesting seeing athletes supporting each other on the playing field. To a large extent, an athletes' enjoyment in sport is associated with how she relates to other athletes on her team, as well as to the coaches and support personnel.

LEVELS OF MOTIVATION

Level of motivation depends on seven factors, and they are:

A motivation

The least self-deterministic kind of motivation is no motivation at all. This is referred to as a motivation. A motivation refers to behaviours that are neither internally nor externally based. For example, an amotivated athlete might say that he is not sure why he plays a particular sport, and he does not see any benefits related to the sport. It is the relative absence of motivation.

Intrinsic Motivation

The kind of motivation that exhibits that highest level of self determinism is referred to as being intrinsic or internal in nature. Intrinsic motivation is motivation that comes from within. Intrinsic motivation is believed to be multidimensional. The three aspects or manifestations of intrinsic motivation are towards knowledge, towards accomplishment, and towards experiencing stimulation. Intrinsic motivation towards knowledge reflects an athlete's desire to learn new skills and ways of accomplishing a task. Intrinsic motivation towards accomplishing reflects an athlete's desire to gain mastery over a particular skill and the pleasure that comes from reaching a personal goal for mastery. And similarly intrinsic motivation towards experiencing stimulation reflects the feeling that an athlete gets from physically experiencing a sensation innate to a specific task.

Extrinsic Motivation

By definition, extrinsic motivation refers to motivation that comes from an external as opposed to an internal source. Extrinsic motivation comes in many forms, but common examples include awards, trophies, money, praises, social approval, and fear of punishment.

External Regulation

A behaviour that is performed only to obtain an external reward or to avoid punishment is said to be externally regulated. For example, an externally regulated runner takes part in a 10-kilometer race because of the promise of a trophy and a cash reward. Now this behaviour leads to self determination and the perception of being in personal control. The athlete in this example is pawn in terms of exercising personal control of their behaviour.

Introjected Regulation

Extrinsic motivation that has undergone introjected regulation is only partially internalized. For example the degree to which an athlete feels that he practices daily to please his coach, as opposed to practicing to become a better player because he wants to become a better player.

Identified Regulation

When an athlete comes to identify with an extrinsic motivation to the degree that it is perceived as being her own, it is referred to as being an identified regulation. Identified regulation is present when an athlete engages in an activity that he does not perceive as being particularly interesting, but he does so because he sees the activity as being instrumental for him to obtain another goal that is interesting to him.

Integrated Regulation

The most internalized form of regulation is referred to as integrated regulation. When regulation mechanisms are well integrated, they become personally valued and freely done. At this level, a behaviour previously considered to be externally controlling becomes fully assimilated and internally controlled. For example, an athlete will perceive his coach's controlling behaviour as being completely consistent with his own aspirations and goals and no longer perceives them as being externally controlling.

CONSEQUENCES OF MOTIVATION

High levels of intrinsic motivation and internalized extrinsic motivation should lead to positive affect, positive behavioural outcomes, and improved cognition. Research shows that athletes who engage in sport for self determined reasons experience more positive and less negative affect, have greater persistence, and exhibit higher levels of sportspersonship.

MOTIVATIONAL THEORIES IN EDUCATIONAL PSYCHOLOGY

INCENTIVE THEORY

A reward, tangible or intangible, is presented after the occurrence of an action (*i.e.* behaviour) with the intent to cause the behaviour to occur again. This is done by associating positive meaning to the behaviour. Studies show that if the person receives the reward immediately, the effect is greater, and decreases as delay lengthens. Repetitive action-reward combination can cause the action to become habit. Motivation comes from two sources: oneself, and other people. These two sources are called intrinsic motivation and extrinsic motivation, respectively. Reinforcers and reinforcement principles of behaviour differ from the hypothetical construct of reward. A reinforcer is any stimulus change following a response that increases the future frequency or magnitude of that response, therefore the cognitive approach is certainly the way forward as in 1973 Maslow described it as being the golden pineapple. Positive reinforcement is demonstrated by an increase in the future frequency or magnitude of a response due to in the past being followed contingently by a reinforcing stimulus. Negative reinforcement involves stimulus change consisting of the removal of an aversive stimulus following a response. Positive reinforcement involves a stimulus change consisting of the presentation or magnification of a positive stimulus following a response. From this perspective, motivation is mediated by environmental events, and the concept of distinguishing between intrinsic and extrinsic forces is irrelevant.

Applying proper motivational techniques can be much harder than it seems. Steven Kerr notes that when creating a reward system, it can be easy to reward A, while hoping for B, and in the process, reap harmful effects that can jeopardize your goals. Incentive theory in psychology treats motivation and behaviour of the individual as they are influenced by beliefs, such as engaging in activities that are expected to be profitable. Incentive theory is promoted by behavioural psychologists, such as B.F. Skinner and literalized by behaviourists, especially by Skinner in his philosophy of Radical behaviourism, to mean that a person's actions always have social ramifications: and if actions are positively received people are more likely to act in this manner, or if negatively received people are less likely to act in this manner.

Incentive theory distinguishes itself from other motivation theories, such as drive theory, in the direction of the motivation. In incentive theory, stimuli "attract", to use the term above, a person towards them, as opposed to the body seeking to reestablish homeostasis pushing it towards the stimulus. In terms of behaviourism, incentive theory involves positive reinforcement: the stimulus has been conditioned to make the person happier. For instance, a person knows that eating food, drinking water, or gaining social capital will make them happier. As opposed to in drive theory, which involves negative reinforcement: a stimulus has been associated with the removal of the punishment-the lack of homeostasis in the body. For example, a person has come to know that if they eat when hungry, it will eliminate that negative feeling of hunger, or if they drink when thirsty, it will eliminate that negative feeling of thirst.

ESCAPE-SEEKING DICHOTOMY MODEL

Escapism and seeking are major factors influencing decision making. Escapism is a need to breakaway from a daily life routine, turning on the television and watching an adventure film, whereas seeking is described as the desire to learn, turning on the television to watch a documentary. Both motivations have some interpersonal and personal facets for example individuals would like to escape from family problems (personal) or from problems with work colleagues (interpersonal). This model can also be easily adapted with regard to different studies.

DRIVE-REDUCTION THEORY

There are a number of drive theories. The Drive Reduction Theory grows out of the concept that we have certain biological drives, such as hunger. As time passes the strength of the drive increases if it is not satisfied (in this case by eating). Upon satisfying a drive the drive's strength is reduced. The theory is based on diverse ideas from the theories of Freud to the ideas of feedback control systems, such as a thermostat.

Drive theory has some intuitive or folk validity. For instance when preparing food, the drive model appears to be compatible with sensations of rising hunger as the food is prepared, and, after the food has been consumed, a decrease in subjective hunger. There are several problems, however, that leave the validity of drive reduction open for debate. The first problem is that it does not explain how secondary reinforcers reduce drive. For example, money satisfies no biological or psychological needs, but a pay check appears to reduce drive through second-order conditioning.

Secondly, a drive, such as hunger, is viewed as having a "desire" to eat, making the drive a homuncular being-a feature criticized as simply moving the fundamental problem behind this "small man" and his desires. In addition, it is clear that drive reduction theory cannot be a complete theory of behaviour, or a hungry human could not prepare a meal without eating the food before he finished cooking it. The ability of drive theory to cope with all kinds of behaviour, from not satisfying a drive (by adding on other traits such as restraint), or adding additional drives for "tasty" food, which combine with drives for "food" in order to explain cooking render it hard to test.

COGNITIVE DISSONANCE THEORY

Suggested by Leon Festinger, cognitive dissonance occurs when an individual experiences some degree of discomfort resulting from an inconsistency between two cognitions: their views on the world around them, and their own personal feelings and actions. For example, a consumer may seek to reassure himself regarding a purchase, feeling, in retrospect, that another decision may have been preferable. His feeling that another purchase would have been preferable is inconsistent with his action of purchasing the item. The difference between his feelings and beliefs causes dissonance, so he seeks to reassure himself.

While not a theory of motivation, *per se*, the theory of cognitive dissonance proposes that people have a motivational drive to reduce dissonance. The cognitive miser perspective makes people want to justify things in a simple way in order to reduce the effort they put into cognition. They do this by changing their attitudes, beliefs, or actions, rather than facing the inconsistencies, because dissonance is a mental strain. Dissonance is also reduced by justifying, blaming, and denying. It is one of the most influential and extensively studied theories in social psychology.

MOTIVATION THEORY OF PERSONALITY DEVELOPMENT

The history of Man's thought is the real history of mankind. Back of all the events of history are the curious systems of beliefs for which men have lived and died. Struggling to understand himself, Man has built up and discarded superstitions, theologies and sciences.

Early in this strange and fascinating history he divided himself into two parts—a body and a mind. Working together with body, mind somehow was of different stuff and origin than body and had only a mysterious connection with it. Theology supported this belief; metaphysics and philosophy debated it with an acumen that was practically sterile of usefulness. Mind and body "interacted" in some mysterious way; mind and body were "parallel" and so set that thought-processes and brain-processes ran side by side without really having anything to do with one another. With the development of modern anatomy, physiology and psychology, the time is ripe for men boldly to say that applying the principle of causation in a practical manner leaves no doubt that mind and character are organic, are functions of the organism and do not exist independently of it. I emphasize "practical" in relation to causation because it would be idle for us here to enter into the philosophy of cause and effect. Such discussion is not taken seriously by the very philosophers who most earnestly enter into it.

The statement that mind is a function of the organism is not necessarily "materialistic." The body is a living thing and as such is as "spiritualistic" as life itself. Enzymes, internal secretions, nervous activities are the products of cells whose powers are indeed drawn from the ocean of life. To prove this statement, which is a cardinal thesis of this book, I shall adduce facts of scientific and facts of common knowledge. One might start with the statement that the death of the body brings about the abolition of mind and character, but this, of course, proves nothing, since it might well be that the body was a lever for the expression of mind and character, and with its disappearance as a functioning agent such expression was no longer possible. It is convenient to divide our exposition into two parts, the first the dependence upon proper brain function and structure, and the second the dependence upon the proper health of other organs. For it is not true that mind and character are functions of the brain alone; they are functions of the entire organism. The brain is simply the largest and most active of the organs upon which the mental life depends; but there are minute organs, as we shall see, upon whose activity the brain absolutely depends.

Any injury to the brain may destroy or seriously impair the mentality of the individual. This is too well known to need detailed exposition. Yet some cases of this type are fundamental in the exquisite way they prove (if anything can be proven) the dependence of mind upon bodily structure. In some cases of fracture of the skull, a piece of bone pressing upon the brain may profoundly alter memory, mood and character. Removal of the piece of bone restores the mind to normality. This is also true of brain tumor of certain types, for example, frontal endotheliomata, where early removal of the growth demonstrates first that a "physical" agent changes mind and character, and second that a "physical" agent, such as the knife of the surgeon, may act to reestablish mentality.

In cases of hydrocephalus (or water on the brain), where there is an abnormal secretion of cerebro-spinal fluid acting to increase the pressure on the brain, the simple expedient of withdrawing the fluid by lumbar puncture brings about normal mental life. As the fluid again collects, the mental life becomes cloudy, and the character alters (irritability, depressed mood, changed purpose, lowered will); another lumbar puncture and presto!—the individual is for a time made over more completely than conversion changes a sinner,—and more easily.

Take the case of the disease known as General Paresis, officially called Dementia Paralytica. This disease is caused by syphilis and is one of its late results. The pathological changes are widespread throughout the brain but may at the onset be confined mostly to the frontal lobes. The very first change may be—and usually is—a change in character! The man hitherto kind and gentle becomes irritable, perhaps even brutal. One whose sex morals have been of the most conventional kind, a loyal husband, suddenly becomes a profligate, reckless and debauched, perhaps even perverted. The man of firm purposes and indefatigable industry may lose his grip upon the ambitions and strivings of his lifetime and become an inert slacker, to the amazement of his associates. Many a fine character, many a splendid mind, has reached a lofty height and then crumbled before the assaults of this disease upon the brain. Philosopher, poet, artist, statesman, captain of industry, handicraftsman, peasant, courtesan and housewife,—all are lowered to the same level of dementia and destroyed character by the consequences of the thickened meninges, the altered blood vessels and the injured nerve cells.

Now and then one is fortunate enough to treat with success an early case of General Paresis. And then the reversed miracle takes place, unfortunately too rarely! The disordered mind, the altered character, leaps upward to its old place,—after being dosed by the marvelous drug Salvarsan, created by the German Jewish scientist, Paul Ehrlich. Of extraordinary interest are the rare cases of loss of personal identity seen after brain injury, say in war. A man is knocked unconscious by a blow and upon restoration of consciousness is separated from that past in which his ego resides. He does not know his history or his name, and that continuity of the "self" so deeply prized and held by all religions to be part of his immortality is gone. Then after a little while, a few days or weeks, the disarranged neuronic pathways reestablish themselves as usual,—and the ego comes back to the man.

One might cite the feeble-mindedness that results from meningitis, brain tumor, brain abscess, brain wounds, etc., as further evidence of the dependence of mind upon brain, of its status as a function of brain. No philosopher seriously doubts that equilibrium and movement are functions of the brain, and yet to prove this there is no evidence of any other kind than that cited to prove the relationship of mind to brain. And what applies to the intelligence applies as forcibly to character, for purpose, emotion, mood, instinct and will are altered with these diseases. Interesting as is the relationship between mind and character and the brain, it is at the present overshadowed by the fascinating relationship between these psychical activities and the bodily organs. What I am about to cite from medicine and biology is part of the finest achievements of these sciences and hints at a future in which a true science of mind and character will appear.

Certain of the glands of the body are described as glands of internal secretions in that the products of their activity, their secretions, are poured into the blood stream rather than on the surface of the body or into the digestive tract. The most prominent of these glands, all of which are very small and extraordinarily active, are as follows: The Pituitary Body (Hypophysis)—a tiny structure which is situated at the base of the brain but is not a part of that organ.

The Pineal Body (Epiphysis)—a still smaller structure, located within the brain substance, having, however, no relationship to the brain. This gland has only lately acquired a significance. Descartes thought it the seat of the soul because it is situated in the middle of the brain. The Thyroid gland, a somewhat larger body, situated in the front of the neck, just beneath the larynx. We shall deal with this in some detail later on.

The Parathyroids, minute organs, four in number, just behind the thyroid. The Thymus, a gland placed just within the thorax, which reaches its maximum size at birth and then gradually recedes until at twenty it has almost disappeared. The Adrenal glands, one on each side of the body, above and adjacent to the kidney. These glands, which are each made up of two opposing structures, stand in intimate relation to the sympathetic nervous system and secrete a substance called adrenalin. The Sex organs, the ovary in the female and the testicle in the male, in addition to producing the female egg (ovum) and the male seed (sperm), respectively, produce substances of unknown character that have hugely important roles in the establishment of mind, temperament and sex character.

Without going into the details of the functions of the endocrine glands, one may say that they are "the managers of the human body." Every individual, from the time he is born until the time he dies, is under the influence of these many different kinds of elements,—some of them having to do with the development of the bones and teeth, some with the development of the body and nervous system, some with the development of the mind, etc. (and character), and later on with reproduction. These glands are not independent of one another but interact in a marvelous manner so that under or overaction of any one of them upsets a balance that exists between them, and thus produces a disorder that is quite generalized in its effects. The work on this subject is a tribute to medicine and one pauses in respect and admiration before the names and labors of Brown, Sequard, Addison, Graves and Basedow, Horsley, King, Schiff, Schafer, Takamine, Marie, Cushing, Kendal, Sajous and others of equal insight and patient endeavor.

But let us pass over to the specific instances that bear on our thesis, to wit, that mind and character are functions of the organism and have their seat not only in the brain but in the entire organism. How do the endocrines prove this? As well as they prove that physical growth and the growth of the secondary sex characters are dependent on these glands. Take diseases of the thyroid gland as the first and shining example. The thyroid secretes a substance which substantially is an “iodized globulin,”—and which can be separated from the gland products. This secretion has the main effect of “activating metabolism” (Vassale and Generali); in ordinary phrase it acts to increase the discharge of energy of the cells of the body. In all living things there is a twofold process constantly going on: first the building up of energy by means of the foodstuffs, air and water taken in, and second a discharge of energy in the form of heat, motion and—in my belief — emotion and thought itself, though this would be denied by many psychologists. Yet how escape this conclusion from the following facts?

There is a congenital disease called cretinism which essentially is due to a lack of thyroid secretion. This disease is particularly prevalent in Southern France, Spain, Upper Italy and Switzerland. It is characterized mainly by marked dwarfism and imbecility, so that the adult untreated cretin remains about as large as a three or four-year-old child and has the mental level about that of a child of the same age. But, this comparison as to intelligence is a gross injustice to the child, for it leaves out the difference in character between the child and the cretin. The latter has none of the curiosity, the seeking for experience, the active interest, the pliant expanding will, the sweet capacity for affection, friendship and love present in the average child. The cretin is a travesty on the human being in body, mind and character. But feed him thyroid gland. Mind you, the dried substance of the glands, not of human beings, but of mere sheep. The cretin begins to grow mentally and physically and loses to a large extent the grotesqueness of his appearance. He grows taller; his tongue no longer lolls in his mouth; the hair becomes finer, the hands less coarse, and the patient exhibits more normal human emotions, purposes, intelligence. True, he does not reach normality, but that is because other defects beside the thyroid defect exist and are not altered by the thyroid feeding.

There is a much more spectacular disease to be cited, —a relatively infrequent but well-understood condition called myxoedema, which occurs mainly in women and is also due to a deficiency in the thyroid secretion. As a result the patient, who may have been a bright, capable, energetic person, full of the eager purposes and emotions of life, gradually becomes dull, stupid, apathetic, without fear, anger, love, joy or sorrow, and without purpose or striving. In addition the body changes, the hair becomes coarse and scanty, the skin thick and swollen (hence the name of the disease) and various changes take place in the sweat secretion, the heart action, etc.

Then, having made the diagnosis, work the great miracle! Obtain the dried thyroid glands of the sheep, prepared by the great drug houses as a by-product of the butcher business, and feed this poor, transformed creature with these glands! No fairy waving a magical wand ever worked a greater enchantment, for with the first dose the patient improves and in a relatively short time is restored to normal in skin, hair, sweat, etc., and MIND and character! To every physician who has seen this happen under his own eyes and by his direction there comes a conviction that mind and character have their seat in the organic activities of the body,—and nowhere else.

An interesting confirmation of this is that when the thyroid is overactive, a condition called hyperthyroidism, the patient becomes very restless and thin, shows excessive emotionality, sleeplessness, has a rapid heart action, tremor and many other signs not necessary to detail here. The thyroid in these cases is usually swollen. One of the methods used to treat the disease is to remove some of the gland surgically. In the early days an operator would occasionally remove too, much gland and then the symptoms, of myxoedema would occur. This necessitated the artificial feeding of thyroid the rest of the patient's life! With the proper dosage of the gland substance the patient remains normal; with too little she becomes dull and stupid; with too much she becomes unstable and emotional! There are plenty of other examples of the influence of the endocrines on mind, character and personality. I here briefly mention a few of these.

In the disease called acromegaly, which is due to a change in the pituitary gland, amongst other things are noted "melancholic tendencies, loss of memory and mental and physical torpor." A very profound effect on character and personality, exclusive of intelligence, is that of the sex glands. One need not accept the Freudian extravagances regarding the way in which the sex feelings and impulses enter into our thoughts, emotions, purposes and acts. No unbiased observer of himself or his fellows but knows that the satisfaction or non-satisfaction of the sex feeling, its excitation or its suppression are of great importance in the destinies of character. Further, man as herdsman and man as tyrant have carried on huge experiments to show how necessary to normal character the sex glands are. As herdsman he has castrated his male Bos and obtained the ox. And the ox is the symbol of patience, docility, steady labour, without lust or passion,—and the very opposite of his non-castrated brother, the bull. The bull is the symbol of irritability and unteachableness, who will not be easily yoked or led and who is the incarnation of lust and passion. One is the male transformed into neuter gender; and the other is rampant with the fierceness of his sex.

Compare the eunuch and the normal man. If the eunuch state be imposed in infancy, the shape of the body, its hairiness, the quality of the voice and the character are altered in characteristic manner. The eunuch essentially is neither man nor woman, but a repelling Something intermediate.

Enough has been said to show that mind and character are dependent upon the health of the brain and the glands of the body; that somewhere in the interaction of tissues, in the chemistry of life, arises thought, purpose, emotion, conduct and deed. But we need not go so far afield as pathology to show this, for common experience demonstrates it as well.

If character is control of emotions, firmness of purpose, cheerfulness of outlook and vigour of thought and memory, then the tired man, worn out by work or a long vigil, is changed in character. Such a person in the majority of cases is irritable, showing lack of control and emotion; he slackens in his life's purposes, loses cheerfulness and outlook and finds it difficult to concentrate his thoughts or to recall his memories. Though this change is temporary and disappears with rest, the essential fact is not altered, namely, fatigue alters character.

It is also true that not all persons show this vulnerability to fatigue in equal measure. For that matter, neither do they show an equal liability to infectious diseases, equal reaction to alcohol or injury. The feeling of vigour which rest gives changes the expression of personality to a marked degree. It is true that we are not apt to think of the tired man as changed in character; yet we must admit on reflection that he has undergone transformation.

Even a loaded bowel may, as is well known, alter the reaction to life. Among men who are coarse in their language there is a salutation more pertinent than elegant that enquires into the state of the bowels. The famous story of Voltaire and the Englishman, in which the sage agreed to suicide because life was not worth living when his digestion was disordered and who broke his agreement when he purged himself, illustrates how closely mood is related to the intestinal tract. And mood is the background of the psychic life, upon which depends the direction of our thoughts, cheerful or otherwise, the vigour of our will and purpose. Mood itself arises in part from the influences that stream into the muscles, joints, heart, lungs, liver, spleen, kidneys, digestive tract and all the organs and tissues by way of the afferent nerves (sympathetic and cerebro-spinal). Mood is thus in part a reflection of the health and proper working of the organism; it is the most important aspect of the subconsciousness, and upon it rests the structure of character and personality.

This does not mean that only the healthy are cheerful, or that the sick are discouraged. To affirm the dependence of mind upon body is not to deny that one may build up faith, hope, courage, through example and precept, or that one may not inherit a cheerfulness and courage (or the reverse). "There are men," says James, "who are born under a cloud." But exceptional individuals aside, the mass of mankind generates its mood either in the tissues of the body or in the circumstances of life. Children, because they have not built up standards of thought, mood and act, demonstrate in a remarkable manner the dependence of their character upon health.

A child shows the onset of an illness by a complete change in character. I remember one sociable, amiable lad of two, rich in the curiosity and expanding friendliness of that time of life, who became sick with diphtheria. All his basic moods became altered, and all his wholesome reactions to life disappeared. He was cross and contrary, he had no interest in people or in things, he acted very much as do those patients in an insane hospital who suffer from Dementia Praecox.

What is character if it is not interest and curiosity, friendliness and love, obedience and trust, cheerfulness and courage? Yet a sick child, especially if very young, loses all these and takes on the reverse characters. The little lad spoken of became "himself" again when the fever and the pain lifted. Yet for a long time afterwards he showed a greater liability to fear than before, and it was not until six months or more had repaired the more subtle damage to his organism that he became the hardy little adventurer in life that he had been before the illness. There is plenty of chemical proof of this thesis as here set forth.

Men have from time immemorial put things "in their bellies to steal their brains away." The chemical substance known as ethyl alcohol has been an artificial basis of good fellowship the world over, as well as furnishing a very fair share of the tragedy, the misery and the humor of the world. This is because, when ingested in any amount, its absorption produces changes in the flow of thought, in the attitude towards life, in the mood, the emotions, the purposes, the conduct,—in a word, in character. One sees the austere man, when drunk, become ribald; the repressed, close-fisted become open-mouthed and open-hearted; the kindly, perhaps brutal; the controlled, uncontrolled. In the change of character it effects is the regret over its passing and the greatest reason for prohibition.

Alcohol causes several well-defined mental diseases as well as mere drunkenness. In Delirium Tremens there is an acute delirium, with confusion, excitement and auditory and visual hallucinations of all kinds. The latter symptom is so prominent as to give the reason for the popular name of the "snakes." In alcoholic hallucinosis the patient has delusions of persecution and hears voices accusing him of all kinds of wrong-doing. Very frequently, as all the medical writers note, these voices are "conscience exteriorized"; that is, the voices say of him just what he has been saying of himself in the struggle against drink. Then there is Alcoholic Paranoia, a disease in which the main change is a delusion of jealousy directed against the mate, who is accused of infidelity. It is interesting that in the last two diseases the patient is "clear-headed"; memory and orientation are good; the patient speaks well and gives no gross signs of his trouble. As the effects of the alcohol wear away, the patient recovers,—*i.e.*, his character returns to its normal.

It becomes necessary at this point to take up a reverse side of our study, namely, what is often called the influence of "mind over matter." Such cures of disease as seem to follow prayer and faith are cited; such incidents as the great strength of men under emotion or the disturbances of the body by ideas are listed as examples. This is not the place to discuss cures by faith. It suffices to say this: that in the first place most of such cures relate to hysteria, a disease we shall discuss later but which is characterized by symptoms that appear and disappear like magic. I have seen "cured" (and have "cured") such patients, affected with paralysis, deafness, dumbness, blindness, etc., with reasoning, electricity, bitter tonics, fake electrodes, hypnotism, and in one case by a forcible slap upon a prominent and naked part of the body. Hysteria has been the basis of many a saint's reputation and likewise has aided many a physician into affluence. Nor is the effect of coincidence taken into account in estimating cures, whether by faith or by drugs. Many a physician has owed his start to the fact that he was called in on some obscure case just when the patient was on the turn towards recovery. He then receives the credit that belonged to Nature.

Medical men understand this,—that many diseases are "self-limited" and pass through a cycle influenced but little by treatment. But faith curists do not so understand, and neither does the mass of people, so that neither one nor the other separates "post hoc" from "propter hoc." If the truth were told, most of the miracle and faith cures that are not of hysterical origin are due to coincidence. Faith curists report in detail their successes, but we have no statistics whatever of their failures. If thought is a product of the brain activated by the rest of the organism, it would be perfectly natural to expect that thought would influence the organism. That thought is intimately associated with impulses to action is well known. This action largely takes place in the speech muscles but also it irradiates into the rest of the organism. Especially is this true if the thought is associated with some emotion.

Emotion, as we shall discuss it later, is at least in large part a bodily reaction, a disturbance in heart, lungs, abdominal organs, blood vessels, sympathetic nervous system, endocrines, etc. The effect of thought and emotion upon the body, whether to heighten its activity or to lower its activity, is, from my point of view, merely the effect of one function of the organism upon others. We are not surprised if digestion affects thinking and mood, and we need not be surprised if thought and mood disturb or improve digestion. And we may substitute for digestion any other organic function.

As a working basis, substantiated by the kind of proof we use in our daily lives in laboratories and machine shops, we may state that mind, character and personality are organic in their origin and are functions of the entire organism. What a man thinks, does and feels (or perhaps we should reverse this order) is the result of environmental forces playing upon a marvelously intricate organism in which every part reacts on every other part, in which nervous energy influences digestion and digestion influences nervous energy, in which enzymes, hormones, and endocrines engage in an extraordinary game of checks and balance, which in the normal course of events make for the individual's welfare. What a man thinks, does, and feels influences the fate of his organism from one end of life to the other.

We have not adduced in favour of the organic nature of mind, character and personality the facts of heredity. This is a most important set of facts, for if the egg and the sperm carry mentality and personality, they may be presumed to carry them in some organic form, as organic potentialities, just as they carry size, colour, sex, etc. That abnormal mind is inherited is shown in family insanity in the second, third and fourth generation cases of mental disease. Certain types of feeble-mindedness surely are transmitted from generation to generation, as witness the case of the famous (or infamous) Jukes family. In this group vagabondage, crime, immorality and other character abnormalities appeared linked with the feeble-mindedness. But there is plenty of evidence to show that normal character qualities are inherited as well as the abnormal. Galton, the father of eugenics, collected facts from the history of successful families to prove this. It is true that he failed to take into account the facts of SOCIAL heredity, in that a gifted man establishes a place for himself and a tradition for his family that is of great help to his son. Nevertheless, musical ability runs in families and races, as does athletic ability, high temper, passion, etc. In short, at least the potentialities, the capacities for character, are transmitted together with other qualities as part of the capital of heredity.

This means that in studying character and personality, we must start with an analysis of the physical make-up of the individual. We are not yet at the point in science where we can easily get at the activities of the endocrinal glands in normal mentality. We are able to recognize certain fundamental types, but more we cannot do; nor are we able to measure nervous energy except in relatively crude ways, but these crude ways have great value under certain conditions. When there has been a change in personality, the question of bodily disease is always paramount. The first questions to be asked under such circumstances are, “Is this person sick?” “Is the brain involved?” “Are endocrinal glands involved?” “Is there disease of some organ of the body, acting to lower the feeling of well-being, acting to slacken the purposes and the will or to obscure the intelligence?”

INDIGENOUS EDUCATION, LEARNING, AND MOTIVATION

For many indigenous students (such as Native American children), motivation may be derived from social organization; an important factor educators should account for in addition to variations in Sociolinguistics and Cognition. While poor academic performance among Native American students is often attributed to low levels of motivation, Top-down classroom organization is often found to be ineffective for children of many cultures, who depend on a sense of community purpose and competence to effectively engage in material. Horizontally-structured, community-based learning strategies often provide a more structurally supportive environment for motivating indigenous children, who tend to be driven by "social/affective emphasis, harmony, holistic perspectives, expressive creativity, and non-verbal communication." This drive is also traceable to a cultural tradition of community-wide expectations of participation in the activities and goals of the greater group, rather than individualized aspirations of success or triumph. Structure for social learning in indigenous communities also often allows siblings to co-parent younger children in their acquisition of behaviours and traditions, which fosters the dynamic of community-motivated engagement from a young age. Furthermore, it is commonplace for children to assist and demonstrate for their younger counterparts without being prompted by authority figures. Observation techniques are demonstrated in such examples as weaving in Chiapas, Mexico, where it is commonplace for children to learn by "a more skilled other" within the community. The assumption of responsibility amongst children is also apparent within Mayan weaving apprenticeships; often, when the "more skilled other" is tasked with multiple obligations, an older child will step in and guide the learner.

Sibling guidance is supported from early youth, where learning through play encourages horizontally-structured environments through alternative educational models such as "Intent Community Participation." Research also suggests that that formal Westernized schooling can actually reshape the traditionally collaborative nature of social life in indigenous communities This research is supported cross-culturally, with variations in motivation and learning often reported higher between indigenous groups and their national Westernized counterparts than between indigenous groups across international continental divides.

SUDBURY MODEL SCHOOLS' APPROACH

Sudbury Model schools adduce that the cure to the problem of procrastination, of learning in general, and particularly of scientific illiteracy is to remove once and for all what they call the underlying disease: compulsion in schools. They contend that human nature in a free society recoils from every attempt to force it into a mold; that the more requirements we pile onto children at school, the surer we are to drive them away from the material we are trying to force down their throats; that after all the drive and motivation of infants to master the world around them is legendary. They assert that schools must keep that drive alive by doing what some of them do: nurturing it on the freedom it needs to thrive.

Sudbury Model schools do not perform and do not offer evaluations, assessments, transcripts, or recommendations, asserting that they do not rate people, and that school is not a judge; comparing students to each other, or to some standard that has been set is for them a violation of the student's right to privacy and to self-determination. Students decide for themselves how to measure their progress as self-starting learners as a process of self-evaluation: real lifelong learning and the proper educational evaluation for the 21st century, they adduce.

According to Sudbury Model schools, this policy does not cause harm to their students as they move on to life outside the school. However, they admit it makes the process more difficult, but that such hardship is part of the students learning to make their own way, set their own standards and meet their own goals. The no-grading and no-rating policy helps to create an atmosphere free of competition among students or battles for adult approval, and encourages a positive cooperative environment amongst the student body.

BUSINESS

At lower levels of Maslow's hierarchy of needs, such as physiological needs, money is a motivator, however it tends to have a motivating effect on staff that lasts only for a short period (in accordance with Herzberg's two-factor model of motivation). At higher levels of the hierarchy, praise, respect, recognition, empowerment and a sense of belonging are far more powerful motivators than money, as both Abraham Maslow's theory of motivation and Douglas McGregor's theory X and theory Y (pertaining to the theory of leadership) demonstrate.

According to Maslow, people are motivated by unsatisfied needs. The lower level needs such as Physiological and Safety needs will have to be satisfied before higher level needs are to be addressed. We can relate Maslow's Hierarchy of Needs theory with employee motivation. For example, if a manager is trying to motivate his employees by satisfying their needs; according to Maslow, he should try to satisfy the lower level needs before he tries to satisfy the upper level needs or the employees will not be motivated. Also he has to remember that not everyone will be satisfied by the same needs. A good manager will try to figure out which levels of needs are active for a certain individual or employee.

Maslow has money at the lowest level of the hierarchy and shows other needs are better motivators to staff. McGregor places money in his Theory X category and feels it is a poor motivator.

Praise and recognition are placed in the Theory Y category and are considered stronger motivators than money.

- Motivated employees always look for better ways to do a job.
- Motivated employees are more quality oriented.
- Motivated workers are more productive.

The average workplace is about midway between the extremes of high threat and high opportunity. Motivation by threat is a dead-end strategy, and naturally staff are more attracted to the opportunity side of the motivation curve than the threat side. Motivation is a powerful tool in the work environment that can lead to employees working at their most efficient levels of production. Nonetheless, Steinmetz also discusses three common character types of subordinates: ascendant, indifferent, and ambivalent who all react and interact uniquely, and must be treated, managed, and motivated accordingly. An effective leader must understand how to manage all characters, and more importantly the manager must utilize avenues that allow room for employees to work, grow, and find answers independently.

The assumptions of Maslow and Herzberg were challenged by a classic study at Vauxhall Motors' UK manufacturing plant. This introduced the concept of orientation to work and distinguished three main orientations: instrumental (where work is a means to an end), bureaucratic (where work is a source of status, security and immediate reward) and solidaristic (which prioritizes group loyalty). Other theories which expanded and extended those of Maslow and Herzberg included Kurt Lewin's Force Field Theory, Edwin Locke's Goal Theory and Victor Vroom's Expectancy theory. These tend to stress cultural differences and the fact that individuals tend to be motivated by different factors at different times.

According to the system of scientific management developed by Frederick Winslow Taylor, a worker's motivation is solely determined by pay, and therefore management need not consider psychological or social aspects of work. In essence, scientific management bases human motivation wholly on extrinsic rewards and discards the idea of intrinsic rewards.

In contrast, David McClelland believed that workers could not be motivated by the mere need for money-in fact, extrinsic motivation (*e.g.*, money) could extinguish intrinsic motivation such as achievement motivation, though money could be used as an indicator of success for various motives, *e.g.*, keeping score. In keeping with this view, his consulting firm, McBer and Company, had as its first motto "To make everyone productive, happy, and free." For McClelland, satisfaction lay in aligning a person's life with their fundamental motivations.

Elton Mayo found that the social contacts a worker has at the workplace are very important and that boredom and repetitiveness of tasks lead to reduced motivation. Mayo believed that workers could be motivated by acknowledging their social needs and making them feel important. As a result, employees were given freedom to make decisions on the job and greater attention was paid to informal work groups. Mayo named the model the Hawthorne effect. His model has been judged as placing undue reliance on social contacts at work situations for motivating employees.

William Ouchi introduced Theory Z, a hybrid management approach consisting of both Japanese and American philosophies and cultures. Its Japanese segment is much like the clan culture where organizations focus on a standardized structure with heavy emphasis on socialization of its members. All underlying goals are consistent across the organization.

Its American segment retains formality and authority amongst members and the organization. Ultimately, Theory Z promotes common structure and commitment to the organization, as well as constant improvement of work efficacy. In Essentials of Organizational Behaviour, Robbins and Judge examine recognition programmes as motivators, and identify five principles that contribute to the success of an employee incentive programme:

- Recognition of employees' individual differences, and clear identification of behaviour deemed worthy of recognition
- Allowing employees to participate
- Linking rewards to performance
- Rewarding of nominators
- Visibility of the recognition process

CONTROLLING MOTIVATION

The control of motivation is only understood to a limited extent. There are many different approaches of motivation training, but many of these are considered pseudoscientific by critics. To understand how to control motivation it is first necessary to understand why many people lack motivation.

EMPLOYEE MOTIVATION

Workers in any organization need something to keep them working. Most of the time, the salary of the employee is enough to keep him or her working for an organization. An employee must be motivated to work for a company or organization. If no motivation is present in an employee, then that employee's quality of work or all work in general will deteriorate. People differ on a personality dimension called locus of control. This variable refers to individual's beliefs about the location of the factors that control their behaviour.

At one end of the continuum are high internals who believe that opportunity to control their own behaviour rests within themselves. At the other end of the continuum there are high externals who believe that external forces determine their behaviour. Not surprisingly, compared with internals, externals see the world as an unpredictable, chancy place in which luck, fate, or powerful people carol their destinies. When motivating an audience, you can use general motivational strategies or specific motivational appeals. General motivational strategies include soft sell versus hard sell and personality type. Soft sell strategies have logical appeals, emotional appeals, advice and praise.

Hard sell strategies have barter, outnumbering, pressure and rank. Also, you can consider basing your strategy on your audience personality. Specific motivational appeals focus on provable facts, feelings, right and wrong, audience rewards and audience threats.

Job Characteristics Model

The Job Characteristics Model (JCM), as designed by Hackman and Oldham attempts to use job design to improve employee motivation. They have identified that any job can be described in terms of five key job characteristics;

1 Skill Variety - the degree to which a job requires different skills and talents to complete a number of different activities
2 Task Identity - this dimension refers to the completion of a whole and identifiable piece of work versus a partial task as part of a larger piece of work
3 Task Significance - is the impact of the task upon the lives or work of others
4 Autonomy - is the degree of independence or freedom allowed to complete a job
5 Task Feedback - individually obtaining direct and clear feedback about the effectiveness of the individual carrying out the work activities

The JCM links these core job dimensions to critical psychological states which results in desired personal and work outcomes. This forms the basis of this 'employee growth-need strength." The core dimensions listed above can be combined into a single predictive index, called the Motivating Potential Score.

MOTIVATING POTENTIAL SCORE

The motivating potential score (MPS) can be calculated, using the core dimensions discussed above, as follows;

$$MPS = \frac{Skill\,Variety + Task\,Identity + Task\,Significance}{3} \times Autonomy \times Feedback$$

Jobs that are high in motivating potential must be high on at least one of the three factors that lead to experienced meaningfulness, and also must be high on both Autonomy and Feedback. If a job has a high MPS, the job characteristics model predicts that motivation, performance and job satisfaction will be positively affected and the likelihood of negative outcomes, such as absenteeism and turnover, will be reduced.

DRUGS

Some authors, especially in the transhumanist movement, have suggested the use of "smart drugs", also known as nootropics, as "motivation-enhancers". These drugs work in various ways to affect neurotransmitters in the brain. It is generally widely accepted that these drugs enhance cognitive functions, but not without potential side effects. The effects of many of these drugs on the brain are emphatically not well understood, and their legal status often makes open experimentation difficult.

PSYCHOLOGY OF MOTIVATION FOR ATHLETES

What is motivation? What motivates us, me, and you? And I don’t just mean the basic definitions, but the science of how motivation works and the psychology behind motivation. In this article, we will go into what motivates us in life and in sports.

DEFINED: MOTIVATION

Motivation is broadly defined as all factors that cause humans and other animals to behave the way they do. Scientists believe the hypothalamus provides the physical basis for pleasure in humans, which plays a large part in motivation.

The Hypothalamus and Limbic System

Hypothalamus means “below the thalamus.” The hypothalamus is involved in controlling certain body states including hunger, thirst, circadian rhythms, fluctuation of hormone release throughout the day, aspects of sexual activity, and body temperature. The hypothalamus is part of the limbic system.

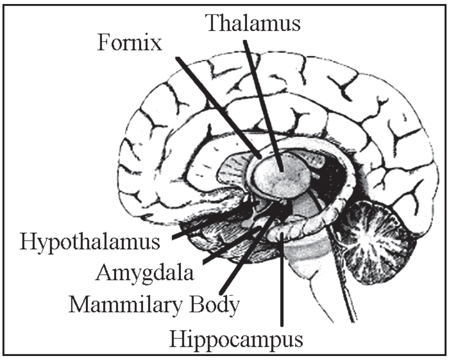


Fig. Limbic system structures.

The limbic system has connections to what motivates us based off memories and other stimuli. The system consists of the following structures: amygdala, fornix, hippocampus, thalamus, hypothalamus, and cingulate gyrus. Experience of emotions and the interpretation of emotionally laden events or stimuli appear as a result of these areas being stimulated and their communications with one another. In addition, the limbic system may be important in our overall consciousness.

Achievement-Based Motivation

Achieving goals can become a powerful motivator in a person's life. Some people live for reaching their goals. The need for achievement is their motivation to accomplish a challenging task quickly and effectively.

David McClelland, his students, and colleagues spent forty years studying this type of motivation. They analysed the personalities of those who demonstrated this need for achievement in entire societies. Those with a high need of achievement tend to work harder than those without it, are more future oriented, and will delay gratification longer. An example of achievement-based motivation comes from research done on medical students. A study using focus group discussions with ten male and nine female students who scored higher than an 85 per cent on all their tests was used on this study. The results established that the students were dedicated to going to lectures, prioritization, self-learning, learning in groups, mind-mapping, learning in skill labs, learning from mistakes, time management, and having family support.

Non-academic factors such as sleep deprivation and homesickness also played a role in academic success. Achievement-based people tend to make sacrifices and make their goal their "life," so they work around the things that may not be related to their sport, academia, or whatever they are motivated to do. They spend their time learning how to get better and be effective. Achievement makes some people content and happy. On the other hand, failure can produce pain, sadness, and unhappiness, which can lead to avoidance behaviours. In athletes we typically find two types of successful motivations:

Intrinsic Motivation: Defined as a construct and desire to be competent and self-determining. These athletes are usually self-starters because of their love of the game. Intrinsically motivated athletes are more likely to maintain effort and consistency across practice and competition. Achievement Motivation: These athletes wish to engage in competition or social comparison. All things being equal between two athletes, whoever has the higher achievement motivation will be the better athlete because of the desire for competition.

COACHES: UNDERSTANDING MOTIVATION

The understanding and utilization of different types of reinforcement and punishment will help coaches motivate their athletes. As far as which type of motivation is used, that depends on the coach's ability to understand the athlete, his or her emotions, and his or her personality.

Reinforcement

Positive reinforcement is the act of increasing the probability of occurrence of a given behaviour (a target behaviour such as correct footwork), which is termed the operant, by following it with an action, object, or event such as praise.

Negative Reinforcement also increases the probability of occurrence of a given operant by removing an act, object, or event that is typically aversive. For example, if you have a CrossFit competition team you want to get to regionals and they do well on the clean and jerk, you don't add additions to their WOD on that day (such as a miserable cash out like 100 double unders or 100 pull ups).

Punishment

Punishment is designed to decrease the occurrence of a given operant, that is, negative behaviours such as mistakes or lack of effort. Positive punishment is the presentation of an act, object, or event following a behaviour that could decrease the behaviour occurrence. For example, some gyms reprimand their athletes for dropping barbells with no weight on them from overhead because it prevents thisbehaviour from becoming habitual (not to mention preserves the equipment).

Negative punishment is the removal of something valued, which can take the form of revoking privileges or playing time. Coaches will use both forms of motivation, but the positive approach is arguably better because it focuses on what athletes should do and what they are doing right. Reinforcement increases task-relevant focus rather than worry focus.

A task-relevant focus facilitates reaction time and decision time. A successful experience colours the athlete's view as positive, which can lead to approach behaviours. Approach behaviours or approach motivation indicates the propensity to move towards a desired stimulus.

Prior research has already shown that positive affect (or positive motivation/ reinforcement) promotes cognitive flexibility. In a study published in *Psychological Science*, they extended motivational dimensional model to the domain of cognitive control by examining both low-and high-motivated positive affect on the balance between cognitive flexibility (our ability to adjust to behaviour in response to a changing environment) and cognitive stability (our ability to change behaviour in the face of distraction). Low and high approach-motivated positive affect would indicate the intensity of the positive affect on a selected individual in regard to approach motivation. Results concluded low approach-motivated positive affect promoted cognitive flexibility but also caused higher distractibility, whereas high approach-motivated positive affect enhanced perseverance but simultaneously reduced distractibility.

THE THEORIES OF MOTIVATION IN SPORTS

Fig. Tailor your motivational techniques to each athlete's personality.

Athletes participate in sports for various reasons, from a hunger for physical activity and competition to the joy of belonging to a team. Coaches can improve the team's performance by finding the right motivation for each situation and player.

Specific motivational theories exist that apply psychological concepts to sports for increased drive and performance.

EXTRINSIC MOTIVATION

Extrinsic motivation is motivation that comes from an outside source. Some of it is tangible, such as financial or other material rewards, including trophies or medals. Tangible extrinsic motivation is not necessarily ideal for athletes who become too focused on materialism at the expense of other aspects of sports. Intangible extrinsic motivation includes praise, recognition and achievement, which can often be enough to motivate athletes.

INTRINSIC MOTIVATION

Intrinsic motivation comes from within the athlete or player. It includes a natural desire to overcome challenges and enjoyment in the repetition of a skill. These factors can remind athletes why they participate in a certain sport—especially during grueling practices. Intrinsic motivation is often best supported by a series of goals, whether they're enhanced skill sets or victories in competition.

THEORY OF VITALITY

The theory of vitality dictates that vitality influences the future capacity for performance. An athlete has a baseline vitality with which to work and won't stray far from that point. Actions or effects affect that vitality and either thwart or satisfy the player's needs. For example, if a player is extrinsically motivated and praise isn't forthcoming, the player's vitality sinks and he loses motivation. Similarly, if a player loves a game and keeps winning at it, her intrinsic enjoyment is satisfied, her vitality rises and she is motivated to continue.

SANDWICH THEORY

The sandwich theory motivates athletes to correct or improve without destroying their sense of enjoyment, pride or inclusion as an equal team member. You can use this theory on yourself by noticing your positive contributions to your team, too. When crafting criticism, sandwich the need between positive reinforcement. Doing so motivates athletes to put forth the necessary effort for improvement because their larger extrinsic or intrinsic needs are being met.

A MOTIVATION

A motivation occurs when players lack motivation, which happens for a few reasons. Sometimes, the player has no sense of capacity and truly doesn't believe he is capable of performing the way that's required. Other times, the player doesn't understand the connection between the actions required and the desired outcome. In these instances, coaches and trainers can build self-esteem by carefully building skill sets. Another solution is conditioning athletes to understand how their improvements in technique can benefit their overall performance or their team.

8

Health Education

Health Education comprises consciously constructed opportunities for learning involving some form of communication designed to improve health literacy, including improving knowledge, and developing life skills which are conducive to individual and community health. Health education is not only concerned with the communication of information, but also with fostering the motivation, skills and confidence (self-efficacy) necessary to take action to improve health. Health education includes the communication of information concerning the underlying social, economic and environmental conditions impacting on health, as well as individual risk factors and risk behaviors, and use of the health care system. Thus, health education may involve the communication of information, and development of skills which demonstrates the political feasibility and organizational possibilities of various forms of action to address social, economic and environmental determinants of health. In the past, health education was used as a term to encompass a wider range of actions including social mobilization and advocacy. These methods are now encompassed in the term health promotion, and a more narrow definition of health education is proposed here to emphasize the distinction.

THE ROLE OF THE HEALTH EDUCATOR

From the late nineteenth to the mid-twentieth century, the aim of public health was controlling the harm from infectious diseases, which were largely under control by the 1950s. By the mid 1970s it was clear that reducing illness, death, and rising health care costs could best be achieved through a focus on health promotion and disease prevention.

At the heart of the new approach was the role of a health educator A health educator is "a professionally prepared individual who serves in a variety of roles and is specifically trained to use appropriate educational strategies and methods to facilitate the development of policies, procedures, interventions, and systems conducive to the health of individuals, groups, and communities".

In January 1979 the Role Delineation Project was put into place, in order to define the basic roles and responsibilities for the health educator. The result was a Framework for the Development of Competency-Based Curricula for Entry Level Health Educators (NCHEC, 1985). A second result was a revised version of A Competency-Based Framework for the Professional Development of Certified Health Education Specialists (NCHEC,1996). These documents outlined the seven areas of responsibilities which are shown below.

Responsibility I: Assessing Individual and Community Needs for Health Education

- Provides the foundation for programme planning
- Determines what health problems might exist in any given group
- Includes determination of community resources available to address the problem
- Community Empowerment encourages the population to take ownership of their health problems
- Includes careful data collection and analysis.

Responsibility II: Plan Health Education Strategies, Interventions, and Programmes:

- Actions are based on the needs assessment done for the community
- Involves the development of goals and objectives which are specific and measurable
- Interventions are developed that will meet the goals and objectives
- According to Rule of Sufficiency, strategies are implemented which are sufficiently robust, effective enough, and have a reasonable chance of meeting stated objectives.

Responsibility III: Implement Health Education Strategies, Interventions, and Programmes:

- Implementation is based on a thorough understanding of the priority population
- Utilize a wide range of educational methods and techniques.

Responsibility IV: Conduct Evaluation and Research Related to Health Education:

- Depending on the setting, utilize tests, surveys, observations, tracking epidemiological data, or other methods of data collection
- Health Educators make use of research to improve their practice

Responsibility V: Administer Health Education Strategies, Interventions, and Programmes:

- Administration is generally a function of the more experienced practitioner
- Involves facilitating cooperation among personnel, both within and between programmes.

Responsibility VI: Serve as a Health Education Resource Person:
- Involves skills to access needed resources, and establish effective consultive relationships

Responsibility VII: Communicate and Advocate for Health and Health Education:
- Translates scientific language into understandable information
- Address diverse audience in diverse settings
- Formulates and support rules, policies and legislation
- Advocate for the profession of health education.

MOTIVATION

Education for health begins with people. It hopes to motivate them with whatever interests they may have in improving their living conditions. Its aim is to develop in them a sense of responsibility for health conditions for themselves as individuals, as members of families, and as communities. In communicable disease control, health education commonly includes an appraisal of what is known by a population about a disease, an assessment of habits and attitudes of the people as they relate to spread and frequency of the disease, and the presentation of specific means to remedy observed deficiencies.

Health education is also an effective tool that helps improve health in developing nations. It not only teaches prevention and basic health knowledge but also conditions ideas that re-shape everyday habits of people with unhealthy lifestyles in developing countries. This type of conditioning not only affects the immediate recipients of such education but also future generations will benefit from an improved and properly cultivated ideas about health that will eventually be ingrained with widely spread health education. Moreover, besides physical health prevention, health education can also provide more aid and help people deal healthier with situations of extreme stress, anxiety, depression or other emotional disturbances to lessen the impact of these sorts of mental and emotional constituents, which can consequently lead to detrimental physical effects.

CREDENTIALING

Credentialing is the process by which the qualifications of licensed professionals, organizational members or an organization are determined by assessing the individuals or group background and legitimacy through a standardized process. Accreditation, licensure, or certifications are all forms of credentialing.

In 1978, Helen Cleary, the president of the Society for Public Health Education (SOPHE) started the process of certification of health educators. Prior to this, there was no certification for individual health educators, with exception to the licensing for school health educators. The only accreditation available in this field was for school health and public health professional preparation programmes.

Her initial response was to incorporate experts in the field and to promote funding for the process. The director if the Division of Associated Health Professions in the Bureau of Health Manpower of the Department of Health, Education, and Welfare, Thomas Hatch, became interested in the project. To ensure that the commonalities between health educators across the spectrum of professions would be sufficient enough to create a set of standards, Dr. Cleary spent a great amount of time to create the first conference called the Bethesda Conference. In attendance were interested professionals who covered the possibility of creating credentialing within the profession.

With the success of the conference and the consensus that the standardization of the profession was vital, those who organized the conference created the National Task Force in the Preparation and Practice of Health Educators. Funding for this endeavor became available in January 1979, and role delineation became a realistic vision for the future. They presented the framework for the system in 1981 and published entry-level criteria in 1983. Seven areas of responsibility, 29 areas of competency and 79 sub-competencies were required of health education professionals for approximately 20 years for entry-level educators.

In 1986 a second conference was held in Bethesda, Maryland to further the credentialing process. In June 1988, the National Task Force in the Preparation and Practice of Health Educators became the National Commission for Health Education Credentialing, Inc. (NCHEC). Their mission was to improve development of the field by promoting, preparing and certifying health education specialists. The NCHEC has three division boards that included preparation, professional development and certification of health educator professionals. The third board, which is called the Division Board of Certification of Health Education Specialist (DBCHES), has the responsibility of developing and administering the CHES exam. An initial certification process allowed 1,558 individuals to be chartered into the programme through a recommendation and application process. The first exam was given in 1990.

In order for a candidate to sit for an exam they must have either a bachelor's, master's, or doctoral degree from and accredited institution, and an official transcript that shows a major in health education, Community Health Education, Public Health Education, or School Health Education, etc. The transcript will be accepted if it reflects 25 semester hours or 37 quarter hours in health education preparation and covers the 7 responsibilities covered in the framework.

In 1998 a project called the Competencies Update Project (CUP) began. The purpose of the CUP project was to up-date entry-level requirements and to develop advanced-level competences. Through research the CUP project created the requirements for three levels, which included entry-level, Advanced I and Advanced II educators.

Recently the Master Certified Health Education Specialist (MCHES) is in the process of being created. It is an exam that will measure the knowledge of the advanced levels and sub levels of the Seven Areas of Responsibilities. The first MCHES exam is expected to be given in October 2011. In order to be eligible to take the MCHES exam you must have at least a Master's degree in health education or related discipline along with a least 25 credit hours related to health education. In addition, five years of documented information of practice in health education and two recommendations of past/present supervisors must be provided. A vitae/resume must also be submitted.

The Competency Update Project (CUP), 1998-2004 revealed that there were higher levels of health education practitioners, which is the reasoning for the advancements for the MCHES. Many health educators felt that the current CHES credential was an entry-level exam. There will be exceptions made for those who have the Certification of Health Education Specialist, that have been active for several consecutive years. They will be required to participate in the MCHES Experience Documentation Opportunity that will omit them from taking the exam.

TEACHING

In the United States some forty states require the teaching of health education. A comprehensive health education curriculum consists of planned learning experiences which will help students achieve desirable attitudes and practices related to critical health issues. Some of these are: emotional health and a positive self image; appreciation, respect for, and care of the human body and its vital organs; physical fitness; health issues of alcohol, tobacco, drug use and abuse; health misconceptions and myths; effects of exercise on the body systems and on general well being; nutrition and weight control; sexual relationships and sexuality, the scientific, social, and economic aspects of community and ecological health; communicable and degenerative diseases including sexually transmitted diseases; disaster preparedness; safety and driver education; factors in the environment and how those factors affect an individual's or population's Environmental health (ex: air quality, water quality, food sanitation); life skills; choosing professional medical and health services; and choices of health careers.

NATURE AND SCOPE OF PRESENT TRENDS IN IT AND HEALTH

Health care needs cannot be met in most countries for the vast majority of the world's population from within the prevailing structures of resource allocation. There are several potential sources of conflict among countries for making and observing rules of international trade in health care goods and services. However, promoting trade and protecting health need not be viewed as incompatible goals. Shared commitments on the use of global health care resources can be a powerful uniting force in the world. Consensus is constrained by glaring disparities in different regions of the world on quality and reliability of basic infrastructure, health care resources, differences in exposure to disease and disease burdens, shares of world trade in health services and inadequate awareness, that adversity anywhere affects prosperity everywhere as amply demonstrated during the SARS epidemic. The enthusiasm for global convergence on health care trade is dampened by differences in national regulations and comes bundled with efficiency, cost and equity considerations.

Information and communication technologies (IT) could revolutionise health care. With the right policy choices, IT is able to promote new thresholds of human connectivity and is a powerful tool of global convergence through cross-border supply of services. Firstly, IT enables new opportunities for production of knowledge, the only factor of production not subject to the economic laws of diminishing returns, and to trade in it for direct economic benefits. Secondly, international diffusion of new knowledge and best practice opens new ways of improving the performance of health systems. This paper focuses its Enquiry into health care problems and solutions impacted by IT with a view to help evaluate policy choices. To put this another way: What difference can IT make to worldwide production and trade in health care commodities and services? Can IT's role in designs of health care trade raise revenues and improve health care in developing countries? Will commercial interests claim priority over people's health?

The most significant feature of IT is its capacity to enable reliable storage, retrieval and instantaneous transfer of text, sounds, images and numbers as audio, visual and data communications. This substitutes and supplements face-to-face contact and transportation. Due to this, spatially distant resources and needs connect easily and add new links to productive capacity in all aspects of health care-development, delivery and administration. When this productive capacity is recognised as a shareable and exchangeable resource to be accounted for, IT products and processes acquire a currency akin to money. Producers of health care products and services – regardless of location – can design the outcome of their efforts to be combinable with each other's efforts and their contributions can also be electronically traded.

In this manner, IT reinforces incentives for trade and investment and for sharing and distribution of work in ways, for purposes, and with consequences that cannot fully be envisaged in advance. Thus, IT raises hopes but it also triggers anxieties over how world health resources and architectural dimensions of cross-border transactions of the international economy would enmesh with each other.

Cross-border supply is the least developed mode of health care services trade whereas the movement of patients across borders, foreign commercial presence and the movement of natural persons have hitherto been, in that order, the three more important ways of services supply in the sector. IT was first introduced in health care at medical facilities and in medical systems at a time when efficiency at the delivery-end mattered the most. Trade in IT-assisted health care (health resources that use, apply or are facilitated by IT) and in health care-related IT (IT that arises from, in or for health care commodities and services including powerful diagnostics and investigative tools) expanded rapidly in the 1990s. Much of this expansion occurred in health care development and health care administration pointing to the need to systematically explore the two-way linkage between IT and health care. However, intangible and invisible services are not easily tracked because (a) they usually come bundled with products, and, (b) due to differences in national accounting conventions.

The linkages remain under-researched at a time when both IT and health sectors are in the process of restructuring to converge in important respect. The role of information technology in health care is not well documented and several important questions arise which are not easily answered.

- Under what conditions does IT diffusion promote health, trade and development?
- How may policies ensure that higher health care costs attributable to IT are commensurate with benefits ?
- Would trade based on IT diffusion for health care development in business-to-business mode and for health care delivery to patients be efficient and equitable in developing countries ?
- Does IT diffusion in health care mitigate disease burdens by reductions in cost per diagnosis or cost per treated/cured illness?
- What lessons can be learnt from the experience of uses and abuses of IT in health care in developed and developing countries?

NEW RISKS AND VULNERABILITIES REQUIRE NEW POLICY SAFEGUARDS

This section is a modest attempt to fill some of this void with a special emphasis on research and policy issues of particular relevance to less developed countries.

IT touches all aspects of health care and at many different points of contact. However, this paper limits its scope to six of the most significant issues arising from IT's role in health care trade. These are

- IT's role in design of health care products and services
- IT's impact on design of health care systems
- Information power effects
- IT's facilitating role in strengthening public-private partnerships
- IT-induced vulnerabilities and IT-aggravated risks, and
- Need for global governance.

In the post-genome era, the search for new therapies based on the study of genes and proteins linked to diseases has caused an explosion in genetic data. IT-enabled tools are indispensable for testing large portions of DNA and protein quickly. The new requirements of speed, economies of scale, standards, simulation and security have rapidly transformed health care into an information-based science driven by IT which is global in scope. New medicines and treatments develop faster and better with cross-border sharing of work across time-zones and simultaneous clinical testing in trials covering a wider geographical area-all of which is IT enabled. Trade in genomic databases involves high stakes ever since the human genome was decoded. The correspondence between the capacity of IT and the economic use of IT lies at the heart of the matter since information easily loses meaning, validity, relevance and the taste for it changes rapidly. It is noteworthy that even at the cutting edge of IT in health care development in genomics, leading firms like Celera could not sustain a business model based on priced information through online subscriber services for genome databases and were forced to diversify into more value-adding services.

In 2001, the size of IT-enabled health care services was about $3.1 billion, of which about 80 per cent were in developed countries. On-line consultations by patients and doctors through web sites and e-mail, distance referrals, emergency evacuations, and advance transmission of images and data of patients from ambulances can reduce lead times of intervention in emergency wards of hospitals. Treatments administered in a shared mode, distance surveillance of convalescing patients, risk assessments by insurers, distance consultations on symptoms revealed by patients and observations noted by doctors with the aid of transferable data from image scanning machines, processing of medico-legal documents, and cross-border networking for TeleEducation are some of the ways IT's contribution to new products and services is expanding rapidly.

In less developed countries, such innovations are mainly offered at urban locations in the quality conscious segment of private health care. Some aspects of TeleHealth such as TelePathology (requiring special cameras to digitalise specimen slides) and TelePsychiatry (requiring two-way interactive video conferencing) would remain limited in the near future, to developed country locations where sophisticated equipment can be maintained and its costs afforded. Applications like TeleEducation for medical degrees of renowned universities in partnership with local institutions and for continuing medical education of professionals in hospitals, medical colleges and health care centres have great demand in developing countries. Health care delivery is not the most significant cross-border trade contribution of IT's role, constrained as it is by lack of widespread harmonisation of medical curricula and qualifications, approved drug lists, benchmarked standards, market access barriers such as economic needs tests (ENTs) and portability of health care insurance. Cross-border trade has been left "unbound" (*i.e.* countries are free to regulate entry of foreign services) by World Trade Organisation (WTO) members in commitments under the General Agreement on Trade in Services (GATS). Outsourced health care administration made up of backroom operations including medical transcriptions, invoicing, collections, purchases and inventory management, accounting, payroll management, continuing education networks linking hospitals and medical schools, surveys, and research networks presently are a bigger segment of domestic business and trade where IT's role is significant and growing.

These opportunities represent potential services exports not only among developed countries (for example between Norway and Sweden for prenatal chekups and nursing care in Lapland or between Australia and Canada for TeleConsultations among physicians), but also from developed to developing countries (for instance, Japan to Thailand, U.S.A to Mexico, U.K. to Kenya, France to Ivory Coast and Singapore to Indonesia for TeleDiagnostics and TeleRadiology), from developing countries to developed countries (for example, Srilanka/India to U.K., U.S.A, Australia and New Zealand for health care administration services which could in future be extended to TeleDiagnostics when professional qualifications are mutually recognised or harmonised) and among developing countries (Bhutan, Thailand, Argentina, Nigeria, India, Brazil, China, Nepal, Bangla Desh, Tanzania for a range of telemedicine applications). Improved career prospects of doctors in short supply in developing countries enabled to export their services without movement as natural persons or foreign commercial presence would reduce the brain drain

IT has enabled doctors and paramedics to use diagnostic packages and decision support tools based on the system of international classification of diseases (latest is ICD- 10). Diagnostic measures on ICD-10 were among the first services to be e-commercialised in Europe. In Japan and Finland, early warning signalling of infectious disease epidemics is done from patterns of database searches of physicians trying to diagnose from symptoms at the time of the outbreak of disease. This kind of IT application could be transported to any developing country or rendered in a distance mode enabling quicker intervention when epidemics break out although safeguards would be needed to ensure it does not trigger panics or false alarms.

HEALTH EDUCATION CODE OF ETHICS

The Health Education Code of Ethics has been a work in progress since approximately 1976, begun by the Society of Public Health Education (SOPHE). Various Public Health and Health Education organizations such as the American Association of Health Education (AAHE), the Coalition of National Health Education Organizations (CNHEO), SOPHE, and others collaborated year after year to devise a unified standard of ethics that health educators would be held accountable to professionally. In 1995, the National Commission for Health Education Credentialing, Inc. (NCHEC) proposed a profession-wide standard at the conference: Health Education Profession in the Twenty-First Century: Setting the Stage. Post-conference, an ethics task force was developed with the purpose of solidifying and unifying proposed ethical standards. The document was eventually unanimously approved and ratified by all involved organizations in November 1999 and has since then been used as the standard for practicing health educators.

“The Code of Ethics that has evolved from this long and arduous process is not seen as a completed project. Rather, it is envisioned as a living document that will continue to evolve as the practice of Health Education changes to meet the challenges of the new millennium.”

HEALTH EDUCATION CODE OF ETHICS FULL TEXT

PREAMBLE The Health Education profession is dedicated to excellence in the practice of promoting individual, family, organizational, and community health. The Code of Ethics provides a framework of shared values within which Health Education is practiced.

The responsibility of each Health Educator is to aspire to the highest possible standards of conduct and to encourage the ethical behavior of all those with whom they work.

Article I: Responsibility to the Public A Health Educator's ultimate responsibility is to educate people for the purpose of promoting, maintaining, and improving individual, family, and community health. When a conflict of issues arises among individuals, groups, organizations, agencies, or institutions, health educators must consider all issues and give priority to those that promote wellness and quality of living through principles of self-determination and freedom of choice for the individual.

Article II: Responsibility to the Profession Health Educators are responsible for their professional behavior, for the reputation of their profession, and for promoting ethical conduct among their colleagues.

Article III: Responsibility to Employers Health Educators recognize the boundaries of their professional competence and are accountable for their professional activities and actions.

Article IV: Responsibility in the Delivery of Health Education Health Educators promote integrity in the delivery of health education. They respect the rights, dignity, confidentiality, and worth of all people by adapting strategies and methods to the needs of diverse populations and communities.

Article V: Responsibility in Research and Evaluation Health Educators contribute to the health of the population and to the profession through research and evaluation activities. When planning and conducting research or evaluation, health educators do so in accordance with federal and state laws and regulations, organizational and institutional policies, and professional standards.

Article VI: Responsibility in Professional Preparation Those involved in the preparation and training of Health Educators have an obligation to accord learners the same respect and treatment given other groups by providing quality education that benefits the profession and the public.

IMPLICATIONS FOR SCHOOL HEALTH PSYCHOLOGY

The concept of service integration has several implications for psychology as a profession and psychologists as health-service providers in schools. Implications for psychological training, practice, research, and leadership, are discussed below.

Training: Service integration has major implications for both graduate and in-service training of psychologists since this integration will require greater breadth and flexibility among practitioners. A school health psychologist with expertise in behavioral health who serves an elementary school, for example, will need to be competent in a broad number of skills and approaches, ranging from typical developmental concerns and issues, to guidelines for monitoring commonly used child psychotropic medications, family interventions, and community consultation. Professionals who are "generalists" in human services will have greater possibilities for employment in an integrated service system than those whose background is limited specifically to traditional psychological practice specialties. There will, of course, always be some need for specialization, particularly with respect to low-incidence or highly technical problems. Psychologists will need more systematic training in collaborative and consultation-based approaches to practice.

Practice: Psychological services within an integrated services framework will look and feel substantially different. Practitioners will be able to exercise greater flexibility in the range of activities in which they engage, and will not be as constrained in regard to funding source and eligibility considerations. They will spend more time working as part of a team, in concert with a variety of providers, caregivers, and community members. In addition to the school-based services they provide, school-health psychologists are likely to spend more time in homes and other community settings. These psychologists will routinely work across interdisciplinary boundaries among various social systems that impinge on children and families to coordinate activities, manage conflict, and insure focus and quality of services.

Leadership: Psychologists should be trained and encouraged to assume leadership roles within integrated service programmes. In addition to the more traditional aspects of programme administration and supervision, leadership activities should focus on establishing an integrative strategic vision for child-serving organizations, building collaborative teams, and facilitating planned organizational change in the direction of more integrated services.

Research: Psychological research on the efficacy of integrated service delivery approaches for children and families represents a unique contribution for psychology. Such research is distinct from traditional controlled experimentation, in that the array of target problems is vast, treatment programmes are diverse and multifaceted, and outcome measurements complicated. Practicing psychologists need to become proficient in a broader range of methods and procedures (*e.g.*, quasi-experimental design, multivariate analysis, programme evaluation techniques, qualitative research) in order to conduct such social policy and programme-related investigations. Psychologists would also be in a unique position to help service systems develop and validate information systems to allow for on-going programme monitoring and management.

HEALTH SERVICES IN THE SCHOOLS: BUILDING INTERDISCIPLINARY PARTNERSHIPS

There are two essential social systems with which virtually all children and families have routine, significant contact: school and health care settings. The school is an environment wherein children not only engage in academic learning and growth, but where they also experience social and emotional interactions with adults and peers so as to build self-esteem and social competence. These essential experiences can serve to increase later prospects for success in relationships, the work place, and personal pursuits. It is vital that schools support the broad developmental needs of children and families.

Schools are being asked to address the needs of children and youth at a time when fundamental transformations of schooling structures and outcome expectations are also being demanded (Children's Defence Fund, 1992). Restructured schools alone cannot satisfactorily address the multidimensional needs of children and youth. Schools and other child-and family-service organizations must collaborate to enhance the likelihood of educational and personal success for all children. Recent legislative and policy initiatives, such as Healthy People 2000 (1990), a blueprint for disease prevention and health promotion, highlight the important role that schools must play in assuring the well-being of our nation's children and youth.

In order to address the developmental needs of children and families in a comprehensive and preventive manner, schools and communities must coordinate services. Therefore, a service integration perspective that recognizes the central role that schools play in the lives of children should guide efforts to establish an empowering, healthy climate for them and their families within the community at large (Institute for Educational Leadership, 1992; Oomes and Herendeen, 1989). Such a view acknowledges the complex, reciprocal interaction among social systems, including families, when problems are conceptualized and service systems are designed. It assumes that children and families are most likely to benefit from collaborative, focused efforts among the various systems responsible for addressing their needs, both formal and informal.

Interdisciplinary Partnerships in Health Care

There is an emerging consensus among professionals and consumers that the current health care service delivery system is not meeting the needs of children and families (Knitzer, 1982; National Commission on Child Welfare and Family Preservation, 1990). Solutions must move beyond adding resources (*e.g.*, more funding, more programmes) and towards fundamental changes in how the system operates. Social and political institutions have not considered the needs of children and families as funding priorities (Melaville and Blank, 1991, 1993). Individual service delivery systems for children (*e.g.*, health care, education, social service, mental health) are funded and designed to address isolated and crisis-oriented needs, rather than to promote healthy development for all children and families in a comprehensive fashion. In addition, some parents either have no knowledge of how to access available services, or they may not value them. Thus, services provided to children and families frequently are not comprehensive, responsive, or integrated. Key sources of difficulty in the current service delivery system are the lack of clarity, coordination, and comprehensiveness, resulting in inflexible patterns of funding, training, and service provision.

Since the cognitive, social, emotional, educational, and physical needs of children are complex, an integrated services model provides for a more coherent, needs-based response to these complex problems (Chaudry, Maurer, Oshinsky and Mackie, 1993; Dunst, Trivette, Gordon, and Pletcher, 1989).

Health Care in Schools Through Service Integration

The efficacy of services to children and families can be viewed from the perspective of the families themselves. When examined in this manner, emphasis is placed upon the nature of service delivery events or episodes that occur, and the impact these events have on children and families. Within a well-integrated programme, typically:

1. services are available in close proximity and are accessible without reference to physical, psychological, social, linguistic, sexual orientation, or other barriers;
2. services are comprehensive and appropriate, in that they possess features that address priority needs the family has identified, at a level of service sufficient to their need;
3. services are formulated and delivered at a high level of quality such that the family perceives them as an organized whole and can participate in a consistent and effective manner;
4. services promote psychological competence and self sufficiency rather than focusing exclusively on dysfunction and pathology;
5. services are oriented towards full participation, partnership, and empowerment of family members;
6. services are sensitive to cultural, gender, racial, linguistic, class, disability, and sexual orientation issues; and,
7. interventions are driven by concern for the needs and desires of the consumers (*i.e.*, children and families) and emphasize explicit outcomes stated in a positive manner. (Paavola et al., 1995).

Features of an Integrated Service System

Relative to the definition offered previously, there is a continuum of integrated services that varies as a function of need, service availability, problem severity, and related dimensions. From the perspective of children and families, many opportunities and services can best be accessed through a single provider and implemented in an integrative manner. An integrated services model also assumes that the greater the number of providers involved (*e.g.*, psychologists, nurses, teachers, social workers, physicians, day care workers), the greater the need for effective collaboration. Timely and responsive interventions on behalf of children and families therefore rely on effective communication, coordination, and collaboration among service providers, agencies and organizations, and the consumers of services (children and families).

Thus, "coordinated and collaborative services" should be the essential standard by which effective services are delivered. The service system must respond to the multiplicity of needs exhibited by children and families through carefully orchestrated teamwork. At a minimum, this collaboration takes the form of different providers (from independent agencies) communicating regularly by phone regarding a child or family. Or, it may involve regular face-to-face meetings and case conferences among providers. Ideally, providers and family members would work as an integrated team to provide needed services. The net result of the integrated team concept could be service delivery models such as "one-stop shopping" or more staff sharing and programme development activity.

The service delivery system should also allow for both "ease of entry and flexibility of movement." For example, if the point of initial contact in a community is a school setting, there should be a clear connection between the school and the array of community services that the family needs, regardless of categorical restrictions. This requires that individual providers and agencies see themselves as part of a much larger ecology that is community-wide and geared to aiding the overall climate within which children grow and develop. The point of initial entry into such a system should be less critical than the fact that child and family needs are considered paramount in responding to the concerns presented. The flexibility of movement concept allows for a child or family to enter such a system at any point and move flexibly between services as their needs dictate without having to confront barriers.

The service delivery system must be organized for both "maximum development of the child and for accountability," first to the family, and also to the community within which the child lives. This means that providers need to be re-trained within a consumer-oriented model, with children and families seen as customers with whom one must collaborate, rather than as patients or adversaries. Community accountability refers to concern for improving the quality of life in communities through community resource development, advocacy, and related activities.

"Funding for coordinated and collaborative service needs to be both flexible and shared" (where possible) among agencies, such that different agencies can be encouraged to develop together programmes that serve children and families holistically. Funding and programme decisions need to be made from the "bottom up" and those providers who have on-going contact and communication with the family should be the major decision makers about how pooled and/or flexible funds can be utilized most effectively, with direct input from the consumers of services. "Interdisciplinary interaction and training" for providers needs to be a top priority in an integrated service model.

It will be necessary for providers from different disciplines to know what other disciplines can contribute to solutions for issues confronting families. Collaborative effort outside of traditional disciplinary lines creates opportunities for true communication and integration among providers.

PUBLIC HEALTH/HEALTH EDUCATION

American Public Health Association (APHA) APHA is the main voice for public health advocacy that is the oldest organization of public health sine 1872. The American Public Health Association aims to "protect all Americans and their communities from preventable, serious health threats and strives to assure community-based health promotion and disease preventions." Any individual can become a member and benefit in online access and monthly printed issues of The Nation's Health and the American Journal of Public Health Society for Public Health Education (SOPHE) The mission of SOPHE is to provide global leadership to the profession of health education and health promotion and to promote the health of society through advances in health education theory and research, excellence in professional preparation and practice, and advocacy for public policies conducive to health, and the achievement of health equity for all. Membership is open to all who have an interest in health education and or work in health education in schools, medical care settings, worksites, community based organizations, state/local government, and international agencies. Founded in 1950, SOPHE publishes 2 indexed, peer-reviewed journals, *Health Education and Behavior* and *Health Promotion Practice*.

American School Health Association (ASHA) The American School Health Association was founded in 1972 by a group of physicians that already belonged to the American Public Health Association. This group specializes in school-aged health specifically. Over the years it has snowballed and now includes any person that can be a part of a child's life, from dentists, to counselors and school nurses. The American School Health Association mission "is to protect and promote the health of children and youth by supporting coordinated school health programmes as a foundation for school success."

American Association of Health Education/American Alliance for Health, Physical Education, Recreation, and Dance (AAHE/AAHPERD) The AAHE/AAHPERD is said to be the largest organization of professionals that supports physical education; which includes leisure, fitness, dance, and health promotion. That is only a few; this incorporates all that is physical movement. This organization is an alliance with five national associations and six districts and is there to provide a comprehensive and coordinated array of resources to help support practitioners to improve their skills and always be learning new things. This organization was first stated in November 1885. William Gilbert Anderson had been out of medical school for two years and was working with many other people that were in the gymnastic field. He wanted them to get together to discuss their field and this organization was created. Today AAHPERD serves 25,000 members and has its headquarters in Reston, Virginia. Eta Sigma Gamma (ESG) The Eta Sigma Gamma is a national health education organization founded in 1967 by three professor from Ball tate University. The mission of the ESG to promote public health education by improving the standards, ideals, capability, and ethics of public health education professionals. The three key points of the organization are to teach, research, and provide service to the members of the public health professionals. Some of the goals that the Eta Stigma Gamma targets are support planning and evaluation of future and existing health education programmes, support and promote scientific research, support advocacy of health education issues, and promote professional ethics.

American College Health Association (ACHA) The American College Health Association originally began as a student health association in 1920, but then in 1948 the association changed the name to what its known today. The principal interest of the ACHA is to promote advocacy and leadership to colleges and universities around the country. Other part of the mission's association is to encourage education, communication, and services to students and campus community in general. The association also promotes advocacy and research. The American College Health Association has three types of membership: institutions of higher education, individual members who are interested in the public health profession, and susbtain members which are profitable and non-profitable organization. The ACHA is connected to 11 organizations located in six regions around the country. Currently, the American College Health Association serves 900 educative institutions and about 2400 individual members in the United States.

Directors of Health Promotion and Education (DHPE) Founded in 1946 as one of the professional groups of the Health Education Profession. The main goal of the HEPE is to improve the health education standards in any public health agency. As well, build networking opportunities among all public health professionals as a media to communicate ideas for implementing health programmes, and to keep accurate information about the latest health news. The DHPE also focus to increase public awareness of health education and promotion by creating and expanding methods of existing health programmes that will improve the quality of health. The Directors of Health Promotion and Education is linked to the Association of State and Territorial Health Officials (ASTHO) to "work on health promotion and disease prevention".

Health Education Career Opportunities

The terms Public Health Educator, Community Health Educator or Health Educator are all used interchangeable to describe an individual who plans implements and evaluates health education and promotion programmes. These individuals play a crucial role in many organizations in various settings to improve our nations health. Just as a Community health educator works work towards population health, a school Health educator generally teaches in our Schools. A community health educator is typically focused on their immediate community striving to serve the public. Health Care Settings: these include hospitals (for-profit and public), medical care clinics, home health agencies, HMOs and PPOs. Here, a health educator teaches employees how to be healthy. Patient education positions are far and few between because insurance companies do not cover the costs.

Public Health Agencies: are official, tax funded, government agencies. They provide police protection, educational systems, as well as clean air and water. Public health departments provide health services and are organized by a city, county, state, or federal government.

School Health Education: involves all strategies, activities, and services offered by, in, or in association with schools that are designed to promote students' physical, emotional, and social development. School health involves teaching students about health and health related behaviours. Curriculum and programmes are based on the school's expectations and health.

Non Profit Voluntary Health Agencies: are created by concerned citizens to deal with health needs not met by governmental agencies. Missions include public education, professional education, patient education, research, direct services and support to or for people directly affected by a specific health or medical problem. Usually funded by such means as private donations, grants, and fund-raisers.

Higher Education: typically two types of positions health educators hold including academic, or faculty or health educator in a student health service or wellness center. As a faculty member, the health educator typically has three major responsibilities: teaching, community and professional service, and scholarly research. As a health educator in a university health service or wellness center, the major responsibility is to plan, implement, and evaluate health promotion and education programmes for programme participants.

Work site Health Promotion: is a combination of educational, organizational and environmental activities designed to improve the health and safety of employees and their families. These work site wellness programmes offer an additional setting for health educators and allow them to reach segments of the population that are not easily reached through traditional community health programmes. Some work site health promotion Some work site health promotion activities include; smoking cessation, stress management, bulletin boards, newsletters, and much more.

Independent Consulting and Government Contracting: international, national, regional, sate, and local organizations contract with independent consultants for many reasons. They may be hired to assess individual and community needs for health education; plan, implement, administer and evaluate health education strategies; conduct research; serve as health education resource person; and or communicate about and advocate for health and health education. Government contractors are often behind national health education programmes, government reports, public information web sites and telephone lines, media campaigns, conferences, and health education materials.

HEALTH EDUCATION FOR YOUNG PEOPLE

Recent decades have seen major changes in health issues. This development has been marked by the relative increase in diseases associated with lifestyles or with behaviours considered to be "at risk" for negative health consequences and by skyrocketing health costs that exhaust the funds available.

Prevention, health education, health promotion—these are some of the diverse labels of the numerous activities involved in reducing risks and modifying behaviours with the aim of improving the quality of life and prolonging it. In 1986, in the Ottawa Charter, the World Health Organisation (WHO) defined health promotion as "the process of enabling people to increase control over, and to improve their health". In France, the National conference on health for the year 2000, stressing the importance of developing prevention and education as an approach to health promotion, emphasized the need to work more deeply on the behavioural and environmental determinants of health. It observed that France lacked a legislatively imposed legal basis for setting the boundaries for health education activities and for establishing a minimum of quantitative or qualitative requirements.

In French, two expressions—literally translated as "education for health" and "education to health"—are used interchangeably in official texts and publications. In fact, however, they cover two different practices. Health professionals, who consider health to be a process of permanent adaptation, prefer education "for" health to stress the maintenance of this process. On the other hand, education professionals, use the preposition "to", by analogy with education "to" citizenship and the environment, to underline the educational dimension of this mission. The 1998 establishment of committees of health and civic education inside schools and the introduction of health education into school curricula marked an important step forward. Non-etheless, both the means and the skills available are frequently inadequate.

The multiplicity of bodies involved in these programmes and interventions does not facilitate the system's consistency or its ability to capitalise on its experience. The lack of coordination between various public agencies and the lack of relationships between researchers and field workers are reflected in the difficulty of developing evaluation methods and of the minimal visibility of most activities in this domain. For example, there is no specialist journal publishing research and innovative actions implemented in health education in the countries of the European Community, although it would enrich the literature in this field, a literature that today is essentially North American.

CANAM sought information and advice from INSERM, through the expert advisory group procedure, on the recent scientific data about the quality, consistency, and effectiveness of health education methods for young people, both nationally and internationally, with particular attention to the methods intended to prevent risk behaviours in the areas of sex and psychoactive substance use.

To respond to canam, inserm established a multidisciplinary group of experts that brought together scholars with expertise in the domains of health education, public health, public law, epidemiology, psychosociology, and pedagogy.

The group's analysis was structured around the following questions:

- What are currently the principal concepts in health education? What observations have structured their development?
- How is health education implemented within the educational system? How do the institutional solutions in France compare to those in other countries?
- What legislative and regulatory framework circumscribes health education for youth in France?
- How do different methods take into account the background items, objective or subjective, that justify particular activities in health education?
- What elements are recognised to determine the quality and effectiveness of interventions in health education? Are there any forms of health education intervention that have been reported to have attained all or part of their objectives? Under what conditions can these activities be perpetuated? Under what conditions are various activities transferrable?
- What are the specific factors involved in the effectiveness of educational activities in preventing "at-risk" behaviours that endanger health, in the domains of sex and psychoactive substance use?

Querying databases and searching for unpublished documents allowed us to construct a corpus of approximately 1,400 documents, including articles published in scientific journals, reference works, reports of interventions, the gray literature, and official texts. Approximately 900 documents, more specifically focused on activities in school settings, were analysed by the expert advisory group.

During six working sessions organised between November 1999 and September 2000, the panel members presented a critical analysis and synthesis of the works published in their fields of expertise. The last two sessions were devoted to the collective validation of the synthesis and to drafting the recommendations.

SYNTHESIS

According to the French Treatise on Public Health, there are three categories of health education: primary, secondary, and tertiary. Primary education is that aimed at reinforcing the students' good health. Secondary education involves measures intended to avoid accidents to health or, if such an accident has already occurred, to restore good health as rapidly as possible. Tertiary education is any educational intervention aimed at rehabilitation and adaptation to the sequelae of an accident. Health education thus intervenes both before and after any disease or injury. Its justification is found today in public health data stressing the importance of behaviour as an explanatory factor in most deaths that are considered premature and avoidable, particularly among the young.

The cost of health education is generally agreed to be quite low in relation to the potential savings and is trivial compared with the costs of other sectors of the health care system. In France each year we spend, on average, per inhabitant: 10 F for health information and education, 250 F for preventive medicine, and 11,000 F for treatment of diseases.

Moreover, the value of health education is not only collective and economic; it also serves definite individual and personal interests. It allows each individual to develop his or her capacity to improve both longevity and quality of life, in the holistic vision of health as defined by WHO: "Health is a state of complete physical, mental and social well-being and does not consist only in an absence of disease or infirmity."

Health education is not the monopoly of government: it concerns all the players in the health care system, and, when young people are involved, all those in the educational system. Because it is everyone's business and involves a mission of general interest, it is performed by a multitude of participants often ignorant of one another, and it raises legal, ethical, and economic questions. We note, however, that in France the few scattered dispositions mentioning health education in the overall legislative and regulatory scheme provide a badly structured and poorly defined framework—helpful neither to its credibility nor its relevance.

A curative or treatment approach and a preventive approach, although different, are naturally complementary. The ethical codes of the medical and paramedical professions make health education a professional obligation, as do the regulatory texts. Moreover, the Public Health Code contains very explicit provisions endowing all health facilities, public and private, whether or not they participate in the public service of hospitals, with a mission of health education in addition to their primary treatment mission. In the last decade, some health insurance funds have created departments of health education and promotion, in accordance with the priorities defined by their Boards of Directors.

The role of schools was redefined in the 1989 framework law on education that first introduced health education into the school setting. A set of provisions in 1998 inserted health education into the nationally mandated curricula of primary and middle schools. Within schools, Committees for health and civics education were charged with the mission of health education and with organising the prevention of substance abuse, other risk behaviours, and violence within the framework of the school's project or plan.

Traditional health education is the set of educational interventions intended to provide individuals with information about health and to induce them to adopt attitudes and behaviours that are good for their health. More recently, health education has broadened to include social and environmental aspects. The concept of "health promotion" was formalised in 1986 in the Ottowa Charter, still the international reference; it enlarged the educational approach by focusing on collective responsibility. It involves not only educating individuals but promoting collective mobilisation and changes, while bearing in mind the psychosocial and societal determinants at the origin of behaviours and attitudes unfavourable to health. Health promotion includes health education, which is one of its essential components. The ethical question of whether changes, of attitude or behaviour, should be promoted is central to the debate about health education. In principle, the desire to change others can be considered ethical if the individual or group is conscious of this influence and if the change benefits the individual or group. Most health educators are averse to normalising behaviour, inducing guilt about health-related subjects, and relying on individual responsibility as the sole motor of change. A conference on health education and ethics held in January 2000 reviewed the four general principles used to guide health education interventions in North America: respect for social justice, respect for individual autonomy, the requirement that the programme or intervention be beneficial, and the requirement that it not do harm. The debate on the construction of an appropriate ethic here has begun.

SCOPE OF THE SCHOOL HEALTH PROGRAMME

The school health programmes is that phase of the educational process which is concerned with developing an understanding of health and providing those experiences and services which play an active part in maintaining and improving the health of both pupils and school personnel. It includes teaching for health, living healthfully at school, and services for health maintenance and improvement. Teaching for health refers to those experiences provided for the purpose of influencing understanding, attitudes, and practices relating to both personal and group health.

Living healthfully at school includes a healthful physical environment as well as such aspects of the non-physical environment as teacher-pupil rapport, organization of the school day, and other aspects of life in school which affect the social and emotional health of pupils. Services for health improvement refer to the six procedures which are vital to a desirable school health programme: (1) appraisal of the physical, mental, emotional, and social health status of children and school personnel, (2) procedures used by teachers, physicians, nurses and others to counsel pupils and parents in regard to health appraisal findings, (3) follow-through procedures for correcting remediable health defects, (4) provision for the exceptional child, (5) procedures for prevention and control of communicable diseases, and (6) emergency care in case of sickness and injury. The teacher is only one of the many persons responsible for accomplishing these important aspects of the school health programme. Some schools have a health coordinator who coordinates all aspects of the programme. School health councils have been set up in some schools to meet the health needs of pupils and school personnel. Such groups are comprised of people who devote their efforts to studying health problems and planning a course of action for their solution. Members of the team include such individuals as the school nurse, dental hygienist, classroom teacher, school principal, physician, custodian, and others who are interested and can contribute to developing a sound school health programme.

RESPONSIBILITIES OF THE CLASSROOM TEACHER

The classroom teacher is the key school person involved in the health of the elementary school child. The organization of the school with the self-contained classroom enables her to observe the pupils continually and to note any deviations from normal. Continuous contact with the same children over a long period of time also makes it possible for her to know a great deal about their physical, social, emotional, and mental health. She can help them to develop the right knowledges, attitudes, and practices.

Below are listed some of the responsibilities which fall to the classroom teacher in regard to the health of her pupils:

1. Possess an understanding of what constitutes a well-rounded school health programme and the teacher's part in the programme.
2. Meet with the school physician, nurse, and others in order to determine how she can best contribute to the total health programme.
3. Become acquainted with the parents and homes of her students: establish parent-school cooperation.
4. Discover the health needs and interests of her pupils.
5. Organize health teaching units which are meaningful and are in terms of the health needs and interests of her students.
6. See that children needing special care are referred to the proper place for help.
7. Be versed in first-aid procedures.
8. Participate in the work of the school health council. If none exists, interpret the need for one.
9. Provide an environment for children while at school which is conducive to healthful living.
10. Be continually on the alert for children with deviations from normal behaviour and signs of communicable disease.
11. Provide experiences for living healthfully at school.
12. Help pupils assume an increasing responsibility for their own health as well as for the health of others.
13. Set an example for the child of what constitutes healthful living.
14. Motivate the child to be well and happy.
15. If possible, be present at pupils' health examinations and contribute in any way which is helpful to physician in charge.
16. Follow through in cooperation with the nurse to see that remediable health defects are corrected.
17. Interpret the school health programme to the community and enlist its support in solving health problems.
18. Provide a well-rounded physical education programme for her class.
19. Help supervise various activities which directly affect health, such as school lunch, rest periods, etc.

20. Become familiar with teaching aids and school and community resources for enhancing the health programme.
21. Be aware of the individual differences of her pupils.

EVALUATING THE EFFECTIVENESS OF HEALTH EDUCATION

The methods for evaluating the effectiveness of health education have been the object of intense debate. One school, which uses an "epidemiological" approach, measures the degree to which objectives, determined in advance, have been reached for a given population (for example, increase the non-smoker rate in a given student population).

A second school, using a "community-based" approach, argues that the objectives and the means used to reach them, and even the evaluation methods, should be determined by the population itself. It is inherent in this approach that no intervention or assessment protocol can be defined in advance. Accordingly, the evaluation process is indissociable from the program's approach.

Principal characteristics of evaluations as a function of the type of programme

The first school uses the experimental method, comparing the development of a population that underwent an educational programme with that of a population that either had no or a different programme. The second school stresses the evaluation of the process, that is, the detailed management of activities, in particular by qualitative evaluations that do not measure changes but instead help to understand how the programme was applied, understood, experienced, and accepted. One way to express the difference might be to say that the first school demands that effectiveness be scientifically determined, and the second, democratically.

The first school considers the comparison of experimental and control groups as the ideal model; its gold standard remains a pure experimental protocol with random assignment of subjects into the different programmes. In many studies, however, it is the site that is randomly selected, whereas the analysis concerns individuals. Analyses of results grouped by (a small number of) sites would have little statistical power. Various solutions can compensate for these disadvantages (increasing the number of sites by diminishing their size, controlling for the sources of variation between sites, taking a group effect into account).

As in every form of education, the effects of interventions, unless they are regularly reinforced, tend to fade in the long run. The observation of this phenomenon has led: for evaluations, to considering a result invalid if its only evaluation occurred immediately after the intervention, and to favouring long-term follow-ups; for processes, to promoting programmes spaced out over time with "review" sessions. In this situation, however, the higher number of cases "lost to follow-up" offsets the increased duration of the follow-up, and those "lost" are often the youth at the highest risk of social or school problems. Several strategies are possible for dealing with the "lost to follow-up" problem: eliminate them from the cohort if they are not different from the active cohort, or attribute a replacement value for them in the post-test.

The adaptation of the intervention protocol to the particular target population and the consistency of its application are now studied in advance. A programme may fail simply because it was inappropriate or not applied. Discussion groups (in particular, focus groups) are organised simultaneously to understand the position of the target youth and also to test the proposed educational material.

The effect measures are most often assessed from a self-administered questionnaire. The change indicators are objective but somewhat less than totally reliable because they are based on self-report. Non-etheless, specific studies on the validity of responses, including every study that could confirm responses with laboratory tests, have shown that responses to these questionnaires are, on the whole, reliable. The literature produced by the second school is often framed as argument or general recommendations about interventions and evaluation. These two schools are non-etheless beginning to converge. Some teams that advocate the community-based approach non-etheless try whenever possible to use epidemiologic evaluation tools. Moreover, those using the "epidemiologic" approach are adapting their approaches according to their target populations.

PREVENTION OF RISKS RELATED TO SEXUAL BEHAVIOUR

The literature about the prevention of sexual risk-taking behaviours in adolescents is particularly abundant and clearly dominated by North American researchers. The most widespread concern is primary prevention of AIDS, which is the major objective of educational programmes about sex-related risks aimed at the young. The answer to the question "Why teach adolescents about sex-related risks?" seems to be considered self-evident in many publications that present "young people" as a population at risk. Adolescence, the life stage when sexuality is discovered, is traditionally described as an unstable period, psychologically, socially, and even sexually; for this reason, it is a preferential target for prevention. Moreover, the young, as a captive population in school, are easy to reach by educational programmes.

In the United States, the risks associated with sexually transmitted diseases (STDs) and unwanted pregnancies in young girls are indeed very high: one adolescent in four contracts an STD during secondary school; 10 per cent of the girls 15–19 years of age become pregnant, for a total of 1 million pregnancies a year in this age group. In France, on the other hand, persons younger than 18 years do not appear to be a particularly exposed group. According to the ACSJ (analysis of sexual behaviour in youths 15 to 18 years of age) survey, carried out in 1994 and published in 1997: 1.1 per cent of the young people in this age bracket had an STD other than a yeast infection; 3.3 per cent of the girls in this age group became pregnant. Currently, the number of annual pregnancies among those younger than 18 years is estimated at approximately 10,000, of which 6,500 are terminated by an elective abortion. In Western nations, no notable increase has been observed in the risk factors that may be associated with earlier sexual activity: the mean age of first intercourse is around 17 years (in France, 17 years 3 months for boys and 17 years 6 months for girls). Sexual risk, however, differs according to social class. Like the risk of sexual violence, it is clearly greater in situations of vulnerability due to social problems. The use of drugs, including excessive alcohol intake, appears to multiply sexual risk.

There is overwhelming recognition that school is the place for sex education and its progression to education about sexual risks (this does, however, raise questions about the young people who have been excluded from the school system). Some publications recommend that education about sex-related risks begin in primary school. Non-etheless, if it is to be understood, sex education must be appropriate to the age and interests of the children; the issue of risks should of course be covered before the adolescent begins sexual activity. This education is included in numerous school curricula, where it is not limited to biology classes; the question, however, is: who teaches it? The teachers are not always well prepared to do so; they are often reticent, more at ease discussing the question of the risks associated with AIDS and other STDs than in dealing with sexuality.

Table. Data on the Risk behaviours of Sexually Active Youths 15–18 Years of Age – According to the ACSJ Survey (Analysis of Sexual Behaviour in Youths).

	Age			
	15–16 years	**17 years**	**18 years**	**All***
Frequency (%) of unprotected intercourse (last penetration, regardless of partner)				
Boys	25.0	31.8	41.1	32.5
Girls	33.2	52.7	67.4	49.8
Frequency (%) of STD other than yeast infections				
Boys	0.0	2.4	0.5	1.1
Girls	0.3	1.5	1.5	1.1
Frequency (%) of pregnancies among girls				
Followed by an elective abortion	1.8	3.2	1.8	2.3
Followed by childbirth	0.8	0.0	0.3	0.4
Total pregnancies	2.8	4.1	2.8	3.3

Note:

* 1,883 boys and 1,384 girls who have had sexual intercourse at least once in their lives.

In some countries, the school health department, school dispensaries, physicians, nurses, and other members of the medical community are brought in. Other experiments and players have been tried in the area of AIDS prevention: a variety of organisations and associations, people with AIDS or HIV, peer groups, medical students. Peer education benefits from favourable preconceptions: it is presented as simultaneously allowing adaptation of risk information to meet the expectations of the peer group and direct action on the norms that influence sexual conduct. Research does not justify this prejudice, however, and the peer approach cannot replace other educational approaches. It can be considered only as a complementary strategy.

Most interventions are based upon a theoretical framework. These can be divided into two main categories: the individualistic approaches that use the Health Belief Model or the theory of reasoned action, and the interactional and comprehensive approaches. The individualistic approaches, which predominate, are based on models of learning and individual decision-making. In the interventions based on these models described in the North American literature, the different objectives all involve, depending on the programme, teaching the teen abstinence, or waiting, or saying no, or, finally, discussing condom use with a sexual partner. These procedures do not really take account of the facts that a sexual situation is a social interaction and that the decision is not individual only. For this reason, they have proved inadequate in terms of stabilising behaviour. They define sexuality only functionally, thereby neglecting its affective aspects, its relation to feelings. Comprehensive approaches have replaced these hortatory approaches. These involve starting from what teens say and express and thus being able to capture the social and affective experience of sexuality, understand its normative dimension, anchor it in a social context, and provide responses to the questions that emerge, at the level at which they are expressed.

The existing literature pays little attention to the associations between sex education and gender identity. The very simple idea of conducting risk education separately with girls and boys, thereby taking into account the ideologies associated with male and female roles in sexuality, is recommended by the authors of a very recent North American study, performed jointly by a prevention group and a research center. Another frequent question involves pedagogical tools, including games, marionettes, comic books, audiovisual media, and computer programmes. The usefulness of these media does not mean that we can dispense with a serious reflection about the type of programme or the framework in which they are used. Regardless of the tools, they should never be used outside of a comprehensive strategy with explicit objectives, nor without pre-testing.

Too many programmes have been set up but not controlled, with free rein left to their staff's preconceptions, of whatever flavour. Non-etheless, the consistency of the data, over time and in different countries, shows changes towards better prevention of sexual risk-taking by teens. It is difficult, though, to attribute these behavioural changes to the effectiveness of any specific programmes, to the prevention programmes aimed at the general public or to the prevention messages broadcast by the media and relayed at many levels. Adolescents are the population group that has best adapted to the AIDS threat. The rate of condoms use at the first act of intercourse is climbing regularly. In France, according to the ACSJ survey, 78.9 per cent of the boys and 74.4 per cent of the girls 15 to 18 years of age who reported sexual activity had used a condom at their first act of intercourse. The corresponding data for the 97/98 Youth Health Barometer were 88.6 per cent and 85.4 per cent, respectively, in boys and girls 15 to 19 years of age. The decline, consistently observed, of condom use at last intercourse must be related, among other things, to a stabilisation of young couples.

It is expected that educational interventions about the risks of AIDS transmission will delay the onset of sexual activity or diminish it, induce greater selectivity of partners, and induce condom use. We might also wonder about the possible adverse effects of some prevention programmes. Sometimes based on the anticipation of regretting starting sexual activity too early, they end up increasing fear and even intolerance of others' non-conformist actions, without necessarily inducing a more rational attitude towards protection, contraception in particular.

Teen pregnancy and motherhood are generally considered to be a failure of sexual risk prevention or of contraception and are associated with immaturity and academic and social problems. Non-etheless, these precocious pregnancies are sometimes desired and can be structuring for some young women. This fact, however, must not mask the need to transmit to young women the resources that enable them to avoid unwanted pregnancies—a much more real risk than AIDS among the young. What is needed is contraceptive education that takes into account their expectations and their sexual trajectories.

One of the principal lacunae of sex education programmes concerns teaching children and adolescents about the risk of sexual violence. The frequency of forcible "sexual relations", reported by 15.4 per cent of the girls questioned in the ACSJ survey, is worrisome. These girls, however, were more often those who were no longer in school or were in low-prestige tracks; it is therefore associated with social vulnerability.

PREVENTION OF THE RISKS ASSOCIATED WITH SMOKING

In the domain of risks associated with the consumption of psychoactive substances, most interventions aimed at young people began by focusing on a specific product (tobacco, alcohol, drugs). Slowly, experimental findings led to a comprehensive approach to prevention aimed at changing behaviour. In all cases, the timing of the prevention activity, its participants, and the type of intervention are all important factors in its effectiveness.

International data indicate that smoking prevention should have higher priority than other types: smoking is the behaviour that causes the most deaths in the long run; tobacco is also considered an introduction to other products, especially cannabis and alcohol. Moreover, the percentage of girls and women smoking is increasing, for reasons not yet explained. Smoking prevention interventions, which began in the 1950s, generally discuss smoking from the perspective of preventing cardiovascular diseases and cancer. Only recently has the prevention of drug addiction has been added.

The process of nicotine addition can be broken down into five stages: a preparatory stage, initiation, experimentation, a phase of regular use without dependence, and a phase of dependence with daily use.

The objectives of most prevention activities are to avoid or delay smoking initiation or to help in smoking cessation. In the first case, the activity may be at the individual, family, or collective level. The factors influencing the beginning of the process differ according to sex, academic status, and period. For smoking cessation, the help may involve the youth or the parents; in any case, it should occur before the youth becomes addicted to nicotine. The study of the subject's motivation is an essential prerequisite. Alongside these programmes, a newer emerging objective is the prevention of regular smoking, that is, reducing risks.

The objectives of the first interventions were simply to provide information on the dangers of smoking, and then, starting in the 1960s, to teach students how to resist social influences. Since the 1970s, programmes have been based upon the reinforcement of "general social skills" or "life skills", including cognitive components, decision-making, coping skills, and assertiveness. All of these strategies were developed to prevent smoking initiation. Currently, two other types of activities are being developed: one to reduce the risk of progression from occasional to regular use, and one to promote teens' psychological well-being as a means of preventing any kind of substance use. Prevention of smoking initiation is often a failure in the long term, regardless of the technique used: after 4 years, no differences are observed between those who did and did not attend a prevention programme. Currently being promoted are activities that take cognitive and social development into account, including the youth's experiments with the products throughout adolescence. Non-etheless the contents of the "pitch" of the intervention continue to be addressed more to boys than girls. Programmes based on ability to cope with stress can only be effective when they take place before initiation: once smoking has become a habit, the physiological process seems to gain the upper hand over the psychological process.

Of the players in prevention, teachers have an important role, and their training is essential. An antismoking intervention, by class or grade level, includes at least five sessions, occurring between the end of primary school and the first 2 years of secondary school. The intervention of clinicians among at-risk youth and families and among school officials for maintaining antismoking programmes in schools has proved to be more effective than the school programmes themselves. Combined interventions by peers and adults yield better results than intervention by peers alone. Interventions involving parents appear to be the most effective.

Public policies have proved more effective in stopping smoking initiation when they raise prices for and limit access to tobacco than when they ban smoking. Interventions at the school, family, and community levels have proved more effective than isolated activities in preventing initiation and in promoting cessation.

PREVENTION OF RISKS ASSOCIATED WITH ALCOHOL USE

The prevention of problem drinking is more complex than the prevention of smoking because excessive alcohol consumption, even only occasional, can have harmful social, medical, and personal effects and because alcohol is more widely used and more prestigious than tobacco. It is also a more feared product because of the accidents and the violence it can cause. Individual factors of sensitivity to the effects of alcohol play a role, as do sex, weight, and genetic polymorphisms.

Several theories of how problem drinking develops have been proposed: one for "normally socialised" adolescents, and another for "problem" adolescents, who progress more rapidly towards more severe alcohol abuse. Psychic factors are often underestimated in explaining the beginning of problem drinking. These factors are associated with diminished skills, especially academic skills. Accordingly, early academic problems and low academic expectations by both parents and children are important indicators of risk.

Prevention activities more often involve the prevention of risks than of drinking itself. Their aim is to limit the risks associated with drunkenness (traffic accidents, sexual conduct), promote responsible drinking, and reduce juvenile alcohol abuse. Interventions that focus on alcohol initiation have been directed towards the reinforcement of general social skills, but have rarely involved the family. Because moderate consumption is considered to be a criterion of social integration, few programmes try to prevent all alcohol use.

Most school interventions involving youths 10 to 18 years of age are too late, according to the authors, because they take place after initiation has occurred. Interventions in primary school have not been evaluated. For high-risk groups, school does not provide adequate guarantees of confidentiality. Television and other mass media can be considered to be an effective channel of information in that they help modify social standards.

Among those involved in prevention, school nurses are recognised as positive sources of individual preventive activity against excess drinking. The family must be truly involved and informed about teaching "responsible drinking" at home. The improvement of intrafamily relations remains an important means of prevention. An essential point in programme success is the understanding by adults that young people are mature and responsible and therefore can be partners in real discussions.

Because girls and boys have different ways of drinking and different reasons for drinking, interventions must take these specificities into account to modify consumption. Interventions among "high-risk" groups have proven more effective than those among more mixed populations. Teens prefer brief interventions that take place in stages. Programmes that include individual treatment have not been especially successful; this must be proposed later, when the adolescents have grown aware of their problem with alcohol.

Of the interventions directed towards prevention of accidents associated with excess drinking, one of the more promising is special training for various night workers to identify signs of early drunkenness. Concrete measures, such as a “zero” blood alcohol level for youth, higher drink prices in bars, and the organisation of rides home at the end of the evening, have yielded good results.

PREVENTION OF THE RISKS ASSOCIATED WITH “DRUG” USE

The term “drugs” most often includes alcohol and illegal drugs, that is, consciousness-altering products. Several interventions target illegal drugs exclusively, and more specifically, cannabis. For a long time, the objective of anti-drug prevention was abstinence. Because the consumption of drugs has not stopped increasing, other objectives related to risk reduction, such as preventing abuse or promoting risk management, have been advanced. The process of drug use is rarely defined. In studies, as in prevention programmes, the terms use, abuse, and dependence are often confused and used interchangeably. A new trend takes into account the experience young people have with drugs and the difficulties they have encountered.

The various types of prevention activities have two objectives: diminish drug use by direct or indirect strategies, or improve the quality of life. The most widespread strategies are those based on information. The KAB (Knowledge, Attitudes, Behaviour) model is used to inform young people about the negative consequences of drugs to induce changes in their behaviour. It has often been combined with programmes about life choices. Around 1970, a psychosocial model, the DARE (Drug Abuse Resistance Education) project was used by more than 50 per cent of schools in the United States: its objective was to train young people to resist pressure to use drugs from those close to them (peers, siblings, adult family members or friends) and the media. Programmes with a more comprehensive perspective, such as Life Skills Training, were established to teach the young to communicate, to resolve interpersonal conflicts, and to cope with the difficulties of daily life.

Prevention activities must be timed according to the usual age at which young people begin using each of these drugs. The optimum age is considered to be between 12 and 14 years. From 10 to 60 sessions seem necessary, distributed over several years. The Life Skills Training programme is composed of 15 sessions the first year, 10 the second, and 8 the third. Teachers and other school professional staff intervene in most of these programmes. The better trained the teachers are, the more capable of intervening they feel. Mobilisation of health care professionals at the school is important for this strategy to succeed. The participation of parents is not sought in these drug prevention strategies, although the importance of parents in education is well known. Interventions by police officers (DARE) appear ineffective in reducing drug use. Interventions by physicians questioning young patients during medical consultations have proved effective. Programmes that use interactive methods and enable teens to acquire general skills have proved more effective than those programmes offering information and values. Because all activities include some information, it is difficult to say that it is useless. It is, however, clearly insufficient by itself. Generally, the youth with the lowest use levels have a more positive opinion of the strategies than those who use more.

PREVENTION TO HEALTH EDUCATION AND PROMOTION

There is no single unequivocal definition of the concept of health, which has several senses: absence of disease; desirable biological state; biological, psychological, and social well-being; individual capacity to manage one's own life and environment; and more. Health can be defined by physiological criteria and thus assessed by objective (or objectivated) indicators of this type. These are expressed essentially in terms of normal or at risk, as are the objectives of acts involving treatment and care. From this viewpoint, health is especially the concern of health care professionals.

Perceived simultaneously as a state and as a capacity, health is assessed in terms of power to mobilise and of social interactions. Health education therefore is not limited to learning which behaviours are risky and which protective but integrates other items, including an understanding of the place health has in life and the power one has over one's own health. Health care professionals and educators thus have a joint role.

Prevention has specific objects, disease and risk; it is thus related to a particular concept of health: the absence of disease. As such, it has two important advantages: it focuses attention on the problem that must be solved, and it has a prospective viewpoint (foresee and prevent). The disadvantage of this concept is that the subject matter is restricted to risks, that is, to behaviours judged to be negative and to their dangers. What health education requires, however, is as much the promotion and maintenance of health as the prevention of disease (and risk). Positive or protective health behaviours, such as physical exercise and good nutrition, are adopted more often for reasons of pleasure or health than to prevent risks.

During the last quarter century, the concept of prevention has slowly been enriched by the concept of "health promotion". Accordingly, prevention has grown progressively, from the avoidance of harmful agents in the biophysical environment to that of associated individual behaviours. This trend, which has shown that, beyond individual behaviours, social conditions play an important role, has been accompanied by a reduction in the incidence of infectious diseases, due to vaccination and improved hygiene. A "positive" vision of health has arisen, in particular, as a resource for life. The aim of health promotion is to increase this "health potential" or "health capital," individually and collectively.

Generically, health education can be defined as a set of intentional activities designed to transfer or construct knowledge about health to or for a person, a social group, or a community. Two concepts of health education prevail today. The first perceives health as involving the successful operation of the human organism in all its aspects—biological, mental, and social. This position is held by the health sciences, whose legitimacy in education rests on this definition. In the second concept, health education is considered one aspect of general education—education in or training for life. This concept is held by those in the field of educational science (or pedagogy): for them health is one of the components and themes of education. It is important to stress that these two approaches are complementary. The first is more biological but also more immediate: it corresponds to concerns about existing risks. The second involves long-term education: alone, it cannot respond to immediate risk situations.

All those who play a role in the lives of children and adolescents are concerned about health education: parents, teachers, youth workers, members of youth movements, family physicians, pediatricians, paramedical staff, and school doctors and nurses. At the intersection of these two fields (education and health), we find the professionals of health education. "Health educators" are defined by their training, their experience, and especially by their ability to go beyond professional divides, both occupational and disciplinary, because of their skills in the domains of education, health, communication, psychology, and sociology.

They are the principal "interface" between the other participants. In addition to these direct participants, others have indirect contact with the young: health and prevention agencies, patient and consumer associations, and others who transmit messages about and influence health behaviours. These different groups of players represent different stakes and the possibilities of various kinds of interventions. Their respective responsibilities, roles, and boundary lines must therefore be defined.

The role of parents and other family members is highlighted today. Parents and other family members must be perceived not only as a possible target audience for health education—but secondary to children, who are the principal target audience—but also as co-participants, players in their own right, and even in some cases, as the principal players in their children's health education. Several experiences have shown that children's health problems can be prevented or solved through interventions conducted only with the parents.

The approach known as "health promotion," defined in the Ottawa Charter, offers a theoretical framework, and interventions are intended to be comprehensive and consistent; it designs strategies (that is, convergent and concerted interventions) synergistically with an interdisciplinary approach that takes into consideration the multicausality of health determinants. The reality of interventions, however, is often more fragmented and accordingly less conclusive. It takes a long time to modify professional practices. Non-etheless, support for the validity of the Charter principles comes from the changes in individual and collective practices that can be seen nearly everywhere.

The principles invoked for health promotion involve the concept of "environments" or life "settings" (cities, communities, schools, workplaces, health care facilities, prisons, etc.). Intervention in such settings is facilitated, not only by the existence of a "captive" population, but also because the community is structured by a strong shared identity, vigorous interactions, and communications between members; it also has networks to finance it. For interventions planned in these settings and in particular in schools, the process developed to reach the objective is also important. Because the aim is to increase individuals' capacity for self-direction, the approach must not be directive: it must accompany the individual's development. Therefore, the subjects participate in the very planning of the project, which aims to create conditions favourable to the emergence of changes in their skills and eventually, their behaviour. This approach, summarised by many authors by the terms "enabling" and "empowerment", is appropriate for the educational enterprise in general and for health education in particular.

Health education contains an individual and a collective component, which ought not be separated: the learning of health and lifestyle behaviours must be approached from both angles at the same time. To develop interventions relevant to health education, it is first necessary to understand the factors that cause and those that influence health behaviours, as well as the processes of health learning. We must therefore analyse the educational needs and reach an "educational and/or behavioural diagnosis".

Any method for analysing these needs must be based on a model or theory that explains health behaviours. More than 20 models have been developed or used in the field of health education. They can be sorted into eight major categories according to their principal characteristics, as summarised in the following table.

Principal categories of psychosocial theories and models explaining health and lifestyle behaviours

Biomedical model	It explains individuals' health behaviour by: • Their psychological predispositions (personality, motivation, comprehension skills) • Social and demographic profile (age, sex, education, etc.) • And by some characteristics • Of expected behaviour (complexity, duration, etc.) • Of the risk to be avoided (prevalence, seriousness, etc.)
Theories of information and communication	The factors considered are: characteristics of the sender, the receiver, the message, the channel and the code: who said what to whom by what means and with what effect?
Personality theories	The principal factors considered are: • Health locus control perceived individually in

	terms of power and duty and which can be internal (individuals themselves), external (others) or luck (random, God, etc.), or a combination of all three • The health logics of individuals leading to behaviour by which they manage or ignore their health
Value-expectation theories	The principal application of this is the Health Belief Model, which considers • Threat perception (vulnerability and seriousness of consequences) • Belief in the efficacy of preventive behaviour to reduce the threat (advantage-disadvantage ratio between preventive behaviour and risk)
Precede model	Predisposing, Reinforcing, Enabling Causes in Educational Diagnosis and Evaluation. This first multifactorial model used several successive diagnoses: social, epidemiologic, behavioural, educational, and then administrative.
Social learning theories	These are applied from social cognitive theory (Self Efficacy Theory), which fills out the preceding theories by taking into account • The individuals' belief in their own personal efficacy: mastery and perpetuation of the desired behaviour • Belief in the efficacy of the given behaviour in obtaining the hoped-for result
Social representation theory	This postulates that social representations of health and of other objects related to health are the principal factors influencing the construction, adoption, and changes in health behaviours.
Integration models	These structure the contents of earlier models and theories into a more comprehensive whole.

We can add to these models those that touch upon the processes of individual change but do not constitute explanatory or comprehensive frameworks for understanding health behaviours. These models are most often elaborated by specialists from a single discipline, based upon observations and experimentation, without any interdisciplinary connections. Although their principal use is the deterministic prediction of health behaviours, they can be used to understand influencing factors (social representations, lay skills) adaptable to each audience.

Because health education is a discipline oriented towards practice with the living human being as its subject matter, research in this field is applied: research in development (programmes, interventions), evaluation research and research-action are its principal trends. Whereas the subject of evaluation research is the processes and effects of educational activities, the other two types simultaneously touch upon the analysis of needs and the setting up of interventions and programmes.

The first programmes were designed from a prevention perspective, and accordingly the method of their evaluation was most commonly the quantitative experiment (randomised trial with control groups) and the quasi experiment (without randomisation). Non-etheless, controlling variables reduces the complexity of the reality and, by that very fact, modifies the object of study. Later, sociology, pedagogy, and psychology contributed their methods and tools (qualitative approaches with semi-structured interviews and discussion groups). The current trend is to combine these approaches, thereby introducing the difficulties inherent in any truly interdisciplinary study.

Moreover, evaluation and evaluation research have long focused on changes in knowledge, health and epidemiologic results, and economic aspects, while ignoring the role played by the overall set of factors, processes, and organisational aspects in the quality and effectiveness of health education. The concept of health promotion with a goal of increasing "health potential" implies that indicators of individuals' "action skill levels" be defined and used in evaluation research.

DETERMINING HEALTH NEEDS AND INTERESTS OF CHILDREN

There is no single procedure that can be followed or set formula applied to determine the health needs and interests of children. The procedure followed will vary with the particular school and community involved. It should be pointed out, however, that needs and interests should be determined in planning and administering a sound health education programme.

Some of the materials and procedures for determining children's interests and needs that have been found successful in certain schools are as follows:

1. *Pupil Observation.* Observing the pupil and watching his habits will reflect his knowledge, attitudes, and health practices.
2. *Records.* Pupils' general health history, attendance, communicable disease, and other health records give information as to certain diseases, health defects, behaviour, and abnormalities.
3. *Tests.* Diagnostics health knowledge, attitude and habit tests, and health interest inventories may be of value for older elementary pupils.
4. *Class Discussions.* Many of the interests and needs of children are revealed during classroom discussions.
5. *Surveys.* Utilizing check sheets, questionnaires, etc. for school and community use will give information. For example, a survey of food habits may suggest needs.

6. *Consultations.* School physician, family physi-cians, nurses, parents, public health workers, and others can provide important health information.
7. *Visits.* Observing practices in other schools may indicate needs not previously revealed.
8. *Health Conferences.* Discussions of school health problems in general frequently focus attention on needs and interests that have been overlooked.
9. *Professional Literature.* Examining material concerned with needs and interests of children can be of value.
10. *Parent-Teacher Association Meetings.* Those meetings can sometimes provide valuable health data.

LIVING HEALTHFULLY AT SCHOOL

The school should be conducive to healthful living. First of all, this means that the physical environment is attractive, sanitary, and functional. The buildings have been constructed with health and safety standards in mind. Secondly, the mental and emotional environment is conducive to good health. This is as important as the physical environment. These two phases of living healthfully at school are discussed in this section.

The Physical Environment: The elementary school classroom teacher should be familiar with certain aspects of the physical environment.

1. *School Site.* In urban communities the school should be situated near transportation facilities, but at the same time away from industrial plants, railroads, noise, heavy traffic, fumes, and smoke. It should be attractive and well landscaped. In rural communities the school should be located in an attractive setting where the physical environment is conducive to healthful school living.
2. *Play AND recreation Area.* Standards recommend at least 5 acres of land for elementary schools. Play area should consist of a minimum of 100 square feet for each child.
3. *Buildings.* There is a trend towards one-story construc-tion, where possible, with stress on functional planning rather than ornamental structure. In addition to an adequate number of classrooms, school buildings should contain such facilities as health suite; special rooms for activities like music, dancing, arts, crafts; gymnasium; cafeteria; and, if possible, a swimming poll.
4. *Lighting.* It is important to conserve vision, prevent fatigue, and improve morale. Light intensity in most classrooms varies from 15 to 50 foot-candles for reading and close work.
5. *Heating and Ventilation.* Heating standards vary according to the activities and the clothing worn by the participants. The following specifications are generally

accepted. Classrooms, offices, and cafeterias: 68-72 degrees (30 inches above the floor); kitchens, closed corridors, shops, and laboratories; 65-68 degree (60 inches above floor); gymnasiums and activity rooms; 55-65 degrees (60 inches above floor). Ventilation should provide from 8 to 21 cubic feet of fresh air per minute per occupant. The recommended humidity ranges from 35 to 60 per cent. The type and amount of ventilation will vary with the specific needs of the particular area to served.

6. *Furniture.* Seats and desks which are adjustable and movable are recommended by most educators. Desks should be of proper height and adjusted to fit the pupil comfortably and properly. This helps to avoid fatigue and faulty posture habits.
7. *School Plant Senitation.*
   (a) Water supply should be safe and adequate. One authority suggests that at least 25 gallons per pupil per day is needed for all purposes.
   (b) Drinking fountains at various height should be recessed in corridor walls. Approximately one drinking fountain should be provided for every seventy-five pupils. The stream of water should flow from the fountain in a manner that does not require the drinker to get his mouth too near the opening.
   (c) Water closets, urinals lavatories, and washroom equipment such as soap dispensers, toilet paper holders, waste containers, mirrors, bookshelves, and hand-drying facilities should be provided as needed.
   (d) Waste disposal should be adequate.
8. *Other Essentials for a Healthful Environment*
   (a) A well-planned school lunch programme should include good, wholesome food, adequate time for eating, and a pleasant, quiet, attractive atmosphere.
   (b) Necessary custodial help and cleaning supplies should be secured for maintaining sanitary conditions in all classrooms, lavatories, gymnasiums, cafeterias, etc.
   (c) Safe, uncrowded transportation to and from school and on all trips sponsored by the school should be provided.
   (d) Special arrangements should be made for handicapped children, such as seats near the front of the room for auditory handicapped individuals.

Mental and Emotional Environment: The mental and emotional environment is equally important if pupils are to live healthfully at school. Some of the factors that the elementary school teacher should recognize as contributing to good mental and emotional health are:

1. The teacher, school official, and pupils work harmoniously and happily together.
2. The teacher exemplifies good physical, mental, emotional, and social health.
3. The educational curriculum is adapted to the needs and interests of the child.
4. An atmosphere of friendliness and consideration of the rights of others exists in all personal school relationships.

5. The teacher gives each child a feeling of belonging and security.
6. The teacher does not use sarcasm or ridicule children.
7. Frequent opportunities are provided for relaxation and play.
8. Pupils are encouraged to improve themselves rather than to excel over their classmates.
9. A permissive climate prevails.
10. Pupils help in making decisions.
11. A Variety of teaching methods is used.
12. Individual differences are recognized.
13. The length of the school day is in conformance with the age of the child.
14. Classes are scheduled in a way that does not result in excessive fatigue.
15. Excessive emphasis is not placed on marks.
16. School attendance is not overemphasized to the point of disregarding the health of the individual.
17. As much freedom as possible is given to the pupils.
18. The teacher takes a personal interest in and loves each child.
19. The teacher has a pleasing personality.
20. There is good rapport between the teacher and the parents of each child.

PHYSICAL EDUCATION, SPORT AND ACADEMIC ACHIEVEMENT

Bailey discussed physiological changes relevant to a prospective relationship between Physical Education and sport and academic performance. He explained that it has been suggested that by increasing blood flow to the brain, Physical Education and sport may enhance mood, mental alertness and self-esteem. 'The evidence base of such claims is varied and more research is still required. However, existing studies do suggest a positive relationship between intellectual functioning and regular physical activity, both for adults and children'. Effects rather than the underlying mechanisms are arguably of greater interest here.

Results of a sustained study undertaken in the United States of the relationship between the time students spent in Physical Education and academic performance were published in 2008. Carlson et al identified that grade-point averages, scores on standardised tests can be regarded as direct indicators of academic achievement and that grades in specific courses; measures of concentration, memory, and classroom behaviour can be deemed indirect indicators. Their study was a longitudinal study of students in kindergarten through to fifth grade, and involved a nationally representative sample group. Measurement of academic achievement was a standardised test administered at 5 time points. Time spent in physical education was ascertained and used as a basis for categorising children as involved in low, medium, or high amounts of physical education.

Academic achievement in terms of performance in mathematics and reading tests was scored on an item response theory scale. Gender differences were observed in this research. For girls, there was some evidence of a positive association between time engaged in physical education and academic achievement: Girls in all grades who were in the low physical education category had the lowest IRT scale scores for mathematics and reading, although only in kindergarten and first grade were these differences significant for reading and mathematics.

In fifth grade differences were significant for reading only. No association between time engaged in physical education and academic achievement was found for boys. Similar results arose when the researchers pursued longitudinal associations. From kindergarten through fifth grade, girls with the highest exposure to physical education scored on average 2,4 points higher on the IRT reading scale and on average 1,5 points higher on the IRT mathematics scale than did those in the low physical education category. Again, no association was found for boys.

Carlson *et al* concluded that their study supported other research in identifying that time spent in physical education does not adversely affect academic achievement and that 'fear of negatively affecting academic achievement does not seem to be a legitimate reason for reducing or eliminating programmes in physical education'. Further, the findings for girls suggest that more time in physical education, may have a positive impact on academic performance.

Similarly, Bailey concluded that: Overall, the available research evidence suggests that increased levels of physical activity in school-such as through increasing the amount of time dedicated to PES-does not interfere with pupils' achievement in other subjects and in many instances is associated with improved academic performance. The QCA's report on the PESS project in England has provided further positive indications in relation to these issues.

Specifically, the QCA reported that the targeted investments in Physical Education and Sport have positively impacted upon student attainment in Physical Education at the end of primary schooling and in terms of examination course results in Physical Education and dance.

As the QCA have acknowledged, attributing broader academic improvement to participation in PESS is problematic, but the indications of impact arising in the project are nevertheless encouraging. Specifically, all of the schools involved in the PESS investigation from the outset have seen improvements in their national curriculum test and GCSE results.

Many school principals feel that PESS has had a significant impact on learning achievements across the curriculum and teachers have reported improvements in students' confidence, concentration and achievement. Improvements in all of these respects are clearly fundamental to advancing student learning and are relevant to students' social well-being.

THE FORMATION OF NATIONAL IDENTITY

In addition to the social practices that contribute actively to a nation's image, national cultures are characterized by competing discourses through which people construct meanings that influence their self-conception and behaviour. These discourses often take the form of stories that are told about the nation in history books, novels, plays, poems, the mass media, and popular culture. Memories of shared experiences-not only triumphs but also sorrows and disasters-are recounted in compelling ways that connect a nation's present with its past. The construction of a national identity in large part involves reference to an imagined community based on a range of characteristics thought to be shared by and specific to a set of people. Stories and memories held in common contribute to the description of those characteristics and give meaning to the notion of nation and national identity. Presented in this way, nationalism can be used to legitimize, or justify, the existence and activities of modern territorial states.

Sports, which offer influential representations of individuals and communities, are especially well placed to contribute to this process of identity formation and to the invention of traditions. Sports are inherently dramatic. They are physical contests whose meanings can be "read" and understood by everyone. Ordinary citizens who are indifferent to national literary classics can become emotionally engaged in the discourses promoted in and through sports. Sometimes the nationhood of countries is viewed as indivisible from the fortunes of the national teams of specific sports. Uruguay, which hosted and won the first World Cup football championship in 1930, and Wales, where rugby union is closely woven with religion and community to reflect Welsh values, is prime examples. In both cases national identity has been closely tied to the fortunes of male athletes engaged in the "national sport." England's eclipse as a cricket power is often thought, illogically, to be symptomatic of a wider social malaise. These examples highlight the fact that a sport can be used to support, or undermine, a sense of national identity. Clifford Geertz's classic study of Balinese cockfighting, Deep Play: Notes on the Balinese Cockfight, illustrates another case in point. Although Balinese culture is based on the avoidance of conflict, men's identification with their birds allows for the vicarious expression of hostility.

PATRIOT GAMES

By the beginning of the final decades of the 19th century, sports had become a form of "patriot games" in which particular views of national identity were constructed. Both established and outsider groups used and continue to use sports to represent, maintain, and challenge identities. In this way sports can either support or undermine hegemonic social relations. The interweaving of sports and national identity politics can be illustrated with several telling examples. In 1896 a team of Japanese schoolboys soundly defeated a team of Americans from the Yokohama Athletic Club in a series of highly publicized baseball games. Their victories, "beating them at their own game," were seen as a national triumph and as a repudiation of the American stereotype of the Japanese as myopic weaklings. Similarly, the "bodyline" controversy of the 1932-33 crickets Test series between Australia and England exemplifies the convergence of sports and politics. At issue were the violent tactics employed by the English bowlers, who deliberately threw at the bodies of the Australian batsmen in order to injure or intimidate them. The bowlers' "unsporting" behaviour raised questions about fair play, good sportsmanship, and national honour. It also jeopardized Australia's political relationship with Great Britain. So great was the resulting controversy that the Australian and British governments became involved. Arguably, one consequence was the forging of a more independent attitude in Australians' dealings with the British in the political, economic, and cultural realms. The Soviet Union's military suppression of reformist efforts to create "socialism with a human face" in Hungary and in Czechoslovakia were followed by famous symbolic reenactments of the conflicts in the form of an Olympic water-polo match and an ice hockey encounter. In both cases, sports were invested with tremendous political significance, and the Soviet team's defeat was seen as a vindication of national identity.

NATIONAL CHARACTER

In each of these examples, a historical legacy was invoked, past glories or travesties were emphasized, and the players were faced with maintaining or challenging a set of invented traditions. This link between sports, national culture, and identity can be extended further. Some sports are seen to encompass all the qualities of national character. In the value system of upper-class Englishmen, for example, cricket embodies the qualities of fair play, valour, graceful conduct, and steadfastness in the face of adversity. Seen to represent the essence of England, the game is a focus of national identification in the emotions of upper-class males. Moreover just as Englishness is represented as an indefinable essence too subtle for foreigners to comprehend so too are the mysteries of cricket deemed to be inscrutable to outsider. In a similar manner, bullfighting has been portrayed in the visual and the verbal arts as a material embodiment of the Spanish soul, Gaelic football is thought to be an expression of an authentic Irishness, and sumo wrestling is said to represent the indefinable uniqueness of Japanese culture.

TRADITIONS AND MYTHS

National culture and identity are also represented by an emphasis on origins, continuity, tradition, and timelessness. For most English people, for example, the origins of their culture and national identity seem to be lost in antiquity. Englishness is taken for granted as the result of centuries of uninterrupted tradition. This emphasis on continuity is strikingly evident in sports contests between nations. Accordingly, when teams from England and Scotland compete, they are characterized as "auld enemies." That political institutions are also imbued with a sense of venerable tradition is easily exemplified in the pageantry that surrounds the English monarchy. Yet the traditions associated with both the monarchy and sports are not as old as claimed. Indeed, both appear to be based on foundational myths-that is, on myths that seek to locate the origins of a nation, a people, or a national character much earlier in time and place than the evidence supports.

Baseball, which for a century was considered to be the "national game" of the United States, is a case in point. Instead of tracing the origins of the game to its English roots in children's games such as cat and rounders, Americans accepted the addled recollections of a lone octogenarian and credited Abner Doubleday with having invented a game that he may never have played. Similarly, Italians use the word calcio to describe the sport known to the rest of the world as "association football," as "soccer," or simply as "football". The use of calcio implies that the origins of modern football can be traced to Renaissance Italy. Sumo provides another striking example of invented tradition. The colourful traditional costume worn by sumo officials suggests that the sport has evolved almost unchanged since the 11th century, but the costume was actually devised in 1909 during a period of intense nationalism. The role sports play in the interaction of culture and national identity is sometimes viewed as inherently conservative.

Some believe that the association of sports with nationalism goes mere patriotism and becomes chauvinistic and xenophobic. The behaviour of football hooligans at international matches lends support to the argument. On the other hand, sports also have contributed to liberal nationalist political struggles. One frequently cited example is the 19th-century Slavic gymnastics movement known as Sokol. Gymnastic clubs in what is now the Czech Republic, Slovakia, and Poland were in the forefront of the struggle for national liberation from Austrian and Russian rule. A similar role was played by Algerian football clubs when they became centres of resistance to French colonialism. Sports-through the use of nostalgia, mythology, invented traditions, flags, anthems, and ceremonies-contribute greatly to the quest for national identity. Sports serve to nurture, refine, and develop the sense that nations have of themselves. Yet, in the context of global sports, this role has become increasingly contradictory. In introducing people to other societies, global sports strengthen cosmopolitanism even as they feed ethnic defensiveness and exclusiveness. For example, the development of cricket in South Asia reflects that region's imperial past and postcolonial present, but the game has taken on uniquely Indian, Pakistani, and Sri Lankan attributes far removed from the pastoral values associated with the English village green.

9

Physical Education, Sport and Physical Activity

In both the United States and United Kingdom, initiatives have acknowledged that the contribution that Physical Education lessons can make in relation to recommended physical activity for children is important but needs to be acknowledged as one of a number of contributions needed. Recess, lunchtime and activities organised before and/or after school have been positioned with Physical Education in 'whole school' approach to enhancing physical activity time through the school day and week. Coordinated efforts are undoubtedly crucial if the prospective benefits of participation in Physical Education and school sport are to be realised. In 2007 Evenson, Ballard, Ginny and Ammerman undertook a research project specifically focusing on the North Carolina State Board of Education's 2005 update to the Healthy Active Children Policy, to include a requirement that all kindergarten through to eighth-grade children receive at least 30 minutes of moderate-to-vigourous physical activity each school day through physical education, recess, and other creative approaches.

Their study provided important insights into the ways in which an extent to which the new requirement was being met and the successes and challenges experienced in implementation. All NC school districts were asked to complete an online survey after the first full year of implementation of the policy in 2007. Of the 106 school districts that completed the survey, 83% reported that the Healthy Active Children Policy was incorporated with their wellness policies and 67% reported it as integral to their school improvement plans.

The 30 minutes of daily moderate to vigourous physical activity required by the NC Healthy Active Children Policy was achieved primarily through recess, physical education, and classroom Energisers for elementary schools. For middle schools, these three components were also the main contributors, with more reliance on physical education and less on recess and classroom Energisers. Based on strategies the survey queried, these data indicate that some districts did not meet the 30-minute physical activity recommendation.

The study also generated notable reports of positive effects of the policy in both elementary and middle schools; specifically:

- Ggreater student focus on studies;
- Physical activity participation of students;
- Awareness of healthy habits,
- Student alertness;
- Student enjoyment; and
- Staff involvement.

These findings align with recent work in England that has sought to capitalise on the capacity of integrated Physical Education and school sport initiatives to have multiple benefits for children's education and development. Addressing mental health and attaining enhanced social and emotional well-being amongst young people clearly remains a major contemporary challenge in Australia. This is another area in which further research is certainly needed–particularly to pursue, whether improvements seen to arise from school-based interventions and curriculum initiatives can be sustained beyond them and over time. The development and implementation of the Mind Matters resources has undoubtedly provided welcome support for curriculum provision directed towards mental health and well being in many schools. A commissioned evaluation of the classroom implementation of the curriculum resource 'Understanding Mental Illness" pointed to improvements in students' knowledge, attitudes and behavioural intentions that could arise from the teaching of the UMI module, particularly immediately post-teaching. In relation to the more sustained benefits, it was noted that: At delayed post-teaching, with cautionary interpretation due to a small number of participant responses, students' knowledge showed a predictable decline whilst attitude and behavioural intentions continued to improve.

Encouragingly, Bailey's review identified that while we may not fully understand the mechanisms that underpin effects, 'there is now fairly consistent evidence that regular activity can have a positive effect upon the psychological well-being of children and young people'. More specifically, he reported that 'the evidence is particularly strong with regards to children's self-esteem' and that research has also associated regular activity with reduced stress, anxiety, and depression. In his view, evidence is thus growing to support claims that 'well-planned and presented PES can contribute to the improvement of psychological health in young people'.

AIMS AND OBJECTIVES OF PHYSICAL EDUCATION

AIMS

The ultimate goal is final end aim of physical education, it the way and achieved some certain objectives. The general education like aim of physical education is to develop human personality in its totality well planned activity programmes. In physical education aim at the all round development of the personality of an individual or some development of human personality and it includes physical, mental, social, emotional and moral aspects to make an individual a good citizen. The physical education means at making an individual physical fit, mentally alert, emotionally balanced, socially well adjusted, morally true and spiritually uplifted.

The objectives of Physical Education are steps considered towards the attainment of the aim and the particular and precise means to realise an aim. An aim is achieved it become an objective in the action that goal on continuing. There some objectives of physical education are the objective of physical fitness is essential to leading a happy, vigourous and abundant life. The second is the objective of social efficiency is concerned with one proper adaptation to group living and all these qualities help a person to make him a good citizen. Another objectives of physical education is culture, it aims at developing an understanding and appreciation of one own local environment as well as the environment and a person understand the history, culture, religious practices etc and the aesthetic values associated with these activities. So find the best step the ultimate goal in aims of physical education.

OBJECTIVES

Physical education is an important part of every school curriculum and a class every pupil awaits. Physical education is that segment of the daily timetable that every student eagerly waits to attend, as it is the only official time when the students can be on the grounds, engaged in their favourite sports. One of the main objectives of physical education is to bring in this element of joy to the academic orientation of schools. Physical education aims at dedicating a daily time for some physical activity for the students. The physical training class, as it is also called, involves sports, games, exercise and most importantly, a break from the sedentary learning indoors.

One of the other important objectives of physical education is to instill in the students the values and skills of maintaining a healthy lifestyle. Daily physical activity promotes an awareness of health and well being among students. It boosts them to engage in physical activities on a daily basis. It promotes them to lead a healthy life in adulthood. Physical education classes constitute programmes to promote physical fitness in students, train them in sports, help them understand rules and strategies in playing and teach them to work as a team. A very vital factor in physical education is to develop interpersonal skills in children. Sports aim at making them team players, developing a sportsman spirit in them and enhancing their competitive spirit. Sports that form a part of physical education classes help the students invest time in fruitful and competitive activities.

One of the other important objectives of physical education is to inculcate in the minds of the students, the importance of personal hygiene and cleanliness. Physical education classes aim at teaching the students, the habits of personal cleanliness and the importance of the maintenance of personal hygiene in life. Physical education classes also impart sex-education to the students, help them clarify their doubts and find answers to all the questions that occur to their minds. The sports, which are a part of the physical education class, help in developing motor skills in children. The ability to hold a racket or a bat, the ability to catch a ball and the ability to swing a bat are some examples of the motor abilities that can develop with the help of sports. The physical activity that is involved in physical education helps the students in bringing discipline to body posture and body movements.

Hitting a ball with a bat or a shuttle with a racket as also aiming a ball for a goal or catching it to get the opponent team out, are some of the commonly observed actions in sports and are extremely beneficial in improving hand-eye coordination. The very important objective of physical education is to encourage the upcoming sportsmen and women of the crowd. Physical education gives the budding sports people a platform to exhibit their talents. Those with a flair for sports get an opportunity to display their talent. Their small step on the school playground can eventually turn into a huge leap in the field of sports. Moreover, sports refresh the students' minds. Physical education class becomes enjoyable for the kids while proving helpful for their overall growth and development. Physical education is indeed one of the most fruitful activities of a school schedule.

PHYSICAL ACTIVITY: A KEY FOR THE PRECLUSION OF OBESITY IN CHILDREN

Obesity is emerging as one of the most serious problem discern of the present century especially childhood obesity which is the major public health crunch globally that are on rise. According to WHO, 22 million children (under 5 years of age) are overweight. The prevalence has increased at an alarming rate. Globally, in 2010 the number of overweight children under the age of five is estimated to be over 42 million.

Close to 35 million of these are living in developing countries. Obesity is defined as a BMI equal to or greater than 95th percentile for age and sex. It is normally measured in children by plotting body mass index on a standard growth chart using a defined cut-off point and generally caused by imbalance between calories consumption and consumed. Overweight and obese children are likely to stay obese into adulthood and more likely to develop non-communicable diseases like diabetes and cardiovascular diseases at a younger age. Prevention of childhood obesity therefore needs high priority. Children represent the future, and ensuring their healthy growth and development ought to be a prime concern of all societies.

Physical inactivity has been identified as the fourth leading risk factor for global mortality causing an estimated 3.2 million deaths globally. Physical activity is defined as any bodily movement produced by skeletal muscles that require energy expenditure. Regular moderate intensity physical activity – such as walking, cycling, or participating in sports – has significant benefits for health. Nearly half of American youths 12-21 years of age are not vigorously active on a regular basis. Moreover, physical activity declines dramatically during adolescence. Therefore the purpose of this chapter is to highlight the importance of physical activity and nutrition in controlling the childhood obesity along with the assessment techniques.

EPIDEMIOLOGY OF CHILDHOOD OBESITY

Obesity was first included in the international classification of diseases in 1948. Since then, an epidemic has developed internationally, affecting all age groups. The combination of our genetic propensity to store fat, the ready availability of calorie dense foods, and sedentary lifestyle promotes overweight. The child's food environment at home and parental obesity are strong determinants.

Over the past 40 years, rates of obesity have doubled in 2- to 5-year-olds, quadrupled in 6-to 11-year-olds, and tripled in 12- to 19-year-olds. The causes of childhood obesity are complex and interconnected. Puberty and the following adolescent period are acknowledged as particularly vulnerable times for the development of obesity due to sexual maturation and, in many individuals, a concomitant reduction in physical activity. Childhood obesity and the accompanying health consequences are growing at a fast rate in India.

A new study carried out by American researchers suggests that community involvement is important in the fight against childhood obesity. Maternal instinct can be a dicey, since mothers want their child to be well fed, which is not a bad thing. But now-a-days home theatre, play station, unlimited internet has locked up kids at home. Lack of exercise or activity has expanded our children's waste line, making them prone to lifestyle diseases and health problems.

Childhood obesity builds a ticking time bomb of sickness in later life - and action to avert ill health should start before the kids attain school going age. A new AIIMS study on 10,000 Delhi schoolchildren has found that 3-4 per cent of them suffered from hypertension, the most common cause of heart-related deaths. Among these were children as young as five year's old.

PHYSICAL ACTIVITY AND CHILDHOOD OBESITY

Current research suggests that physical inactivity is inversely correlated with the risk of obesity. The World Health Organisation, the U.S. Dept. of Health and Human Services, and other authorities recommend that for good health, adults should get the equivalent of two and a half hours of moderate-to-vigorous physical activity each week. Worldwide, people are less active today than they were decades ago. While studies find that sports and leisure activity levels have remained stable or increased slightly.

Regular physical activity 45-60 min per day prevents unhealthy weight gain and obesity, whereas sedentary behaviours such as watching television promote them. Regular exercise can markedly reduce body weight and fat mass without dietary caloric restriction in overweight individuals. Local health departments (LHDs) play a crucial role in the identification, management and prevention of obesity.

Physical activity and health has been late to emerge as a scientific field and as a target of research, beyond the early impetus provided by medical epidemiologists. The continuum of physical activity behaviour ranges from being inactive to being very physically active. Some individuals can be considered physically active if they are meeting the current recommendations for moderate to vigorous physical activity, yet they may also be very sedentary throughout the rest of their day.

The reality is that attaining a healthy weight is difficult if physical activity is not a part of child’s lifestyle. One probable cause of childhood overweight and obesity is decreased daily energy expenditure. Evidences clearly showed that regular physical activity helps in reduction of body fat percentage. Children are more active than adults, but their activity levels decline as they move towards adolescence, and significant numbers of young people do not participate in recommended levels of physical activity. Inactivity among children has likely increased because of factors such as dependence on cars for transportation, increased screen time (*e.g.*, television, videogames, internet), and the constraints of the built environment (*e.g.*, urban sprawl, lack of recreational facilities, neighbourhood safety).

Physical activity also has several other aspects that are positive for the obese child’s health, such as improving the metabolic profile and psychological wellbeing. In recent years, physical education and recess have been reduced in many schools, resulting in long periods of inactivity during the school day. Without opportunities for physical activity in school, many children will fail to meet minimal activity requirements. Physical education provides youths with meaningful amounts of daily physical activity. Exercise has long been considered an integral component of weight management, but available evidence suggests that exercise alone is a relatively inefficient means for losing weight. In contrast, regular exercise appears crucial in the prevention of weight gain and successful maintenance of weight loss, and in the fostering of cardiovascular health.

ENDURANCE EXERCISE AND OBESITY

Endurance is the ability to do a single movement under the condition of fatigue. Research suggests that endurance play an important role in preventing the obesity. Endurance exercise helps in reducing skeletal muscle-specific and systemic oxidative damage while improving insulin resistance and cytokine profile associated with obesity, independent of weight loss. Endurance exercise training decreased plasma leptin levels independently of changes in plasma insulin levels and body fat percentage.

The relationship between fat mass and leptin, which is also found in humans, might be affected by mechanisms acting on fat mass. The insulin hormone is thought to be important in this control mechanism. A continuous exercise training protocol can elicit high rates of fat oxidation, increases the contribution of fat to substrate oxidation during exercise. One of the fundamental mechanisms for the positive effects of exercise training in obesity may be related to its effects on fat utilisation. Researches proved that an endurance exercise programme is more effective in increasing fat oxidation during exercise.

NUTRITION AND OBESITY

Good nutrition is just like a good antibiotic or vaccine in preventing illness. Changes in diet and sedentariness are fuelling the obesity epidemic. Good nutrition play an essential parts of a person's overall health and well-being. Insufficient nutrient intake during childhood has been linked to physical and mental health problems as well as emotional and behavioural problems, learning deficiencies, and lower grades. Fundamental causes of the obesity epidemic are sedentary lifestyles and high-fat energy-dense diets, both resulting from the profound changes taking place in society and the behavioural patterns of communities as a consequence of increased urbanisation and industrialisation and the disappearance of traditional lifestyles.

According to the study overweight children, particularly girls, reported eating less bread, cake, and cream, adding less sugar to beverages, and eating sweets and ice cream less frequently than thin and normal-weight children. A higher percentage of the obese group reported skipping one meal and eating no snack at school. Overweight teen-agers appear to be more conscious of their food intake than under- and normal-weight children.

As schools have been proposed worldwide as a major setting for tackling childhood obesity it is essential that future policy evaluations measure the long term effectiveness of a range of school food policies in tackling both dietary intake and overweight and obesity. People living in urban areas consume diets distinctly different from those of their rural counterparts. One of the more profound effects is the accelerated change in the structure of diet, only partially explained by economic factors. A second is the emergence of a large proportion of families with both currently malnourished and overweight members. Low birth-weight babies whose families were food insecure in early childhood are almost 28 times more likely than their peers to be overweight or obese at age 4 ½. Gregory et al. reported that the average fat intake of children aged 4-18 years in the UK is close to the government recommendation of 35 per cent energy.

On the other hand, some cross-sectional studies have found a positive relationship between fat intake and adiposity in children even after controlling for confounding factors. Soft drink intake has been also associated with the prevalence of obesity and type II diabetes among children.

While it is possible that drinking soda instead of milk would result in higher intake of total energy, it cannot be concluded definitively that sugar containing soft drinks promote weight gain because they displace dairy products. Fast foods are one of the most advertised products on television and children are often the targeted market. Reducing the huge volume of marketing of energy-dense foods and drinks and fast-food restaurants to young children, particularly through the powerful media of television, is a potential strategy that has been advocated

ANTHROPOMETRIC VARIABLES FOR THE ASSESSMENT OF OBESITY IN CHILDREN

Obesity can be assess by a variety of anthropometric variables namely Body mass index, Waist Circumference, hip circumference, Waist to height ratio, waist to hip ratio, different skinfolds of the body, standing height and weight measurement and predictive equations. In addition to this there are some more variables that may also help in assessing the childhood obesity namely upper arm and leg length, head and mid arm circumference. Anthropometry is a key component of nutritional status assessment in children as well as in adults.

Comparing anthropometric data from children of dierent ages is complicated by the fact that children are still growing (We do not expect the height of a 5-year-old to be the same as the height of a 10-year-old!). But the question is which anthropometric variable is more reliable in comparison to others. One of study suggested that the validity of the anthropometric skinfold thickness in the obese children is low and BMI provides the best estimate of body fat.

The childhood obesity working group of the international obesity taskforce recommended the use of body mass index (BMI) cut off points to categorize children as normal weight, overweight, or obese based on age, gender, and BMI. A new function for egen has been developed to allow transformation of child anthropometric data to z-scores using the LMS method. An additional function allows for children to be categorised according to body mass index (weight/height^2) using international cut o points recommended by the Childhood Obesity working group of the international obesity taskforce.

MOLECULAR MECHANISM OF OBESITY

The command centre for communication between the brain and the body is the hypothalamus. It controls all homeostatic systems, including circadian rhythms, sleep, body temperature, stress regulation, sexual behaviour, and water balance. The hypothalamus also regulates food intake. Hormonal regulators of energy homeostasis can also act on brain reward circuits, most notably on the mesoaccumbens dopamine system, to increase or decrease the incentive value of food depending on energy requirements.

However, electrical or chemical stimulation of brain areas that regulate food reward can trigger binge-like overeating even in recently fed animals in which homeostatic satiety signals have been engaged. Recent research has identified specific molecular mediators of the link between inflammation and insulin resistance in obesity. Study of these mediators and the specific mechanisms underlying inflammation and insulin resistance in obesity holds the promise for novel pharmacotherapy for obesity-related metabolic disease. Leptin, a key appetite-regulating hormone derived from the white adipose tissue, primarily acts on hypothalamic neurons to activate catabolic pathway and inhibit anabolic pathway, which can result in anorexia and weight reduction.

THE CONTRIBUTIONS OF PHYSICAL ACTIVITY

While awareness of the importance of physical activity to health has certainly increased, what most people do not understand is how important physical activity is to the well-being of children and youth in areas other than health. Although these contributions may not receive as much press, they are equally important.

GROWTH AND DEVELOPMENT

Children and adolescents are growing and developing as physical beings. Regular physical activity is essential to their growth and development. Regular physical activity helps build strong bones and muscles, helps control weight, and may play a major role in improving blood pressure and cholesterol levels. Strong bones develop as a result of weight-bearing activities and those that stress the bones. Active children have a higher bone mass and are less likely to have problems (osteoporosis) later in life. The development of all systems of the body is affected by the level of physical activity of children and adolescents.

SOCIAL AND EMOTIONAL WELL-BEING

Play is an important human behaviour. While the forms of play change throughout the life span, the need for playful activity does not. There is evidence that motor skills used in play and learned early in life enhance a child's ability to participate in activities later in life. Physical play is a critical contributor to the development of children's social skills and the well-being of adults. Elementary schools that have eliminated recess, as well as home environments that do not provide children opportunities to go out and play, deprive students not only of the opportunity to be physically active but also of the opportunity to develop the social skills they will need as an adult.

Physical play is important to our emotional well-being. Studies show that regular physical activity in childhood and adolescence reduces stress and improves self-esteem. We are physical beings, and as such we need to move. Each culture has accepted forms of play. Children who do not learn to participate in the accepted forms of play of their culture are at a disadvantage socially as children and later as adults. Such learning not only takes care of our physical body but also facilitates emotional well-being.

COGNITIVE FUNCTIONING AND ACADEMIC PERFORMANCE

A common misperception of educators is that if they take time out to provide students with the physical activity they need, the students will not do well academically. Actually there is more and more evidence that physical activity enhances cognitive functioning; time spent in increased physical activity during the school day does not decrease academic performance but instead actually increases it. Children need breaks from sedentary activity. Physical activity is a great medium for learning other content areas and should be used to actually teach academic content. When physical activity is provided for children during the school day, they are more attentive and teachers have fewer behaviour problems. Beginning research with adolescent-aged students also indicated that students who participated in vigorous activity for at least 20 minutes three times a week had higher grades. Higher grades are attributed to the increased attention that students have when they are not bored and not forced to spend an entire day in sedentary activities.

AN EXPLANATION OF ACTIVITY THEORY

This section presents a short introduction to activity theory, and some brief comments on human creativity in activity theory and the implications of activity theory for tacit knowledge and learning.

ACTIVITIES

Activity theory begins with the notion of activity. An activity is seen as a system of human “doing” whereby a subject works on an object in order to obtain a desired outcome. In order to do this, the subject employs tools, which may be external (*e.g.*, an axe, a computer) or internal (*e.g.*, a plan). As an illustration, an activity might be the operation of an automated call centre. As we shall see later, many subjects may be involved in the activity and each subject may have one or more motives (*e.g.*, improved supply management, career advancement or gaining control over a vital organisational power source). A simple example of an activity within a call centre might be a telephone operator (subject) who is modifying a customer's billing record (object) so that the billing data is correct (outcome) using a graphical front end to a database (tool). Kuutti formulates activity theory in terms of the structure of an activity. "An activity is a form of doing directed to an object, and activities are distinguished from each other according to their objects. Transforming the object into an outcome motivates the existence of an activity. An object can be a material thing, but it can also be less tangible."

Kuutti then adds a third term, the tool, which 'mediates' between the activity and the object. "The tool is at the same time both enabling and limiting: it empowers the subject in the transformation process with the historically collected experience and skill 'crystallised' to it, but it also restricts the interaction to be from the perspective of that particular tool or instrument; other potential features of an object remain invisible to the subject...". As Verenikina remarks, tools are "social objects with certain modes of operation developed socially in the course of labour and are only possible because they correspond to the objectives of a practical action."

THE LEVELS OF ACTIVITY THEORY

An activity is modelled as a four-level hierarchy. Kuutti schematises processes in activity theory as a four-level system.

Verenikina paraphrases Leont'ev as explaining that "the non-coincidence of action and operations... appears in actions with tools, that is, material objects which are crystallised operations, not actions nor goals. If a person is confronted with a specific goal of, say, dismantling a machine, then they must make use of a variety of operations; it makes no difference how the individual operations were learned because the formulation of the operation proceeds differently to the formulation of the goal that initiated the action."

The levels of activity are also characterised by their purposes: "Activities are oriented to motives, that is, the objects that are impelling by themselves. Each motive is an object, material or ideal, that satisfies a need. Actions are the processes functionally subordinated to activities; they are directed at specific conscious goals... Actions are realised through operations that are determined by the actual conditions of activity."

Engeström developed an extended model of an activity, which adds another component, community ("those who share the same object"), and then adds rules to mediate between subject and community, and the division of labour to mediate between object and community.

Kuutti asserts that "These three classes should be understood broadly. A tool can be anything used in the transformation process, including both material tools and tools for thinking. Rules cover both explicit and implicit norms, conventions, and social relations within a community. Division of labour refers to the explicit and implicit organisation of the community as related to the transformation process of the object into the outcome."

Activity theory therefore includes the notion that an activity is carried out within a social context, or specifically in a community. The way in which the activity fits into the context is thus established by two resulting concepts:

- *Rules:* These are both explicit and implicit and define how subjects must fit into the community;
- *Division of labour:* This describes how the object of the activity relates to the community.

The Internal Plane of Action

Activity theory provides a number of useful concepts that can be used to address the lack of expression for 'soft' factors which are inadequately represented by most process modelling frameworks. One such concept is the internal plane of action. Activity theory recognises that each activity takes place in two planes: the external plane and the internal plane. The external plane represents the objective components of the action while the internal plane represents the subjective components of the action. Kaptelinin defines the internal plane of actions as "[...] a concept developed in activity theory that refers to the human ability to perform manipulations with an internal representation of external objects before starting actions with these objects in reality."

The concepts of motives, goals and conditions also contribute to the modelling of soft factors. One principle of activity theory is that many activities have multiple motivation ('polymotivation'). For instance, a programmer in writing a programme may address goals aligned towards multiple motives such as increasing his or her annual bonus, obtaining relevant career experience and contributing to organisational objectives. Activity theory further argues that subjects are grouped into communities, with rules mediating between subject and community and a division of labour mediating between object and community. A subject may be part of several communities and a community, itself, may be part of other communities.

Human Creativity

Human creativity plays an important role in activity theory, that "human beings... are essentially creative beings" in "the creative, non-predictable character". Tikhomirov also analyses the importance of creative activity, contrasting it to routine activity, and notes the important shift brought about by computerisation in the balance towards creative activity.

Learning and Tacit Knowledge

Activity theory has an interesting approach to the difficult problems of learning and, in particular, tacit knowledge. Learning has been a favourite subject of management theorists, but it has often been presented in an abstract way separated from the work processes to which the learning should apply. Activity theory provides a potential corrective to this tendency. For instance, Engeström's review of Nonaka's work on knowledge creation suggests enhancements based on activity theory, in particular suggesting that the organisational learning process includes preliminary stages of goal and problem formation not found in Nonaka. Lompscher, rather than seeing learning as transmission, sees the formation of learning goals and the student's understanding of which things they need to acquire as the key to the formation of the learning activity.

Of particular importance to the study of learning in organisations is the problem of tacit knowledge, which according to Nonaka, "is highly personal and hard to formalise, making it difficult to communicate to others or to share with others." Leont'ev's concept of operation provides an important insight into this problem. In addition, the key idea of internalisation was originally introduced by Vygotsky as "the internal reconstruction of an external operation." Internalisation has subsequently become a key term of the theory of tacit knowledge and has been defined as "a process of embodying explicit knowledge into tacit knowledge." Internalisation has been described by Engeström as the "key psychological mechanism" discovered by Vygotsky and is further discussed by Verenikina.

COMPONENTS OF PHYSICAL FITNESS

Physical fitness is the ability to function effectively throughout your workday, perform your usual other activities and still have enough energy left over to handle any extra stresses or emergencies which may arise.

The components of physical fitness are:

- *Cardiorespiratory endurance*: The efficiency with which the body delivers oxygen and nutrients needed for muscular activity and transports waste products from the cells.
- *Muscular strength*: The greatest amount of force a muscle or muscle group can exert in a single effort.
- *Muscular endurance*: The ability of a muscle or muscle group to perform repeated movements with a sub-maximal force for extended periods of times.
- *Flexibility*: The ability to move the joints or any group of joints through an entire, normal range of motion.
- *Body composition*: The percentage of body fat a person has in comparison to his or her total body mass.

Improving the first three components of fitness listed above will have a positive impact on body composition and will result in less fat. Excessive body fat detracts from the other fitness components, reduces performance, detracts from appearance, and negatively affects your health. Factors such as speed, agility, muscle power, eye-hand coordination, and eye-foot coordination are classified as components of "motor" fitness.

These factors most affect your athletic ability. Appropriate training can improve these factors within the limits of your potential. A sensible weight loss and fitness programme seeks to improve or maintain all the components of physical and motor fitness through sound, progressive, mission specific physical training.

Principles of Exercise

Adherence to certain basic exercise principles is important for developing an effective programme. The same principles of exercise apply to everyone at all levels of physical training, from the Olympic-calibre athlete to the weekend jogger.

These basic principles of exercise must be followed:

- *Regularity*: To achieve a training effect, you must exercise often. You should exercise each of the first four fitness components at least three times a week. Infrequent exercise can do more harm than good. Regularity is also important in resting, sleeping, and following a sensible diet.
- *Progression*: The intensity and/or duration of exercise must gradually increase to improve the level of fitness.
- *Balance*: To be effective, a programme should include activities that address all the fitness components, since overemphasizing any one of them may hurt the others.
- *Variety*: Providing a variety of activities reduces boredom and increases motivation and progress.
- *Specificity*: Training must be geared towards specific goals. For example, people become better runners if their training emphasizes running. Although swimming is great exercise, it does not improve a 2-mile-run time as much as a running programme does.
- *Recovery*: A hard day of training for a given component of fitness should be followed by an easier training day or rest day for that component and/or muscle group(s) to help permit recovery. Another way to allow recovery is to alternate the muscle groups exercised every other day, especially when training for strength and/or muscle endurance.
- Overload: The work load of each exercise session must exceed the normal demands placed on the body in order to bring about a training effect.

PHYSICAL FITNESS

Physical fitness refers to the ability to function as healthy human beings with energy and alertness in all daily activities. There are many places to learn about fitness, but the most common resources are easily found and can be used by anyone. These resources include libraries, the Internet, a physical fitness instructor, or a health teacher. Libraries contain vast amounts of information about health in general and physical fitness in particular. Libraries have medical journals, magazines, and many books that can inform people about the ways to become fit and to maintain their fitness levels. Information sources can provide information about the physical and psychological benefits of being fit, as well as the benefits for a persons self-esteem and emotions. Libraries also may offer an opportunity to obtain audio and video materials that can provide the same types of information as magazines or books. Libraries also give low-income people a chance to see fitness materials that they may not readily have.

The Internet represents an excellent source of information about physical fitness. It is like a combination of many articles, books, libraries, and information from other people. If a search is performed on the word fitness using one of the popular search engines on the Internet, a list of many, many resources will be returned.

Search engines generally provide their lists in terms of the relevance of the material to the initial questions, so users should keep that in mind and ask relevant questions. With any search engine, the first few pages will provide the most useful links to information about the topic. Using a search engine, individuals can find information about local fitness centers and instructors in their area who offer specialized, one-to-one fitness training programmes. Local schools are also good sources of information. A schools physical education instructors and health teachers represent valuable sources of fitness information. They understand and are educated about the major issues related to physical fitness. Their expertise can help individuals find other reliable sources of fitness data. Many articles that are found may not always be totally accurate, so having the ability to ask teachers and health instructors specific questions is very useful to those interested in pursuing a physical fitness routine.

Additionally, the federal government publishes lots of information about fitness issues in the United States from a variety of perspectives. The United States Department of Agriculture, which is responsible for setting the daily recommended allowances of various vitamins and other food substances for human consumption, collects and provides information about fitness alternatives, the status of fitness in the country, and how well Americans participate in fitness programmes. The local fitness centre also has considerable information to share about the subject of physical fitness. These centers may provide posted information or current magazines that address the most current fitness questions. The centers often let people observe fitness programmes in action as well.

DEFINITION OF PHYSICAL FITNESS

Definition of Physical Fitness is AgelessWhen you think of physical fitness you may think of athletic feats such as Lance Armstrong riding endlessly through the Swiss Alps. Maybe you think of professional football players sprinting past their competition? You may even think of body builders flexing their ripped muscles on stage. What about the contortionists performing on stage bending their bodies as far as they can bend? What about someone who is confined to a wheel chair? Do you think physical fitness means the same thing to them as it dos to you or a professional athlete?

Physical Fitness Vastly Differs from Person to Person

What you consider physically fit others may not. You may be proud of your 7 minute mile, but professional track stars would think it is slow. You may feel tremendously proud for your 200 pound bench press but a professional power lifter would not be so happy. Athletic feats are part of physical fitness but there is far more reasons why physical fitness is important. It is best to look at physical fitness as a hierarchy to attempt an accurate general definition.

Base Level: The base level of the physical fitness pyramid highlights the primary reason why physical fitness is important. This level of physical fitness directly relates to quality of life. A physically fit body has less chance of acute health problems and chronic disease. The base level of physical fitness represents your ability to efficiently perform the anaerobic and aerobic activities of daily living (ADLs) which exist in your environment on a daily basis in addition to novel tasks which may arise during day to day living.

Mid Level: Only after your body has grown accustomed to the base level of physical fitness you can advance to the mid level. The middle level of the pyramid is about adaptation and improvement. Most people remain at the mid level for their entire lifetime as they adapt and progress only to regress and start the process over and over again. In order to improve mid level physical fitness it is important to have a balanced exercise programme. Exercise selection should focus on all 5 components of fitness.

Top Level: Very few are able to reach the top level of the physical fitness hierarchy. Elite athletes often spend their entire lifetime aiming to reach the very top level. The top level of physical fitness tends to mesh physiology and psychology. In order to reach the top level of physical fitness one must reach their full potential.

FITNESS

AEROBIC CIRCUIT TRAINING

Students perform a "circuit" of 10-15 exercises using jump ropes, steps, exercise bands/balls and hand weights designed to improve cardiovascular and muscular endurance, flexibility, and muscular strength/toning.

Students learn

1) The purpose and principles of warm-up, work-out, and cool-down;
2) The use of estimated target heart rates for cardiovascular improvement; and
3) Prior techniques for a variety of aerobic and strength-building exercises.

A primary objective of this class is to teach the student proper and safe exercise techniques along with the principles of aerobic conditioning. This class meets indoors with very few exceptions.

AEROBIC CONDITIONING

This class provides instruction and participation in a variety of progressively challenging indoor/outdoor aerobic activities (choreographed callisthenic movements, high/low impact aerobics, jump rope, power walking, jog/run, etc.) to develop, enhance and maintain cardiovascular endurance. Muscular endurance training using light-moderate weights for upper body strength and toning, floor work for back, abdominals and legs, and stretch/relaxation are included.

CORE

This class has an emphasis and instruction in isolated and compound exercises/movements responsible for core strength in human movement and activity. Focus is on the abdominal, back and gluteal muscle groups. Exercises/movements/stretches are included to enhance flexibility.

Elementary

Sequential instruction, drills and periodised training in isolated/compound/functional movements, asymmetrical exercises, and instability/balance exercises. Training protocol for strength, flexibility, endurance, balance, power, agility, reaction time and explosiveness are emphasized for skill acquisition and comprehensive core development/conditioning utilising body weight, medicine balls, dumbbells.

Low Intermediate

Sequential instruction and advanced training in exercises, drills and prescribed workouts to aggressively but progressively challenge core strength and conditioning with a variety of training modalities using body weight, medicine balls, dumbbells, and body bars. Prerequisite: demonstrated proficiency or recent experience in strength, flexibility and endurance factors.

PHYSICAL EDUCATORS, COACHES, AND ALL PROFESSIONALS IN FITNESS

Models can influence attitudes and behaviors in many ways, including health practices, motor skill acquisition, and the adoption of physical activity patterns. Physical educators, coaches, and all professionals in fitness and physical activity have strong modelling status for many children and youth. Participation in regular physical activity is an essential behaviour of physical educators and other physical activity-related professionals as role models as well as for personal health-related wellness. Since a primary goal of the profession is to promote an active healthy lifestyle (AAHPERD, AAHPERD vision statement, 1998) for everyone, those involved in professions related to physical activity and fitness should teach and model the most current “established” behaviors and processes for improving health and physical fitness. Presently, those behaviors include participating in a variety of physical activities as noted in the Physical Activity Pyramid (Corbin and Lindsey, 1997; Heyward, 1998).

The foundation of this model calls for an accumulation of at least 30 minutes of moderate "lifestyle physical activities" on all or most days of the week. To further guide the selection of physical activities, the model recommends moderate to vigorous aerobic activities 3 to 6 times per week, muscle fitness exercises 2 to 3 times per week, and flexibility or range of motion exercises 3 to 7 times per week. Physical activity professionals should demonstrate a personal understanding and appreciation of the basic physiological training principles such as warm-up, cool-down, gradual overload, progression, and the application of the principles of frequency, intensity, and time as they relate to improving and maintaining fitness (Howley and Franks, 1997; AAHPERD, 1999). Modelling these principles in their own fitness endeavors will have a positive influence on those who expect fitness and exercise leaders to be leaders in their profession and to set a positive example for young people and the community.

Studies by Cardinal and Sachs (1995, 1996) have found that most people are more likely to adopt moderate intensity "lifestyle physical activities" as opposed to traditional, more structured forms of exercise. This finding suggests that encouragement and modelling of "lifestyle physical activities" by fitness and exercise professionals and physical educators may be especially important to young people and a community that is already inclined to adopt moderate physical activity to improve the quality of life.

In addition to the importance of a physical activity professional's potential influence on others as a model, engaging in a physically active lifestyle is very important for personal reasons. There are studies that show that participation in organized fitness programmes (*e.g.*, corporate fitness programmes) results in greater productivity, reduced absenteeism, lower health care costs, and greater job satisfaction among employees (Opatz, 1994).

It is reasonable to assume that physical educators and other physical activity professionals who exhibit active lifestyles, similar to those of corporate employees involved in physical activity programmes, can expect to experience similar benefits. This was confirmed in one study conducted among 117 schoolteachers (Blair et al., 1984). Those involved in the school district's experimental wellness programme demonstrated significantly greater improvements compared to those not involved in the wellness programme on a number of physiological variables, as well as general well-being, level of job satisfaction, and self-concept. Moreover, both self-and principal-ratings of the teachers' stress management and performance were higher among those in the wellness programme compared to those not in the programme.

When children and youth see physical activity professionals participating in accordance with established activity guidelines, applying sound physiological training principles, and actively modelling a physically active lifestyle, it reinforces student learning and will likely lead some to adopt similar activity patterns. The active lifestyle of professionals promotes credibility among parents and colleagues and exemplifies evidence of the value of a physically active lifestyle.

Furthermore, the physical activity professionals who exercise regularly provide examples of the positive physiological, health-related effects of an active lifestyle. Achievement and maintenance of health-related physical fitness (based on accepted criterion referenced standards) is an appropriate expectation for all professionals in the field of physical activity.

Throughout the years, leaders in our discipline have claimed that physical educators need to maintain acceptable physical fitness levels to be totally effective teachers (Cadinal, 2001; Corbin, 1984; McCloy, 1940; Melville, 1999; Sargeant, 1900; Staffo and Stier, 2000; Wilmore, 1982). Physical educators also need to maintain and exhibit acceptable physical fitness levels to be totally effective fitness role models. Professionals and pre-service professionals are adamant that fitness is important and that the components of health-related fitness, cardio-vascular endurance, muscular endurance, muscular strength, flexibility, and body composition (Howley and Franks, 1997) should be a measure of that fitness level.

Cardinal and Cardinal (2001) found that physical education professionals strongly believe, "it is important for health, physical education, recreation, and dance professionals to maintain a healthy body fat percentage." Those same people strongly concurred (4.4 on a 5-point scale) with the statement "involvement in regular physical activity at a level sufficient to promote health-related physical fitness is a desirable and recommended behaviour for physical education teachers." Such models are worthy of commendation if they are careful to keep long-term health considerations ahead of shorter-term performance ones.

Administrators have also expressed that they value teacher fitness. School administrators have identified lack of teacher fitness as a barrier to implementing quality elementary physical education programmes (Sallis, McKenzie, Kolody, and Curtis, 1996). Surveys of individual's responsible for hiring physical educators have shown that applicants perceived to be unfit have a much-reduced chance of employment (Melville and Cardinal, 1997). Although these studies present a topic that requires further study, it should be noted that there has been little research undertaken to determine whether or not a teacher's fitness level actually affects student learning and behaviour or teacher effectiveness.

Implications: Although the physical activity professional's physical fitness level is only one of many characteristics which contribute to the learning process, modelling an appropriate fitness level and an active lifestyle needs to be considered a significant factor in encouraging young people, colleagues, and communities to do the same.

The level of physical fitness we model may have a powerful influence on youth and our success in the advocacy for our profession to the public.

STRETCH, STRENGTH AND ALIGNMENT

ELEMENTARY

Beginning and intermediate skill levels are welcome in this course which focuses on gaining strength and flexibility. Intensive floor work, use of Dynabands, proper body placement and specific use of breath is emphasized.

LOW INTERMEDIATE

Intermediate and advance students are welcome in this continuation of Level 1 Stretch, Strength and Alignment. In addition to intensive progressive floorwork, students will also be introduced to exercises at the barre, standing and sitting. All exercises are designed for those with good flexibility, dance training and either instructor approval or a prerequisite of Level 1 Stretch, Strength and Alignment.

GENERAL CONDITIONING

This course combines Aerobic Circuit Training with Resistance Training. Two days a week, students will perform a "circuit" of 10 exercises using jump ropes, exercise balls/bands and hand weights designed to improve cardiovascular and muscular endurance, flexibility and muscular strength/toning. The other two days a week, students will receive instruction in the safe and appropriate use of weight training equipment, including machines and free weights, and participate in an individualised weight training programme.

INTERVAL SPORTS CONDITIONING

Students will perform sports-inspired conditioning drills in small groups. Exercises will consist of movements and pace that are designed to mimic the rigors of sport. This class will get students in better shape for their favourite sport, such as football, basketball, tennis, soccer, as well as many others.

CIRCUIT WEIGHT TRAINING

Circuit weight training is a fast-paced circuit where students will participate in a balanced strength training programme that includes upper body, lower body, and core exercises using free weights, machine weights, resistance bands, and their own body weight. A student's progress will be evaluated on improvement in strength testing with push-ups, pull-ups/ supine rows, and the number of abdominal crunches completed in one minute.

RESISTANCE TRAINING (WEIGHT TRAINING)

This class will provide instruction in the safe and appropriate use of weight training equipment, including machines and free weights, and the design of different weight training routines. Students will be introduced to the concepts of correct body position, concentric and eccentric contraction, complete range of motion of each joint, and basic muscle and skeletal anatomy.

SPORTS PERFORMANCE TRAINING

Students work on developing power and proper speed and agility biomechanics found in all ground based sports (*e.g.* football, basketball, tennis, soccer, lacrosse, ultimate Frisbee). Linear and lateral speed techniques will be taught with emphasis placed on first step acceleration and change of direction.

STEP AEROBICS

This class will introduce basic stepping movements at low to moderate intensities using adjustable platforms that are performed continuously to music. Hand-held weights and stretch bands are used to provide resistance during the workout.

YOGA

Yoga is the Sanskrit term, "to yoke." The classes teach and embrace this 3,000 year old tradition and practice of harmonising mind, body and spirit through three styles.

Vinyasa Yoga

This is a multi-faceted introduction to the study, practice and discipline of Ashtanga-based yoga. Students will be taught the components of the practice including breathing (ujjayi prananyama), gaze (drishti), poses (asanas), meditation, and mantra. The history, roots, and paths of yoga as well as the application of yoga to daily life will also be covered.

Elementary

Fundamental foundations of Ashtanga-based yoga/Yoga Chikitsa (Primary Series) are established with reference to core principles from the Yoga Korunta, Eight Limbs of Ashtanga, Yoga Sutras of Patanjali, and related texts. Ujjayi breathing, drishti (gaze points), bandhas ("locks" used to control energy) and vinyasa (synchronisation of the pose with the breath) are emphasized to develop a strong practice in beginning-level standing, seated, twisting, and preparatory-inverted poses. Instruction included basic anatomy and biomechanics to assure comprehension, acquisition, and proficiency of technique; japa (recitation of mantra); and Sanskrit.

Low Intermediate

A continued practice from a strong beginner-level foundation with instruction in intermediate-level standing, seated, twisted, inverted, backbend, and arm-balancing poses including binding of the poses. Strict emphasis on the foundation of the practice: vinyasa, bandha, and drishti (collectively known as Tristhana, the three places of attention or action). A Mysore (self-led) setting is integrated to foster independence, self-reliance, and self-responsibility for one's personal practice. Prerequisite: recent yoga experience.

Intermediate

Instructor-led and Mysore style practice. Progressive instruction in advanced variations of intermediate-level poses and introduction to advanced poses. Prerequisite: solid fundamentals and consistency with Tristhana; comfort with backbends, headstands, and arm-balancing.

GENTLE/RESTORATIVE YOGA

This course introduces the study and practice of yoga sequences to reduce overall stress from daily life. Students will learn set-up techniques and how to modify the basic poses for individual needs. No previous yoga experience is necessary.

Using props such as blankets, bolsters, blocks and straps, restorative yoga encourages the student to reach a state of deep relaxation. The poses will gently stretch the body in various supported backbends, twists, forward bends, inversions and 'neutral' poses. The main goal of this type of yoga is to encourage a clear mind and calm emotions.

After becoming acquainted with each of the restorative poses and how to sequence them, students will be encouraged to make choices as to what poses to do and how long to hold the pose. Topics to be incorporated include restorative poses that may relieve or reduce the symptoms of headaches, jet lag, fatigue and insomnia.

YOGA SPORTS CONDITIONING

A holistic multi-disciplinary training approach to optimise physical performance and mindfulness for sport, exercise and yoga. Performance parameters (strength, power, speed, endurance, quickness, agility and reaction time) are addressed, examined and developed. Instruction and training integrate the foundations of exercise and sport science with the principles and practice of vinyasa yoga.

HATHA YOGA

Hatha Yoga is an ancient form of mind-body training that develops strength, flexibility, and coordination of the physical body while simultaneously nurturing a calm mind, and emotional equanimity. Students will learn traditional standing and seated yoga poses as well as breath awareness exercises. All levels of fitness are encouraged to participate.

FITNESS STATUS OF U.S. YOUTH

Despite the beliefs and assertions discussed above, too many boys and girls in our schools and colleges show clear indications of inadequate organic power, as reflected in early fatigue under a normal demand of muscular effort and muscular inefficiency reflected in poor motor skills, low strength indices, and lack of endurance. If proof is needed, consider the following summary made from many studies available.

Kraus reported that 57.9 per cent of eastern seaboard school children failed to meet even a minimum standard of muscular fitness. These children were far below similar accomplishments of children in Switzerland, Austria, and Italy, only 8.7 per cent of whom failed to meet this standard. He indicated further that this situation is not being alleviated, since American boys and girls leave their schools in much the same condition as when they enter. High failure rates on this test by American children have also been found by other investigators from many parts of the country. Kraus also produced clinical evidence to show that adult patients with failures on two or more of his simple strength and flexibility tests were prone to lower back pain and manifested constant strain and frequently, emotional stress. These symptoms disappeared when patients performed prescribed exercises to strengthen the muscles of the abdomen and back and to stretch the muscles of the back and legs.

The AAHPER Youth Fitness Tests have been used to contrast the fitness of American children with their counterparts in other countries. In all instances, the American children were revealed with much lower typical performance on most of the items composing this test battery. For example, the American boys and girls exceeded the means of the Japanese only on the test involving abdominal endurance; in the 600-yard run-walk, 98 per cent of the Danish boys and 99 per cent of the Danish girls exceeded the American averages for boys and girls; and British boys on the average for all tests and for all ages were at the 64th percentile for United States performance scores.

Utilizing Rogers' Physical Fitness Index, Clarke reported that male students entering the University of Oregon with four years of high school physical education had a considerably higher average than did those with two years or less. Typically, large numbers of students enter college with low Physical Fitness Indices, not only in Oregon, but generally throughout the country. Whittle found that 12-year-old boys in elementary schools with good physical education programmes had much higher scores on fitness and motor ability tests than did boys of the same age who had little or no physical education.

After administering physical fitness tests, Karpovich and Weiss concluded that personnel entered the Army Air Forces during World War II in fairly poor physical condition. Despite relatively high acceptance standards, these men were deficient in running speed, endurance of the abdominal muscles, and arm and shoulder strength, Other branches of the armed forces had and continue to have similar experience.

Cureton studied the fitness of 1000 young men upon entrance to the University of Illinois shortly before World War II with these negative results: 64 per cent could not swim 50 yards, 26 per cent could not chin four times, and 24 per cent could not jump an obstacle waist high. Similar results were obtained one year later with a larger sample of 3099 Illinois freshmen.

PHYSICAL FITNESS COUNCIL

As a general rule, unfit students should be re-tested at intervals of approximately six weeks in order to reveal the progress—or lack of progress or retrogression—made by each. If fair improvement is recorded, the physical educator may feel that the treatment regimen adopted is effective, at least for the moment. However, as indicated earlier, it is necessary to initiate case studies, if this has not already been done, for students who fail to make significant gains. Case studies would be proper, also, for any unfit boy or girl who register's an initial gain, but whose subsequent progress towards adopted fitness standards stops short of their realization.

Where progress towards fitness goals does not occur after case studies have been initiated, a more comprehensive approach to the student's fitness problem will be necessary. Rechecks on living habits should be made, additional interviews with the unfit student should be held, supplementary tests may be used to help locate causative factors, and quite generally, the school health should be consulted. There are several other specialists in the school or college who may aid in the search for causes of unfitness in individual cases.

Depending on the school system or institution, help may come from the guidance counsellor, psychologist, social worker, home economist, nutritionist, health educator, administrator, and classroom teacher. If not available in the school, the services of certain of these specialists may be obtained in the community through the Public Health Service or other agencies.

The formation of a Physical Fitness Council to assist with the follow-up of unfitness cases is usually desirable. The permanent membership of the council might consist of the school physician, school nurse, guidance counsellor, health-educator, school administrator, and the physical educator. If the school or college has a psychologist, social worker, or nutritionist, these individuals might also be included. The classroom or home-room teacher should be invited to meet with the council when the case involves one of his her students.

As a matter of course, the physical educator should thoroughly acquaint the council members with the nature and progress of the programme for physically unfit students. Obviously, this orientation is essential for the full co-operation of all members and for maximal effectiveness of the Council's efforts.

The Fitness Council members are likely to be busy people. Therefore, the physical educator should solicit their help only in essential cases. In each instance, a brief of the case study as it is known should be prepared in advance and presented at the meeting. This brief should include an account of the steps taken to improve the student's fitness and the success or failure of these steps. Following this presentation, the council should discuss the case, each member contributing as he can from his knowledge of the student. A recommended action proposal should result.

The student under consideration does not necessarily need to be present at the council session. However, there may be instances when his presence would be desirable in order for him to answer questions and to help find a solution to his difficulty. In such cases, a friendly and helpful attitude towards the student should prevail; and, he should be made to feel as comfortable and relaxed as possible.

The confidential nature of the case study and the interview has been stressed in the case-study approach to meeting the needs of unfit boys and girls. Taking this information to the council, then, could well be construed as a breach of this agreement. Consequently, it should be done only with permission of the student. If rapport has been thoroughly established and if the student understands that this action is in his best interest, such permission will be readily given. However, it must now be stressed that all deliberations of the council on individual case studies must remain confidential within the group—and this confidence must be strictly upheld. Perhaps it might be well to inject a word of warning. Caution should be exercised to avoid making students so conscious of their physical condition that ill health becomes a phobia with them. A frank discussion of his physical condition should be helpful for any student. But if he is constantly met on every side with reminders of his unfitness he is apt to react with apprehension, which in itself could cause lack of fitness, and "inferiority complex," or an attitude antagonistic to the programme. On the other hand, the ignoring of students' unfitness is obviously an even worse expedient. Common sense and propriety must always prevail when the physical educator discusses the problems of unfitness with a student. The physical fitness programme presented, however, has a distinct advantage in that progress can be measured and proved to the boy or girl, a fact that will counteract to a great extent unfavourable psychological reactions. The student can see his progress and enjoy the emotional satisfaction of work successfully accomplished. Nevertheless, good judgement should be exercised in the follow-up approach, especially in regard to those few students not responding to treatment.

DEVELOPMENT OF THE PHYSICAL FITNESS

The development of the physical fitness of school children is proposed as a fundamental objective of physical education. In the realization of this objective, standard operating procedure should be the identification of those individuals who are deficient in this basic quality, so that appropriate steps may be taken to improve their condition. The process of insuring for each student a body that has adequate physical strength and is capable of prolongede effort, without efficiency-destroying fatigue, therefore, should constitute a primary responsibility of the physical educator. Assuming a body free from organic drains and handicapping defects, physical education is concerned with three basic physical fitness components—muscular strength, muscular endurance, and circulatory respiratory endurance. The construction of tests to measure these components, and others related to physical fitness, has occupied the efforts of researchers in physical education for a century. During World War II, motor fitness tests, so-called, were developed and used by physical educators in evaluating the fitness status of military personnel.

STRENGTH TESTS

Strength tests, although they do not measure all aspects of fitness as the physical educator views the problem, do evaluate basic elements of the individual's general physical status. They have been used successfully in practical school and college situations as a means of selecting students for Physical Education: Values and Promotion. A great deal of research has been done in this area, dating from 1880.

The strength test most widely used in physical education to select boys and girls who are sub-part in this quality is the Rogers Physical Fitness Index (PFI). The PFI battery consists of four muscular strength tests (right and left grips, back and leg lifts), two muscular endurance tests (pull-ups, push-ups), and lung capacity. The Strength Index is the gross score from these tests. The PFI is derived from dividing the achieved Strength Index by the normal Strength Index and multiplying by 100; norms are based upon sex, age, and weight. A score of 100 is average; scores of 85 and 115 indicate the first and third quartiles respectively. A number of modifications of this test battery have been posed. The Oregon simplification predicts the Strength Index for boys from a reduced number of test items. Kennedy has successfuly proposed substitution of the tensiometer for the dynamometer in back and leg testing.

The selection of individuals with unsatisfactory PFI's for Physical Education: Values and Promotion includes at least three groups.

1. *Low scores:* Definitely weak individuals obviously need attention. Many choose 85 as the point below which pupils should be selected; others prefer the national average for the test; still others are governed by practical limitations of time and personnel for follow-up work and include as many of the low PFI's as the local situation permits.
2. *Declining scores:* With the exception of individuals with very high scores, drops in PFI (approximately 10 points or more), regardless of their level, may be a definite danger sign. They usually indicate undesirable changes in physical condition which need to be investigated.
3. *Extremely high scores:* Boys and girls may develop unusual musculatures through extended training in various types of strength-building activities, and, consequently, possess very high PFI scores, which would be a satisfactory status for them. However, in others, an extremely high PFI (150 and above) may indicate that they are high-strung, over-stimulated, and highly nervous. As a high degree of strength *may* thus be a symptom of such conditions, a PFI reduction becomes an indicator of the success of programmes designed to achieve the relaxation of individuals so afflicted.

Circulatory-Respiratory Endurance

Physical education has long been concerned with exercise for the development of circulatory-respiratory endurance. Such exercise requires moderate contraction of large-muscle groups for relatively long periods of time, involving major adjustments of the circulatory and respiratory systems to the activity, as in distance running and swimming. Efforts to measure this component of physical fitness have taken two directions: (*i*) tests of circulatory-respiratory functions based on the premise that the endurance of the body is dependent upon the efficiency of these systems, hence, a measure of one is a measure of the other; (*ii*) tests of running endurance.

Extensive research has been conducted on the Harvard Step Test, which measures the pulse recovery following stepping up and down on a bench 20 inches high for five minutes at a rate of 30 steps per minute. Adaptations of this test have been made for secondary school boys and for college women and high school girls. Techniques for administering and scoring these tests have also been described by Clarke. The research evidence shows that trained athletes obtain better scores on this test than do untrained athletes and non-athletes.

The best correlations between a step test and motor-fitness criteria were achieved by Russell with college men as subjects. He utilized a 17-inch high bench for stepping and a cadence of 40 steps per minute. Stepping was continued until the subject was no longer able to maintain the cadence due to fatigue. His correlations were 70 with the University of Illinois Motor Fitness Test, 64 with the Air Force's Physical Fitness Test and with the Navy Standard Physical Fitness Test, and 61 with the Indiana Motor Fitness Test. He also obtained the high correlation of.85 between the length of time the step test was continued and the gross oxygen intake on a treadmill run to exhaustion.

PLANNING EXERCISE

In planning exercise regimens for unfit individuals, the specific physical status of each student should be known. No formulae exist to guide physical educators in determining the specific types and amounts of exercise which should be prescribed at each stage of the unfit individual's progress towards the realisation of adequate standards of fitness. The following principles may be used as guides in the preparation of exercise regimens.

Overloading should be applied to induce a higher level of performance: In overloading, the individual's exercise is increased in intensity or extended for a longer time than normally. Thus, overload is a relative term; a slight overload exceeds normal activity to a small degree, while a heavy overload equals the maximal performance of which the individual is capable at the moment. In order to develop either strength or endurance effectively, therefore, the individual must be pushed beyond his customary performance. Athletic coaches use his principle routinely. In track, work-outs are planned to extend the runner more and more as his exercise tolerance increases until his maximum is reached during the competitive season; in football and basketball, scrimmages and other procedures are utilized to accomplish the same end. For the unfit individual, too, overloading means increased intensity and dosage within his tolerance level.

The exercise plan should provide for progression: Progression is intimately involved with the first two principles. The exercise plan starts with an understanding of the individual's exercise tolerance, then, within this tolerance level, an exercise regimen is prepared to provide for overloading the muscles to develop strength or for increasing the demands on the circulatory-respiratory systems to improve cardiovascular endurance. If the exercise regimen stopped at this point, some improvement in fitness elements could reasonably be expected, but it would soon cease as the body adjusted to the new requirements in output. Both normal exercise and exercise tolerance levels have risen. Progression must now be effected by increasing exercise in some logical way, thus keeping its demands ahead of the improvement made.

Progression may be accomplished by increasing either the intensity or the duration of exercise. In strength development, the common method of intensifying exercise is by adding to the resistance against which muscles work, as by increasing the amount of weight in weight training. Greater intensity, however, can be achieved by increasing the cadence (speed) of the lift and leaving the load unchanged. Progression in duration for this form of development is accomplished by requiring more repetitions of the same load. In circulatory-respiratory endurance, intensity is increased by stepping up the speed at which a cardiovascular activity, such as running is performed; duration is enhanced by prolonging the time the activity is continued at the previous pace. Untrained persons should not be overloaded by increasing both intensity and duration at the same time. It is probably best first to increase duration a bit; later, to increase intensity; then, repeat the pattern as often as desired in keeping with the individual's progress.

Exercise should be adapted to the individual's exercise tolerance: Exercise tolerance refers to the ability of the individual to execute a given exercise, series of exercises, or activities involving exercise, in accordance with a specified dosage without undue discomfort or fatigue. An exercise performed as specified which is easy for the individual falls short of his exercise tolerance; on the other hand, an exercise which is either impossible for the individual to perform or leaves him in a distressful state exceeds a reasonable interpretation of exercise tolerance.

For the most part, exercise tolerance must be judged by the physical educator through observation of the student, although some rough estimates are possible from his physical or motor fitness scores; the lower the scope, of course, the less his exercise tolerance. Furthermore, the scores on the different items composing the test battery may give information related to the student's exercise tolerance. For example, if the student cannot chin himself, the inclusion of chins in his exercise plan would be useless. Such indicators as the degree of discomfort during exercise and the amount of breathlessness following exercise are helpful in judging the tolerance level. Also, the presence of exhaustion, slow recuperation, and excessively sore muscles would indicate that the exercise assigned was too severe. With unfit groups, exercise tolerance will be low at the start, but should gradually rise as the fitness programme continues and is effective.

BREATHING EXERCISES

We give below three forms of breath, quite popular among the Yogis. The first is the well-known Yogi Cleansing Breath, to which is attributed much of the great lung endurance found among the Yogis. They usually finish up a breathing exercise with this Cleansing Breath, and we have followed this plan in this book. We also give the Yogi Nerve Vitalising Exercise, which has been handed down among them for ages, and which has never been improved on by Western teachers of Physical Culture, although some of them have "borrowed" it from teachers of Yogi. We also give the Yogi Vocal Breath, which accounts largely for the melodious, vibrant voices of the better class of the Oriental Yogis. We feel that if this book contained nothing more than these three exercises, it would be invaluable to the Western student. Take these exercises as a gift from your Eastern brothers and put them into practice.

THE YOGI CLEANSING BREATH

The Yogis have a favourite form of breathing which they practice when they feel the necessity of ventilating and cleansing the lungs. They conclude many of their other breathing exercises with this breath, and we have followed this practice in this book. This Cleansing Breathing ventilates and cleanses the lungs, stimulates the cells and gives a general tone to the respiratory organs, and is conducive to their general healthy condition. Besides this effect, it is found to greatly refresh the entire system. *Speakers, singers, etc., will find this breath especially restful, after having tired the respiratory organs:*

(I) Inhale a complete breath.

(2) Retain the air a few seconds.

(3) Pucker tip the lips as if for a whistle (but do not swell out the cheeks), then exhale a little air through the opening, with considerable vigour. Then stop for a moment, retaining the air, and then exhale a little more air. Repeat until the air is completely exhaled. Remember that considerable vigour is to be used in exhaling the air through the opening in the lips.

This breath will be found quite refreshing when one is tired and generally "used up.' A trial will convince the student of its merits. This exercise should he practiced until it can be performed naturally and easily, as it is used to finish up a number of other exercises given in this book, and it should be thoroughly understood.

THE YOGI NERVE VITALISING BREATH

This is an exercise well known to the Yogis, who consider it one of the strongest nerve stimulants and invigorants known to man. Its purpose is to stimulate the Nervous System, develop nerve force, energy and vitality.

This exercise brings a stimulating pressure to bear on important nerve centers, which in turn stimulate and energise the entire nervous system, and send an increased flow of nerve force to all parts of the body.

- Stand erect.
- Inhale a Complete Breath, and retain same.
- Extend the arms straight in front of you, letting them be somewhat limp and relaxed, with only sufficient nerve force to hold them out.
- Slowly draw the hands back towards the shoulders, gradually contracting the muscles and putting force into them, so that when they reach the shoulders the fists will be so tightly clenched that a tremulous motion is felt.
- Then, keeping the muscles tense, push the fists slowly out, and then draw them back rapidly (still tense) several times.
- Exhale vigorously through the mouth.
- Practice the Cleansing Breath.

The efficiency of this exercise depends greatly upon the speed of the drawing back of the fists, and the tension of the muscles, and, of course, upon the full lungs. This exercise must be tried to be appreciated. It is without equal as a “bracer,” as our Western friends put it.

THE YOGI VOCAL BREATH

The Yogis have a form of breathing to develop the voice. They are noted for their wonderful voices, which are strong, smooth and clear, and have a wonderful trumpet-like carrying power. They have practiced this particular form of breathing exercise which has resulted in rendering their voices soft, beautiful and flexible, imparting to it that indescribable, peculiar floating quality, combined with great power.

The exercise given below will in time impart the above-mentioned qualities, or the Yogi Voice, to the student who practices it faithfully. It is to be understood, of course, that this form of breath is to be used only as an occasional exercise, and not as a regular form of breathing.

- Inhale a Complete Breath very slowly, but steadily, through the nostrils, taking as much time as possible in the inhalation.
- Retain for a few seconds.
- Expel the air vigorously in one great breath, through the wide opened mouth.
- Rest the lungs by the Cleansing Breath.

Without going deeply into the Yogi theories of sound-production in speaking and singing, we wish to say that experience has taught them that the timbre, quality and power of a voice depends not alone upon the vocal organs in the throat, but that the facial muscles, etc., have much to do with the matter. Some men with large chests produce but a poor tone, while others with comparatively small chests produce tones of amazing strength and quality. Here is an interesting experiment worth trying: Stand before a glass and pucker up your mouth and whistle, and note the shape of your mouth and the general expression of your face.

Then sing or speak as you do naturally1 and see the difference. Then start to whistle again for a few seconds, and then, without changing the position of your lips or face, sing a few notes and notice what a vibrant, resonant, clear and beautiful tone is produced. The following are the seven favourite exercises of the Yogis for developing the lungs, muscles, ligaments, air cells, etc. They are quite simple but marvelously effective. Do not let the simplicity of these exercises make you lose interest, for they are the result of carefuI experiments and practice on the part of the Yogis, and are the essence of numerous intricate and complicated exercises, the non-essential portions being eliminated and the essential features retained.

The Retained Breath

This is a very important exercise which tends to strengthen and develop the respiratory muscles as well as the lungs, and its frequent practice will also tend to expand the chest. The Yogis have found that an occasional holding of the breath, after the lungs have been filled with the Complete Breath, is very beneficial, not only to the respiratory organs but to the organs of nutrition, the nervous system and the blood itself.

They have found that an occasional holding of the breath tends to purify the air which has remained in the lungs from former inhalations, and to more fully oxygenate the blood. They also know that the breath so retained gathers up all the waste matter, and when the breath is expelled it carries with it the effete matter of the system, and cleanses the lungs just as a purgative does the bowels.

The Yogis recommend this exercise for various disorders of the stomach, liver and blood, and also find that it frequently relieves bad breath, which often arises from poorly ventilated lungs. We recommend students to pay considerable attention to this exercise, as it has great merits.

The following directions will give you a clear idea of the exercise:

- Stand erect.
- Inhale a Complete Breath.
- Retain the air as long as you can comfortably.
- Exhale vigorously through the open mouth.
- Practice the Cleansing Breath.

At first you will be able to retain the breath only a short time, but a little practice will also show a great improvement. Time yourself with a watch if you wish to note your progress.

Lung Cell Stimulation

This exercise is designed to stimulate the air cells in the lungs, but beginners must not overdo it, and in no case should it be indulged in too vigorously.

Some may find a slight dizziness resulting from the first few trials, in which case let them walk around a little and discontinue the exercise for a while:

- Stand erect, with hands at sides.
- Breathe in very slowly and gradually.
- While inhaling, gently tap the chest with the finger tips, constantly changing position.
- When the lungs are filled, retain the breath and pat the chest with the palms of the hands.
- Practice the Cleansing Breath.

This exercise is very bracing and stimulating to the whole body, and is a well-known Yogi practice. Many of the air cells of the lungs become inactive by reason of incomplete breathing, and often become almost atrophied. One who has practiced imperfect breathing for years will find it not so easy to stimulate all these ill-used air cells into activity all at once by the Complete Breath, but this exercise will do much towards bringing about the desired result, and is worth study and practice.

Rib Stretching

We have explained that the ribs are fastened by cartilages, which admit of considerable expansion. In proper breathing, the ribs play an important part, and it is well to occasionally give them a little special exercise in order to preserve their elasticity. *Standing or sitting in unnatural positions, to which many of the Western people are addicted, is apt to render the ribs more or less stiff and inelastic, and this exercise will do much to overcome same:*

- Stand erect.
- Place the hands one on each side of the body, as high up under the armpits as convenient, the thumbs reaching towards the back, the palms on the side of the chest and fingers to the front over the breast.
- Inhale a Complete Breath.
- Retain the air for a short time.
- Then gently squeeze the sides, at the same time slowly exhaling.
- Practice the Cleansing Breath.

Use moderation in this exercise and do not overdo it.

Chest Expansion

The chest is quite apt to be contracted from bending over one's work.

This exercise is very good for the purpose of restoring natural conditions and gaining chest expansion:

- Stand erect.

- Inhale a Complete Breath.
- Retain the air.
- Extend both arms forward and bring the two clenched fists together on a level with the shoulder.
- Then Swing back the fists vigorously until the arms stand out straight sideways from the shoulders.
- Then bring back to Position 4, and swing to Position 5. Repeat.
- Exhale vigorously through the opened mouth.
- Practice the Cleansing Breath.

Use moderation and do not overdo this exercise.

Walking Exercise

- Walk with head up, chin drawn slightly in, shoulders back, and with measured tread.
- Inhale a Complete Breath, counting (mentally) I, 2, 3, 4, 5, 6, 7, 8, one count to each step, making the inhalation extend over the eight counts.
- Exhale slowly through the nostrils, counting as before—I, 2, 3, 4, 5, 6, 7, 8—one count to a step.
- Rest between breaths, continuing walking and counting, I, 2, 3, 4, 5, 6, 7, 8, one count to the step.
- Repeat until you begin to feel tired. Then rest for a while, and resume at pleasure. Repeat several times a day.

Some Yogis vary this exercise by retaining the breath during a 1, 2, 3, 4, count, and then exhale in an eight-step count. Practice whichever plan seems most agreeable to you.

Morning Exercise

- Stand erect in a military attitude, head up, eyes front, shoulders back, knees stiff, hands at sides.
- Raise body slowly on toes, inhaling a Complete Breath, steadily and slowly.
- Retain the breath for a few seconds, maintaining the same position.
- Slowly sink to the first position, at the same time slowly exhaling the air through the nostrils.
- Practice Cleansing Breath.
- Repeat several times, varying by using right leg alone, then left leg alone.

Stimulating Circulation

- Stand erect.
- Inhale a Complete Breath and retain.
- Bend forward slightly and grasp a stick or cane steadily and firmly, and gradually exerting your entire strength upon the grasp.

- Relax the grasp, return to first position, and slowly exhale.
- Repeat several times.
- Finish with the Cleansing Breath.

This exercise may be performed without the use of a stick or cane, by grasping an imaginary cane, using the will to exert the pressure. The exercise is a favourite Yogi plan of stimulating the circulation by driving the arterial blood to the extremities, and drawing back the venous blood to the heart and lungs that it may take up the oxygen which has been inhaled with the air.

In cases of poor circulation there is not enough blood in the lungs to absorb the increased amount of oxygen inhaled, and the system does not get the full benefit of the improved breathing. In such cases, particularly, it is well to practice this exercise, occasionally with the regular Complete Breathing exercise.

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